

# SIMPLE ZEROS ON THE CRITICAL LINE FOR A WEIGHTED FAMILY OF DIRICHLET $L$ -FUNCTIONS, FROM POLYLOGARITHMIC TO POLYNOMIAL HEIGHT

NIC JOHNS

ABSTRACT. Let  $Q \rightarrow \infty$  and weight each primitive Dirichlet character  $\chi$  of conductor  $q \in [\eta Q, Q]$  by  $Q^2/(q\varphi(q))$  or a smoothed variant of it. We prove unconditionally that for every  $\varepsilon > 0$  and every sufficiently small  $\eta = \eta(\varepsilon) > 0$ , for all large  $Q$ , the weighted proportion of nontrivial zeros  $\rho = \beta + i\gamma$  with  $T < \gamma \leq 2T$ , counted with multiplicity, that are simple and lie on the critical line is at least  $0.9322 - \varepsilon$ , uniformly for  $(\log Q)^{a_0} \leq T \leq (\log Q)^{A_0}$ . This constant is approached only as  $\eta \rightarrow 0$ ; at  $\eta = 10^{-4}$  the certified bound is 0.9241 (smoothed weight). It is the value produced by the form factor  $\min(\alpha, 1)$  at support 2; under the Generalised Riemann Hypothesis Sono obtained it for a differently weighted statistic of low-lying zeros, for which Chirre, Gonçalves and de Laat later obtained 0.9350. The proof carries the finite-compression method of Alpöge–Furman and Hua–Yang from bandwidth 1 to bandwidth 2. The required upper bound for the family pair-correlation form factor comes from a flattening of the von Mangoldt function, a time localisation, and a sharp large sieve bound, valid on all vectors, for a positive semidefinite Toeplitz form that dominates the family form on sifted vectors and whose constant tends to 1 for logarithmically wide modulus weights. A Gauss-sum transfer and a weighted Farey large sieve give an independent bound  $0.8859 - \varepsilon$ . The method extends to heights polynomial in the conductor: uniformly for  $(\log Q)^{a_0} \leq T \leq Q^{\kappa_1}$  the proportion is at least the value produced by  $\min(\alpha, 1)$  at support  $(2 + \kappa)/(1 + \kappa)$  minus  $\varepsilon$ , where  $T = Q^\kappa$ ; for instance  $0.8656 - \varepsilon$  for  $T \leq Q$ , decreasing to  $0.6725 \dots$  as  $\kappa \rightarrow \infty$ . All statements concern weighted averages over the family; nothing is asserted about any individual  $L$ -function.

## CONTENTS

|                                                        |    |
|--------------------------------------------------------|----|
| 1. Introduction                                        | 2  |
| 2. Setup and notation                                  | 10 |
| 3. The positivity certificate                          | 14 |
| 4. The zero side                                       | 15 |
| 5. The prime side                                      | 19 |
| 6. The two large-sieve constants                       | 27 |
| 7. The variational problem and the certified constants | 43 |
| 8. Remarks                                             | 45 |
| 9. Zeros at polynomial height                          | 47 |
| AI-use statement                                       | 64 |
| Appendix A. Numerical computations                     | 64 |
| Appendix B. Proofs of the ported lemmas                | 71 |
| References                                             | 73 |

---

*Date:* Pre-publication draft.

*2020 Mathematics Subject Classification.* Primary 11M26; Secondary 11M06, 11N35, 15A42.

*Key words and phrases.* Dirichlet  $L$ -functions, simple zeros, critical line, pair correlation, large sieve, Farey fractions, Weil explicit formula.

## 1. INTRODUCTION

**1.1. The result.** Let  $Q$  be a large parameter and put  $\ell := \log Q$ . Fix  $\eta \in (0, 1/2)$  and a weight  $w : (0, 1] \rightarrow [0, \infty)$  of bounded variation with  $\text{supp } w \subset [\eta, 1]$  and  $\int_0^1 uw(u) du > 0$  (an *admissible* weight, Definition 2.1; the condition  $w \not\equiv 0$  alone would allow, e.g., the indicator function of a single point). Let  $\mathcal{F} = \mathcal{F}(Q, w)$  be the set of primitive Dirichlet characters  $\chi$  modulo  $q$  with  $w(q/Q) > 0$  (so  $\eta Q \leq q \leq Q$ ), and attach to  $\chi \bmod q$  the weight

$$(1.1) \quad \omega_\chi = \omega(q) := w(q/Q) \frac{q}{\varphi(q)}.$$

The *log-wide* weights are

$$(1.2) \quad w_\eta(u) = u^{-2} \mathbf{1}_{[\eta, 1]}(u), \quad w_\eta^{\text{sm}}(u) = u^{-2} \sin^2\left(\frac{\pi \log u}{\log \eta}\right) \mathbf{1}_{[\eta, 1]}(u).$$

For  $w_\eta$  the character  $\chi \bmod q$  receives weight  $Q^2/(q\varphi(q))$ , and every dyadic range of conductors in  $[\eta Q, Q]$  carries the same total weight.

Fix  $0 < a_0 < A_0$  and let  $\ell^{a_0} \leq T \leq \ell^{A_0}$ ,  $I = (T, 2T]$ . For  $\chi \in \mathcal{F}$  let  $N_\chi$  be the number of nontrivial zeros  $\rho = \beta + i\gamma$  of  $L(s, \chi)$  with  $\gamma \in I$ , counted with multiplicity; let  $N_{0, \chi}^s$  be the number of those that are simple and satisfy  $\beta = 1/2$ ,  $N_{0, \chi}^*$  the number of distinct zeros with  $\gamma \in I$  and  $\beta = 1/2$ , and  $N_{d, \chi}$  the number of distinct zeros with  $\gamma \in I$ . Write

$$N := \sum_{\chi \in \mathcal{F}} \omega_\chi N_\chi, \quad N_0^s := \sum_{\chi \in \mathcal{F}} \omega_\chi N_{0, \chi}^s, \quad N_0^* := \sum_{\chi \in \mathcal{F}} \omega_\chi N_{0, \chi}^*, \quad N_d := \sum_{\chi \in \mathcal{F}} \omega_\chi N_{d, \chi}.$$

We call  $N_0^s/N$  the *weighted proportion* of simple zeros on the critical line (at height  $T$ , in the family  $\mathcal{F}$ ). By Lemma 4.1,  $N = (1 + o(1))HT\ell/(2\pi)$ , where  $H = \sum_{\chi \in \mathcal{F}} \omega_\chi \asymp_w Q^2$ .

Let  $F_C(\alpha) = \alpha$  for  $0 \leq \alpha \leq 1$  and  $F_C(\alpha) = C$  for  $1 < \alpha \leq 2$ , and define

$$(1.3) \quad p(C) := 2 - \inf \left\{ \int f^2 + \iint f(x)f(y)F_C(|x-y|) dx dy : f \geq 0 \text{ even, } \text{supp } f \subset [-1, 1], \int f = 1 \right\}.$$

(For  $C = 1$ , and numerically for  $C \leq 1.6$ , the restriction to even  $f$  is immaterial; see Remark 7.4.) By Proposition 7.3,  $p(1) \geq 0.932282$  and  $p(1.2688) \geq 0.885912$ ; these are exact rational evaluations at explicit step functions.

**Theorem 1.1** (Main theorem). *Fix  $0 < a_0 < A_0$ . For every  $\varepsilon > 0$  there is  $\eta_0(\varepsilon) > 0$  such that for every  $\eta \in (0, \eta_0]$  and  $w \in \{w_\eta, w_\eta^{\text{sm}}\}$ ,*

$$\liminf_{Q \rightarrow \infty} \inf_{\ell^{a_0} \leq T \leq \ell^{A_0}} \frac{N_0^s}{N} \geq p(1) - \varepsilon \geq 0.9322 - \varepsilon,$$

*and the same lower bound holds for  $N_0^*/N$  (trivially, as  $N_0^* \geq N_0^s$ ), while  $\liminf \inf_T N_d/N \geq (1 + p(1))/2 - \varepsilon \geq 0.9661 - \varepsilon$ .*

The constant 0.9322 is reached only in the limit  $\eta \rightarrow 0$ , and the convergence in  $\eta$  is slow. The proved rate  $C_T^+(w_\eta^{\text{sm}}) \leq 1 + 83/\log(1/\eta)$  of Lemma 6.22, combined with Lemma 7.1, guarantees a proportion  $\geq 0.93$  once  $\log(1/\eta) \geq 3.7 \cdot 10^4$  (for  $w_\eta$ , once  $\log(1/\eta) \geq 1.9 \cdot 10^4$ ). The best value certified at a fixed  $\eta$  is 0.9241, at  $\eta = 10^{-4}$  (Theorem 1.3). No effective value of  $Q$  is claimed either: the thresholds implicit in  $\liminf_{Q \rightarrow \infty}$  are astronomically large in terms of  $\eta^{-1}$ .

The number  $p(1) = 0.93228 \dots$  is what the GUE form factor  $\min(\alpha, 1)$  gives on  $|\alpha| \leq 2$  in the Montgomery–Taylor extremal problem; under GRH it was obtained by Sono [38], improving the 11/12 of Chandee, Lee, Liu and Radziwiłł [8]. It is not the best GRH constant for that statistic: Chirre, Gonçalves and de Laat [9, Theorem 5] obtained 0.9350 under GRH, with test functions that are not supported in  $|\alpha| \leq 2$  (see §8.8). Theorem 1.1 obtains Sono’s constant without any

hypothesis, for a related but different statistic: the results of [8, 38, 9] weight zeros near the central point by  $|\widehat{\Phi}(i\gamma)|^2$ , whereas we count zeros with  $T < \gamma \leq 2T$  at polylogarithmic height  $T \rightarrow \infty$ , with the modulus weights (1.1). The constants coincide because the same form factor  $\min(\alpha, 1)$  on  $[0, 2]$  enters the same extremal problem.

The proof of Theorem 1.1 uses a large sieve inequality for a positive Toeplitz majorant of the family form (§6.2); §1.3 states which parts of it are new and which are classical. A second, independent route replaces it by a Gauss-sum transfer and a weighted Farey large sieve, with a worse constant:

**Theorem 1.2** (Gauss-transfer bound). *Let  $C_G = 6/(\pi^2 \mathcal{E}) \approx 1.2687739$ , where  $\mathcal{E} = \prod_p (1 - p^{-2} - p^{-3})$ . For every  $\varepsilon > 0$  there is  $\eta_0(\varepsilon) > 0$  such that for  $\eta \leq \eta_0$  and  $w \in \{w_\eta, w_\eta^{\text{sm}}\}$  the conclusions of Theorem 1.1 hold with  $p(1)$  replaced by  $p(C_G) \geq 0.885912$ ; that is,  $N_0^s/N, N_0^*/N \geq 0.8859 - \varepsilon$  and  $N_d/N \geq 0.9429 - \varepsilon$ . More generally, for every admissible  $w$ ,*

$$\liminf_{Q \rightarrow \infty} \inf_T \frac{N_0^s}{N} \geq p(C_G R_w),$$

where  $R_w \geq 1$  is the explicit Farey constant (6.2).

For concrete  $\eta$  the constants are smaller, and the convergence in  $\eta$  is slow. For fixed  $\eta$  the constant  $C_T^+(w)$  is reduced to a finite computation (Proposition 6.23), and  $R_w$  is the essential supremum of an explicit finite sum (6.2). Upper bounds for both are established by a computer-assisted computation in interval arithmetic (Appendix A; ancillary files `ctplus_arb.py` and `rw_arb.py`, with their output logs). These bounds and the exact rational evaluations of Proposition 7.3 (which also enter Theorems 1.1 and 1.2) are the computer-assisted inputs of the following theorem.

**Theorem 1.3** (Fixed  $\eta$ ). *Let  $w = w_\eta^{\text{sm}}$ .*

- (i) *For  $\eta = 10^{-3}$ :  $C_T^+(w) \leq 1.0911$ , and hence  $\liminf_Q \inf_T N_0^s/N \geq 0.9155$  and  $\liminf_Q \inf_T N_d/N \geq 0.9577$ .*
- (ii) *For  $\eta = 10^{-4}$ :  $C_T^+(w) \leq 1.0434$ , and hence  $\liminf_Q \inf_T N_0^s/N \geq 0.9241$  and  $\liminf_Q \inf_T N_d/N \geq 0.9620$ .*
- (iii) *(Gauss route.) For  $\eta = 10^{-3}, 10^{-4}, 10^{-5}$ :  $R_w \leq 1.0599, 1.0289, 1.0159$ , and hence  $\liminf_Q \inf_T N_0^s/N \geq 0.8744, 0.8802, 0.8827$  respectively.*

*The same bounds hold for  $N_0^*/N$ . The lower bounds for  $p$  used here are exact rational certificates (Proposition 7.3).*

*Heights polynomial in  $Q$ .* The zero counts  $N_\chi, N_{0,\chi}^s, N_{0,\chi}^*, N_{d,\chi}$  and their weighted sums are defined in the same way for every  $T > 1$ , with  $I = (T, 2T]$ . For  $T > 1$  put

$$(1.4) \quad \ell_* := \log(QT), \quad \bar{\lambda}(\kappa) := \frac{2 + \kappa}{1 + \kappa} \quad (\kappa \geq 0), \quad \bar{\lambda}_T := \bar{\lambda}\left(\frac{\log T}{\log Q}\right) = \frac{\log(Q^2 T)}{\log(QT)}.$$

At height  $t \asymp T$  the zeros of  $L(s, \chi)$ ,  $\chi \in \mathcal{F}$ , have density  $\frac{1}{2\pi} \log \frac{qt}{2\pi} = \frac{\ell_*}{2\pi} + O(1 + \log \frac{1}{\eta})$ , so  $\ell_*$  is the relevant log-conductor, and  $\bar{\lambda}_T$  is the length  $Q^2 T$  of Gallagher's hybrid large sieve [16] measured in units of  $\ell_*$ . For  $\bar{\lambda} \in [1, 2]$  and a nondecreasing  $F : [0, \bar{\lambda}] \rightarrow [0, \infty)$  put

$$(1.5) \quad p(\bar{\lambda}; F) := 2 - \inf \left\{ \mathcal{Q}_F(f) : f \geq 0 \text{ even}, f \in L^2, \text{supp } f \subset [-\frac{\bar{\lambda}}{2}, \frac{\bar{\lambda}}{2}], \int f = 1 \right\},$$

where  $\mathcal{Q}_F(f) := \int f^2 + \iint f(x)f(y)F(|x-y|) dx dy$ , as in Proposition 5.13. Thus  $p(2; F_C) = p(C)$  is the constant (1.3), and  $p(1; F_1) = 0.6725\dots$  is the Montgomery–Taylor constant [29, 30];  $\bar{\lambda} \mapsto p(\bar{\lambda}; F_C)$  is nondecreasing and continuous (Lemma 9.21).

**Theorem 1.4** (Polynomial height). *Fix  $a_0 > 0$  and  $\kappa_1 > 0$ .*

- (a) (Sharp route.) *For every  $\varepsilon > 0$  there is  $\eta_0(\varepsilon) > 0$ , independent of  $a_0$  and  $\kappa_1$ , such that for every  $\eta \in (0, \eta_0]$  and  $w \in \{w_\eta, w_\eta^{\text{sm}}\}$*

$$\liminf_{Q \rightarrow \infty} \inf_{\ell^{a_0} \leq T \leq Q^{\kappa_1}} \left( \frac{N_0^s}{N} - p(\bar{\lambda}_T; F_1) \right) \geq -\varepsilon,$$

*and the same holds for  $N_0^*/N$ ; for  $N_d/N$  it holds with  $p(\bar{\lambda}_T; F_1)$  replaced by  $\frac{1}{2}(1 + p(\bar{\lambda}_T; F_1))$ .*

- (b) (Gauss route.) *For fixed  $\eta$  and admissible  $w$ ,  $\liminf_{Q \rightarrow \infty} \inf_{\ell^{a_0} \leq T \leq Q^{\kappa_1}} (N_0^s/N - p(\bar{\lambda}_T; F_{C_G R_w})) \geq 0$ ; likewise for  $N_0^*$ , and for  $N_d$  with  $\frac{1}{2}(1 + p(\bar{\lambda}_T; F_{C_G R_w}))$ .*

- (c) (Classical route.) *The conclusion of (b) holds with  $C_G R_w$  replaced by  $C_{\text{cl}}(w) := w_{\text{max}}/(\mathcal{E} I_w)$ . Its proof uses, besides the zero side, only flattening, time localisation, the classical large sieve (4.1) and the mass asymptotic of Lemma 6.1; neither Lemma 6.5 nor §6.2 is used.*

*In particular, for fixed  $\kappa > 0$  and  $T = Q^\kappa$ , (a) gives  $\liminf_{Q \rightarrow \infty} N_0^s/N \geq p(\frac{2+\kappa}{1+\kappa}; F_1) - \varepsilon$ .*

For  $w = \mathbf{1}_{[\eta, 1]}$  one has  $C_{\text{cl}}(w) = 2/(\mathcal{E}(1 - \eta^2)) \leq 4.175$  for  $\eta \leq 0.0146$  (with the certified bound  $\mathcal{E} \geq 0.47914533$  of Table 3 the threshold is 0.01469...; ancillary file `classical_constant_check.py`); for the dyadic weight  $w = \mathbf{1}_{[1/2, 1]}$ ,  $C_{\text{cl}}(w) = 8/(3\mathcal{E}) = 5.56546... \leq 5.5657$ .

**Corollary 1.5** (The dyadic family). *Fix  $a_0 > 0$ . For the family of primitive characters of conductor  $q \in [Q/2, Q]$  with the weights  $q/\varphi(q)$ , uniformly for  $\ell^{a_0} \leq T \leq Q^\kappa$ , the weighted proportion of simple zeros on the critical line with  $T < \gamma \leq 2T$  is at least  $0.7224 - o(1)$  for  $\kappa = 3$ ,  $0.7203 - o(1)$  for  $\kappa = 5$  and  $0.7102 - o(1)$  for  $\kappa = 10$ , where  $o(1) \rightarrow 0$  as  $Q \rightarrow \infty$  uniformly in  $T$ .*

The constants in Theorem 1.4 and Corollary 1.5 are exact rational certificates (Proposition 9.24). Since  $\bar{\lambda}(\kappa)$  decreases and  $p(\bar{\lambda}; F)$  is nondecreasing in  $\bar{\lambda}$ , a value in the row  $\kappa$  holds uniformly for all  $T \leq Q^\kappa$ :

| $\kappa$ | $\bar{\lambda}(\kappa)$ | $p(\bar{\lambda}; F_1) \geq$<br>(a) | $\frac{1}{2}(1 + p(\bar{\lambda}; F_1)) \geq$<br>(a), distinct | $p(\bar{\lambda}; F_{1.2688}) \geq$<br>(b), $\eta \rightarrow 0$ | $p(\bar{\lambda}; F_{4.175}) \geq$<br>(c), $\mathbf{1}_{[\eta, 1]}$ | $p(\bar{\lambda}; F_{5.5657}) \geq$<br>Cor. 1.5 |
|----------|-------------------------|-------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------|
| 0        | 2                       | 0.932282                            | 0.966141                                                       | 0.885912                                                         | 0.738253                                                            | 0.722418                                        |
| 1/4      | 9/5                     | 0.912574                            | 0.956287                                                       | 0.876420                                                         | 0.738255                                                            | –                                               |
| 1/2      | 5/3                     | 0.894985                            | 0.947492                                                       | 0.865967                                                         | 0.738256                                                            | 0.722422                                        |
| 1        | 3/2                     | 0.865673                            | 0.932836                                                       | 0.845721                                                         | 0.738258                                                            | 0.722424                                        |
| 2        | 4/3                     | 0.824355                            | 0.912177                                                       | 0.813189                                                         | 0.738246                                                            | 0.722426                                        |
| 3        | 5/4                     | 0.797213                            | 0.898606                                                       | 0.790061                                                         | 0.736919                                                            | 0.722416                                        |
| 5        | 7/6                     | 0.764149                            | 0.882074                                                       | 0.760485                                                         | 0.730269                                                            | 0.720310                                        |
| 10       | 12/11                   | 0.727484                            | 0.863742                                                       | 0.726227                                                         | 0.714688                                                            | 0.710252                                        |

The row  $\kappa = 0$  is the polylogarithmic range. The column  $F_{1.2688}$  is the limit  $\eta \rightarrow 0$  of Theorem 1.4(b) for the log-wide weights. Each entry is a lower bound from an explicit step function with 400 cells at its own support; finer step functions (800 cells) give values larger by at most  $10^{-6}$  in the columns for (a) and (b), and by at most  $9 \cdot 10^{-6}$  in the column for (c); the entries need not be monotone in  $\kappa$  in the sixth decimal (discretisation), although  $\kappa \mapsto p(\bar{\lambda}(\kappa); F)$  is nonincreasing. As  $\kappa \rightarrow 0$ , Theorem 1.4(a) contains Theorem 1.1, and its proof in §9 specialises to a proof of that theorem with a different localisation scale. As  $\kappa \rightarrow \infty$  the sharp constant decreases to the Montgomery–Taylor value  $p(1; F_1) = 0.6725...$ , the value obtained for a single primitive character as  $T \rightarrow \infty$  in [1, Theorem B];  $\bar{\lambda}_T$  is the limit of worst-case large sieve bounds at height  $T$  (Proposition 9.30). As for Theorem 1.1, no effective value of  $Q$  is claimed, and the

thresholds are larger still: for  $\kappa_1 = 10$ , (9.3) gives  $\varepsilon_5 \leq 1/(32(1 + \kappa_1)^2) \approx 2.6 \cdot 10^{-4}$ , and the tail term  $T^{-\varepsilon_5/2} \log T$  of Corollary 9.14 is below  $10^{-2}$  only once  $\log T \geq 1.27 \cdot 10^5$ .

**1.2. Context.** Table 1 places Theorem 1.1 among related results. The normalisations differ (see the notes), so the table compares mechanisms rather than identical statements.

| Reference                         | Object                                          | Statistic                      | Proportion                   | Hypothesis |
|-----------------------------------|-------------------------------------------------|--------------------------------|------------------------------|------------|
| Montgomery [29]                   | $\zeta$                                         | simple                         | 2/3                          | RH         |
| Montgomery–Taylor [30]            | $\zeta$                                         | simple                         | 0.6725                       | RH         |
| Chirre–Gonçalves–de Laat [9]      | $\zeta$                                         | simple                         | 0.6792                       | RH         |
| Alpöge–Furman [1]                 | $\zeta$                                         | simple, on line                | 0.6725                       | none       |
| Lamzouri [25]                     | $\zeta$                                         | simple, on line                | 0.6725                       | none       |
| Wang [42]                         | $\zeta$                                         | simple, on line                | $0.6725 + \delta_0^{(c)}$    | none       |
| Hua–Yang [22, 21]                 | $\chi \bmod q$ , $q$ prime                      | simple, on line                | 0.6725                       | none       |
| Prüzelius [32]                    | $\chi \bmod q$ , $q$ prime                      | simple, on line                | 2/3                          | none       |
| Fabbian [15]                      | $S_2(N)$ newforms, $N$ prime                    | simple, on line                | 0.6725                       | none       |
| Dickinson [14]                    | primitive $\chi \bmod q$                        | simple, on line <sup>(e)</sup> | 0.382                        | none       |
| [1, Rem. 7.2] (announced)         | primitive $\chi$ , $q \leq Q$                   | simple, on line                | $0.811^{(a)}$                | none       |
| Conrey–Iwaniec–Soundararajan [12] | primitive $\chi$ , $q \sim Q$                   | simple, on line <sup>(b)</sup> | 0.56                         | none       |
| Sono [39]                         | even primitive $\chi$ , $q \sim Q$              | simple, on line <sup>(b)</sup> | 0.6044                       | none       |
| Wu [45]                           | primitive $\chi$ , $q \sim Q$                   | simple <sup>(b)</sup>          | 0.6024                       | none       |
| Conrey et al. [11]                | $\mathrm{GL}_2$ form $\times \chi$ , $q \leq Q$ | simple, on line <sup>(b)</sup> | 1/3                          | none       |
| Özlük [31]                        | all $\chi \bmod q$ , $q \sim Q$                 | simple                         | 0.8688                       | GRH        |
| Chandee–Lee–Liu–Radziwiłł [8]     | primitive $\chi$                                | simple                         | 11/12                        | GRH        |
| Sono [38]                         | primitive $\chi$                                | simple                         | 0.9322                       | GRH        |
| Chirre–Gonçalves–de Laat [9]      | primitive $\chi$                                | simple                         | 0.9350                       | GRH        |
| Theorem 1.2                       | primitive $\chi$ , weights (1.1)                | simple, on line                | $0.8859 - \varepsilon$       | none       |
| Theorem 1.1                       | primitive $\chi$ , weights (1.1)                | simple, on line                | $0.9322 - \varepsilon$       | none       |
| Theorem 1.4, $T \leq Q$           | primitive $\chi$ , weights (1.1)                | simple, on line                | $0.8656 - \varepsilon^{(d)}$ | none       |

TABLE 1. Proportions of simple and critical zeros. <sup>(a)</sup> Announced without details, for smooth weights  $(1 - q/Q)^2$  at bandwidth  $3/2$ ; see §8.6. <sup>(b)</sup> By mollified moments and the asymptotic large sieve [10]: [11] for the twists by primitive  $\chi$ ,  $(q, q_\Pi) = 1$ , of a fixed  $\mathrm{GL}_2$  form and zeros with  $|\gamma| \leq Q^\varepsilon$ ; [12] for  $(\log Q)^6 \leq T \leq (\log Q)^A$  and zeros with  $|\gamma| \leq T$ , and [45] in the same range (also 0.8012 for distinct zeros); [39] for  $(\log Q)^2 \leq T \leq (\log Q)^A$  and  $T \leq \gamma < 2T$ , with 0.6107 for critical zeros. The GRH results [31, 8, 38, 9] weight low-lying zeros near the central point by  $|\widehat{\Phi}(i\gamma)|^2$ ; the results of this paper and of [22, 21, 32, 15] count zeros with  $T < \gamma \leq 2T$ , at height  $T$  at most polylogarithmic in the conductor in Theorems 1.1–1.3 and [22, 32, 15], at every sub-polynomial height  $T = q^{o(1)}$  with  $(\log q)^s = o(T)$  for some fixed  $s > 1$  in [21], and at height up to  $Q^{\kappa_1}$  in Theorem 1.4. The bandwidth of [22, 21, 32, 15] is at most 1. In [15] the newforms of weight 2 and prime level  $N$  are averaged with the harmonic (Petersson) weights. <sup>(c)</sup> Here 0.6725 stands for the Montgomery–Taylor constant  $0.67250\dots$ , and  $\delta_0 = 6.66\dots \cdot 10^{-8}$ . <sup>(d)</sup> For heights  $T \leq Q$ ; for  $T \leq Q^\kappa$  the constant is  $p(\frac{2+\kappa}{1+\kappa}; F_1) - \varepsilon$ , e.g.  $0.7641 - \varepsilon$  for  $T \leq Q^5$ . <sup>(e)</sup> By Levinson’s method, with a twisted second moment uniform in  $q$  and  $t$ : for every modulus  $q$ , on average over the primitive  $\chi \bmod q$ , zeros with  $0 < \gamma < T$  as  $qT \rightarrow \infty$  with  $T \gg q^\varepsilon$ , so heights polynomial in  $q$  are included.

*Polynomial height.* Theorems 1.1–1.3 concern polylogarithmic heights, and the other unconditional results for Dirichlet families in Table 1 with a proportion above 1/2 (apart from the announcement [1, Rem. 7.2], which gives no details) concern heights that are sub-polynomial in the conductor, except that the bound of [21] at height  $T = q^\theta$  (see below) stays above 1/2 for  $\theta$  below about 0.23, and above 2/3 for  $\theta$  below about 0.0085: [21] obtains the Montgomery–Taylor constant at every height  $T = q^{o(1)}$  with  $(\log q)^s = o(T)$  for some fixed  $s > 1$ . The closest unconditional result at height polynomial in the conductor that we know for Dirichlet  $L$ -functions is due to Dickinson [14, Theorem 1.4, Corollary 1.1]: for every modulus  $q$ , on average over the primitive characters modulo  $q$ , at least 38.2% of the zeros with  $0 < \gamma < T$  are simple and on the critical line, as  $qT \rightarrow \infty$  with  $T \gg q^\varepsilon$ ; this comes from Levinson’s method and a twisted second moment that is uniform in  $q$  and  $t$ . At height  $T = q^{\theta+o(1)}$  the bandwidth of [21], for the characters modulo one prime  $q$ , is  $1/(1+\theta)$  [21, Remark 2.4], which gives 0.601 at  $\theta = 0.1$ , 0.486 at  $\theta = \frac{1}{4}$ , 0.279 at  $\theta = \frac{1}{2}$ , and nothing for  $\theta \gtrsim 0.8$ . By Levinson’s method, Conrey, Kwan, Lin and Turnage-Butterbaugh [11, Corollaries 1.13, 1.14] show, for the twists of a fixed  $\mathrm{GL}_2$  form by primitive  $\chi$ ,  $(q, q\pi) = 1$ , that on average over the family a positive proportion of the zeros with  $|\gamma| \leq Q^\varpi$  lie on the critical line for every  $\varpi < 1$  (at least  $(1-\varpi)/(3(1+\varpi))$ ), and that a positive proportion of those with  $|\gamma| \leq Q^{1/2-\varepsilon}$  are simple and on the line; for the twists of a self-dual  $\mathrm{GL}_3$  form they obtain at least 1/9 on the line for  $|\gamma| \leq Q^\varepsilon$ , and a smaller positive proportion for  $|\gamma| \leq Q^\varpi$ ,  $\varpi < \frac{1}{3}$ . These are  $L$ -functions of degree 2 and 3, and the proportions are about 1/3 or less (in a footnote they note that Conrey’s modification of Levinson’s method raises the 1/3 slightly, to 36.58%). Garunkštis and Paliulionytė [17] prove an unconditional pair-correlation theorem for individual Dirichlet  $L$ -functions. In a targeted search of arXiv (math.NT, announcements through 28 September 2026) and Zenodo we found no statement of Theorem 1.4, and, apart from [21] at heights  $T = q^\theta$  with  $\theta$  below about 0.0085, no unconditional bound above 2/3 for a family of Dirichlet  $L$ -functions at height polynomial in the conductor.

**1.3. Relation to previous large-sieve work.** The positivity certificate is that of [1, 22] (§1.5), and the constant  $p(1) = 0.93228\dots$  was obtained by Sono under GRH for a different statistic [38] (for that statistic [9] obtained the larger constant 0.9350 under GRH; see §8.8). Theorem 1.1 is thus an unconditional theorem that reaches an existing extremal constant; the constant itself is not new. What is new is the large-sieve input of §6 and the way it is applied. Several of its ingredients are classical.

- (a) *Counting Farey points near a rational.* Write  $S_x(\theta) = \sum_n x_n e(n\theta)$ , with  $e(\theta) = e^{2\pi i\theta}$ , for  $x$  supported on  $N$  consecutive integers. Wolke [44] proved  $\sum_{p \leq Q} \sum_{b=1}^{p-1} |S_x(b/p)|^2 \leq c(1-\delta)^{-1} Q^2 \frac{\log \log Q}{\log Q} \|x\|^2$  for  $Q \geq 10$  and  $N = Q^{1+\delta}$ ,  $0 < \delta < 1$ , with an absolute constant  $c$ , by counting Farey fractions with prime denominator in short intervals. Baier [2, §4] developed Wolke’s method for general sparse sets of moduli. The large sieve is bounded through the maximal number of Farey points in a short interval; the centre of the interval is approximated by a rational (Dirichlet’s theorem), and the Farey points near it are counted through moduli in arithmetic progressions. The constants there are unspecified absolute constants. Proposition 6.4 and Lemma 6.20 use the same geometry: a Dirichlet approximation  $u/v$  of the centre, and the spokes  $k = vb - uq$ . They replace the maximal count by a Fejér majorant, duality and Schur’s test (Lemma 6.2), and they evaluate the main term exactly. This is what produces the explicit local profiles  $P_w$  and  $R_S$  instead of bounds up to a constant.

- (b) *The identity on sifted vectors.* Put  $G_q(Q) := \frac{q}{\varphi(q)} \sum_{\ell \leq Q/q, (\ell, q)=1} \frac{\mu^2(\ell)}{\varphi(\ell)}$ . If  $x$  is supported on integers with no prime factor  $\leq Q$ , then

$$\sum_{q \leq Q} G_q(Q) \sum_{\chi \bmod q}^* \left| \sum_n x_n \chi(n) \right|^2 = \sum_{q \leq Q} \sum_{a \bmod q}^* |S_x(a/q)|^2.$$

This is the case of the invertible residue classes in Ramaré's identity [33, Thm. 2.1]; see the proof of [33, Thm. 2.3], where it yields the inequality of Bombieri and Davenport [7]. Lemma 6.13 is the same computation for general weights  $\omega(q)$ : the weights  $\omega(q) = G_q(Q)$  give  $\Omega(e) = \mathbf{1}_{e \leq Q}$  (Remark 6.14(d)). For the weights (1.1),  $\Omega(e)$  has a negative part, and the Toeplitz form is not positive semidefinite on all vectors.

- (c) *Large sieves for sifted sequences.* Ramaré [33, Thms. 5.2, 5.3] improves the large sieve inequality for sequences carried by a sifted set. For instance, if  $x_n = 0$  whenever  $n$  has a prime factor  $< \sqrt{N}$ , and  $N \geq 100$ , then

$$\sum_{q \leq Q_0} \sum_{a \bmod q}^* |S_x(a/q)|^2 \leq 7N \frac{\log Q_0}{\log N} \|x\|^2 \quad (Q_0 \leq \sqrt{N}).$$

Ramaré's recent papers [36, 35] derive arithmetic forms of the weighted large sieve for sifted sets from Parseval's identity and the Farey dissection. These results concern additive forms, or sums, of sifted sequences. They do not treat the primitive-character family with the weights (1.1), and they contain no analogue of the normalised constant  $C_T^+(w)$  below.

- (d) *The approximant  $\Lambda_R$*  goes back to Selberg and was used by Goldston and by Goldston and Yıldırım (§5.6). Vaughan [40] uses it, mainly at small levels  $R \leq (\log x)^A$ , for moments of primes in arithmetic progressions. We use it at the levels  $R_j \asymp N_j^{1-\varepsilon_3}/(QT)$  and need only the upper bound of Lemma 5.7.

Accordingly, we do not claim as new the Toeplitz identity, the duality and Schur-test argument, the rational approximation and spoke counting, the Gauss-sum transfer, or the approximant  $\Lambda_R$ . We claim as new the following two statements.

- (i) *A positive Toeplitz envelope, defined on all vectors, whose constant tends to 1.* This consists of three parts, for the weights (1.1).
- The explicit normalised local profile  $R_S$  of the signed Farey measure  $\mu_\Omega$ , uniformly over finite sets  $S$  of primes (Definition 6.18, Lemma 6.19).
  - An explicit positive measure  $\mu^\natural \geq \mu_\Omega^+$ , whose Toeplitz form  $T^\natural$  is positive semidefinite and defined on all vectors (Lemma 6.17).
  - On intervals of  $K \leq Q^{2-\varepsilon}$  integers,  $\lambda_{\max}(T_I^\natural) \leq (C_T^+(w) + o(1))H$  (Proposition 6.21). Here  $C_T^+(w) \leq 1 + 42/\log(1/\eta)$  for  $w_\eta$  and  $C_T^+(w) \leq 1 + 83/\log(1/\eta)$  for  $w_\eta^{\text{sm}}$ , if  $\eta \leq e^{-3}$  (Lemma 6.22).

On vectors supported on  $Q$ -rough integers (all of whose prime factors exceed  $Q$ ),  $T^\natural$  dominates the family form. The normalised family large-sieve constant on such vectors is therefore at most  $C_T^+(w)$ , and the value 1 is optimal. The error term of Proposition 6.21 grows like  $\eta^{-2}$ , so these statements are for fixed  $\eta$  as  $Q \rightarrow \infty$ .

- (ii) *The transfer to the flattened vector.* The flattened prime vector is not supported on  $Q$ -rough integers. Above the band around  $\alpha = 1$ , the localised prime vector (not the flattened one) is split into its  $Q$ -rough part and prime powers. The identity and Lemma 6.17 are applied to the rough part. Only then, inside  $T^\natural$ , is the rough part rewritten through the flattened coefficients and the localised subtracted vector. The subtracted vector is resonant at low denominators, where the additive large sieve bounds it with a saving in length; at high denominators Poisson summation makes it negligible (Proposition 6.24).

The Farey constant  $R_w$  of the Gauss route (Lemmas 6.5 and 6.10) is a result of the same kind as (i), for the additive Farey measure; it enters only Theorem 1.2. We found neither (i) nor (ii) in the sources above. Our literature search was targeted rather than exhaustive, and we claim no priority beyond it.

#### 1.4. What is claimed and what is not.

- (1) *Weighted averages.* Theorems 1.1–1.4 bound ratios of *weighted family totals*. The weights  $\omega_\chi$  are part of the statement. They are positive and, for fixed  $\eta$ , bounded by  $\eta^{-2} q/\varphi(q) \ll \eta^{-2} \log \log Q$ . Every dyadic range of conductors carries comparable total weight; this spreading over logarithmic scales is essential to the constants (§2.2).
- (2) *Heights.* The zeros counted have ordinates in  $(T, 2T]$  with  $(\log Q)^{a_0} \leq T \leq (\log Q)^{A_0}$ . We call them *low-lying* only in this sense. In units of the mean spacing  $2\pi/\log q$  the window contains  $\asymp T\ell \rightarrow \infty$  zeros of each  $L(s, \chi)$ ; this is not the one-level-density regime of a fixed normalised window at the central point, and it is also not the regime  $T \rightarrow \infty$  with  $q$  fixed. Theorem 1.4 allows  $(\log Q)^{a_0} \leq T \leq Q^{\kappa_1}$  with  $\kappa_1$  fixed; those zeros are at height polynomial in the conductor and are not low-lying in any sense. Heights  $T > Q^{\kappa_1}$  with  $\kappa_1 \rightarrow \infty$  are *not* covered (§9.6).
- (3) *No individual  $L$ -functions.* Nothing is asserted about any single  $L(s, \chi)$  or about a single modulus. Characters with few simple critical zeros are possible; they carry a small proportion of the total weight.
- (4) *Lower bounds only.* We do not show that the remaining zeros lie off the line. Off-line zeros and multiple zeros are not distinguished by the method.
- (5) *Unconditional.* No form of GRH, no zero-free region for any  $L(s, \chi)$  and no hypothesis on exceptional zeros is used (the prime number theorem for  $\zeta$  is used in Lemma 5.7); the only information about zeros off the line is Montgomery’s classical zero-density theorem [26], used to discard a negligible set of characters. An exceptional zero merely removes its character from the certificate, at negligible cost. The same holds for Theorem 1.4: Montgomery’s theorem is used there exactly as here, at heights at most  $e^{1/2}Q^{1/4}$  (§9.2).
- (6) *Not RH-moving.* Theorems 1.1–1.3 say nothing about zeros at heights beyond  $(\log Q)^{A_0}$ , Theorem 1.4 nothing about heights beyond  $Q^{\kappa_1}$ , and none of them anything about an individual zero; for  $\zeta$  the method stops at bandwidth 1 (§8.1).
- (7) *Kinds of verification.* Three kinds are kept apart (§A.4). The numerical inputs are certified by exact rational or interval arithmetic. Theorem 1.1 is also proved formally in Lean 4, with no hypotheses and only Lean’s standard axioms; what the machine checks is the Lean statement, and its agreement with Theorem 1.1 as displayed above is a correspondence of definitions that the reader can check. Theorems 1.2 and 1.3 and Lemma 6.5 are not formalised; they rest on the written proofs and on the certified numerics. Theorem 1.4(a) is also proved formally in Lean 4, with no hypotheses and only Lean’s standard axioms; again the machine checks the Lean statement, and its agreement with Theorem 1.4(a) as displayed above is a correspondence of definitions that the reader can check. Theorem 1.4(b), (c) and Corollary 1.5 are not formalised; they rest on the written proofs of §9 and on the certified numerics of Proposition 9.24.

**1.5. The method.** The starting point is the finite compression of Weil’s Hermitian form introduced in [1] (the argument of [1] is due to Claude, as recorded there) and ported to the family of characters modulo a large prime in [22, 21] (see also [32]), where the bandwidth is at most 1; it has also been transferred to the weight-2 newforms of prime level with Petersson weights [15], again at bandwidth at most 1. For each  $\chi$  one builds a real symmetric  $d \times d$  matrix  $G_\chi$  by sampling the explicit formula against a Gabor system of bandwidth  $L = \lambda\ell$ . Its zero side is a

sum of rank-one blocks over zeros; critical-line zeros give positive semidefinite blocks and off-line pairs give blocks with at most one positive eigenvalue. A rank–trace inequality (Lemma 3.2) turns this block structure into a lower bound for  $N_{0,\chi}^s$  in terms of the trace and the Frobenius norm of the part  $\widehat{A}_\chi$  of the normalised matrix that comes from zeros with ordinates in  $(T, 2T]$  (Proposition 3.3). After the exterior zeros and a negligible set of characters are removed (§4), these are replaced by  $\text{tr } \widehat{G}_\chi$  and  $\|\widehat{G}_\chi\|_{\mathbb{F}}^2$ , which are computed from the prime side of the explicit formula. The certificate is valid character by character, so it may be summed with any nonnegative weights.

The trace is easy. The Frobenius norm is a pair-correlation sum: after the explicit formula it becomes

$$\sum_{\chi} \omega_{\chi} \iint_{J^2} \Phi(t-t')^2 \nu_{\chi}(t) \nu_{\chi}(t') dt dt',$$

whose prime part is governed by the quadratic form  $\Delta(n, m) = \sum_{\chi} \omega_{\chi} \chi(n) \overline{\chi}(m)$  restricted to  $n, m \leq Q^\lambda$ . For  $\lambda < 1$  it is essentially diagonal on the relevant range (for a single prime modulus  $q > Q^\lambda$  exactly so, by orthogonality); this is the situation of [1, 22], and it gives the Montgomery–Taylor constant 0.6725. (For  $\zeta$  the corresponding unconditional input in [1, 25] is the pair-correlation theorem of Baluyot, Goldston, Suriajaya and Turnage-Butterbaugh [3].) The certificate only needs an *upper* bound for this form, and an upper bound for a positive semidefinite form is exactly what a large sieve inequality provides. We use three devices to make the upper bound sharp enough to be useful up to  $\lambda = 2$ .

- (i) *Flattening.* Subtracting from  $\Lambda(n)$  a truncated divisor sum  $\Lambda_R(n)$ , at a level  $R = R_j \leq N_j^{1-\varepsilon_3}/(QT)$  fixed on each dyadic block  $n \asymp N_j$ , changes no character sum  $S_\chi(t)$ ,  $\chi \in \mathcal{F}$ ,  $|t| \leq 3T$ , by more than  $O(Q^{-4})$  (Lemma 5.6), but lowers the mean square of the coefficients from  $\log n$  to about  $\log Q$  (Lemma 5.7). This is where the form factor acquires the plateau  $\min(\alpha, 1)$ .
- (ii) *Time localisation.* The time integral in  $\mathcal{K}(n, m)$  couples  $n$  and  $m$  with weight decaying like  $1/(|T| \log(n/m))$ ; localising to  $|\log(n/m)| \lesssim T^{-1/2}$  costs a relative error  $O(T^{-1/4})$  (Lemmas 5.9–5.11). Hence only large sieve constants for vectors supported on intervals of length about  $n/\sqrt{T}$  are needed. For  $n \leq Q^{1-\varepsilon}$  the Gram matrix is trivially diagonally dominant on such intervals, which gives the exact form factor  $\alpha$  on  $[0, 1)$ .
- (iii) *A sharp constant on  $(1, 2)$ .* For  $\alpha > 1$  we need the large sieve constant of the primitive family on intervals of length  $K \in [Q^{1-\varepsilon}, Q^{2-\varepsilon}]$ , normalised by  $H$ . The Gauss-sum transfer followed by a sharp weighted Farey large sieve (Lemma 6.5) gives  $C_G R_w$ , and  $R_w \rightarrow 1$  for log-wide weights; this yields Theorem 1.2. For vectors supported on integers free of prime factors  $\leq Q$ , the family form is *exactly* a Toeplitz form attached to a signed Farey measure (Lemma 6.13). Adding a positive majorant of the negative part of this measure gives a positive semidefinite Toeplitz form  $T^\natural$  defined on *all* vectors (Lemma 6.17). Its constant  $C_T^+(w)$ , on intervals of length at most  $Q^{2-\varepsilon}$ , is a supremum of explicit local densities, and  $C_T^+(w) \rightarrow 1$  for log-wide weights (Lemma 6.22). Above the band around  $\alpha = 1$  the flattened vector, which is not sifted, enters only through  $T^\natural$ , after a swap inside that form (Proposition 6.24). This yields Theorem 1.1.

The bandwidth barrier at  $\lambda = 2$  is the classical one: for intervals longer than  $Q^2$  the length term of the large sieve dominates the family mass  $H \asymp Q^2$  (§8.1).

*Polynomial height.* At height  $T$  the bandwidth is measured in units of the log-conductor  $\ell_* = \log(QT)$ , and three changes carry the argument up to every bandwidth  $\lambda < \bar{\lambda}_T$  (§9). Every occurrence of  $\ell$  that measures the density of zeros, the bandwidth or the flattening becomes  $\ell_*$ ; in particular the flattened diagonal density is  $\min(\alpha, 1)$  in units of  $\ell_*$ . The time localisation is

at scale  $T^{-1+\varepsilon_5}$  instead of  $T^{-1/2}$ ; then the localised vectors live on intervals of about  $e^u/T^{1-\varepsilon_5}$  integers, so the family large sieve on intervals of at most  $Q^{2-\varepsilon}$  integers reaches  $n \leq Q^{2-\varepsilon}T^{1-\varepsilon_5}$ , i.e.  $\alpha < \bar{\lambda}_T$ : this converts height into bandwidth. And the tails of the localisation, which were bounded above through  $\Lambda_{\text{mult}}(Y) \leq 2w_{\text{max}}Q^2$  (valid only for  $Y \leq Q^2$ ), are bounded by dyadic shells and the classical large sieve (Lemma 9.13). The large sieve constants of §6 concern intervals of at most  $Q^{2-\varepsilon}$  integers, independently of  $T$ , and are used verbatim; on the zero side nothing new is needed. A worst-case bound cannot go beyond  $\bar{\lambda}_T$  (Proposition 9.30).

**1.6. Organisation.** Section 2 fixes notation, the explicit formula and the Gabor matrix. Section 3 states the per-character certificate and reduces everything to an upper bound for the weighted second moment. Section 4 carries out the zero-side analysis for the weighted family, following [22]. Section 5 evaluates the second moment up to the ratio terms and contains the flattening and time-localisation lemmas. Section 6 proves the two large-sieve constants. Section 7 treats the variational problem and assembles the proofs. Section 8 discusses limitations. Appendix A lists the computations, separately from the proofs. Appendix B proves the explicit formula and the rank–trace inequality; together with the proofs in §§2–4, this makes the paper independent of [1, 22] for correctness, which are cited for attribution. Section 9 extends the results to heights polynomial in  $Q$  (Theorem 1.4, Corollary 1.5). Table 2 records which results each main theorem depends on.

| Theorem   | Inputs                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| all three | Proposition 3.6 (Sections 3–4: Lemmas 3.1, 3.2, 4.1, 4.2, 4.3, 4.7 with [26]; Propositions 3.3, 4.4, 4.6, 4.8; the explicit formula of Appendix B); Proposition 5.13 (Lemmas 5.1, 5.2, 5.4, 5.6, 5.7); Lemmas 5.9–5.11; Theorem 5.16; Lemmas 7.2, 7.1; Proposition 7.3.                                                                                                                                                      |
| 1.1       | Proposition 6.24, through Lemmas 6.13, 6.15, 6.17, 6.19, 6.20, 6.22 and Proposition 6.21; from Section 6.1 only Lemmas 6.1, 6.2, 6.9, Proposition 6.4 and the layer-cake argument of Lemma 6.10; the classical large sieve (4.1) on the band. Not used: Lemmas 6.5, 6.7, Corollary 6.8.                                                                                                                                      |
| 1.2       | Corollary 6.8 (Lemmas 6.7, 6.5, 6.1, 6.2, Proposition 6.4); Proposition 5.12; Lemmas 6.9, 6.10; $C_G \leq 1.2688$ (certified).                                                                                                                                                                                                                                                                                               |
| 1.3       | (i), (ii): as Theorem 1.1 at fixed $\eta$ , with Proposition 6.23 and the interval certificates <code>ctplus.arb.py</code> . (iii): as Theorem 1.2, with Lemma 6.11 and the interval certificates <code>rw.arb.py</code> .                                                                                                                                                                                                   |
| 1.4, 1.5  | Proposition 9.9 (Lemmas 9.2, 9.3, 9.4; Propositions 9.5, 9.6, 9.7; and the results of Sections 2–4 listed in §9.1); Lemma 9.1; Lemmas 9.10, 9.11, 9.12, 9.13, Corollary 9.14; Proposition 9.19; Theorem 9.20; Lemmas 9.21, 9.22, 9.23; Proposition 9.24. (a): Proposition 9.18, with the results of §6.2 used for Theorem 1.1. (b): Proposition 9.17 with Corollary 6.8. (c) and Corollary 1.5: Proposition 9.17 with (4.1). |

TABLE 2. Logical dependencies of the main theorems (numerical inputs: Appendix A).

## 2. SETUP AND NOTATION

**2.1. Conventions.** Throughout,  $Q \rightarrow \infty$ ,  $\ell = \log Q$ , and  $\varphi$  is Euler’s function.  $\sum_{\chi \bmod q}^*$  denotes a sum over primitive characters and  $\varphi^*(q)$  their number. The Fourier transform is  $\widehat{f}(t) = \int_{\mathbb{R}} f(u)e^{itu} du$ , so that  $\int \widehat{f}\widehat{g} = 2\pi \int fg$  (as in [22, (2.11)]). For a matrix  $M$ ,  $\|M\|_F$  and  $\|M\|_1$  are the Frobenius and trace norms and  $n_+(M)$  is the number of positive eigenvalues of a Hermitian  $M$ .

## 2.2. The family and its masses.

**Definition 2.1.** A weight  $w : (0, 1] \rightarrow [0, \infty)$  is *admissible* if it has bounded variation  $V_w$ ,  $\text{supp } w \subset [\eta, 1]$  for some  $\eta \in (0, 1/2)$ , and  $\int_0^1 uw(u) du > 0$ . Put  $w_{\max} = \sup w$ .

Since  $q \geq \eta Q \geq 3$  for  $Q$  large, every  $\chi \in \mathcal{F}$  is nonprincipal of conductor  $q$ . The family is closed under  $\chi \mapsto \bar{\chi}$  and  $\omega_{\bar{\chi}} = \omega_{\chi}$ . Define

$$(2.1) \quad H := \sum_{\chi \in \mathcal{F}} \omega_{\chi} = \sum_q w(q/Q) \frac{q\varphi^*(q)}{\varphi(q)}, \quad W := \sum_q w(q/Q) \varphi(q).$$

By Lemma 6.1,

$$(2.2) \quad H = \mathcal{E} Q^2 \int_0^1 uw(u) du + O_w(Q^{3/2}), \quad W = \frac{6}{\pi^2} Q^2 \int_0^1 uw(u) du + O_w(Q \log Q),$$

where  $\mathcal{E} = \prod_p (1 - p^{-2} - p^{-3})$ , so that  $W/H \rightarrow C_G := 6/(\pi^2 \mathcal{E}) \approx 1.2687739$ , independently of  $w$ . (The proof, which is elementary, is given in §6.1;  $\mathcal{E}$  is the mean value of  $\varphi^*(q)/\varphi(q)$ .) Numerically  $\mathcal{E} = 0.4791453444\dots$ , and  $\mathcal{E} \in [0.47914533, 0.47914535]$ ,  $C_G < 1.26877393$  are certified in interval arithmetic (Appendix A). The factor  $q/\varphi(q)$  in (1.1) is chosen so that the Gauss-sum transfer of §6.1 produces the smooth additive weights  $w(q/Q)$ ; it also makes the Toeplitz measure of §6.2 locally uniform for log-wide  $w$ . We note  $\omega_{\chi} \leq w_{\max} q/\varphi(q) \ll w_{\max} \log \log Q$  and  $H \asymp_w Q^2$ .

*Remark 2.2.* The weight  $Q^2/(q\varphi(q))$  has, up to the factor  $Q$ , the shape  $W(q/Q)/\varphi(q)$  of the weights of [8, 38], with  $W(u) = u^{-1}$ . The identification is only formal: in [8] the function  $W$  is supported in  $(1, 2)$ , so the family there is dyadic, whereas the spreading of the weights over many dyadic ranges of conductors is essential here (§8.4). Likewise  $\mathcal{E}$  is the Euler product in the constant  $A = \widehat{W}(1) \prod_p (1 - p^{-2} - p^{-3})$  of [8, (1.1)].

**2.3. Parameters.** We fix, in this order:  $0 < a_0 < A_0$ ;  $\eta$  and an admissible  $w$ ; the bandwidth  $\lambda \in (0, 2)$  and a margin  $\varepsilon_1 > 0$  with  $\lambda(1 + \varepsilon_1) \leq 2 - \varepsilon_1$  (possible for every  $\lambda < 2$ );  $\theta \in (0, 1/4)$ ; a real even window  $\psi \in C_c^\infty((-\frac{1}{2}, \frac{1}{2}))$ ,  $\psi \not\equiv 0$ ; a flattening margin  $\varepsilon_3 \in (0, 1/4)$ ; the band parameter  $\varepsilon \in (0, \frac{1}{3})$  of Propositions 5.12 and 6.24 and §7.3; the constants  $B_1 = 30$  and  $B$  of §4.4; and the auxiliary smooth cut-offs:  $\Upsilon_0$  (§2.4; it depends on  $\varepsilon_1$ ), the dyadic partition of unity  $(\psi_j)$  with its derivative bounds  $(c_i)$  (§5.7) and the localisation cut-off  $\rho$  (Lemma 5.11). All of these are fixed while  $Q \rightarrow \infty$ , and  $T = T(Q)$  varies in  $[\ell^{a_0}, \ell^{A_0}]$ ; all  $o(1)$  terms are uniform in  $T$ . Implied constants may depend on the fixed data.

*Order of limits.* This order is used throughout the paper. First  $Q \rightarrow \infty$ , with all the data above fixed; then  $\theta, \varepsilon, \varepsilon_3 \rightarrow 0$ ; then  $\lambda \rightarrow 2$  (with  $\varepsilon_1 = \varepsilon_1(\lambda) \rightarrow 0$ ) and  $\psi$  tends to an optimiser; and only at the very end  $\eta \rightarrow 0$ . In particular  $\eta$  is always fixed before  $Q \rightarrow \infty$ : Theorems 1.1 and 1.2 bound  $\liminf_{Q \rightarrow \infty}$  for each fixed small  $\eta$ , and Theorem 1.3 is the case of a fixed  $\eta$ . The precise choices are made in the proof of Theorem 5.16 (§7.3); see also Remark 6.25. Put

$$L = \lambda \ell, \quad X = e^L = Q^\lambda, \quad Y = X^{1+\varepsilon_1} \leq Q^{2-\varepsilon_1}, \quad I = (T, 2T], \quad J = [(1+\theta)T, (2-\theta)T].$$

**2.4. Window, lattice and the explicit formula.** Let  $\psi_L(u) = \psi(u/L)$ ,  $v = \psi^2$ ,  $a = \int v$ ,  $b = \int v^2$ ,  $\mathcal{J}(v) = \iint |s - s'| v(s) v(s') ds ds'$ , and

$$g := \psi_L^2 * \psi_L^2 = L(v * v)(\cdot/L), \quad \Phi := \widehat{\psi_L^2},$$

so that  $\Phi^2 = \widehat{g}$ ,  $\int \Phi^2 = 2\pi bL$  and  $\int |r| \Phi(r)^2 dr = O_\psi(1)$ . The function  $g \geq 0$  is smooth, even, supported in  $(-L, L)$ , with  $\|g\|_\infty = g(0) = bL$ . Let  $\tau_k = \tau_0 + 2\pi k/L$  ( $k \in \mathbb{Z}$ ,  $\tau_0$  fixed),

$\mathbf{K} = \{k : \tau_k \in J\}$ ,  $d = \#\mathbf{K} = |J|L/(2\pi) + O(1)$ , and

$$p_k(z) := \widehat{\psi}_L(z - \tau_k) \quad (z \in \mathbb{C}).$$

Each  $p_k$  is entire; it is real and even about  $\tau_k$  on  $\mathbb{R}$ , and  $\overline{p_k(z)} = p_k(\bar{z})$ .

**Lemma 2.3** (Envelope). *For every  $A \geq 0$  there is  $C_A = C_A(\psi)$  such that for  $r, \delta \in \mathbb{R}$ ,*

$$|\widehat{\psi}_L(r - i\delta)| \leq C_A L e^{L|\delta|/2} (1 + L|r|)^{-A}.$$

Consequently, if  $A > \frac{1}{2}$ , then with  $u(z) := (p_k(z))_{k \in \mathbf{K}}$  and  $D := L \operatorname{dist}(r, J)$ ,  $\|u(r - i\delta)\| \|u(r + i\delta)\| \ll_A L^2 e^{L|\delta|} (1 + D)^{1-2A}$ .

*Proof.*  $\widehat{\psi}_L(w) = L \int \psi(s) e^{iws} ds$  and  $|e^{iws}| \leq e^{L|\operatorname{Im} w|/2}$  for  $|s| \leq 1/2$ ; integrate by parts  $\lceil A \rceil$  times. For the second claim, the first gives  $\|u(r \mp i\delta)\|^2 \leq C_A^2 L^2 e^{L|\delta|} \Sigma$  with  $\Sigma := \sum_{k \in \mathbf{K}} (1 + L|r - \tau_k|)^{-2A}$ . The points  $L\tau_k$ ,  $k \in \mathbf{K}$ , lie in  $LJ$  and are  $2\pi$ -spaced, and  $L|r - \tau_k| \geq D$ ; so on each side of  $Lr$  the  $m$ -th nearest of them ( $m \geq 0$ ) is at distance  $\geq D + 2\pi m$ . As  $2A > 1$ , comparison with an integral gives

$$\Sigma \leq 2 \sum_{m \geq 0} (1 + D + 2\pi m)^{-2A} \leq 2(1 + D)^{-2A} + \frac{(1 + D)^{1-2A}}{\pi(2A - 1)} \ll_A (1 + D)^{1-2A}.$$

(For  $A \leq \frac{1}{2}$  this lattice bound fails: for  $A = 0$  and  $r \in J$ ,  $\Sigma = d \asymp TL$ . The second claim is only used with  $A > 1$ .)  $\square$

**Lemma 2.4** (Poisson–Gabor identity; cf. [22, Lemma 5.1]). *For real  $t, t'$ ,  $\sum_{k \in \mathbb{Z}} p_k(t) p_k(t') = L\Phi(t - t')$ . In particular  $\sum_{k \in \mathbb{Z}} p_k(t)^2 = aL^2$ , and  $\|u(t)\|^2 \leq aL^2$  for real  $t$ .*

*Proof.* For real  $x$ ,  $\widehat{\psi}_L(x) = \int \psi_L(u) \cos(xu) du$  is real, since  $\psi$  is real and even; so each  $p_k(t)$  is real. As  $\operatorname{supp} \psi_L \subset (-\frac{L}{2}, \frac{L}{2})$ , for real  $t$

$$p_k(t) = \int_{-L/2}^{L/2} \psi_L(u) e^{i(t-\tau_0)u} e^{-2\pi iku/L} du = L c_k(f_t), \quad f_t(u) := \psi_L(u) e^{i(t-\tau_0)u},$$

where  $c_k(f) := L^{-1} \int_{-L/2}^{L/2} f(u) e^{-2\pi iku/L} du$  is the  $k$ -th Fourier coefficient on  $\mathbb{R}/L\mathbb{Z}$ . With  $\tilde{f}(u) := \overline{f_t(-u)}$  one has  $c_k(f_{t'}) = \overline{c_k(\tilde{f})}$ , so by Parseval's identity

$$\sum_{k \in \mathbb{Z}} p_k(t) p_k(t') = L^2 \sum_k c_k(f_t) \overline{c_k(\tilde{f})} = L \int_{-L/2}^{L/2} f_t(u) f_{t'}(-u) du = L \int_{\mathbb{R}} \psi_L^2(u) e^{i(t-t')u} du,$$

which is  $L\Phi(t - t')$ ; we used  $\psi_L(-u) = \psi_L(u)$ . At  $t' = t$  this is  $L\Phi(0) = L \int \psi_L^2 = aL^2$ , and  $\|u(t)\|^2 = \sum_{k \in \mathbf{K}} p_k(t)^2 \leq aL^2$ .  $\square$

For  $\chi \in \mathcal{F}$  let  $\mathfrak{a}_\chi = (1 - \chi(-1))/2$  and

$$(2.3) \quad \mu_\chi(t) = \frac{1}{2\pi} \log \frac{q}{\pi} + \frac{1}{2\pi} \operatorname{Re} \frac{\Gamma'}{\Gamma} \left( \frac{1/2 + \mathfrak{a}_\chi + it}{2} \right).$$

Fix  $\Upsilon_0 \in C^\infty(\mathbb{R})$  with  $0 \leq \Upsilon_0 \leq 1$ ,  $\Upsilon_0 = 1$  on  $[0, 1]$  and  $\operatorname{supp} \Upsilon_0 \cap [0, \infty) \subset [0, 1 + \varepsilon_1]$ , and put  $\Upsilon(n) = \Upsilon_0(\log n/L)$ ,

$$(2.4) \quad a_n := \frac{\Lambda(n)}{\sqrt{n}} \Upsilon(n), \quad S_\chi[x](t) := \sum_n x_n \chi(n) n^{-it}, \quad P_\chi(t) := -\frac{1}{2\pi} (S_\chi[a](t) + \overline{S_\chi[a](t)}),$$

and  $\nu_\chi := \mu_\chi + P_\chi$ . Note that  $a$  is supported on  $n \leq Y$ .

**Definition 2.5** (Gabor matrix). For  $\chi \in \mathcal{F}$  and  $k, l \in \mathbb{K}$  let

$$(2.5) \quad G_{\chi,kl} := \sum_{\rho} m_{\rho} p_k(z_{\rho}) p_l(z_{\rho}), \quad z_{\rho} := \frac{\rho - 1/2}{i} = \gamma - i\delta_{\rho}, \quad \delta_{\rho} = \beta - \frac{1}{2},$$

the sum running over the nontrivial zeros of  $L(s, \chi)$  with multiplicity  $m_{\rho}$ . Put  $\widehat{G}_{\chi} := G_{\chi}/(aL^2)$ .

The entries use the holomorphic product  $p_k(z)p_l(z)$ , as in [22, (4.6)]. Since  $p_l(z) = \overline{p_l(\bar{z})}$ , a zero contributes  $m_{\rho} u(z_{\rho}) u(\bar{z}_{\rho})^*$  to  $G_{\chi}$ . The series converges absolutely, even in trace norm, for each fixed  $\chi$ , by Lemma 2.3 with  $A = 2$  (every zero has  $|\delta_{\rho}| < \frac{1}{2}$ , so  $e^{L|\delta_{\rho}|} \leq e^{L/2}$ ) and the classical local zero count

$$(2.6) \quad \#\{\rho : |\gamma - t| \leq 1\} \ll \log(q(|t| + 3)) \quad (t \in \mathbb{R})$$

for primitive  $\chi \bmod q$  [13, Ch. 16] (explicitly in [4]).

**Lemma 2.6** (Explicit formula on the test class; cf. [22, Prop. 4.1, Lemma 4.2]). *For  $\chi \in \mathcal{F}$  and  $k, l \in \mathbb{K}$ ,*

$$G_{\chi,kl} = \int_{\mathbb{R}} p_k(t) p_l(t) \nu_{\chi}(t) dt.$$

*In particular  $G_{\chi}$  is real symmetric. The value does not depend on the choice of  $\Upsilon_0$  subject to  $\Upsilon_0 = 1$  on  $[0, 1]$ .*

*Proof.* This is the case  $h = f_k * f_l$ ,  $f_k(u) := \psi_L(u) e^{-i\tau_k u}$ , of Weil's explicit formula (Proposition B.1), which is proved in Appendix B; the details are given there. The inverse Fourier transform  $h$  of  $p_k p_l$  is supported in  $\text{supp } \psi_L + \text{supp } \psi_L \subseteq (-L, L)$ , so the terms  $n \geq X$  of the prime sum vanish; hence any cut-off equal to 1 on  $[2, X]$  gives the same value. The smooth cut-off  $\Upsilon$  is used only to allow Poisson summation in Lemma 5.6.  $\square$

**2.5. The second-moment kernels.** Define the *family Gram kernel* and the *time kernel*

$$(2.7) \quad \Delta(n, m) := \sum_{\chi \in \mathcal{F}} \omega_{\chi} \chi(n) \overline{\chi(m)}, \quad \mathcal{K}(n, m) := \iint_{J^2} \Phi(t - t')^2 n^{-it} m^{it'} dt dt'.$$

Both are positive semidefinite:  $\Delta = \sum_{\chi} \omega_{\chi} v_{\chi} v_{\chi}^*$  with  $v_{\chi} = (\chi(n))_n$ , and, writing  $\Phi(t - t')^2 = \int g(u) e^{i(t-t')u} du$ ,

$$(2.8) \quad \sum_{n,m} x_n \overline{x_m} \mathcal{K}(n, m) = \int_{\mathbb{R}} g(u) \left| \int_J \sum_n x_n n^{-it} e^{itu} dt \right|^2 du \geq 0.$$

Since  $\mathcal{F}$  is closed under conjugation with equal weights,  $\Delta$  is real and symmetric. On the diagonal,  $\Delta(n, n) = \sum_q \omega(q) \varphi^*(q) \mathbf{1}_{(n,q)=1} \leq H$ , and

$$(2.9) \quad \mathcal{K}(n, n) = \int_{\mathbb{R}} \Phi(r)^2 n^{-ir} (|J| - |r|)_+ dr = 2\pi |J| g(\log n) + O_{\psi}(1).$$

For  $K \geq 1$  let  $\Lambda_{\text{mult}}(K)$  be the supremum of  $x^* \Delta x$  over unit vectors  $x$  supported on an interval of at most  $K$  consecutive positive integers (for instance  $[1, Y]$ , with  $K = Y$ ). Since  $\Delta(p, p) = H$  for primes  $p > Q$ ,  $\Lambda_{\text{mult}}(K) \geq H$ .

## 3. THE POSITIVITY CERTIFICATE

The certificate is a statement about a single character; the family enters only through the summation.

**Lemma 3.1** (Block structure; cf. [22, §8], [1, Lemma 3.1, Prop. 4.1]). *Let  $\chi \in \mathcal{F}$ . The multiset of nontrivial zeros of  $L(s, \chi)$  is invariant under  $\rho \mapsto 1 - \bar{\rho}$ , preserving multiplicity; in the coordinate  $z_\rho$  this is  $z \mapsto \bar{z}$ . Consequently:*

- (a) *a zero  $1/2 + i\gamma$  of multiplicity  $m$  contributes the positive semidefinite block  $m u(\gamma)u(\gamma)^*$ , of rank at most one and normalised trace  $m \|u(\gamma)\|^2 / (aL^2) \leq m$ ;*
- (b) *an off-line orbit  $\{\rho, 1 - \bar{\rho}\}$  ( $\rho \neq 1 - \bar{\rho}$ ) of common multiplicity  $m$  contributes  $m B \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} B^*$  with  $B = [u(z_\rho) \ u(\bar{z}_\rho)]$ , which has at most one positive eigenvalue. Both points of the orbit have the same ordinate.*

*Proof.* Let  $\xi(s, \chi) = (q/\pi)^{(s+a_\chi)/2} \Gamma(\frac{s+a_\chi}{2}) L(s, \chi)$ . The functional equation  $\xi(s, \chi) = \epsilon(\chi) \xi(1 - s, \bar{\chi})$ ,  $|\epsilon(\chi)| = 1$  [13, Ch. 9], maps a zero  $\rho$  of  $L(s, \chi)$  of multiplicity  $m$  to the zero  $1 - \rho$  of  $L(s, \bar{\chi})$  of the same multiplicity, and  $L(s, \bar{\chi}) = \bar{L}(\bar{s}, \chi)$  maps that to the zero  $1 - \bar{\rho}$  of  $L(s, \chi)$ , again of multiplicity  $m$ . In the coordinate  $z_\rho = -i(\rho - \frac{1}{2})$  one has  $z_{1-\bar{\rho}} = \bar{z}_\rho$ , and both points have ordinate  $\gamma$ . Since  $\overline{p_l(\bar{z})} = p_l(z)$ , a zero contributes  $m_\rho u(z_\rho)u(\bar{z}_\rho)^*$  to  $G_\chi$ . For  $z_\rho = \gamma$  real this is  $m_\rho u(\gamma)u(\gamma)^*$ , and its normalised trace is at most  $m_\rho$  by Lemma 2.4. For an off-line orbit the two contributions add up to  $m(u(z)u(\bar{z})^* + u(\bar{z})u(z)^*) = m B \Pi B^*$ ,  $\Pi = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ . If  $B \Pi B^*$  were positive definite on a subspace  $V$  of dimension 2, then  $B^*$  would be injective on  $V$  and  $\Pi$  positive definite on the 2-dimensional space  $B^*V$ , which is impossible as  $n_+(\Pi) = 1$ . Hence  $n_+(B \Pi B^*) \leq 1$ .  $\square$

**Lemma 3.2** (Rank–trace inequality; [1, Lemma 3.2], [22, Lemma 8.2]). *Let  $P \succeq 0$  be Hermitian of rank at most  $r$  and  $R$  Hermitian with  $n_+(R) \leq b$ , of the same size. Then  $\|P + R\|_F^2 \geq 2 \operatorname{tr} P - r + 4 \operatorname{tr} R - 4b$ .*

The proof, which follows [1, 22], is given in Appendix B.

For  $\chi \in \mathcal{F}$  let  $A_\chi$  be the sum of the blocks of Lemma 3.1 whose common ordinate lies in  $I$ ,  $E_\chi := G_\chi - A_\chi$ , and  $\hat{A}_\chi = A_\chi / (aL^2)$ ,  $\hat{E}_\chi = E_\chi / (aL^2)$ .

**Proposition 3.3** (Per-character certificate; cf. [22, Prop. 8.3]). *For every  $\chi \in \mathcal{F}$ , with  $\mathcal{M}_\chi := 4 \operatorname{tr} \hat{A}_\chi - \|\hat{A}_\chi\|_F^2$ ,*

$$N_{0,\chi}^s \geq \mathcal{M}_\chi - 2N_\chi, \quad N_{0,\chi}^* \geq \mathcal{M}_\chi - 2N_\chi, \quad N_{d,\chi} \geq \frac{1}{2}(\mathcal{M}_\chi - N_\chi).$$

(The second inequality also follows from the first, since  $N_{0,\chi}^* \geq N_{0,\chi}^s$ .)

*Proof.* Let  $s_1$  be the number of simple zeros  $\frac{1}{2} + i\gamma$  with  $\gamma \in I$ , let  $s_2$  be the number of distinct multiple zeros  $\frac{1}{2} + i\gamma$  with  $\gamma \in I$ , of multiplicities  $m_1, \dots, m_{s_2} \geq 2$ , and let  $p$  be the number of off-line orbits  $\{\rho, 1 - \bar{\rho}\}$  with ordinate in  $I$ , of common multiplicities  $n_1, \dots, n_p \geq 1$ . A point  $\rho$  with  $\rho = 1 - \bar{\rho}$  has  $\operatorname{Re} \rho = \frac{1}{2}$ , so every off-line orbit consists of two distinct zeros, both with ordinate in  $I$  (Lemma 3.1). Hence

$$N_\chi = s_1 + \sum_j m_j + 2 \sum_\ell n_\ell, \quad N_{0,\chi}^s = s_1, \quad N_{0,\chi}^* = s_1 + s_2, \quad N_{d,\chi} = s_1 + s_2 + 2p.$$

Let  $P$  be the normalised sum of the blocks of the  $s_1$  simple critical zeros, and  $R := \hat{A}_\chi - P$ . By Lemma 3.1(a),  $P \succeq 0$ ,  $\operatorname{rank} P \leq s_1$  and  $\operatorname{tr} P \leq s_1$ . Each block in  $R$  has at most one positive eigenvalue (Lemma 3.1), and  $n_+$  is subadditive: if  $n_+(R_i) \leq b_i$ , then  $R_i \preceq 0$  on a subspace of

codimension  $b_i$ , so  $R_1 + R_2 \preceq 0$  on a subspace of codimension  $\leq b_1 + b_2$ , and  $n_+(R_1 + R_2) \leq b_1 + b_2$  by the min-max principle. Hence  $n_+(R) \leq s_2 + p$ , and Lemma 3.2 gives

$$\mathcal{M}_\chi = 4 \operatorname{tr}(P + R) - \|P + R\|_{\mathbb{F}}^2 \leq 2 \operatorname{tr} P + s_1 + 4(s_2 + p) \leq 3s_1 + 4s_2 + 4p.$$

Since  $m_j \geq 2$  and  $n_\ell \geq 1$ ,

$$\begin{aligned} N_{0,\chi}^s + 2N_\chi &= 3s_1 + 2 \sum_j m_j + 4 \sum_\ell n_\ell \geq 3s_1 + 4s_2 + 4p, \\ N_{0,\chi}^* + 2N_\chi &= 3s_1 + s_2 + 2 \sum_j m_j + 4 \sum_\ell n_\ell \geq 3s_1 + 4s_2 + 4p, \\ 2N_{d,\chi} + N_\chi &= 3s_1 + 2s_2 + 4p + \sum_j m_j + 2 \sum_\ell n_\ell \geq 3s_1 + 4s_2 + 4p, \end{aligned}$$

and the three inequalities follow.  $\square$

*Remark 3.4* (Lamzouri's inequality). Lamzouri [25, Prop. 2.1] gives a Hilbert-space form of the same mechanism: for a finite multiset  $\mathcal{Z} \subset \mathbb{C}$  invariant under conjugation and a kernel  $K = \widehat{\mathbf{g}}^2$  with  $\mathbf{g} \in L^2(\mathbb{R})$  real, even and compactly supported and  $\widehat{\mathbf{g}}^2(0) = 1$ , the number of simple real elements is at least  $2|\mathcal{Z}| - \sum_{z,s \in \mathcal{Z}} K(z-s)^2$ . Applied to  $\mathcal{Z}_\chi = \{z_\rho : \rho \in I\}$  for each  $\chi$  and summed with the weights  $\omega_\chi$ , it reduces the problem to an upper bound for a weighted pair-correlation sum over complex zeros. We use the matrix form because the exterior zeros are then removed in trace norm (§4.4) and the second moment is computed exactly from the explicit formula, with no weight  $4/(4 - (\rho - \rho')^2)$  to remove.

*Remark 3.5* (Only an upper bound is needed). The certificate enters through  $-\sum_\chi \omega_\chi \|\widehat{A}_\chi\|_{\mathbb{F}}^2$ . We therefore need the trace asymptotically and the second moment only from *above*. By Lemma 2.6 and the finite-centre reduction (Proposition 4.4), the second moment is, up to negligible errors,  $(aL^2)^{-2} L^2 \sum_\chi \omega_\chi \iint_{J^2} \Phi^2 \nu_\chi \nu_\chi$ . Its prime part is the quadratic form  $x \mapsto \sum_{n,m} x_n \overline{x_m} \Delta(n,m) \mathcal{K}(n,m)$ , which is monotone in  $\Delta$  for the positive semidefinite order (Schur product theorem). Any large sieve inequality valid on *all* vectors supported on the relevant intervals, i.e. any form bound  $\Delta \preceq CH$  there, therefore feeds directly into the certificate (Theorem 5.16). A bound that holds only on a subclass of vectors does not: the exact Toeplitz identity of §6.2 holds only on sifted ( $Q$ -rough) vectors, and it enters through a positive semidefinite Toeplitz form  $T^\natural$  that dominates the family form on those vectors and is defined and bounded on all vectors; the flattened coefficients are swapped into  $T^\natural$  (Proposition 6.24). No positivity of a kernel at complex arguments, no lower bound for pair correlation, and no information on individual zeros is used.

We summarise the reduction. Put

$$(3.1) \quad \mathfrak{M} := \sum_{\chi \in \mathcal{F}} \omega_\chi \|\widehat{G}_\chi\|_{\mathbb{F}}^2.$$

**Proposition 3.6** (Zero-side reduction). *For fixed data as in §2.3 and every  $B > 0$ , uniformly for  $\ell^{a_0} \leq T \leq \ell^{A_0}$ ,*

$$\begin{aligned} N_0^s, N_0^* &\geq (2 - 8\theta)N - \mathfrak{M} - o(N) - O_B(\ell^{-B}(H\mathfrak{M})^{1/2}), \\ N_d &\geq \tfrac{1}{2}((3 - 8\theta)N - \mathfrak{M}) - o(N) - O_B(\ell^{-B}(H\mathfrak{M})^{1/2}). \end{aligned}$$

The proof occupies §4. Combined with the upper bound  $\mathfrak{M} \leq (1 - 2\theta)\mathcal{Q}_F(f_v)N + o(N)$  of §5, it gives  $N_0^s \geq ((1 - 2\theta)(4 - \mathcal{Q}_F(f_v)) - 2 - o(1))N$ .

#### 4. THE ZERO SIDE

This section follows the zero-side analysis of [22, §§3,5,6,9,10] for one prime modulus, adapted to the weighted family; all proofs are given here or in Appendix B. Two things change. The

pointwise bound for  $\sum_{\chi} \omega_{\chi} \nu_{\chi}(t)^2$  comes from the multiplicative large sieve instead of orthogonality modulo one prime, and exceptional characters are removed with Montgomery's zero-density theorem instead of the Hiary–Zhao estimate. Because  $T \geq \ell^{a_0} \rightarrow \infty$ , the finer inputs of [22] needed near  $T \asymp (\log \log q)^{1+\eta}/\log q$  (Selberg's averaged argument bound and Gevrey windows) are not needed.

#### 4.1. Weighted Riemann–von Mangoldt.

**Lemma 4.1.** *Uniformly for  $\ell^{a_0} \leq T \leq \ell^{A_0}$ ,*

$$N = \frac{HT\ell}{2\pi} \left( 1 + O\left( \frac{\log T + \log(1/\eta)}{\ell} + \frac{1}{T} \right) \right).$$

*Proof.* For primitive  $\chi \bmod q$  and  $t \geq 2$  let  $N^{\text{sym}}(t, \chi)$  count the nontrivial zeros with  $|\gamma| \leq t$ . Then  $N^{\text{sym}}(t, \chi) = \frac{t}{\pi} \log \frac{qt}{2\pi e} + O(\log q(t+2))$  [13, Ch. 16], [4, Thm. 1.1]; the error term absorbs the jumps, so no endpoint convention is needed. Complex conjugation maps zeros of  $L(s, \chi)$  with  $\gamma \in [-2T, -T)$  bijectively (with multiplicity) to zeros of  $L(s, \bar{\chi})$  with  $\gamma \in (T, 2T]$ . Since  $\mathcal{F}$  is closed under conjugation and  $\omega_{\bar{\chi}} = \omega_{\chi}$ ,

$$N = \frac{1}{2} \sum_{\chi} \omega_{\chi} (N^{\text{sym}}(2T, \chi) - N^{\text{sym}}(T, \chi)) = \frac{1}{2} \sum_{\chi} \omega_{\chi} \left( \frac{T}{\pi} \log \frac{qT}{2\pi e} + \frac{2T \log 2}{\pi} + O(\log qT) \right),$$

where the first equality is exact, since  $N^{\text{sym}}(2T, \chi) - N^{\text{sym}}(T, \chi)$  counts the zeros with  $T < |\gamma| \leq 2T$  (the symmetrisation follows [22, Lemmas 3.1, 3.2]). Since  $\log q = \ell + O(\log(1/\eta))$  on  $\mathcal{F}$ , the claim follows.  $\square$

#### 4.2. The family pointwise bound and finite centres.

**Lemma 4.2.** *Let  $\mathfrak{F}(t)^2 := \sum_{\chi \in \mathcal{F}} \omega_{\chi} \nu_{\chi}(t)^2$ . Then  $\mathfrak{F}(t) \ll_w H^{1/2}(\ell + \log(|t| + 2))$  for all real  $t$ .*

*Proof.* By the digamma asymptotic,  $|\mu_{\chi}(t)| \ll \ell + \log(|t| + 2)$ . Since  $|P_{\chi}| \leq |S_{\chi}[a]|/\pi$  and  $\omega(q) \leq w_{\max} q/\varphi(q)$ , the multiplicative large sieve [27], [23, Thm. 7.13],

$$(4.1) \quad \sum_{q \leq Q} \frac{q}{\varphi(q)} \sum_{\chi \bmod q}^* \left| \sum_{M < n \leq M+K} x_n \chi(n) \right|^2 \leq (Q^2 + K) \|x\|^2,$$

gives  $\sum_{\chi} \omega_{\chi} |P_{\chi}(t)|^2 \leq \pi^{-2} w_{\max} (Q^2 + Y) \sum_n a_n^2 \ll w_{\max} Q^2 L^2$ , uniformly in  $t$ . As  $H \asymp_w Q^2$ , the claim follows.  $\square$

In particular  $\sum_{\chi} \omega_{\chi} |\nu_{\chi}(t) \nu_{\chi}(t')| \leq \mathfrak{F}(t) \mathfrak{F}(t')$  and  $\sum_{\chi} \omega_{\chi} |\nu_{\chi}(t)| \leq H^{1/2} \mathfrak{F}(t)$ .

We use the envelope  $\varpi_A(x) := C_A L(1 + L|x|)^{-A}$  of Lemma 2.3 ( $|p_k(t)| \leq \varpi_A(t - \tau_k)$  for real  $t$ ), and for measurable  $U \subset \mathbb{R}$  put  $\Xi_k(U) := \int_U \varpi_A(t - \tau_k) \mathfrak{F}(t) dt$ .

**Lemma 4.3** (Endpoint tails; cf. [22, Lemma 5.3]). *Let  $A > 5/2$ . Uniformly in  $T$  and in the lattice offset  $\tau_0$ ,*

$$\sum_{k \notin \mathbf{K}} \Xi_k(J)^2 + \sum_{k \in \mathbf{K}} (\Xi_k(\mathbb{R}) \Xi_k(J^c) + \Xi_k(J^c)^2) \ll_A H \ell^2,$$

and, with  $\mathcal{L}(t) := \ell + \log(|t| + 2)$ ,  $\int_J \sum_{k \notin \mathbf{K}} \varpi_A(t - \tau_k)^2 \mathcal{L}(t) dt + \int_{J^c} \sum_{k \in \mathbf{K}} \varpi_A(t - \tau_k)^2 \mathcal{L}(t) dt \ll_A L \ell$ .

*Proof.* By Lemma 4.2,  $\mathfrak{F}(t) \ll H^{1/2} \mathcal{L}(t)$ . For  $k \notin \mathbf{K}$  put  $D_k := L \text{dist}(\tau_k, J)$ , and for  $k \in \mathbf{K}$  put  $E_k := L \text{dist}(\tau_k, J^c)$ , the scaled distance to the nearer endpoint of  $J$ . For  $t \in J$  we have  $\mathcal{L}(t) \ll \ell$  (as  $T \leq \ell^{A_0}$ ) and, if  $k \notin \mathbf{K}$ ,  $L|t - \tau_k| \geq D_k$ ; the substitution  $x = L(t - \tau_k)$  gives  $\Xi_k(J) \ll H^{1/2} \ell \int_{|x| \geq D_k} (1 + |x|)^{-A} dx \ll H^{1/2} \ell (1 + D_k)^{1-A}$ . For  $k \in \mathbf{K}$  we have  $|\tau_k| \leq 2T$ , so  $\mathcal{L}(\tau_k + x/L) \leq \ell + \log(2 + 2T + |x|) \ll \ell(1 + |x|)^{1/2}$ ; hence  $\Xi_k(\mathbb{R}) \ll H^{1/2} \ell$  and, since  $t \in J^c$  forces  $L|t - \tau_k| \geq E_k$ ,  $\Xi_k(J^c) \ll H^{1/2} \ell (1 + E_k)^{3/2-A}$ . The scaled centres  $L\tau_k$  are  $2\pi$ -spaced, so

on each side of each endpoint of  $J$  the  $m$ -th nearest centre ( $m \geq 0$ ) is at scaled distance  $\geq 2\pi m$ ; hence for every  $\sigma > 1$

$$\sum_{k \notin K} (1 + D_k)^{-\sigma} + \sum_{k \in K} (1 + E_k)^{-\sigma} \leq 4 \sum_{m \geq 0} (1 + 2\pi m)^{-\sigma} \ll_{\sigma} 1.$$

The three sums in the first claim are therefore  $\ll H\ell^2$  times sums of this form with  $\sigma = 2A - 2$ ,  $A - \frac{3}{2}$  and  $2A - 3$ , all  $> 1$  for  $A > 5/2$ . The second claim follows in the same way from  $\varpi_A^2 \leq C_A^2 L^2 (1 + L|t - \tau_k|)^{-2A}$ , with exponents  $2A - 1$  and  $2A - \frac{3}{2}$ .  $\square$

**Proposition 4.4** (Finite-centre replacement; cf. [22, Props. 5.2, 5.4]). *Uniformly in  $T$ ,*

$$(4.2) \quad \sum_{\chi \in \mathcal{F}} \omega_{\chi} \operatorname{tr} \hat{G}_{\chi} = \sum_{\chi \in \mathcal{F}} \omega_{\chi} \int_J \nu_{\chi}(t) dt + O(H),$$

$$(4.3) \quad \sum_{\chi \in \mathcal{F}} \omega_{\chi} \operatorname{tr} ((G_{\chi}/L)^2) = \sum_{\chi \in \mathcal{F}} \omega_{\chi} \iint_{J^2} \Phi(t - t')^2 \nu_{\chi}(t) \nu_{\chi}(t') dt dt' + O(H\ell^2).$$

*Proof.* Take  $A = 3$  in Lemmas 2.3 and 4.3. By Lemma 2.6,  $\operatorname{tr} G_{\chi} = \int (\sum_{k \in K} p_k^2) \nu_{\chi}$  (a finite sum of absolutely convergent integrals). By Lemma 2.4,  $\sum_{k \in K} p_k(t)^2 = aL^2 \mathbf{1}_J(t) + e_J(t)$  with  $e_J(t) := -\mathbf{1}_J(t) \sum_{k \notin K} p_k(t)^2 + \mathbf{1}_{J^c}(t) \sum_{k \in K} p_k(t)^2$ , and  $|e_J(t)|$  is at most the integrand of the second claim of Lemma 4.3 divided by  $\mathcal{L}(t)$ . By Lemma 4.2 and Cauchy–Schwarz,  $\sum_{\chi} \omega_{\chi} |\nu_{\chi}(t)| \leq H^{1/2} \mathfrak{F}(t) \ll H\mathcal{L}(t)$ . Hence  $\sum_{\chi} \omega_{\chi} |\int e_J \nu_{\chi}| \ll H\ell$ , and (4.2) follows after division by  $aL^2$ .

For (4.3):  $G_{\chi}$  is real symmetric, so, by finite-dimensional Fubini,

$$\begin{aligned} \operatorname{tr} ((G_{\chi}/L)^2) &= L^{-2} \sum_{k, l \in K} G_{\chi, kl}^2 = L^{-2} \iint K_d(t, t')^2 \nu_{\chi}(t) \nu_{\chi}(t') dt dt', \\ K_d(t, t') &:= \sum_{k \in K} p_k(t) p_k(t'). \end{aligned}$$

Let  $K_{\infty}(t, t') := \sum_{k \in \mathbb{Z}} p_k(t) p_k(t') = L\Phi(t - t')$  (Lemma 2.4). By Cauchy–Schwarz and Lemma 2.4,  $|K_d|, |K_{\infty}| \leq aL^2$  on  $\mathbb{R}^2$ . Hence on  $J^2$

$$|K_d^2 - L^2 \Phi(t - t')^2| = |K_{\infty} - K_d| |K_{\infty} + K_d| \leq 2aL^2 \sum_{k \notin K} \varpi_A(t - \tau_k) \varpi_A(t' - \tau_k),$$

and off  $J^2$ ,  $K_d^2 \leq aL^2 \sum_{k \in K} \varpi_A(t - \tau_k) \varpi_A(t' - \tau_k)$ . Multiply by  $\sum_{\chi} \omega_{\chi} |\nu_{\chi}(t) \nu_{\chi}(t')| \leq \mathfrak{F}(t) \mathfrak{F}(t')$  and integrate: since  $\iint_{(J^2)^c} f(t) f(t') = \Xi_k(\mathbb{R})^2 - \Xi_k(J)^2$  for  $f = \varpi_A(\cdot - \tau_k) \mathfrak{F}$ , the difference of the two sides of (4.3) is at most

$$2a \sum_{k \notin K} \Xi_k(J)^2 + a \sum_{k \in K} (\Xi_k(\mathbb{R})^2 - \Xi_k(J)^2) \leq 2a \sum_{k \notin K} \Xi_k(J)^2 + 2a \sum_{k \in K} (\Xi_k(\mathbb{R}) \Xi_k(J^c) + \Xi_k(J^c)^2),$$

which is  $O(H\ell^2)$  by Lemma 4.3.  $\square$

### 4.3. The trace.

**Lemma 4.5** (Family first moment). *For  $n \geq 2$  let  $\sigma_q(n) := \sum_{\chi \bmod q}^* \chi(n)$ . Then  $\sum_q \omega(q) |\sigma_q(\pm n)| \ll w_{\max} Q \tau(n \mp 1)$ .*

*Proof.* We have  $\sigma_q(n) = \mathbf{1}_{(n, q)=1} \sum_{d|(q, n-1)} \varphi(d) \mu(q/d)$  [23, (3.9)]. Hence  $\sum_q \omega(q) |\sigma_q(n)| \leq w_{\max} \sum_{d|n-1} \varphi(d) \sum_{e \leq Q/d} \frac{de}{\varphi(de)} \leq w_{\max} \sum_{d|n-1} d \sum_{e \leq Q/d} \frac{e}{\varphi(e)} \ll w_{\max} Q \tau(n-1)$ . For  $-n$  replace  $n-1$  by  $n+1$ .  $\square$

**Proposition 4.6** (Trace). *Uniformly in  $T$ ,  $\sum_{\chi \in \mathcal{F}} \omega_{\chi} \operatorname{tr} \hat{G}_{\chi} = (1 - 2\theta)N + o(N)$ .*

*Proof.* By Stirling,  $\mu_\chi(t) = \frac{1}{2\pi} \log \frac{qt}{2\pi} + O(1/t)$  for  $t \geq 1$ , so  $\sum_\chi \omega_\chi \int_J \mu_\chi = \frac{H|J|\ell}{2\pi} (1 + O(\frac{\log T + \log(1/\eta)}{\ell}))$ , which is  $(1 - 2\theta)N(1 + o(1))$  by Lemma 4.1. (The terms  $\log T$  do not cancel between the trace and  $N$ ; they are  $o(\ell)$  because  $\log T \leq A_0 \log \ell$ . Equivalently, the effective bandwidth in units of the analytic conductor is  $\lambda(1 + o(1))$ .) The prime part is  $-\frac{1}{2\pi} \sum_n a_n \sum_\chi \omega_\chi (\chi(n) \int_J n^{-it} dt + \overline{\chi}(n) \int_J n^{it} dt)$ . Since  $|\int_J n^{\mp it} dt| \leq 2/\log n$ , Lemma 4.5 bounds it by

$$\ll w_{\max} Q \sum_{2 \leq n \leq Y} \frac{\Lambda(n) \tau(n-1)}{\sqrt{n} \log n} \ll w_{\max} Q Y^{1/2} \log Y,$$

which is  $O(N \cdot Y^{1/2} \log Y / (QT\ell)) = o(N)$ . Finally apply (4.2); its error  $O(H)$  is  $O(N/(T\ell))$ .  $\square$

**4.4. Exterior zeros.** Put  $\delta_0 := B_1 \log \ell / \ell$  with  $B_1 := 6(B_0 + 52)/13$ , where  $B_0$  is the absolute exponent in (4.4) below. Call  $\chi \in \mathcal{F}$  *bad* if  $L(s, \chi)$  has a zero  $\rho = 1/2 + \delta + i\gamma$  with  $|\delta| \geq \delta_0$  and  $|\gamma| \leq Q^{|\delta|/2}$ , and *good* otherwise. The condition is invariant under  $\rho \mapsto 1 - \bar{\rho}$ , and it involves only  $Q$ : the set of bad characters in  $\mathcal{F}$  does not depend on  $T$ , nor on  $\lambda, \theta, \psi, \varepsilon, \varepsilon_3$ . A real exceptional zero makes its character bad.

We use Montgomery's zero-density theorem for the family of all primitive characters of conductor at most  $Q$  [26, Thm. 1]: for  $Q \geq 1$ ,  $T' \geq 2$ ,

$$(4.4) \quad \sum_{q \leq Q} \sum_{\chi \bmod q}^* N(\sigma, T', \chi) \ll \begin{cases} (Q^2 T')^{3(1-\sigma)/(2-\sigma)} (\log QT')^{B_0}, & \frac{1}{2} \leq \sigma \leq \frac{4}{5}, \\ (Q^2 T')^{2(1-\sigma)/\sigma} (\log QT')^{B_0}, & \frac{4}{5} \leq \sigma \leq 1, \end{cases}$$

where  $N(\sigma, T', \chi)$  counts the zeros of  $L(s, \chi)$  with  $\beta \geq \sigma$  and  $|\gamma| \leq T'$ , and  $B_0$  is an absolute constant ([26] gives explicit powers of  $\log QT'$ , which may differ in the two ranges; we take  $B_0$  to be the larger, and its value plays no role). Only the uniform consequence (4.5) is used; any bound  $(Q^2 T')^{1-c\delta} (\log QT')^{C_0}$  with  $c > 0$  would do after adjusting  $B_1$  and replacing the height  $Q^{|\delta|/2}$  in the definition of bad characters by  $Q^{a|\delta|}$  with  $0 < a < 2c$  (Proposition 4.8 only uses  $e^{L|\delta|} < |\gamma|^{2/a}$  for good  $\chi$ ). For  $\sigma = \frac{1}{2} + \delta$  both exponents are at most  $1 - \frac{4}{3}\delta$ : in the first range  $3(\frac{1}{2} - \delta)/(\frac{3}{2} - \delta) = 1 - 2\delta/(\frac{3}{2} - \delta) \leq 1 - \frac{4}{3}\delta$ , and in the second range ( $\delta \geq \frac{3}{10}$ ) the inequality  $(1 - 2\delta)/(\frac{1}{2} + \delta) \leq 1 - \frac{4}{3}\delta$  is equivalent to  $8\delta^2 - 14\delta + 3 \leq 0$ , i.e. to  $\frac{1}{4} \leq \delta \leq \frac{3}{2}$ , which holds since  $[\frac{3}{10}, \frac{1}{2}] \subset [\frac{1}{4}, \frac{3}{2}]$ . Hence, uniformly for  $0 \leq \delta \leq \frac{1}{2}$ ,

$$(4.5) \quad \sum_{q \leq Q} \sum_{\chi \bmod q}^* N(\frac{1}{2} + \delta, T', \chi) \ll (Q^2 T')^{1-4\delta/3} (\log QT')^{B_0}.$$

**Lemma 4.7** (Exceptional characters).  $\#\{\chi \in \mathcal{F} \text{ bad}\} \ll Q^2 \ell^{-52}$ , and  $\sum_{\chi \text{ bad}} \omega_\chi \ll_w Q^2 \ell^{-51}$ .

*Proof.* By the symmetry  $\rho \mapsto 1 - \bar{\rho}$  we may take  $\delta \geq \delta_0$ . Put  $\delta_j = \delta_0 + j/\ell$  ( $j \geq 0$ ) and  $T_j = Q^{\delta_{j+1}/2} \geq \ell^{B_1/2} \geq 2$ . A bad character with a witnessing zero with  $\delta \in [\delta_j, \delta_{j+1})$  has a zero with  $\beta \geq \frac{1}{2} + \delta_j$  and  $|\gamma| \leq Q^{\delta/2} \leq T_j$ , so it is counted in  $\sum_{q \leq Q} \sum_{\chi}^* N(\frac{1}{2} + \delta_j, T_j, \chi)$ . Only shells with  $\delta_j < \frac{1}{2}$  occur, since  $\beta < 1$  for every zero, so (4.5) applies. Since  $Q^2 T_j = e^{1/2} Q^{2+\delta_j/2}$ ,  $\log QT_j \leq 2\ell$  and  $(2 + \frac{\delta_j}{2})(1 - \frac{4}{3}\delta_j) \leq 2 - \frac{13}{6}\delta_j$ , (4.5) bounds this by  $\ll Q^{2-13\delta_j/6} \ell^{B_0}$ . Summing over  $j \geq 0$  gives  $\ll Q^2 \ell^{B_0} Q^{-13\delta_0/6} \sum_j e^{-13j/6} \ll Q^2 \ell^{B_0-13B_1/6} = Q^2 \ell^{-52}$ . The second claim follows from  $\omega_\chi \ll w_{\max} \log \log Q$ .  $\square$

**Proposition 4.8** (Exterior tail for good characters). *For every  $B > 0$ , uniformly for good  $\chi \in \mathcal{F}$  and  $\ell^{a_0} \leq T \leq \ell^{A_0}$ ,  $\|\hat{E}_\chi\|_1 \ll_B \ell^{-B}$ .*

*Proof.* Since  $\|xy^*\|_1 = \|x\| \|y\|$ , we have  $\|E_\chi\|_1 \leq \sum_{\gamma \notin I} m_\rho \|u(z_\rho)\| \|u(\bar{z}_\rho)\|$ , the sum running over all zeros of  $L(s, \chi)$  with ordinate outside  $I$ . For a good  $\chi$ , a zero with  $|\delta_\rho| < \delta_0$  has

$e^{L|\delta_\rho|} \leq Q^{2\delta_0} = \ell^{2B_1}$ , and a zero with  $|\delta_\rho| \geq \delta_0$  has  $|\gamma| > Q^{|\delta_\rho|/2}$ , so that  $e^{L|\delta_\rho|} \leq Q^{2|\delta_\rho|} < |\gamma|^4$ . By Lemma 2.3, for any  $A > \frac{1}{2}$ ,

$$\|\widehat{E}_\chi\|_1 \ll_A \ell^{2B_1} \sum_{\gamma \notin I} m_\rho (1 + |\gamma|)^4 (1 + L \operatorname{dist}(\gamma, J))^{1-2A}.$$

For  $\gamma \notin I$  we have  $\operatorname{dist}(\gamma, J) \geq \theta T$ , and  $\operatorname{dist}(\gamma, J) \geq |\gamma|/2$  if  $|\gamma| > 4T + 1$ . The zeros with  $|\gamma| \leq 4T + 1$  number  $\ll T\ell$  and contribute  $\ll \ell^{2B_1} T\ell T^4 (\theta TL)^{1-2A}$ . The zeros with  $|\gamma| > 4T + 1$  contribute  $\ll \ell^{2B_1} \sum_{n \geq 4T} (\ell + \log(n+3))(n+1)^4 (1 + Ln/2)^{1-2A}$ . Since  $T \geq \ell^{a_0}$ , both are  $O(\ell^{-B})$  once  $A = A(B, B_1, a_0)$  is large.  $\square$

#### 4.5. Deletion and assembly of the zero side.

*Proof of Proposition 3.6.* For every  $\chi$  and every eigenvalue  $x$  of the real symmetric matrix  $\widehat{G}_\chi$ ,  $4x - x^2 \leq 4$ ; hence  $4 \operatorname{tr} \widehat{G}_\chi - \|\widehat{G}_\chi\|_F^2 \leq 4d$ . For good  $\chi$ , write  $\widehat{A}_\chi = \widehat{G}_\chi - \widehat{E}_\chi$ . Then  $|\operatorname{tr} \widehat{A}_\chi - \operatorname{tr} \widehat{G}_\chi| \leq \|\widehat{E}_\chi\|_1$  and  $|\|\widehat{A}_\chi\|_F^2 - \|\widehat{G}_\chi\|_F^2| \leq 2\|\widehat{G}_\chi\|_F \|\widehat{E}_\chi\|_1 + \|\widehat{E}_\chi\|_1^2$ . Using Proposition 3.3 for good  $\chi$  and  $N_{0,\chi}^s \geq 0$  for bad  $\chi$ ,

$$\begin{aligned} N_0^s &\geq \sum_{\chi \text{ good}} \omega_\chi (4 \operatorname{tr} \widehat{A}_\chi - \|\widehat{A}_\chi\|_F^2 - 2N_\chi) \\ &\geq \sum_{\chi \in \mathcal{F}} \omega_\chi (4 \operatorname{tr} \widehat{G}_\chi - \|\widehat{G}_\chi\|_F^2 - 2N_\chi) - 4d \sum_{\chi \text{ bad}} \omega_\chi - \sum_{\chi \text{ good}} \omega_\chi (4\|\widehat{E}_\chi\|_1 + 2\|\widehat{G}_\chi\|_F \|\widehat{E}_\chi\|_1 + \|\widehat{E}_\chi\|_1^2). \end{aligned}$$

By Lemma 4.7,  $4d \sum_{\chi \text{ bad}} \omega_\chi \ll T\ell \cdot Q^2 \ell^{-51} = o(N)$ . By Proposition 4.8 and Cauchy–Schwarz, the last sum is  $\ll \ell^{-B} (H + (H\mathfrak{M})^{1/2})$ . By Proposition 4.6 the first sum is  $4(1-2\theta)N - \mathfrak{M} - 2N + o(N)$ . The same argument applies to  $N_0^*$ , and to  $N_d$  with  $\frac{1}{2}(\mathcal{M}_\chi - N_\chi)$  and the cap  $2d$ .  $\square$

### 5. THE PRIME SIDE

**5.1. Decomposition.** By (4.3),  $\mathfrak{M} = (a^2 L^2)^{-1} (\mathcal{M} + O(H\ell^2))$  with  $\mathcal{M} := \sum_\chi \omega_\chi \iint_{J^2} \Phi(t - t')^2 \nu_\chi(t) \nu_\chi(t')$ . Expanding  $\nu_\chi = \mu_\chi + P_\chi$  and  $P_\chi P'_\chi$ ,

$$(5.1) \quad \mathcal{M} = M_{\mu\mu} + 2M_{\mu\Lambda} + M_{\Lambda\Lambda}^{\text{rat}} + M_{\Lambda\Lambda}^{\text{ss}},$$

where  $M_{\mu\mu} = \sum \omega_\chi \iint \Phi^2 \mu_\chi \mu'_\chi$ ,  $M_{\mu\Lambda} = \sum \omega_\chi \iint \Phi^2 \mu_\chi(t) P_\chi(t')$ ,

$$(5.2) \quad \begin{aligned} M_{\Lambda\Lambda}^{\text{rat}} &= \frac{1}{2\pi^2} \sum_{n,m} a_n a_m \Delta(n, m) \mathcal{K}(n, m), \\ M_{\Lambda\Lambda}^{\text{ss}} &= \frac{1}{2\pi^2} \operatorname{Re} \sum_\chi \omega_\chi \iint_{J^2} \Phi(t - t')^2 S_\chi[a](t) S_\chi[a](t') dt dt'. \end{aligned}$$

(The two ratio terms  $S\overline{S'}$  and  $\overline{S}S'$  are equal after  $t \leftrightarrow t'$ , since  $\Phi^2$  is even.)

#### 5.2. The archimedean term.

**Lemma 5.1.**  $M_{\mu\mu} = \frac{H|J|bL\ell^2}{2\pi} \left(1 + O\left(\frac{\log T + \log(1/\eta)}{\ell}\right)\right) + O(H\ell^2)$ .

*Proof.* On  $J$ ,  $\mu_\chi(t) = \ell/(2\pi) + O(\log T + \log(1/\eta))$  uniformly in  $\chi \in \mathcal{F}$ , by the digamma asymptotic. Moreover

$$\begin{aligned} \iint_{J^2} \Phi(t - t')^2 dt dt' &= \int_{\mathbb{R}} \Phi(r)^2 (|J| - |r|)_+ dr \\ &= |J| \int \Phi^2 - \int \Phi(r)^2 \min(|r|, |J|) dr = 2\pi bL|J| + O_\psi(1), \end{aligned}$$

since  $\int \Phi^2 = 2\pi \int \psi_L^4 = 2\pi bL$  and  $\int |r|\Phi(r)^2 dr = \int |s|\widehat{v}(s)^2 ds$  (because  $\Phi(r) = L\widehat{v}(Lr)$ ). As  $\Phi^2 \geq 0$  and  $\sum_\chi \omega_\chi = H$ , the claim follows.  $\square$

### 5.3. The mixed term.

**Lemma 5.2.**  $M_{\mu\Lambda} \ll w_{\max} Q L \ell Y^{1/2} \log Y$ . In particular  $M_{\mu\Lambda} = O(H|J|L\ell^2 \cdot Q^{-\varepsilon_1/2})$ .

*Proof.* Write  $\mu_\chi = A_q(t) + \chi(-1)B(t)$  with  $A_q, B$  independent of the character modulo  $q$ ; on  $J$ ,  $A_q \ll \ell$ ,  $B \ll 1$ ,  $A'_q, B' \ll 1/T$ . Then  $\sum_{\chi \bmod q}^* \mu_\chi(t)\chi(n) = A_q(t)\sigma_q(n) + B(t)\sigma_q(-n)$ . For  $m_q(t') := \int_J \Phi(t-t')^2 A_q(t) dt$  we have  $\sup |m_q| \ll L\ell$ , and, after one integration by parts in  $t$ ,  $\int_J |m'_q| \ll L\ell$ . Hence  $|\int_J m_q(t') n^{\mp it'} dt'| \ll L\ell / \log n$ , and similarly for  $B$ . By Lemma 4.5,

$$|M_{\mu\Lambda}| \ll L\ell \sum_{2 \leq n \leq Y} \frac{\Lambda(n)}{\sqrt{n} \log n} \sum_q \omega(q) (|\sigma_q(n)| + |\sigma_q(-n)|) \ll w_{\max} Q L \ell Y^{1/2} \log Y. \quad \square$$

*Remark 5.3.* The bound of Lemma 5.2 is  $o(H|J|L\ell^2)$  as soon as  $Y^{1/2} \log Y = o(QT\ell)$ , which holds even for  $Y = Q^2$  because  $T \rightarrow \infty$ . So the mixed term does *not* force  $\lambda < 2$ . The constraints that do are: (i) the interval length  $K \leq Q^{2-\varepsilon}$  in the large sieve bounds of §6 (it enters through the spoke count in Lemmas 6.5 and 6.20, and in Lemma 6.20 also through the isolated point); (ii) the length term of the large sieve, through  $\Lambda_{\text{mult}}(Y) \ll_w H$ , which needs  $Y \ll Q^2$  (for  $Y > Q^2$  see Lemma 9.13 and Corollary 9.14) and is used in Lemma 4.2 (hence in Proposition 4.4), in Lemma 5.4, and for the tails of the localisation in Propositions 5.12 and 6.24 and in the proof of Theorem 5.16. On intervals of length  $K \gg Q^2$  the form  $\Delta$  has eigenvalues  $\gg_w K$ , which is not  $O(H)$  (§8.1). The condition  $R \leq N^{1/2-\varepsilon_3}$  in Lemma 5.7 is automatic for  $N \leq 2Q^2$ ; see also Remark 5.8.

### 5.4. Same-sign terms.

**Lemma 5.4.**  $M_{\Lambda\Lambda}^{\text{ss}} \ll w_{\max}(Q^2 + Y)L^2$ . In particular  $M_{\Lambda\Lambda}^{\text{ss}} = O(H|J|L\ell^2/(T\ell))$ .

*Proof.* Substitute  $t = t' + r$ , let  $J_r = J \cap (J - r) = [\alpha_r, \beta_r]$ . For  $nm \geq 4$ ,

$$\int_{J_r} (nm)^{-it'} dt' = \frac{(nm)^{-i\beta_r} - (nm)^{-i\alpha_r}}{-i \log nm}, \quad \frac{1}{\log nm} = \int_0^\infty (nm)^{-\sigma} d\sigma.$$

Thus  $\sum_\chi \omega_\chi \int_{J_r} S_\chi[a](t' + r) S_\chi[a](t') dt'$  is a difference of two terms  $i \int_0^\infty \sum_\chi \omega_\chi \mathcal{A}_\chi(\sigma) \mathcal{B}_\chi(\sigma) d\sigma$ , where  $\mathcal{A}_\chi, \mathcal{B}_\chi$  are character sums with coefficients of modulus  $a_n n^{-\sigma}$  on  $[1, Y]$ . By Cauchy-Schwarz and (4.1),

$$\left| \sum_\chi \omega_\chi \mathcal{A}_\chi \mathcal{B}_\chi \right| \leq w_{\max}(Q^2 + Y) \sum_n a_n^2 n^{-2\sigma},$$

and  $\int_0^\infty \sum_n a_n^2 n^{-2\sigma} d\sigma = \sum_n a_n^2 / (2 \log n) \leq \log Y$ . Integrating against  $\Phi(r)^2$ , with  $\int \Phi^2 = 2\pi bL$ , gives the claim.  $\square$

This is the only place where the same-sign terms enter; they carry no factor  $T$ , which is one of the reasons why  $T \rightarrow \infty$  is used.

**5.5. Ratio terms: the Schur product.** Since  $\Delta \succeq 0$  and  $\mathcal{K} \succeq 0$ , for every real vector  $y$  supported on  $[1, Y]$  the Schur product theorem gives

$$(5.3) \quad \sum_{n,m} y_n y_m \Delta(n, m) \mathcal{K}(n, m) \leq \Lambda_{\text{mult}}(Y) \sum_n y_n^2 \mathcal{K}(n, n).$$

With  $y = a$  and the crude bound  $\Lambda_{\text{mult}}(Y) \leq w_{\max}(Q^2 + Y)$  from (4.1), this recovers  $M_{\Lambda\Lambda}^{\text{rat}} \ll H|J|L^3$ , which is of the order of the main terms but with a poor constant and the density  $\log n$  in place of  $\log Q$ . The rest of this section improves both.

**5.6. Flattening and time localisation.** We subtract from  $\Lambda$  a truncated divisor sum; such sums go back to Selberg [37] and were used by Goldston [18] and Goldston–Yıldırım [19],

$$\Lambda_R(n) = \sum_{r \leq R} \frac{\mu(r)}{\varphi(r)} c_r(n),$$

with  $c_r$  the Ramanujan sum, at a level  $R = R_j \leq N_j^{1-\varepsilon_3}/(QT)$  that is fixed on each dyadic block  $n \asymp N_j$ . (The same truncation serves as the approximation to  $\Lambda$  in Vaughan’s work on moments of primes in arithmetic progressions [40, 41].) Two facts make this useful. First, the subtracted vector  $a^\sharp$  is invisible to every character of  $\mathcal{F}$  at every height  $|t| \leq 3T$  (Lemma 5.6), because  $n \mapsto c_r(n)\chi(n)$  has mean zero and period  $[q, r] \leq QR_j \leq N_j^{1-\varepsilon_3}/T$ . Second,  $b = a - a^\sharp$  has mean square density at most  $(1 + o(1))\min(\alpha, 1 + 2\varepsilon_3)\ell$  at  $n = Q^\alpha$ , instead of  $\alpha\ell$  for  $a$  (Lemma 5.7); the asymptotics for  $\sum \Lambda_R^2$  and  $\sum \Lambda \Lambda_R$  go back to Graham [20] and Goldston [18]. The consequence is that  $a$  may be replaced by  $b$  in  $M_{\Lambda\Lambda}^{\text{rat}}$  at the cost  $O(Q^{-A})$ ; see (5.7).

Time localisation rests on the factorisation  $\mathcal{K}(n, m) = \int g(u) \widehat{\mathbf{1}}_J(u - \log n) \widehat{\mathbf{1}}_J(u - \log m) du$  with  $\widehat{\mathbf{1}}_J(\xi) = \int_J e^{it\xi} dt$  (Lemma 5.9). Since  $|\widehat{\mathbf{1}}_J(\xi)| \leq 2/|\xi|$ , after integration in  $u$  the  $n$  with  $|\log n - u| > T^{-1/2}$  contribute only a relative  $O(T^{-1/4})$ , so only large sieve constants for vectors supported on intervals of length  $\leq 5T^{-1/2}e^u$  enter (Lemma 5.11, Proposition 5.12). For  $K \leq Q^{1-\varepsilon}$  one has the trivial but decisive bound  $\Lambda_{\text{mult}}(K) \leq (1 + O(K \log K/Q))H$  (Lemma 5.10), because  $|\Delta(n, m)| \ll w_{\max} Q \tau(|n - m|)$  for  $n \neq m$ . The precise statements and proofs follow.

**5.7. Flattening: Lemma B.** Recall  $a_n = \Lambda(n)n^{-1/2}\Upsilon(n)$ , where  $\Upsilon(n) = \Upsilon_0(\log n/L)$  is smooth,  $\Upsilon = 1$  on  $[2, X]$  and  $\Upsilon(n) = 0$  for  $n > Y = X^{1+\varepsilon_1} \leq Q^{2-\varepsilon_1}$ . Fix a smooth partition of unity  $\sum_{j \geq 0} \psi_j = 1$  on  $[1, \infty)$ ,  $\psi_j \geq 0$ ,  $\text{supp } \psi_j \subset [N_j/2, 2N_j]$ ,  $N_j = 2^j$ , with  $|\psi_j^{(i)}(y)| \leq c_i N_j^{-i}$  for constants  $c_i$  independent of  $j$ . Only blocks with  $N_j \leq 2Y$  meet the support of  $a$ . Fix  $\varepsilon_3 \in (0, \frac{1}{4})$  and put

$$R_j := \lfloor N_j^{1-\varepsilon_3}/(QT) \rfloor, \quad \Lambda_R(n) := \sum_{r \leq R} \frac{\mu(r)}{\varphi(r)} c_r(n),$$

$$a_n^\sharp := \sum_{j: R_j \geq 1} \psi_j(n) \Upsilon(n) n^{-1/2} \Lambda_{R_j}(n), \quad b := a - a^\sharp,$$

where  $c_r(n) = \sum_{h \bmod r}^* e(hn/r)$  is Ramanujan’s sum. No subtraction is made in blocks with  $R_j = 0$ , in particular for  $N_j < QT$ . If  $R_j \geq 1$  then  $N_j \geq QT$ , and, since  $N_j \leq 2Y \leq 2Q^2$ ,

$$(5.4) \quad QR_j \leq N_j^{1-\varepsilon_3}/T, \quad R_j \leq N_j^{1/2-\varepsilon_3} \cdot \frac{N_j^{1/2}}{QT} \leq N_j^{1/2-\varepsilon_3} \quad (T \geq 2).$$

We write  $G(R) := \sum_{r \leq R} \mu^2(r)/\varphi(r)$ ,  $S_\chi[x](t) := \sum_n x_n \chi(n) n^{-it}$  and  $\|\beta\|$  for the distance from  $\beta \in \mathbb{R}$  to  $\mathbb{Z}$ . We use the elementary bounds  $|c_r(n)| \leq \varphi(r)$  and  $G(R) \leq \sum_{r \leq R} 1/\varphi(r) \leq 1.95(1 + \log R)$  ( $R \geq 1$ ; proof of Lemma 6.20).

The Poisson estimates in this subsection and in the proof of Proposition 6.24 all reduce to the following lemma.

**Lemma 5.5** (smooth sums). *Let  $N \geq 1$ ,  $\Theta \geq 1$ ,  $B_0 > 0$  and an integer  $i \geq 2$ . Let  $F : \mathbb{R} \rightarrow \mathbb{C}$  be smooth, supported in  $[N/2, 2N]$ , with  $|F^{(k)}(y)| \leq B_0 N^{-1/2-k} \Theta^k$  for  $0 \leq k \leq i$  and all  $y$ .*

- (a) *For  $\beta \in \mathbb{R} \setminus \mathbb{Z}$ :  $|\sum_{n \in \mathbb{Z}} F(n) e(n\beta)| \leq B_0 N^{1/2} (\Theta/(N\|\beta\|))^i$ .*
- (b) *If  $f : \mathbb{Z} \rightarrow \mathbb{C}$  has period  $M \geq 1$  and mean 0 over a period, then  $|\sum_n F(n) f(n)| \leq B_0 M \|f\|_\infty N^{1/2} (\Theta M/N)^i$ .*

$$(c) \quad \left| \sum_n F(n) - \int_{\mathbb{R}} F \right| \leq 2B_0 N^{1/2} (\Theta/N)^i.$$

*Proof.* Let  $\widehat{F}(\xi) := \int F(y) e(-\xi y) dy$ . Integrating by parts  $i$  times,  $|\widehat{F}(\xi)| \leq (2\pi|\xi|)^{-i} \int |F^{(i)}| \leq \frac{3}{2} B_0 N^{1/2} (\Theta/(2\pi N|\xi|))^i$ , since the support has length  $\frac{3}{2}N$ . As  $F \in C_c^\infty$ , Poisson summation gives  $\sum_n F(n) e(n\beta) = \sum_{m \in \mathbb{Z}} \widehat{F}(m - \beta)$ . (a) Put  $d := \|\beta\| \in (0, \frac{1}{2}]$ . The numbers  $|m - \beta|$ ,  $m \in \mathbb{Z}$ , are  $d + k$  and  $1 - d + k$  ( $k \geq 0$ ), so  $\sum_m |m - \beta|^{-i} \leq d^{-i} + 2^i + 2\zeta(2) \leq 3d^{-i}$ , because  $2^i \leq d^{-i}$  and  $2\zeta(2) < 4 \leq d^{-i}$ . Hence the sum is at most  $\frac{9}{2} (2\pi)^{-i} B_0 N^{1/2} (\Theta/(Nd))^i$ , and  $\frac{9}{2} (2\pi)^{-2} < 1$ . (b) Write  $f(n) = \sum_{h=1}^{M-1} \widehat{f}(h) e(hn/M)$  with  $\widehat{f}(h) = M^{-1} \sum_{a \bmod M} f(a) e(-ah/M)$ ; the term  $h = 0$  is the mean, which vanishes, and  $|\widehat{f}(h)| \leq \|f\|_\infty$ . Apply (a) with  $\beta = h/M$ ,  $\|h/M\| \geq 1/M$ . (c)  $\sum_n F(n) - \int F = \sum_{m \neq 0} \widehat{F}(m)$ , and  $\sum_{m \neq 0} |m|^{-i} \leq 2\zeta(2) < 4$ .  $\square$

For a block  $j$ , a real  $t$  and a real  $u$  put

$$(5.5) \quad F_{j,t}(y) := \psi_j(y) \Upsilon(y) y^{-1/2-it}, \quad F_{j,t,u}(y) := F_{j,t}(y) \rho\left(\frac{\log y - u}{\delta}\right),$$

with  $\rho$  and  $\delta = T^{-1/2}$  as in Lemma 5.11 and Proposition 6.24. By the Leibniz rule, since  $\frac{d^k}{dy^k} y^{-1/2-it} \ll_k (1 + |t|)^k N_j^{-1/2-k}$  on  $[N_j/2, 2N_j]$ ,  $\frac{d^k}{dy^k} \Upsilon_0(\log y/L) \ll_k N_j^{-k}$  ( $L \geq 1$ ) and  $\frac{d^k}{dy^k} \rho((\log y - u)/\delta) \ll_k \delta^{-k} N_j^{-k}$ , we have for  $|t| \leq 3T$  and all  $u$

$$(5.6) \quad |F_{j,t}^{(k)}(y)| + |F_{j,t,u}^{(k)}(y)| \leq B_k N_j^{-1/2-k} T^k \quad (k \geq 0),$$

with  $B_k$  nondecreasing in  $k$  and depending only on  $k$ ,  $(c_i)$ ,  $\Upsilon_0$  and  $\rho$  (we used  $\delta^{-1} = T^{1/2} \leq T$  and  $1 + |t| \leq 4T$ ; for  $F_{j,t}$  alone,  $\rho$  does not enter). The partition  $(\psi_j)$  and the cut-offs  $\Upsilon_0$  and  $\rho$  are part of the fixed data of §2.3.

**Lemma 5.6** (B1: invisibility). *For every  $\chi \in \mathcal{F}$ , every  $|t| \leq 3T$  and every  $A > 0$ :  $S_\chi[a^\sharp](t) \ll_{A, \varepsilon_3, (c_i), \Upsilon_0} Q^{-A}$ ; that is, the implied constant depends only on  $A$ ,  $\varepsilon_3$ , the derivative bounds  $(c_i)$  of the partition of unity and the cut-off  $\Upsilon_0$  (through the constants  $B_i$  of (5.6) for  $F_{j,t}$ , in which  $\rho$  does not enter).*

*Proof.* Let  $\chi$  be primitive mod  $q \in [\eta Q, Q]$ , fix  $j$  with  $R_j \geq 1$  and  $r \leq R_j$ , and let  $M := [q, r] \leq QR_j$ . Mean zero. The function  $f_r(n) := c_r(n) \chi(n) = \sum_{h \bmod r}^* e(hn/r) \chi(n)$  has period  $M$  and mean 0 for every  $r \geq 1$ : if  $r \nmid q$ , the mean  $m_h$  of  $\chi(n) e(hn/r)$  satisfies  $m_h = e(hq/r) m_h$  (shift  $n \mapsto n + q$ ) with  $e(hq/r) \neq 1$ ; if  $r \mid q$ ,  $r < q$ , then  $m_h = q^{-1} \tau(\chi, hq/r) = q^{-1} \bar{\chi}(hq/r) \tau(\chi) = 0$  as  $(hq/r, q) > 1$ ; if  $r = q$ , then  $\sum_h m_h = q^{-1} \tau(\chi) \sum_h^* \bar{\chi}(h) = 0$  as  $\chi$  is non-principal. In particular no condition such as  $R_j < \eta Q$  is needed. *Poisson.* By (5.6) and Lemma 5.5(b) with  $\Theta = T$ ,  $\|f_r\|_\infty \leq \varphi(r)$  and (5.4),

$$\left| \sum_n F_{j,t}(n) f_r(n) \right| \leq B_i Q R_j \varphi(r) N_j^{1/2} \left( \frac{T Q R_j}{N_j} \right)^i \leq B_i \varphi(r) N_j^{3/2-i\varepsilon_3}.$$

Since

$$S_\chi[a^\sharp](t) = \sum_{j: R_j \geq 1} \sum_{r \leq R_j} \frac{\mu(r)}{\varphi(r)} \sum_n F_{j,t}(n) f_r(n),$$

we get  $|S_\chi[a^\sharp](t)| \leq B_i \sum_{j: R_j \geq 1} R_j N_j^{3/2-i\varepsilon_3} \leq B_i (2 + \log_2 Y) (QT)^{5/2-i\varepsilon_3}$ , using  $R_j \leq N_j$  and  $N_j \geq QT$ . Take  $i \geq (A + 3)/\varepsilon_3$ .  $\square$

Consequently, with  $M_{\text{rat}} = M_{\Lambda\Lambda}^{\text{rat}}$  the ratio part of the second moment (§2) and  $\mathcal{K}(n, m) = \iint_{J^2} \Phi(t - t')^2 n^{-it} m^{it'} dt dt'$ ,

$$(5.7) \quad M_{\text{rat}} = \frac{1}{2\pi^2} \sum_{n,m} a_n a_m \Delta(n, m) \mathcal{K}(n, m) = \frac{1}{2\pi^2} \sum_{n,m} b_n b_m \Delta(n, m) \mathcal{K}(n, m) + O(Q^{-A}),$$

since the middle expression is  $\frac{1}{2\pi^2} \sum_{\chi} \omega_{\chi} \iint_{J^2} \Phi^2 S_{\chi}[a](t) \overline{S_{\chi}[a](t')}$ ,  $S_{\chi}[a] = S_{\chi}[b] + S_{\chi}[a^{\#}]$ ,  $|S_{\chi}[b](t)| \leq \sum_n |b_n| \ll Q^3$ ,  $\iint_{J^2} \Phi^2 \leq 2\pi b L |J|$  and  $H \ll Q^3$ .

For Lemma 5.7 we use the prime number theorem with the de la Vallée Poussin error term: there are absolute constants  $c_0 \in (0, 1]$  and  $C_0 \geq 1$  such that, for  $x \geq 2$  and  $\vartheta(x) = \sum_{p \leq x} \log p$ ,

$$(5.8) \quad |\vartheta(x) - x| \leq C_0 x e^{-c_0 \sqrt{\log x}}$$

([13, Ch. 18] proves this for  $\psi(x) = \sum_{n \leq x} \Lambda(n)$ , and  $0 \leq \psi(x) - \vartheta(x) \ll \sqrt{x} \log x$ ).

**Lemma 5.7** (B2: flattened norm). *Let  $h : \mathbb{R} \rightarrow [0, \infty)$  be smooth with  $\|h^{(k)}\|_{\infty} \leq C_k \|h\|_{\infty}$  for all  $k \geq 1$ . Then, uniformly in  $\ell^{a_0} \leq T \leq \ell^{A_0}$ ,*

$$\sum_n |b_n|^2 h(\log n) \leq \left(1 + \frac{2 \log T + 2}{\ell}\right) \int_{\mathbb{R}} \ell F_b(s/\ell) h(s) ds + O(\|h\|_{\infty} \ell), \quad F_b(\alpha) := \min(\alpha^+, 1 + \varepsilon_4),$$

with  $\varepsilon_4 = 2\varepsilon_3$ ; the implied constant depends only on  $(C_k)$ ,  $\varepsilon_3$ ,  $(c_i)$  and  $\Upsilon_0$ . More precisely, the proof bounds  $|b_n|^2$  by the convexity majorant  $\Upsilon(n)^2 n^{-1} \sum_j \psi_j(n) (\Lambda(n) - \Lambda_{R_j}(n))^2$  (with  $\Lambda_0 := 0$ ), whose local density at  $n = y$  (in the variable  $\log y$ ) is  $\Upsilon(y)^2$  times the  $\psi_j$ -weighted average  $\sum_j \psi_j(y) (\log y - G(R_j))$  of the block densities  $\log y - G(R_j)$  (with  $G(0) := 0$ ) of the overlapping blocks that contain  $y$ ; it is not claimed that the density of  $|b_n|^2$  is at most that of any single block. Since  $G(R) \geq \log(R+1)$ , the quantity  $\log(N/R)$  is only an upper bound for  $\log N - G(R)$ .

*Proof.* (1) *Reduction to blocks.* Since  $b_n = \sum_j \psi_j(n) \Upsilon(n) n^{-1/2} (\Lambda(n) - \Lambda_{R_j}(n))$  (with  $\Lambda_0 := 0$ ),  $\psi_j \geq 0$  and  $\sum_j \psi_j = 1$ , convexity gives  $|b_n|^2 h(\log n) \leq \sum_j F_j(n) (\Lambda(n) - \Lambda_{R_j}(n))^2$  with  $F_j(y) := \psi_j(y) \Upsilon(y)^2 h(\log y)/y$ . So it suffices to evaluate  $\sum_j F_j(n) (\Lambda(n) - \Lambda_{R_j}(n))^2$  for each  $j$  with  $N_j \leq 2Y$ . The function  $F_j$  is supported in  $[N_j/2, 2N_j]$ ,  $\|F_j\|_{\infty} \leq 2\|h\|_{\infty}/N_j$ , and  $\int |(F_j(y) \log y)'| dy \leq C'_1(1 + C_1)\|h\|_{\infty}(1 + \log N_j)/N_j$  with  $C'_1$  depending only on  $(c_i)$  and  $\Upsilon_0$ . Write  $\mathcal{R}_j := (1 + C_1)\|h\|_{\infty}(1 + \log N_j)e^{-c_0 \sqrt{\log(N_j/2)}}$  for  $N_j \geq 4$ .

(2) *The term  $\Lambda^2$  ( $N_j \geq 4$ ).* The prime powers  $p^k \leq 2N_j$  with  $k \geq 2$  number  $O(N_j^{1/2})$ , so they contribute  $O(\|h\|_{\infty} N_j^{-1/2} \log^2(2N_j))$ . For the primes, by Stieltjes integration by parts against  $d\vartheta(y) = dy + d(\vartheta(y) - y)$  (the boundary terms vanish) and (5.8),

$$\sum_p F_j(p) (\log p)^2 = \int F_j(y) \log y dy - \int (\vartheta(y) - y) d(F_j(y) \log y) = \int F_j(y) \log y dy + O(C_0 C'_1 \mathcal{R}_j).$$

(3) *The term  $\Lambda \Lambda_R$  ( $R := R_j \geq 1$ ).* For a prime  $p > R$  and  $r \leq R$  we have  $c_r(p) = \mu(r)$ , so  $\Lambda_R(p) = G(R)$ ; every  $p \geq N_j/2$  exceeds  $R$  by (5.4). For a prime power  $p^k$  and squarefree  $r$ ,  $c_r(p^k) = \mu(r) + \mathbf{1}[p \mid r] p \mu(r/p)$ , hence  $|\Lambda_R(p^k)| \leq G(R) + \frac{p}{p-1} G(R/p) \leq 3G(R) \ll \ell$ . By the same Stieltjes argument with  $d\vartheta$ ,

$$\sum_n F_j(n) \Lambda(n) \Lambda_R(n) = G(R) \int F_j + O(G(R) \mathcal{R}_j) + O(\ell \|h\|_{\infty} N_j^{-1/2} \log^2(2N_j)).$$

(4) *The term  $\Lambda_R^2$ .*  $\sum_n F_j \Lambda_R^2 = \sum_{r, r' \leq R} \frac{\mu(r) \mu(r')}{\varphi(r) \varphi(r')} \sum_n F_j(n) c_r(n) c_{r'}(n)$ . The product  $c_r c_{r'}$  has period  $M = [r, r'] \leq R^2 \leq N_j^{1-2\varepsilon_3}$  (by (5.4)) and mean  $\varphi(r) \mathbf{1}[r = r']$  over a period (orthogonality of Ramanujan sums). Apply Lemma 5.5(b) to  $f = c_r c_{r'} - \varphi(r) \mathbf{1}[r = r']$  (so  $\|f\|_{\infty} \leq 2\varphi(r) \varphi(r')$ )

and Lemma 5.5(c) to the mean, both with  $F = N_j^{1/2} F_j$  and  $\Theta = 1$  (so  $B_0 \ll_i (1 + C_1 + \dots + C_i) \|h\|_\infty$ ):

$$\sum_n F_j \Lambda_R^2 = G(R) \int F_j + O_i \left( \|h\|_\infty (R^4 N_j^{-2i\varepsilon_3} + G(R) N_j^{-i}) \right) = G(R) \int F_j + O(\|h\|_\infty N_j^{-1}),$$

taking  $i \geq 3/\varepsilon_3$ . This is the only place where  $R \leq N^{1/2-\varepsilon_3}$  is needed (step (3) only needs  $R < N_j/2$ ), and it holds automatically for  $N \leq 2Q^2$ . (These are the classical asymptotics for the Selberg–Goldston truncation [18, 19, 20]; only the prime number theorem is needed here, since  $c_r(p) = \mu(r)$ .)

(5) *Collecting.*  $\Sigma_j = \int F_j(y) (\log y - G(R_j)) dy + \mathcal{E}_j$  with

$$|\mathcal{E}_j| \ll (1 + G(R_j)) \mathcal{R}_j + \ell \|h\|_\infty N_j^{-1/2} \log^2(2N_j) + \|h\|_\infty N_j^{-1},$$

and, for the finitely many  $j$  with  $N_j < 4$ ,  $\Sigma_j \ll \|h\|_\infty$ . Since  $G(R_j) \ll \ell$ ,  $\log N_j = j \log 2$  and  $\sum_{j \geq 2} (1+j) e^{-c_0 \sqrt{(j-1) \log 2}} < \infty$ ,

$$(5.9) \quad \sum_j |\mathcal{E}_j| \ll \|h\|_\infty \ell.$$

(6) *The density.*  $G(R) = \sum_{r \leq R} \frac{\mu^2(r)}{r} \sum_{\text{rad}(m)|r} \frac{1}{m} \geq \sum_{n \leq R} \frac{1}{n} \geq \log(R+1)$ , so  $\log y - G(R_j) \leq \log y - \log R_j$  (an upper bound only). On  $\text{supp } \psi_j$ ,  $R_j \geq N_j^{1-\varepsilon_3}/(2QT)$  gives  $\log y - \log R_j \leq \ell + \varepsilon_3 \log N_j + \log T + 2 \log 2 \leq \ell(1+2\varepsilon_3) + \log T + 2$ , as  $\log N_j \leq \log(2Y) \leq 2\ell$ . Also  $\log y - G(R_j) \leq \log y$ . In blocks with  $R_j = 0$  the density is exactly  $\log y$  (step 2), and there  $N_j^{1-\varepsilon_3} < 2QT$ , so  $\log y \leq \log(2N_j) \leq (1-\varepsilon_3)^{-1}(\ell + \log T + \log 2) + \log 2 \leq \ell(1+2\varepsilon_3) + 2 \log T + 2$ . In all cases the density is at most  $\min(\log^+ y, \ell(1+2\varepsilon_3) + 2 \log T + 2) \leq (1 + \frac{2 \log T + 2}{\ell}) \ell F_b(\log y/\ell)$ . (7) *Summing over  $j$ .*  $\sum_j \int F_j(y) \ell F_b(\log y/\ell) dy \leq \int h(s) \ell F_b(s/\ell) ds$  (substitute  $y = e^s$ ;  $Y \leq 1$ ,  $\sum \psi_j = 1$ ), and the errors are bounded by (5.9).  $\square$

*Remark 5.8* (where  $\lambda < 2$  is used). The mixed term  $M_{\mu\Lambda}$  does *not* force  $\lambda < 2$ : its relative size is  $\ll X^{(1+\varepsilon_1)/2} \log X/(QT\ell)$ , which is  $O(1/T)$  even at  $X^{1+\varepsilon_1} = Q^2$ . The genuine constraints are (i) the window length  $K \leq Q^{2-\varepsilon}$  in Lemmas 6.5 and 6.20 (it enters through the spoke count, and in Lemma 6.20 also through the isolated point  $\Omega(v) + m(v/Q)$ ), and (ii) the  $N$ -term of the large sieve:  $\Lambda_{\text{mult}}(Y) \ll_w H$  (used for  $\mathfrak{F}(t)^2$  in the finite-centre step and for the tails of the localisation in Propositions 5.12 and 6.24) requires  $Y \ll Q^2$  (for  $Y > Q^2$  see Lemma 9.13 and Corollary 9.14). Condition (iii),  $R \leq N^{1/2-\varepsilon_3}$  in Lemma 5.7, is automatic for  $N \leq 2Q^2$ . No condition of the form  $R_j < \eta Q$  is needed anywhere (Lemma 5.6).

**5.8. The time-localised majorant.** Let  $\widehat{\mathbf{1}}_J(\xi) := \int_J e^{it\xi} dt$ , so  $|\widehat{\mathbf{1}}_J(\xi)| \leq \min(|J|, 2/|\xi|)$ . For  $y \in \mathbb{C}^\mathbb{Z}$  finitely supported and  $u \in \mathbb{R}$  put  $x_y(u)_n := y_n \widehat{\mathbf{1}}_J(u - \log n)$ . We write  $x^* \Delta x := \sum_{\chi \in \mathcal{F}} \omega_\chi |\sum_n x_n \chi(n)|^2 \geq 0$  for the family form;  $\Delta \succeq 0$ .

**Lemma 5.9** (M1: time–frequency factorisation).

$$\mathcal{K}(n, m) = \int_{\mathbb{R}} g(u) \widehat{\mathbf{1}}_J(u - \log n) \overline{\widehat{\mathbf{1}}_J(u - \log m)} du.$$

Hence  $\sum_{n,m} y_n \bar{y}_m \Delta(n, m) \mathcal{K}(n, m) = \int g(u) x_y(u)^* \Delta x_y(u) du$ , and  $\mathcal{K}(n, n) = \int g |\widehat{\mathbf{1}}_J(\cdot - \log n)|^2 = 2\pi |J| g(\log n) + O_v(1)$ .

*Proof.* Insert  $\Phi(t - t')^2 = \hat{g}(t - t') = \int g(u) e^{i(t-t')u} du$  into the definition of  $\mathcal{K}$  and integrate in  $t, t'$  first (Fubini:  $g$  is integrable with compact support,  $J$  is bounded). For the diagonal,  $\mathcal{K}(n, n) = \int \Phi(r)^2 n^{-ir} (|J| - |r|)_+ dr$ ,  $\int \Phi(r)^2 n^{-ir} dr = 2\pi g(\log n)$  and  $\int \Phi(r)^2 |r| dr = O_v(1)$ .  $\square$

**Lemma 5.10** (M2: short intervals). *For every interval  $I$  of  $K$  integers and  $x \in \mathbb{C}^I$ :  $x^* \Delta x \leq (H + 4w_{\max} Q K (1 + \log K)) \|x\|^2$ .*

*Proof.*  $\Delta(n, n) = \sum_q \omega(q) \varphi^*(q) \mathbf{1}[(n, q) = 1] \leq H$ . For  $n \neq m$ ,  $\Delta(n, m) = \sum_q \omega(q) \mathbf{1}[(nm, q) = 1] \sum_{d|(q, n-m)} \varphi(d) \mu(q/d)$  [23, (3.9)], so  $|\Delta(n, m)| \leq w_{\max} \sum_{d|n-m} \varphi(d) \sum_{j \leq Q/d} \frac{dj}{\varphi(dj)} \leq 1.95 w_{\max} Q \tau(|n - m|)$ , using  $\frac{dj}{\varphi(dj)} \leq \frac{d}{\varphi(d)} \frac{j}{\varphi(j)}$  and  $\sum_{j \leq x} \frac{j}{\varphi(j)} \leq \frac{\zeta(2)\zeta(3)}{\zeta(6)} x$ . Schur's test and  $\sum_{h < K} \tau(h) \leq K(1 + \log K)$  give the claim. No zero-free region or Möbius cancellation is used.  $\square$

**Lemma 5.11** (M3: localisation). *Let  $\delta \in (0, \frac{1}{4}]$ ,  $\kappa \in (0, 1]$ , and  $\rho \in C_c^\infty((-2, 2))$  with  $0 \leq \rho \leq 1$ ,  $\rho = 1$  on  $[-1, 1]$ . For  $u \in \mathbb{R}$  split  $x = x_y(u)$  as  $x^s + x^t$ ,  $x_n^s := x_n \rho((\log n - u)/\delta)$ . Then: (i)  $x^* \Delta x \leq (1 + \kappa) x^{s*} \Delta x^s + (1 + \kappa^{-1}) x^{t*} \Delta x^t$ ; (ii)  $x^s$  is supported in an interval of length  $\leq 5\delta e^u$ ; (iii)  $\int g(u) \|x^s(u)\|^2 c(u) du \leq \sum_n |y_n|^2 \mathcal{K}(n, n) \sup\{c(u) : |u - \log n| < 2\delta\}$  for every bounded measurable  $c \geq 0$ ; (iv)  $\int g(u) \|x^t(u)\|^2 du \leq 8bL \|y\|^2 / \delta$ .*

*Proof.* (i)  $\Delta \succeq 0$  and  $2|\operatorname{Re}\langle x^s, x^t \rangle_\Delta| \leq \kappa \|x^s\|_\Delta^2 + \kappa^{-1} \|x^t\|_\Delta^2$ . (ii)  $e^{2\delta} - e^{-2\delta} \leq 5\delta$ . (iii)  $\|x^s(u)\|^2 \leq \sum_n |y_n|^2 |\widehat{\mathbf{1}}_J(u - \log n)|^2 \mathbf{1}[|u - \log n| < 2\delta]$ , and  $\int g |\widehat{\mathbf{1}}_J(\cdot - \log n)|^2 = \mathcal{K}(n, n)$  (Lemma 5.9). (iv)  $|1 - \rho| \leq \mathbf{1}_{|\xi| > \delta}$ ,  $g \leq g(0) = bL$  and  $\int_{|\xi| > \delta} 4\xi^{-2} d\xi = 8/\delta$ .  $\square$

We record the norms of the three vectors that are localised. By Mertens' estimate  $\sum_{n \leq x} \Lambda(n)/n = \log x + O(1)$ ,  $\|a\|^2 \leq \sum_{n \leq Y} \Lambda(n)^2/n \leq \log Y (\log Y + O(1)) \ll L^2$ . By convexity (step (1) of the proof of Lemma 5.7) and step (4) of that proof with  $h \equiv 1$ ,  $\|a^\sharp\|^2 \leq \sum_{j: R_j \geq 1} (G(R_j) \int \psi_j \Upsilon^2 \frac{dy}{y} + O(N_j^{-1})) \ll L\ell$ . Hence

$$(5.10) \quad \|a\|^2 + \|a^\sharp\|^2 + \|b\|^2 \ll L^2.$$

**Proposition 5.12** (TI: time-integrated majorant, Gauss-transfer constant). *Let  $\delta := T^{-1/2}$ ,  $\kappa := T^{-1/4}$ . For  $\varepsilon \in (0, \frac{1}{3})$  let  $\tilde{C}_\varepsilon : \mathbb{R} \rightarrow [1, C_w]$  be smooth,  $= 1$  on  $(-\infty, 1 - 2\varepsilon]$  and  $= C_w$  on  $[1 - \frac{3}{2}\varepsilon, \infty)$ . Then, as  $Q \rightarrow \infty$  with all parameters fixed, uniformly in  $T$ ,*

$$\sum_{n, m} b_n b_m \Delta(n, m) \mathcal{K}(n, m) \leq (1 + o(1)) H \sum_n |b_n|^2 \tilde{C}_\varepsilon\left(\frac{\log n}{\ell}\right) \mathcal{K}(n, n) + O_w(T^{-1/4} H |J| L^2 \ell),$$

and consequently  $M_{\text{rat}} \leq (1 + o(1)) \frac{H|J|}{\pi} L^2 \ell \int_0^1 F_\varepsilon(\lambda s) (v * v)(s) ds + o(H|J|L^2 \ell)$  with  $F_\varepsilon(\alpha) := \tilde{C}_\varepsilon(\alpha) \min(\alpha, 1 + \varepsilon_4)$ . As  $\varepsilon, \varepsilon_4 \rightarrow 0$ ,  $F_\varepsilon \rightarrow F_{C_w}$  ( $\alpha$  on  $[0, 1]$ ,  $C_w$  on  $(1, 2]$ ) pointwise off  $\alpha = 1$  and boundedly, so Proposition 5.13 applies with  $F = F_{C_w}$  in the limit.

*Proof.* By (5.7) and Lemma 5.9 it suffices to bound  $\int g(u) x_b(u)^* \Delta x_b(u) du$ ; invisibility is used before localisation. Apply Lemma 5.11(i) at each  $u$ . If  $u \leq (1 - \varepsilon)\ell$ , the short part lies in an interval of at most  $5\delta e^u + 1 \leq 5\delta Q^{1-\varepsilon} + 1 \leq Q^{1-\varepsilon/2}$  integers (Lemma 5.11(ii)), and Lemma 5.10 gives the constant  $1 + 4w_{\max} H^{-1} Q^{2-\varepsilon/2} (1 + \ell) = 1 + o(1)$ , with no condition on  $T$ . If  $u > (1 - \varepsilon)\ell$ , the short part lies in an interval of at most  $5\delta X + 1 \leq Q^{2-\varepsilon_1}$  integers ( $e^u < X$  on  $\operatorname{supp} g$ ), and Corollary 6.8 (with  $\varepsilon_1$  in place of  $\varepsilon$ ) gives  $C_w + o(1)$  (the Gauss-transfer constant is used from  $(1 - \varepsilon)\ell$  on). If  $|u - \log n| < 2\delta$  and  $u > (1 - \varepsilon)\ell$  then  $\log n/\ell \geq 1 - \frac{3}{2}\varepsilon$  for  $Q$  large, so in both cases the constant is  $\leq (1 + o(1)) \tilde{C}_\varepsilon(\log n/\ell)$ , and Lemma 5.11(iii) gives the main term. The tails: by Lemma 5.11(iv),  $\Lambda_{\text{mult}}(Y) \leq w_{\max}(Q^2 + Y) \leq 2w_{\max} Q^2 \ll_w H$  ((4.1),  $Y \leq Q^2$ ) and (5.10), they contribute  $\ll_w (1 + \kappa^{-1}) H bL \|b\|^2 / \delta \ll T^{3/4} H L^3$ , which is  $O(T^{-1/4})$  relative to  $H|J|L^2 \ell$ . For the second display use  $\mathcal{K}(n, n) = 2\pi |J| g(\log n) + O(1)$  and Lemma 5.7 with the smooth weight  $h(s) = g(s) \tilde{C}_\varepsilon(s/\ell)$ , then substitute  $s = L\sigma$ ,  $g(L\sigma) = L(v * v)(\sigma)$ .  $\square$

### 5.9. Assembly of the prime side.

**Proposition 5.13** (Second moment with a profile). *Let  $c : \mathbb{R} \rightarrow [1, \infty)$  be smooth (only its values on  $[-2, 2]$  matter), and suppose that, uniformly in  $T$ ,*

$$(5.11) \quad \sum_{n,m} b_n b_m \Delta(n, m) \mathcal{K}(n, m) \leq (1 + o(1)) H \sum_n b_n^2 c\left(\frac{\log n}{\ell}\right) \mathcal{K}(n, n) + o(H|J|L^2\ell).$$

Put  $F(\alpha) = c(\alpha) \min(\alpha, 1 + \varepsilon_4)$  with  $\varepsilon_4$  as in Lemma 5.7, and  $f_v(x) = v(x/\lambda)/(\lambda a)$ . Then, uniformly in  $T$ ,

$$\mathfrak{M} \leq (1 - 2\theta) \mathcal{Q}_F(f_v) N + o(N), \quad \mathcal{Q}_F(f) := \int f^2 + \iint f(x)f(y)F(|x-y|) dx dy.$$

*Proof.* By (2.9),  $\mathcal{K}(n, n) \leq 2\pi|J|g(\log n) + O(1)$ , and  $\sum_n b_n^2 \ll L^2$ . Apply Lemma 5.7 with the smooth function  $h(y) = c(y/\ell)g(y) \geq 0$ , supported in  $|y| < L$ ; it satisfies  $\|h^{(i)}\|_\infty \ll_i \|h\|_\infty$ , as Lemma 5.7 requires, because  $g^{(i)} \ll_i L^{1-i}$  and  $c$  is a fixed smooth function:

$$\sum_n b_n^2 c\left(\frac{\log n}{\ell}\right) \mathcal{K}(n, n) \leq 2\pi|J|(1 + o(1)) \int_0^L \ell F_b(y/\ell) c(y/\ell) L(v * v)(y/L) dy + O(|J|L\ell + L^2).$$

Substituting  $y = Ls$  and using (5.7),

$$M_{\Lambda\Lambda}^{\text{rat}} \leq \frac{H|J|L^2\ell}{\pi} (1 + o(1)) \int_0^1 F(\lambda s)(v * v)(s) ds + o(H|J|L^2\ell).$$

Add Lemmas 5.1, 5.2 and 5.4:

$$\mathcal{M} \leq \frac{H|J|L\ell^2}{2\pi} \left( b + 2\lambda \int_0^1 F(\lambda s)(v * v)(s) ds \right) + o(H|J|L\ell^2).$$

Divide by  $a^2 L^2$  and use  $L = \lambda\ell$ ,  $|J| = (1 - 2\theta)T$  and  $N = (1 + o(1))HT\ell/(2\pi)$ :  $\mathfrak{M} \leq (1 - 2\theta)N a^{-2} (b/\lambda + 2 \int_0^1 F(\lambda s)(v * v)(s) ds) + o(N)$ . Finally,  $\int f_v^2 = b/(\lambda a^2)$  and  $\iint f_v(x)f_v(y)F(|x-y|) = a^{-2} \int (v * v)(r)F(\lambda|r|) dr = 2a^{-2} \int_0^1 (v * v)(s)F(\lambda s) ds$ , since  $v$  is even.  $\square$

*Remark 5.14* (Normalisation check). With  $F(\alpha) = \alpha$  and  $\lambda \leq 1$ ,  $\mathcal{Q}_F(f_v) = (b + \lambda^2 \mathcal{J}(v))/(\lambda a^2) =: 1/c_\lambda(v)$ , because  $\int_0^1 s(v * v)(s) ds = \frac{1}{2} \int_{\mathbb{R}} |s|(v * v)(s) ds = \frac{1}{2} \iint |x-y|v(x)v(y) dx dy = \mathcal{J}(v)/2$  ( $v$  is even and  $\text{supp } v * v \subset (-1, 1)$ ). This is exactly [22, Prop. 7.2] with  $q$  replaced by  $H$ , and the optimisation of [22, §11] returns 0.6725.

The hypothesis that isolates what the large sieve must supply is the following.

**Hypothesis 5.15** (LS( $C$ )). *For every  $\varepsilon > 0$ , uniformly in  $K \leq Q^{2-\varepsilon}$ ,  $\Lambda_{\text{mult}}(K) \leq (C + o(1))H$ . (Necessarily  $C \geq 1$ .)*

**Theorem 5.16** (Conditional form). *Let  $w$  be admissible and suppose LS( $C$ ) holds for the family  $\mathcal{F}(Q, w)$ . Then*

$$\liminf_{Q \rightarrow \infty} \inf_T \frac{N_0^s}{N} \geq p(C), \quad \liminf_{Q \rightarrow \infty} \inf_T \frac{N_0^*}{N} \geq p(C), \quad \liminf_{Q \rightarrow \infty} \inf_T \frac{N_d}{N} \geq \frac{1 + p(C)}{2}.$$

The proof is given in §7.3. Theorem 1.2 is the case  $C = C_G R_w$ , which is unconditional by §6.1. Theorem 1.1 does not go through LS( $C$ ) with  $C \rightarrow 1$ ; it uses the structured substitute of §6.2, which only concerns the vectors that actually occur.

## 6. THE TWO LARGE-SIEVE CONSTANTS

This section has two parts: the Gauss-sum transfer with a sharp Farey large sieve (§6.1), and the Toeplitz route (§6.2).

**Overview of §6.1.** For a primitive character  $\chi \bmod q$  the Gauss sum expresses  $\chi(n)$  through the additive characters  $e(bn/q)$ . Summing over primitive  $\chi$  and enlarging to all characters bounds the family form by the additive large sieve form of the weighted Farey measure  $\sum_q w(q/Q) \sum_{b \bmod q}^* \delta_{b/q}$  (Lemma 6.7). The factor  $q/\varphi(q)$  in  $\omega$  is chosen exactly so that the additive side carries the smooth weights  $w(q/Q)$ . The price is the ratio of masses  $W/H \rightarrow C_G$ : the transfer bounds the  $\varphi^*(q)$  primitive characters by all  $\varphi(q)$  reduced fractions with denominator  $q$ . The additive form is then bounded sharply on intervals of length  $K \leq Q^{2-\varepsilon}$  by a Fejér majorant, duality and Schur's test, and an  $\mathrm{SL}_2(\mathbb{Z})$  lattice count along the spokes through each rational (in the spirit of Boca–Cobeli–Zaharescu [5]); the constant is the essential supremum  $R_w$  of the Farey profile  $P_w$ , a weighted form of the Davenport–Wolke  $K\delta$  principle for Farey points (cf. [44, 2, 33, 6]; see also [36] for a recent arithmetic form of the weighted large sieve, obtained through Parseval's identity, and §1.3 for the comparison). Numerically  $R_w$  is also the limit of the top eigenvalue of the additive form (Appendix A), but only the upper bound is used. For log-wide weights  $R_w \rightarrow 1$  (Lemma 6.10).

## 6.1. A sharp weighted Farey large sieve on intervals.

Notation. Throughout this subsection  $e(x) := e^{2\pi i x}$ ,  $\mathbb{T} := \mathbb{R}/\mathbb{Z}$ , and  $\|\xi\|$  is the distance from  $\xi$  to the nearest integer. For a finitely supported  $x \in \mathbb{C}^{\mathbb{Z}}$  we write  $S_x(\theta) := \sum_n x_n e(n\theta)$  and  $\|x\|^2 = \sum_n |x_n|^2$ . The weight  $w : (0, 1] \rightarrow [0, \infty)$  is admissible (Definition 2.1): it has bounded variation,  $\mathrm{supp} w \subset [\eta, 1]$  and  $\int_0^1 u w(u) du > 0$ ; we extend it by 0 to  $\mathbb{R}$  and write  $w_{\max} := \sup w$  and  $V_w$  for the total variation of the extended function on  $\mathbb{R}$  (so the jumps at the end points of the support are included, and  $V_w \geq 2w_{\max}$ ). Put

$$I_w := \int_0^1 u w(u) du > 0, \quad c_w := \frac{6}{\pi^2} I_w, \quad \tilde{w}(u) := u^2 w(u).$$

The additive and multiplicative masses are

$$W := \sum_{q \geq 1} w(q/Q) \varphi(q), \quad H := \sum_{\chi \in \mathcal{F}} \omega_\chi = \sum_{q \geq 1} w(q/Q) \frac{q}{\varphi(q)} \varphi^*(q),$$

and the weighted Farey measure is  $\mathfrak{F}_w := \sum_{q \geq 1} w(q/Q) \sum_{b \bmod q}^* \delta_{b/q}$  on  $\mathbb{T}$ , of total mass  $W$ . The *Farey profile* of  $w$  is

$$(6.1) \quad P_w(t) := \frac{t^2}{c_w} \sum_{k \geq 1} \varphi(k) w(kt) = \frac{1}{c_w} \sum_{k \geq 1} \frac{\varphi(k)}{k^2} \tilde{w}(kt) \quad (t > 0),$$

a finite sum for each  $t$  ( $P_w(t) = 0$  for  $t > 1$ ), and

$$(6.2) \quad R_w := \operatorname{ess\,sup}_{t > 0} P_w(t) \quad (\text{with respect to Lebesgue measure}), \quad C_w := C_G R_w.$$

Since  $w \leq w_{\max}$  and  $\sum_{k \leq 1/t} \varphi(k) \leq t^{-2}$  ( $t \leq 1$ ),  $R_w \leq w_{\max}/c_w < \infty$ . Moreover  $c_w P_w(t) = t^2 \sum_k \varphi(k) w(kt)$  is  $t^2 W$  with  $Q = 1/t$ , so Lemma 6.1 below gives  $|P_w(t) - 1| \leq c_0(V_w/c_w) t \log(2/t)$  for  $0 < t \leq \frac{1}{2}$ ; hence  $P_w(t) \rightarrow 1$  as  $t \rightarrow 0^+$ , and  $R_w \geq 1$ . For  $0 < \varsigma < \frac{1}{2}$  let  $k_0(\xi) := \varsigma^{-1}(1 - |\xi|/\varsigma)_+$  on  $\mathbb{R}$  (so  $\int k_0 = 1$ ) and let  $k_\varsigma(\xi) := \sum_{m \in \mathbb{Z}} k_0(\xi + m)$  be its periodisation, a function on  $\mathbb{T}$ .

**Lemma 6.1** (the masses  $W$  and  $H$ ). *Let  $\mathcal{E} = \prod_p(1 - p^{-2} - p^{-3})$ . There is an absolute constant  $c_0$  such that for  $Q \geq 2$*

$$|W - c_w Q^2| \leq c_0 V_w Q \log(2Q), \quad |H - \mathcal{E} I_w Q^2| \leq c_0 V_w Q^{3/2}.$$

*In particular  $W/H = C_G + O(V_w I_w^{-1} Q^{-1/2})$  with  $C_G = 6/(\pi^2 \mathcal{E}) = 1.268773 \dots$  (The proof applies to every real  $Q \geq 2$ .)*

*Proof.*  $W$ . Put  $\Phi(x) := \sum_{q \leq x} \varphi(q) = \frac{3}{\pi^2} x^2 + r(x)$  with  $|r(x)| \leq c_1 x \log(2x)$  for  $x \geq 1$  (classical). Abel summation over  $1 \leq q \leq Q$  gives  $W = w(\frac{N}{Q}) \Phi(N) - \sum_{q < N} \Phi(q) (w(\frac{q+1}{Q}) - w(\frac{q}{Q}))$  with  $N := \lfloor Q \rfloor$ , and the same identity with  $\Phi$  replaced by  $\Phi_0(x) := \frac{3}{\pi^2} x^2$ . The difference of the two right-hand sides is at most  $\max_{x \leq Q} |r(x)| (w_{\max} + V_w) \leq 2c_1 V_w Q \log(2Q)$ . The  $\Phi_0$ -sum equals  $\sum_q w(q/Q) \frac{3}{\pi^2} (2q-1) = \frac{6}{\pi^2} \sum_q w(q/Q) q - \frac{3}{\pi^2} \sum_q w(q/Q)$ . For  $F(x) := x w(x/Q)$  on  $[0, Q]$  we have  $|\sum_{q \in \mathbb{Z}} F(q) - \int F| \leq \text{TV}(F) \leq Q(V_w + w_{\max})$  (the elementary inequality (6.7) below with  $D = 1$ ), and  $\int F = Q^2 I_w$ ; also  $\sum_q w(q/Q) \leq Q w_{\max} + V_w$ . Collecting,  $|W - c_w Q^2| \leq c_0 V_w Q \log(2Q)$ .

$H$ . Let  $\psi(q) := \varphi^*(q)/\varphi(q)$ . It is multiplicative with  $\psi(p) = (p-2)/(p-1)$  and  $\psi(p^k) = (p-1)/p$  ( $k \geq 2$ ), so  $\psi = 1 * g$  with  $g$  multiplicative,  $g(p) = -1/(p-1)$ ,  $g(p^2) = 1/(p(p-1))$ ,  $g(p^k) = 0$  ( $k \geq 3$ ). Hence  $B_g := \sum_d |g(d)| d^{-1/2} < \infty$  and  $\sum_d g(d)/d = \prod_p (1 - \frac{1}{p(p-1)} + \frac{1}{p^3(p-1)}) = \mathcal{E}$ . Then  $\sum_{q \leq x} \psi(q) = \sum_{d \leq x} g(d) \lfloor x/d \rfloor = \mathcal{E} x - x \sum_{d > x} g(d)/d + O(\sum_{d \leq x} |g(d)|) = \mathcal{E} x + O(B_g x^{1/2})$ , and by partial summation  $\sum_{q \leq x} q \psi(q) = \frac{\mathcal{E}}{2} x^2 + O(B_g x^{3/2})$ . Since  $H = \sum_q w(q/Q) q \psi(q)$ , the same Abel-summation argument as for  $W$  gives the claim.  $\square$

**Lemma 6.2** (band-limited duality). *Let  $\mu = \sum_s a_s \delta_{\theta_s}$  be a finite positive measure on  $\mathbb{T}$  ( $a_s > 0$ , the  $\theta_s$  distinct), let  $I \subset \mathbb{Z}$  be an interval containing  $K \geq 1$  integers, let  $0 < \kappa \leq \frac{1}{4}$  and  $\varsigma := \kappa/K$ . Then for every  $x \in \mathbb{C}^I$*

$$\sum_s a_s |S_x(\theta_s)|^2 \leq (1 + \kappa^2) \max_s (\mu * k_\varsigma)(\theta_s) \|x\|^2, \quad (\mu * k_\varsigma)(\theta) := \sum_{s'} a_{s'} k_\varsigma(\theta - \theta_{s'}).$$

*Proof.* *Step 1 (Fejér majorant).* Let  $F(y) := (\sin \pi y / \pi y)^2$ ;  $F$  is even,  $F(0) = 1$ ,  $F$  is decreasing on  $[0, 1]$ , and  $\int_{\mathbb{R}} F(y) e(y\xi) dy = (1 - |\xi|)_+$ . Let  $c \in \frac{1}{2}\mathbb{Z}$  be the midpoint of  $I$ , so  $|n - c| \leq K/2$  for  $n \in I$ , and put  $g(y) := F(\kappa(y - c)/K)/F(\kappa/2)$ . Then  $g \geq 0$  on  $\mathbb{R}$ ,  $g(n) \geq 1$  for  $n \in I$  (because  $\kappa|n - c|/K \leq \kappa/2 < 1$ ),  $g(y) \ll_{\kappa, K} (1 + |y|)^{-2}$ , and

$$\hat{g}(\xi) := \int_{\mathbb{R}} g(y) e(y\xi) dy = \frac{e(c\xi)}{F(\kappa/2)} k_0(\xi),$$

which is continuous and supported in  $[-\varsigma, \varsigma] \subset (-\frac{1}{2}, \frac{1}{2})$ . The 1-periodic function  $G(\xi) := \sum_{m \in \mathbb{Z}} \hat{g}(\xi + m)$  is continuous, its Fourier coefficients are  $\int_0^1 G(\xi) e(-n\xi) d\xi = \int_{\mathbb{R}} \hat{g}(\xi) e(-n\xi) d\xi = g(n)$  (Fourier inversion;  $\hat{g} \in L^1$  and  $g$  is continuous), and  $\sum_n |g(n)| < \infty$ . Hence (Poisson summation)

$$(6.3) \quad \sum_{n \in \mathbb{Z}} g(n) e(n\xi) = G(\xi), \quad |G(\xi)| = \frac{k_\varsigma(\xi)}{F(\kappa/2)} \quad (\xi \in \mathbb{T}),$$

the second identity because for each  $\xi$  at most one  $m$  contributes to  $G(\xi)$  (the support of  $\hat{g}$  has length  $2\varsigma < 1$ ). Thus  $G(\xi) = e(c\xi) k_\varsigma(\xi)/F(\kappa/2)$ , where  $\tilde{\xi} \in (-\frac{1}{2}, \frac{1}{2})$  is the representative of  $\xi$ ; only the modulus  $|G|$  is used below (Remark 6.3).

*Step 2 (duality and the Schur test on  $|G|$ ).* Let  $A : \mathbb{C}^I \rightarrow \mathbb{C}^{\{s\}}$ ,  $(Ax)_s := \sqrt{a_s} S_x(\theta_s)$ . The left-hand side of the lemma is  $\|Ax\|^2 \leq \|A\|^2 \|x\|^2$ , and  $\|A\| = \|A^*\|$  with  $(A^*c)_n = \sum_s c_s \sqrt{a_s} e(-n\theta_s)$

for  $n \in I$ . For any  $c$ , using  $g \geq 0$ ,  $g \geq 1$  on  $I$  and (6.3) (the double series converges absolutely),

$$\|A^*c\|^2 \leq \sum_{n \in \mathbb{Z}} g(n) \left| \sum_s c_s \sqrt{a_s} e(-n\theta_s) \right|^2 = \sum_{s, s'} c_s \bar{c}_{s'} \sqrt{a_s a_{s'}} G(\theta_{s'} - \theta_s) \leq \frac{1}{F(\kappa/2)} \sum_{s, s'} |c_s| |c_{s'}| B_{ss'},$$

where  $B_{ss'} := \sqrt{a_s a_{s'}} k_\zeta(\theta_s - \theta_{s'}) \geq 0$  is symmetric ( $k_\zeta$  is even). This is the entrywise-modulus domination  $\|M\| \leq \| |M| \|$  for the Hermitian matrix  $M_{ss'} = \sqrt{a_s a_{s'}} G(\theta_{s'} - \theta_s)$ . Schur's test with the positive weights  $p_s := \sqrt{a_s}$  (write  $|c_s| |c_{s'}| \leq \frac{1}{2}(|c_s|^2 p_{s'}/p_s + |c_{s'}|^2 p_s/p_{s'})$  and use symmetry) gives

$$\sum_{s, s'} |c_s| |c_{s'}| B_{ss'} \leq \|c\|^2 \max_s \frac{1}{p_s} \sum_{s'} B_{ss'} p_{s'} = \|c\|^2 \max_s (\mu * k_\zeta)(\theta_s).$$

Finally  $F(y) \geq 1 - \pi^2 y^2/3$ , so  $F(\kappa/2)^{-1} \leq (1 - \pi^2 \kappa^2/12)^{-1} \leq 1 + \kappa^2$  for  $\kappa \leq \frac{1}{4}$ .  $\square$

*Remark 6.3* (on the phases). It is tempting to say that the phases  $e(c \cdot)$  conjugate the kernel matrix unitarily, i.e. that  $G(\theta_{s'} - \theta_s) = e(c\tilde{\theta}_{s'}) e(c\tilde{\theta}_s) |G(\theta_{s'} - \theta_s)|$  for fixed lifts  $\tilde{\theta}_s$ . This fails under wrap-around when  $c \notin \mathbb{Z}$ : if  $\tilde{\theta}_{s'} - \tilde{\theta}_s$  differs from the representative of  $\theta_{s'} - \theta_s$  in  $(-\frac{1}{2}, \frac{1}{2}]$  by a nonzero integer  $j$ , the two sides differ by the factor  $e(cj) \neq 1$ . Lemma 6.2 avoids the issue by applying Schur's test to the entrywise modulus (Perron–Frobenius domination  $\|M\| \leq \| |M| \|$ ); nothing is lost, since the Schur bound only ever used  $|G|$ . (Alternatively one may take  $c \in \mathbb{Z}$  at the cost of replacing  $K$  by  $K + 1$ .)

**Proposition 6.4** (local count along spokes). *Let  $\mathfrak{w} : \mathbb{R} \rightarrow [0, \infty)$  be of bounded variation  $V_{\mathfrak{w}}$ , with  $\mathfrak{w} = 0$  outside  $(0, 1]$ , and let  $\mathfrak{m} := \sum_{q \geq 1} \mathfrak{w}(q/Q) \sum_{b \bmod q}^* \delta_{b/q}$ . Let  $Q \geq 1$ ,  $0 < \varsigma < \frac{1}{2}$ ,  $V \geq 1$  and  $\theta \in \mathbb{T}$ . By Dirichlet's theorem choose  $u \in \mathbb{Z}$ ,  $1 \leq v \leq V$ ,  $(u, v) = 1$  with  $|\theta - u/v| \leq 1/(vV)$  for a lift  $\theta \in \mathbb{R}$ , and put  $y := \theta - u/v$ . Then*

$$(6.4) \quad (\mathfrak{m} * k_\zeta)(\theta) = \mathfrak{w}(v/Q) k_0(y) + \int_{\mathbb{R}} k_0(z - y) \rho_{\mathfrak{w}, v}(z) dz + \varrho, \quad |\varrho| \leq 4V_{\mathfrak{w}} \varsigma^{-1} k_* (1 + \log^+ k_*),$$

where  $k_* := Q/V + QV\varsigma$  and, for  $z \neq 0$ ,

$$(6.5) \quad \rho_{\mathfrak{w}, v}(z) := \frac{1}{(vz)^2} \sum_{k \geq 1} \varphi(k) \mathfrak{w}\left(\frac{k}{v|z|Q}\right) = Q^2 t^2 \sum_{k \geq 1} \varphi(k) \mathfrak{w}(kt) \Big|_{t=1/(v|z|Q)}.$$

*Proof. Reduction to lattice points.* Since  $\varsigma < \frac{1}{2}$ , for each  $b \bmod q$  at most one lift of  $b/q$  lies within  $\varsigma$  of  $\theta$ , so  $(\mathfrak{m} * k_\zeta)(\theta) = \sum_{(b, q) \in \mathcal{P}} \mathfrak{w}(q/Q) k_0(b/q - \theta)$  with  $\mathcal{P} := \{(b, q) \in \mathbb{Z}^2 : q \geq 1, (b, q) = 1\}$  (here  $b$  runs over all of  $\mathbb{Z}$ ).

The  $\mathrm{SL}_2(\mathbb{Z})$  change of variables (the “spoke” parametrisation of Farey fractions near  $u/v$ , cf. [5]). Fix  $u', v' \in \mathbb{Z}$  with  $vu' - uv' = 1$ ; note  $(v', v) = 1$ . The map  $(b, q) \mapsto (k, m) := (vb - uq, -v'b + u'q)$  is a bijection of  $\mathbb{Z}^2$  with inverse  $(k, m) \mapsto (u'k + um, v'k + vm)$ , it preserves gcd, and  $b/q - u/v = k/(qv)$ . Thus

$$(6.6) \quad (\mathfrak{m} * k_\zeta)(\theta) = \sum_{k \in \mathbb{Z}} \sum_{\substack{m \in \mathbb{Z}, (m, k) = 1 \\ q := v'k + vm \geq 1}} \mathfrak{w}(q/Q) k_0\left(\frac{k}{qv} - y\right).$$

The isolated point  $k = 0$ .  $(0, m) = 1$  forces  $m = \pm 1$ , and  $q \geq 1$  leaves  $(b, q) = (u, v)$ : the contribution is  $\mathfrak{w}(v/Q) k_0(-y) = \mathfrak{w}(v/Q) k_0(y)$ .

The spokes  $k \neq 0$ . Put  $f_k(q) := \mathfrak{w}(q/Q) k_0(k/(qv) - y)$  for  $q > 0$  and  $f_k(q) := 0$  for  $q \leq 0$ . If  $f_k(q) \neq 0$  then  $0 < q \leq Q$  and  $|k| < qv(|y| + \varsigma) \leq Q/V + QV\varsigma \leq k_*$ , so only  $1 \leq |k| < k_*$  occur.

By Möbius inversion over  $d \mid k$  (only squarefree  $d$  contribute),

$$\sum_{m:(m,k)=1} f_k(v'k + vm) = \sum_{d \mid k} \mu(d) \sum_{j \in \mathbb{Z}} f_k(v'k + vdj).$$

For a function  $f$  of bounded variation with compact support and any  $a \in \mathbb{R}$ ,  $D > 0$ ,

$$(6.7) \quad \left| \sum_{j \in \mathbb{Z}} f(a + Dj) - \frac{1}{D} \int_{\mathbb{R}} f \right| \leq \text{TV}(f) :$$

indeed, with  $F(x) := f(a + Dx)$ , the left side is  $|\sum_j \int_j^{j+1} (F(j) - F(x)) dx| \leq \sum_j \text{Var}_{[j, j+1]} F = \text{TV}(F) = \text{TV}(f)$ . Since  $\sum_{d \mid k} \mu(d)/(vd) = \varphi(|k|)/(v|k|)$ , we get

$$\left| \sum_{m:(m,k)=1} f_k(v'k + vm) - \frac{\varphi(|k|)}{v|k|} \int_{\mathbb{R}} f_k \right| \leq 2^{\omega(k)} \text{TV}(f_k).$$

The map  $q \mapsto k/(qv) - y$  is monotone on  $q > 0$ , so  $\text{TV}(k_0(k/(\cdot)v - y)) \leq \text{TV}(k_0) = 2/\varsigma$ ; with  $\text{TV}(fg) \leq \|f\|_{\infty} \text{TV}(g) + \|g\|_{\infty} \text{TV}(f)$  and  $V_{\mathfrak{w}} \geq 2\|\mathfrak{w}\|_{\infty}$  (the variation of  $\mathfrak{w}$  on  $\mathbb{R}$  includes the rise from 0 and the return to 0 at the ends of its support, jumps included),  $\text{TV}(f_k) \leq (2\|\mathfrak{w}\|_{\infty} + V_{\mathfrak{w}})/\varsigma \leq 2V_{\mathfrak{w}}/\varsigma$ . Summing over  $1 \leq |k| < k_*$  and using  $\sum_{k \leq x} 2^{\omega(k)} \leq \sum_{k \leq x} \tau(k) \leq x(1 + \log x)$  ( $x \geq 1$ ) gives the bound on  $\varrho$ .

*Main term.* For  $k \geq 1$  substitute  $q = k/(vz)$ ,  $z > 0$  (so  $dq = -k dz/(vz^2)$ ), and for  $k \leq -1$  substitute  $q = |k|/(v|z|)$ ,  $z < 0$ . By Tonelli (all terms are  $\geq 0$ )

$$\sum_{k \neq 0} \frac{\varphi(|k|)}{v|k|} \int_0^{\infty} f_k(q) dq = \int_{\mathbb{R}} k_0(z - y) \sum_{k \geq 1} \frac{\varphi(k)}{v^2 z^2} \mathfrak{w}\left(\frac{k}{v|z|Q}\right) dz = \int_{\mathbb{R}} k_0(z - y) \rho_{\mathfrak{w},v}(z) dz. \quad \square$$

**Lemma 6.5** (Lemma A: sharp weighted Farey large sieve on intervals). *Fix  $\varepsilon \in (0, 1)$  and  $w$  as above. There is  $Q_0 = Q_0(\varepsilon, V_w/c_w)$  such that for  $Q \geq Q_0$ , for every interval  $I \subset \mathbb{Z}$  containing  $K \leq Q^{2-\varepsilon}$  integers and every  $x \in \mathbb{C}^I$ ,*

$$\sum_{q \geq 1} w(q/Q) \sum_{b \bmod q}^* |S_x(b/q)|^2 \leq (R_w + \vartheta_Q) W \|x\|^2, \quad \vartheta_Q := 30 \left( R_w + \frac{V_w}{c_w} \right) Q^{-\varepsilon/4} \log Q.$$

For the sharp log-wide weight  $w_{\eta} = u^{-2} \mathbf{1}_{[\eta, 1]}$  one has  $V_w = 2\eta^{-2}$  and  $c_w = \frac{6}{\pi^2} \log \frac{1}{\eta}$ , so  $\vartheta_Q \leq 30(R_w + 3.3 \eta^{-2} / \log \frac{1}{\eta}) Q^{-\varepsilon/4} \log Q$ .

*Proof.* Enlarging  $I$  only increases the left side, so we may assume  $Q \leq K \leq Q^{2-\varepsilon}$ . Take  $\kappa := Q^{-\varepsilon/4}$  ( $\leq \frac{1}{4}$  for  $Q \geq 4^{4/\varepsilon}$ ),  $\varsigma := \kappa/K \leq \kappa/Q < \frac{1}{2}$ , and  $V := Q^{1-\varepsilon/2}$ . By Lemma 6.2 with  $\mu = \mathfrak{F}_w$  the left side is at most  $(1 + \kappa^2) \sup_{\theta} (\mathfrak{F}_w * k_{\varsigma})(\theta) \|x\|^2$ . Apply Proposition 6.4 with  $\mathfrak{w} = w$ . By (6.5) and (6.1),  $\rho_{w,v}(z) = c_w Q^2 P_w(1/(v|z|Q))$ . On each half-line  $z \geq 0$  the map  $z \mapsto 1/(v|z|Q)$  is a  $C^1$ -diffeomorphism onto  $(0, \infty)$ , so it maps null sets to null sets in both directions and  $\rho_{w,v}(z) \leq c_w Q^2 R_w$  for almost every  $z$ . As  $k_0 \geq 0$  and  $\int k_0 = 1$ , the main term of (6.4) is at most  $c_w Q^2 R_w$ . The other two terms:

- *isolated point:*  $w(v/Q)k_0(y) \leq w_{\max}/\varsigma = w_{\max}K/\kappa \leq \frac{1}{2}V_w Q^{2-3\varepsilon/4}$ ; in fact this term vanishes once  $V < \eta Q$  (i.e.  $Q > \eta^{-2/\varepsilon}$ ), since  $v \leq V$  and  $w = 0$  on  $(0, \eta)$ ;
- *spokes* (this is where  $K \leq Q^{2-\varepsilon}$ , the genuine bandwidth constraint, enters; in Lemma 6.20 it also enters through a non-vanishing isolated point):  $k_*/\varsigma = QK/(V\kappa) + QV \leq Q^{2-\varepsilon/4} + Q^{2-\varepsilon/2} \leq 2Q^{2-\varepsilon/4}$  and  $k_* \leq Q^{\varepsilon/2} + Q^{1-3\varepsilon/4} \leq 2Q$ , so  $|\varrho| \leq 4V_w \cdot 2Q^{2-\varepsilon/4} \cdot 3 \log Q$  for  $Q \geq 3$ .

Hence  $\sup_{\theta}(\mathfrak{F}_w * k_{\varsigma})(\theta) \leq c_w Q^2 (R_w + 25 (V_w/c_w) Q^{-\varepsilon/4} \log Q)$ . By Lemma 6.1,  $c_w Q^2 \leq W(1 + 2c_0(V_w/c_w)Q^{-1} \log(2Q))$  once  $Q$  is large in terms of  $V_w/c_w$ ; together with  $1 + \kappa^2 = 1 + Q^{-\varepsilon/2}$  this gives the stated  $\vartheta_Q$  for  $Q \geq Q_0(\varepsilon, V_w/c_w)$ . For  $w_{\eta}$ : the variation is  $\eta^{-2}$  (jump at  $\eta$ )  $+(\eta^{-2} - 1)$  (decrease on  $[\eta, 1]$ )  $+1$  (jump at 1)  $= 2\eta^{-2}$ , and  $\frac{\pi^2}{3} < 3.3$ .  $\square$

*Remark 6.6* (why an essential supremum). For weights with jumps,  $P_w$  has jumps at the points  $t = j/k$  ( $j$  a jump point of  $w$ ), and its pointwise supremum can exceed (6.2): for  $w_{0.1}$  the pointwise supremum is 1.5086 (attained only at  $t = 0.1$ , where both  $k = 1$  and  $k = 10$  are counted) while  $R_{w_{0.1}} = 1.4801$  (Table 5). The proof only sees  $P_w$  through the integral in (6.4), so the essential supremum is the correct constant. All tabulated values of  $R_w$  and  $C_w$  (Table 5) are essential suprema; in particular the sharp log-wide value at  $\eta = 10^{-5}$  is  $C_w = C_G \cdot 1.099642 = 1.3952$  (at this  $\eta$  the pointwise supremum 1.099643 gives the same four digits).

The Gauss-sum transfer. Let  $\Delta(n, m) := \sum_{\chi \in \mathcal{F}} \omega_{\chi} \chi(n) \bar{\chi}(m)$  be the family Gram matrix, so that  $x^* \Delta x = \sum_{\chi \in \mathcal{F}} \omega_{\chi} |\sum_n x_n \chi(n)|^2$ , and recall from §2.5 that  $\Lambda_{\text{mult}}(K)$  is the supremum of  $x^* \Delta x$  over unit vectors  $x$  supported on an interval of at most  $K$  consecutive positive integers.

**Lemma 6.7** (Gauss transfer). *For every finitely supported  $x$ ,*

$$x^* \Delta x \leq \sum_q w(q/Q) \sum_{b \bmod q}^* |S_x(b/q)|^2.$$

*Proof.* For  $\chi$  primitive mod  $q$  and every  $n \in \mathbb{Z}$ ,  $\chi(n) \tau(\bar{\chi}) = \sum_{b \bmod q} \bar{\chi}(b) e(bn/q)$  and  $|\tau(\bar{\chi})|^2 = q$  [23, §3.4]. Hence  $|\sum_n x_n \chi(n)|^2 = q^{-1} |\sum_b^* \bar{\chi}(b) S_x(b/q)|^2$ . Summing over primitive  $\chi$  and enlarging to all  $\chi \bmod q$ , orthogonality gives  $\sum_{\chi \bmod q}^* |\sum_n x_n \chi(n)|^2 \leq \frac{\varphi(q)}{q} \sum_{b \bmod q}^* |S_x(b/q)|^2$ . Multiply by  $\omega(q) = w(q/Q)q/\varphi(q)$  and sum over  $q$ .  $\square$

**Corollary 6.8.** *Fix  $\varepsilon \in (0, 1)$ . For  $Q \geq Q_0(\varepsilon, w)$  and all  $K \leq Q^{2-\varepsilon}$ ,  $\Lambda_{\text{mult}}(K) \leq \frac{W}{H}(R_w + \vartheta_Q)H = (C_w + o(1))H$ , where  $o(1) \rightarrow 0$  as  $Q \rightarrow \infty$  for fixed  $(\varepsilon, w)$ .*

*Proof.* Lemma 6.7, Lemma 6.5, and  $W/H = C_G + o(1)$  (Lemma 6.1).  $\square$

Log-wide weights:  $R_w \rightarrow 1$ .

**Lemma 6.9.** *Let  $B_{\varphi} := \sum_d |\mu(d)| d^{-3/2} = \zeta(3/2)/\zeta(3) < 2.174$  and  $c_{\varphi} := \gamma - \zeta'(2)/\zeta(2) = 1.1471 \dots$ . For  $y \geq 1$ ,*

$$E_{\varphi}(y) := \sum_{k \leq y} \frac{\varphi(k)}{k^2} - \frac{6}{\pi^2} (\log y + c_{\varphi}) \quad \text{satisfies} \quad |E_{\varphi}(y)| \leq 2B_{\varphi} y^{-1/2}$$

(also for left limits  $E_{\varphi}(y^-)$ ). Consequently, for all  $0 < \alpha \leq \beta$ ,  $\sum_{\alpha \leq k \leq \beta} \varphi(k)/k^2 \leq \frac{6}{\pi^2} \log(\beta/\alpha) + 4B_{\varphi}$ .

*Proof.*  $\varphi(k)/k^2 = k^{-1} \sum_{d|k} \mu(d)/d$ , so  $\sum_{k \leq y} \varphi(k)/k^2 = \sum_{d \leq y} \frac{\mu(d)}{d^2} \mathcal{H}(y/d)$  with  $\mathcal{H}(x) := \sum_{j \leq x} 1/j$ . For  $x \geq 1$ ,  $\mathcal{H}(x) = \log x + \gamma + \vartheta(x)$  with  $|\vartheta(x)| < 1/\lfloor x \rfloor \leq 2/x$  (because  $\mathcal{H}(n) - \log n - \gamma \in (0, \frac{1}{2n})$  and  $0 \leq \log(x/n) < \frac{1}{n}$  for  $n = \lfloor x \rfloor$ ). Using  $\sum_d \mu(d)/d^2 = 6/\pi^2$  and  $-\sum_d \mu(d) \log d/d^2 = -\zeta'(2)/\zeta(2)^2$ ,

$$E_{\varphi}(y) = - \sum_{d > y} \frac{\mu(d)}{d^2} \left( \log \frac{y}{d} + \gamma \right) + \sum_{d \leq y} \frac{\mu(d)}{d^2} \vartheta(y/d).$$

For  $d > y$ ,  $|\log(y/d) + \gamma| \leq (\gamma + \log(d/y)) \leq 0.99 (d/y)^{1/2}$  (the maximum of  $(\gamma + \log z)z^{-1/2}$  on  $z \geq 1$  is  $2e^{-(2-\gamma)/2} < 0.99$ ); for  $d \leq y$ ,  $|\vartheta(y/d)| \leq 2d/y \leq 2(d/y)^{1/2}$ . As the two ranges

of  $d$  are disjoint,  $|E_\varphi(y)| \leq 2y^{-1/2} \sum_d |\mu(d)| d^{-3/2}$ . For the consequence: if  $\alpha \geq 1$  the sum is  $\frac{6}{\pi^2} \log(\beta/\alpha) + E_\varphi(\beta) - E_\varphi(\alpha^-)$ ; if  $\alpha < 1$  it is at most  $\frac{6}{\pi^2}(\log \beta + c_\varphi) + 2B_\varphi \leq \frac{6}{\pi^2} \log(\beta/\alpha) + 0.70 + 2B_\varphi$  (and 0 if  $\beta < 1$ ).  $\square$

**Lemma 6.10** ( $R_w \rightarrow 1$  for log-wide weights). *Let  $h : \mathbb{R} \rightarrow [0, 1]$  vanish on  $(-\infty, 0)$ , have  $I_h := \int h < \infty$ , and be quasi-concave (each set  $\{h > \lambda\}$ ,  $0 < \lambda < 1$ , is an interval). Let  $w(u) := u^{-2}h(\log(1/u))$  on  $(0, 1]$ . Then  $I_w = I_h$  and, for every  $t > 0$ ,*

$$P_w(t) \leq 1 + \frac{\pi^2}{6} \cdot \frac{4B_\varphi}{I_h} \leq 1 + \frac{14.31}{I_h}.$$

*In particular, for  $L := \log(1/\eta)$ :  $R_{w_\eta} \leq 1 + 14.31/L$  for  $w_\eta = u^{-2}\mathbf{1}_{[\eta, 1]}$  ( $h = \mathbf{1}_{[0, L]}$ ), and  $R_{w_\eta^{\text{sm}}} \leq 1 + 28.62/L$  for  $w_\eta^{\text{sm}} = u^{-2}\sin^2(\pi \log u / \log \eta)\mathbf{1}_{[\eta, 1]}$  ( $h(y) = \sin^2(\pi y/L)\mathbf{1}_{[0, L]}(y)$ ,  $I_h = L/2$ ). Hence  $C_{w_\eta}, C_{w_\eta^{\text{sm}}} \rightarrow C_G$  as  $\eta \rightarrow 0$ .*

*Proof.*  $I_w = \int_0^1 u^{-1}h(\log \frac{1}{u}) du = \int_0^\infty h = I_h$ . By (6.1),  $c_w P_w(t) = \sum_k \frac{\varphi(k)}{k^2} h(\log \frac{1}{kt})$ . Write  $h(y) = \int_0^1 \mathbf{1}[h(y) > \lambda] d\lambda$  and let  $[a_\lambda, b_\lambda]$  be the closure of the interval  $\{h > \lambda\}$ ; by Fubini  $\int_0^1 (b_\lambda - a_\lambda) d\lambda = I_h$ . By Tonelli and Lemma 6.9,

$$\sum_k \frac{\varphi(k)}{k^2} h\left(\log \frac{1}{kt}\right) \leq \int_0^1 \sum_{e^{-b_\lambda}/t \leq k \leq e^{-a_\lambda}/t} \frac{\varphi(k)}{k^2} d\lambda \leq \int_0^1 \left( \frac{6}{\pi^2} (b_\lambda - a_\lambda) + 4B_\varphi \right) d\lambda = \frac{6}{\pi^2} I_h + 4B_\varphi.$$

Divide by  $c_w = \frac{6}{\pi^2} I_h$ . For the two weights  $h$  is quasi-concave with  $I_h = L$ , resp.  $L/2$ .  $\square$

The same layer-cake argument controls  $P_w$  near  $t = 0$  (the “far field”); this is the analogue for  $R_w$  of Proposition 6.23(a), and it is used in the certified evaluation of  $R_w$  (Remark 6.12).

**Lemma 6.11** (far field of  $P_w$ ). *Let  $h$  be as in Lemma 6.10, and assume moreover that  $h = 0$  on  $(L, \infty)$ , where  $L = \log \frac{1}{\eta}$ ; let  $w(u) = u^{-2}h(\log \frac{1}{u})$ . For  $T_1 \geq 1$  put  $\epsilon_\varphi(T_1) := \sup_{y > T_1} \max(|E_\varphi(y)|, |E_\varphi(y^-)|)$ , with  $E_\varphi$  as in Lemma 6.9. Then for every  $t \in (0, \eta/T_1)$*

$$|P_w(t) - 1| \leq \frac{2\epsilon_\varphi(T_1)}{c_w} \leq \frac{4B_\varphi}{c_w \sqrt{T_1}}.$$

*Proof.* As in the proof of Lemma 6.10,  $c_w P_w(t) = \int_0^1 \Sigma_\lambda d\lambda$  with  $\Sigma_\lambda := \sum \{\varphi(k)/k^2 : \log \frac{1}{kt} \in \{h > \lambda\}\}$ . Fix  $\lambda \in (0, 1)$  with  $\{h > \lambda\} \neq \emptyset$ , let  $a_\lambda \leq b_\lambda$  be the end points of this interval, and put  $\alpha := e^{-b_\lambda}/t$ ,  $\beta := e^{-a_\lambda}/t$ . Then  $\Sigma_\lambda$  lies between the sums of  $\varphi(k)/k^2$  over  $\alpha < k < \beta$  and over  $\alpha \leq k \leq \beta$ . Since  $h = 0$  outside  $[0, L]$ ,  $b_\lambda \leq L$ , so  $\alpha \geq \eta/t > T_1 \geq 1$ . By the definition of  $E_\varphi$ ,

$$\sum_{\alpha \leq k \leq \beta} \frac{\varphi(k)}{k^2} = \frac{6}{\pi^2} \log \frac{\beta}{\alpha} + E_\varphi(\beta) - E_\varphi(\alpha^-), \quad \sum_{\alpha < k < \beta} \frac{\varphi(k)}{k^2} = \frac{6}{\pi^2} \log \frac{\beta}{\alpha} + E_\varphi(\beta^-) - E_\varphi(\alpha),$$

and all four values of  $E_\varphi$  are taken at points  $> T_1$  or as left limits at such points. Hence  $|\Sigma_\lambda - \frac{6}{\pi^2}(b_\lambda - a_\lambda)| \leq 2\epsilon_\varphi(T_1)$ . Integrating over  $\lambda$  and using  $\int_0^1 (b_\lambda - a_\lambda) d\lambda = I_h$  gives  $|c_w P_w(t) - \frac{6}{\pi^2} I_h| \leq 2\epsilon_\varphi(T_1)$ ; divide by  $c_w = \frac{6}{\pi^2} I_h$ . The last inequality is Lemma 6.9.  $\square$

In particular  $P_w(t) \rightarrow 1$  as  $t \rightarrow 0^+$  at the rate  $O(t^{1/2})$  for these weights.

*Remark 6.12* (numerical values). Lemma 6.10 only gives the rate  $O(1/\log \frac{1}{\eta})$  with a poor constant. The actual essential suprema are much closer to 1: for the sharp log-wide weight  $R_{w_\eta} = 1.4801$  at  $\eta = 10^{-1}$ , decreasing to 1.0996 at  $\eta = 10^{-5}$  (COMPUTED; floating point, exact finite sums, essential supremum over the open intervals between breakpoints), and for the

log-smooth weight  $R_w \leq 1.059816, 1.028831, 1.015859$  at  $\eta = 10^{-3}, 10^{-4}, 10^{-5}$  (CERTIFIED); see Table 5. These values enter Theorem 1.2 only in the limit  $\eta \rightarrow 0$ , where Lemma 6.10 is rigorous. The upper bounds used in Theorem 1.3(iii) are CERTIFIED in interval arithmetic (ancillary file `rw_arb.py` and its logs `rw_arb_lsmooth*.log`):  $P_w$  is evaluated in ball arithmetic for  $t \geq \eta/T_1$ , with  $T_1 = 10^7\eta$  and a proved interpolation bound between grid points, and for  $t < \eta/T_1$  Lemma 6.11 is used, with  $\epsilon_\varphi(T_1)$  bounded by a rigorous scan of  $E_\varphi$  on  $[T_1, 10^7]$  and by  $2B_\varphi y^{-1/2}$  beyond  $10^7$  (Lemma 6.9).

*Proof of Theorem 1.2, given Theorem 5.16.* By Corollary 6.8 (Lemmas 6.7, 6.5 and 6.1),  $\text{LS}(C_G R_w)$  holds. Theorem 5.16 gives the general statement. For the log-wide weights,  $C_G R_w \rightarrow C_G$  as  $\eta \rightarrow 0$  by Lemma 6.10, and  $p(C_G R_w) \geq p(C_G) - C_G(R_w - 1)$  by Lemma 7.1. Finally  $C_G < 1.26877393 \leq 1.2688$  (certified in interval arithmetic; Table 4), so  $p(C_G) \geq p(1.2688) \geq 0.885912$  by monotonicity and Proposition 7.3.  $\square$

**Overview of §6.2.** The Gauss transfer throws away cancellation: it replaces the  $\varphi^*(q)$  primitive characters by all  $\varphi(q)$  Farey points. For the vectors that occur in (5.7) above  $Q$ , namely primes  $p > Q$  (and prime powers, whose contribution is negligible), there is an exact alternative. If every prime factor of every  $n \in \text{supp } x$  exceeds  $Q$ , then  $(nm, q) = 1$  for all  $q \leq Q$ , and the orthogonality relation  $\sum_{\chi \bmod q}^* \chi(n) \overline{\chi(m)} = \sum_{d|(q, n-m)} \varphi(d) \mu(q/d)$  turns the family form into a Toeplitz form. Expanding its kernel in Ramanujan sums identifies it with the additive large sieve form of a signed Farey measure  $\mu_\Omega$  of total mass exactly  $H$ . Adding to  $\mu_\Omega$  an explicit positive majorant of its negative part gives a positive measure  $\mu^\natural \geq \mu_\Omega^+$  and a positive semidefinite Toeplitz majorant  $T^\natural$ , defined on *all* vectors (Lemma 6.17); its constant on intervals of length  $K \leq Q^{2-\varepsilon}$  is a supremum  $C_T^+(w)$  of explicit local densities near rationals (Lemma 6.20), and  $C_T^+(w) \rightarrow 1$  for log-wide weights (Lemma 6.22).

**6.2. The exact Toeplitz structure and the sharp constant.** We keep the notation of §6.1:  $w$  has bounded variation,  $\text{supp } w \subset [\eta, 1]$ ,  $\tilde{w}(u) = u^2 w(u)$ ,  $I_w = \int_0^1 u w$ ,  $\omega(q) = w(q/Q)q/\varphi(q)$ ,  $H = \sum_q \omega(q) \varphi^*(q)$ ,  $\Delta(n, m) = \sum_{\chi \in \mathcal{F}} \omega_\chi \chi(n) \bar{\chi}(m)$ ,  $S_y(\theta) = \sum_n y_n e(n\theta)$ , and  $k_0, k_\varsigma$  are the triangle kernels of width  $\varsigma$ . An integer  $n \geq 1$  is  $Q$ -rough if every prime factor of  $n$  exceeds  $Q$ . We write  $L_\eta := \log(1/\eta)$  and  $\|\tilde{w}\|_\infty = \sup_u u^2 w(u)$  ( $= 1$  for log-wide weights).

*The identity.*

**Lemma 6.13** (exact Toeplitz identity). *For  $Q$ -rough  $n, m$  we have  $\Delta(n, m) = k_\omega(n - m)$ , where*

$$k_\omega(h) := \sum_{d|h} \varphi(d) \sum_{j \geq 1} \mu(j) \omega(dj) \quad (h \in \mathbb{Z}; \text{ all } d \geq 1 \text{ if } h = 0).$$

Moreover  $k_\omega(h) = \sum_{e \geq 1} \Omega(e) c_e(h)$  with  $c_e$  the Ramanujan sum and

$$(6.8) \quad \Omega(e) = \frac{\varphi(e)}{e} \sum_{\substack{r \text{ squarefree} \\ (r, e)=1}} \frac{\mu(r) \omega(er)}{r} = \sum_{\substack{r \text{ squarefree} \\ (r, e)=1}} \frac{\mu(r)}{\varphi(r)} w\left(\frac{er}{Q}\right)$$

( $\Omega(e) = 0$  for  $e > Q$ ). Consequently, for every  $y$  supported on  $Q$ -rough integers,

$$(6.9) \quad y^* \Delta y = y^* \mathbf{T}_\omega y = \int_{\mathbb{T}} |S_y|^2 d\mu_\Omega, \quad \mathbf{T}_\omega := [k_\omega(n - m)]_{n, m}, \quad \mu_\Omega := \sum_{e \geq 1} \Omega(e) \sum_{c \bmod e}^* \delta_{c/e},$$

and  $\mu_\Omega(\mathbb{T}) = \sum_e \Omega(e) \varphi(e) = k_\omega(0) = H$ .

*Proof.* (i) For  $(a, q) = 1$ ,  $\sum_{\chi \bmod q}^* \chi(a) = \sum_{d|(q, a-1)} \varphi(d) \mu(q/d)$  [23, (3.9)]. If  $n, m$  are  $Q$ -rough then  $(nm, q) = 1$  for every  $q \leq Q$ ; with  $a \equiv n\bar{m} \pmod{q}$  we have  $d \mid a - 1 \iff d \mid n - m$ . Hence  $\Delta(n, m) = \sum_q \omega(q) \sum_{d|(q, n-m)} \varphi(d) \mu(q/d)$ ; put  $q = dj$ . (ii) Since  $d \mathbf{1}[d \mid h] = \sum_{e|d} c_e(h)$ ,  $k_\omega(h) = \sum_e c_e(h) \sum_q \omega(q) c(q, e)$  with  $c(q, e) := \sum_{d|q} \mu(q/d) \varphi(d)/d$ . Writing  $q = es$ ,  $d = ej$ : the function  $j \mapsto \varphi(ej)/(ej)$  equals  $\frac{\varphi(e)}{e} \prod_{p|j, p \nmid e} (1 - \frac{1}{p})$ , and a prime-by-prime evaluation of the Möbius sum over  $j \mid s$  gives  $c(es, e) = \frac{\varphi(e)}{e} \frac{\mu(s)}{s}$  if  $s$  is squarefree and  $(s, e) = 1$ , and 0 otherwise. (iii) For  $(e, r) = 1$ ,  $\omega(er) = w(er/Q) \frac{\varphi(e)}{\varphi(e)} \frac{r}{\varphi(r)}$ , which gives the second form of (6.8). (iv)  $\sum_{n, m} y_n \bar{y}_m c_e(n - m) = \sum_{c \bmod e}^* |S_y(c/e)|^2$ ; sum over  $e$ . Finally  $k_\omega(0) = \sum_q \omega(q) \sum_{d|q} \varphi(d) \mu(q/d) = \sum_q \omega(q) \varphi^*(q) = H$ .  $\square$

*Remark 6.14.* (a) The  $r = 1$  term of (6.8) is  $w(e/Q)$ , the additive weight of the Gauss transfer (Lemma 6.7); the terms  $r \geq 2$  are exactly the corrections that the transfer discards. They have either sign ( $\mu(r) = \pm 1$ ), and their total mass is  $H - W \leq 0$ , which is why  $\mu_\Omega$  has mass  $H$  rather than  $W$ . (b)  $\mathbf{T}_\omega$  is *not* positive semidefinite on all vectors, and  $\mu_\Omega$  has a negative part; both facts are handled by Lemma 6.17. (c) The identity was checked numerically with explicit characters to relative error  $2 \cdot 10^{-15}$  (ancillary file `toeplitz_identity_check.py`); it fails, as it must, off  $Q$ -rough vectors. (d) Steps (i), (ii) and (iv) of the proof use only that  $\omega$  is supported on  $q \leq Q$ . For  $\omega(q) = G_q(Q) := \frac{q}{\varphi(q)} \sum_{\ell \leq Q/q, (\ell, q)=1} \mu^2(\ell)/\varphi(\ell)$ , the first form of (6.8) gives, with  $m = r\ell$ ,  $\Omega(e) = \sum_{m \leq Q/e, (m, e)=1} \frac{\mu^2(m)}{\varphi(m)} \sum_{r|m} \mu(r) = \mathbf{1}_{e \leq Q}$ . The lemma then becomes the identity  $\sum_{q \leq Q} G_q(Q) \sum_{\chi \bmod q}^* |\sum_n y_n \chi(n)|^2 = \sum_{q \leq Q} \sum_{a \bmod q}^* |S_y(a/q)|^2$  for  $Q$ -rough  $y$ , which is the case of the invertible residue classes in [33, Thm. 2.1] (see the proof of [33, Thm. 2.3], and [7]). Lemma 6.13 is thus a weighted form of a known identity; see §1.3.

*The level weights and a positive majorant for  $\mu_\Omega^-$ .* Let  $\sigma_0^- := \sum_{\mu(r)=-1} \frac{1}{r^2 \varphi(r)} < 0.33$  and  $\sigma_1^- := \sum_{\mu(r)=-1} \frac{\log r}{r^2 \varphi(r)} < 0.27$  (both CERTIFIED; Table 4), and define

$$(6.10) \quad m(u) := \left( \sum_{\substack{r \text{ squarefree} \\ \mu(r)=-1}} \frac{w(ur)}{\varphi(r)} - w(u) \right)_+ \quad (u > 0), \quad \tilde{m}(u) := u^2 m(u) = \left( \sum_{\mu(r)=-1} \frac{\tilde{w}(ur)}{r^2 \varphi(r)} - \tilde{w}(u) \right)_+.$$

**Lemma 6.15** (level weights). *For all  $e \geq 1$  and  $u > 0$ :*

- (a)  $|\Omega(e)| \leq 3 \|\tilde{w}\|_\infty \min\{(Q/e)^2, \eta^{-2}\}$ ;
- (b)  $\Omega^-(e) := \max(-\Omega(e), 0) \leq m(e/Q)$ ;
- (c)  $m(u) \leq 3 \|\tilde{w}\|_\infty \min\{u^{-2}/4, \eta^{-2}\}$ , i.e.  $\tilde{m}(u) \leq 3 \|\tilde{w}\|_\infty \min\{\frac{1}{4}, (u/\eta)^2\}$ , and  $m(u) = 0$  for  $u > \frac{1}{2}$ ;
- (d) if  $\tilde{w}$  is constant on  $[\eta, 1]$  (the sharp log-wide weight), then  $m = 0$  on  $[\eta, \infty)$ ; in particular  $\Omega(e) > 0$  for  $\eta Q \leq e \leq Q$ ;
- (e) if  $\tilde{w}(u) = h(\log \frac{1}{u})$  with  $h$  Lipschitz on  $\mathbb{R}$  (constant  $\|h'\|_\infty$ ), then  $\tilde{m}(u) \leq \|h'\|_\infty \sigma_1^-$  for  $u \geq \eta$ . For  $w_\eta^{\text{sm}}$ ,  $\|h'\|_\infty = \pi/L_\eta$ .

*Proof.* We use  $\sum_{r \geq y} \mu^2(r)/(r^2 \varphi(r)) \leq 3y^{-2}$  ( $y > 0$ ): indeed  $\frac{1}{r^2 \varphi(r)} = \frac{1}{r^3} \sum_{d|r} \frac{\mu^2(d)}{\varphi(d)}$ , so the sum is at most  $\sum_d \frac{\mu^2(d)}{\varphi(d) d^3} \sum_{j \geq y/d} j^{-3} \leq \frac{1.5}{y^2} \sum_d \frac{\mu^2(d)}{d \varphi(d)} = \frac{1.5}{y^2} \frac{\zeta(2)\zeta(3)}{\zeta(6)} \leq \frac{3}{y^2}$ , using  $\sum_{j \geq z, j \geq 1} j^{-3} \leq 1.5z^{-2}$  for all  $z > 0$ . (a) Only  $r$  with  $\eta Q \leq er \leq Q$  contribute, and  $w(er/Q) \leq \|\tilde{w}\|_\infty (Q/er)^2$ ; so  $|\Omega(e)| \leq \|\tilde{w}\|_\infty (Q/e)^2 \sum_{r \geq \max(1, \eta Q/e)} \frac{\mu^2(r)}{r^2 \varphi(r)}$ , and the last sum is  $\leq \min(1.34, 3(e/\eta Q)^2)$  ( $\sum_r \mu^2(r)/(r^2 \varphi(r)) < 1.34$ ; Table 4). (b) In (6.8) drop

the terms with  $\mu(r) = +1$ ,  $r \geq 2$  (they are  $\geq 0$ ) and add the terms with  $\mu(r) = -1$ ,  $(r, e) > 1$  (they are  $\leq 0$ ):  $\Omega(e) \geq w(e/Q) - \sum_{\mu(r)=-1} w(er/Q)/\varphi(r) \geq -m(e/Q)$ . (c)  $m(u) \leq \sum_{\mu(r)=-1, ur \geq \eta} w(ur)/\varphi(r) \leq \|\tilde{w}\|_\infty u^{-2} \sum_{r \geq \max(2, \eta/u)} \frac{\mu^2(r)}{r^2 \varphi(r)}$ ; if  $u > \frac{1}{2}$  every  $r \geq 2$  has  $ur > 1$ . (d) For  $u \geq \eta$ :  $\sum_{\mu(r)=-1} \tilde{w}(ur)/(r^2 \varphi(r)) \leq \|\tilde{w}\|_\infty \sigma_0^- < \tilde{w}(u)$ . (e) For  $u \geq \eta$ ,  $y := \log \frac{1}{u} \in [0, L_\eta]$  and  $h(y - \log r) \leq h(y) + \|h'\|_\infty \log r$ , so the bracket in (6.10) is at most  $h(y)(\sigma_0^- - 1) + \|h'\|_\infty \sigma_1^- \leq \|h'\|_\infty \sigma_1^-$ .  $\square$

*Remark 6.16* (where  $\mu_\Omega^-$  lives). By (d), for the *sharp* weight  $\Omega < 0$  only on levels  $e < \eta Q$ . This is *false* for log-smooth weights: there  $\tilde{w}(e/Q)$  vanishes to second order at  $e = \eta Q$  while the  $r = 2, 3$  terms do not, and 64–70% of the mass of  $\mu_\Omega^-$  lies on levels  $e \in [\eta Q, 2.5\eta Q]$  (numerically, at  $Q = 10^6$ ,  $\eta = 10^{-2}, 10^{-3}$ ; ancillary file `omega_check.c`). So for such weights it is *not* true that  $\Omega < 0$  only below  $\eta Q$ , and the positive part must be handled at levels  $e \geq \eta Q$  as well. Lemma 6.15(e) is what is used there: the negative part is  $O(L_\eta^{-1})$  relative to the local scale  $(Q/e)^2$ .

Define the positive measures  $\mu^m := \sum_{e \geq 1} m(e/Q) \sum_{c \bmod e}^* \delta_{c/e}$  and

$$\mu^\natural := \mu_\Omega + \mu^m \geq \mu_\Omega + \mu_\Omega^- = \mu_\Omega^+ \geq 0$$

(Lemma 6.15(b)). Let  $T^+ := [k^+(n-m)]$  and  $T^\natural := [k^\natural(n-m)]$  be the Toeplitz matrices of  $\mu_\Omega^+$  and  $\mu^\natural$ , i.e.  $k^\bullet(h) = \int_{\mathbb{T}} e(h\theta) d\mu^\bullet(\theta)$ ; for an interval  $I$  write  $T_I^\bullet$  for the restriction to  $\mathbb{C}^I$ .

**Lemma 6.17** (positivity and duality). *For every finitely supported  $y$ ,  $\int |S_y|^2 d\mu_\Omega \leq \int |S_y|^2 d\mu_\Omega^+ \leq \int |S_y|^2 d\mu^\natural$ , i.e.  $\mathbf{T}_\omega \preceq T^+ \preceq T^\natural$  as Hermitian forms, and  $T^+, T^\natural \succeq 0$ . If  $I$  contains  $K$  integers and  $0 < \kappa \leq \frac{1}{4}$ , then with  $\varsigma = \kappa/K$*

$$\lambda_{\max}(T_I^+) \leq \lambda_{\max}(T_I^\natural) \leq (1 + \kappa^2) \sup_{\theta \in \mathbb{T}} (\mu^\natural * k_\varsigma)(\theta).$$

*In particular  $y^* \Delta y \leq y^* T^\natural y$  for  $y$  supported on  $Q$ -rough integers.*

*Proof.*  $\mu_\Omega \leq \mu_\Omega^+ \leq \mu^\natural$  as measures and  $|S_y|^2 \geq 0$ . The duality bound is Lemma 6.2 applied to the positive measure  $\mu^\natural$  (restricted to the points of positive mass). The last claim is (6.9).  $\square$

*Local densities.*

**Definition 6.18** (local densities). For a finite set  $S$  of primes let  $f_S$  be the multiplicative function with

$$f_S(p^a) = 1 - \frac{1}{p} - \frac{1}{p^2} \quad (p \notin S, a \geq 1); \quad f_S(p) = \frac{p-2}{p-1}, \quad f_S(p^a) = \frac{p-1}{p} \quad (a \geq 2) \quad (p \in S).$$

For  $t > 0$  put

$$(6.11) \quad R_S(t) := \frac{1}{\mathcal{E}I_w} \sum_{n \geq 1} \frac{f_S(n)}{n} \tilde{w}(nt), \quad R^m(t) := \frac{1}{\mathcal{E}I_w} \sum_{k \geq 1} \frac{\varphi(k)}{k^2} \tilde{m}(kt), \quad R_S^+ := R_S + R^m,$$

and

$$C_T(w) := \sup_S \operatorname{ess\,sup}_{t>0} R_S(t), \quad C_T^+(w) := \sup_S \operatorname{ess\,sup}_{t>0} R_S^+(t),$$

the suprema over all finite sets of primes.  $R_S^+$  is the local density of the positive measure  $\mu^\natural$ , which dominates  $\mu_\Omega^+$ ; thus  $C_T^+(w)$  bounds the local density of  $\mu_\Omega^+$ .

The sums in (6.11) are finite ( $\tilde{w}(nt) = 0$  unless  $nt \in [\eta, 1]$ ),  $0 \leq f_S \leq 1$ , so  $R_S \geq 0$ .

**Lemma 6.19** (arithmetic of  $f_S$ ; uniformity in  $S$ ). *Let  $G_S(k, r) := \frac{\varphi(k)}{k} \prod_{p|r, p \nmid k, p \notin S} (1 - \frac{1}{p})$ .  $\mathbf{1}[no\ p \in S\ divides\ (k, r)]$  for  $k \geq 1$ ,  $r$  squarefree.*

(i)  $f_S(n) = \sum_{kr=n, r \text{ sqfree}} \mu(r) G_S(k, r) / (r\varphi(r))$ ; hence, with absolute convergence,

$$R_S(t) = (\mathcal{E}I_w)^{-1} t^2 \sum_{k \geq 1} \sum_{r \text{ sqfree}} \frac{\mu(r)}{\varphi(r)} k G_S(k, r) w(krt).$$

(ii)  $f_S = 1 * g_S$  with  $g_S$  multiplicative:  $g_S(p) = -\frac{1}{p} - \frac{1}{p^2}$ ,  $g_S(p^a) = 0$  ( $a \geq 2$ ) for  $p \notin S$ ;  $g_S(p) = -\frac{1}{p-1}$ ,  $g_S(p^2) = \frac{1}{p(p-1)}$ ,  $g_S(p^a) = 0$  ( $a \geq 3$ ) for  $p \in S$ . In all cases  $|g_S(p)| \leq \frac{1}{p-1}$ ,  $|g_S(p^2)| \leq \frac{1}{p(p-1)}$ , and  $\sum_d g_S(d)/d = \mathcal{E}$  for every  $S$ .

(iii) Let  $F_S(s) := \sum_{n \leq e^s} f_S(n)/n$  and  $E_S(s) := F_S(s) - \mathcal{E}(s + c_S)$  for  $s \geq 0$ , where

$$c_\emptyset = \gamma + \sum_p \frac{(p^{-2} + p^{-3}) \log p}{1 - p^{-2} - p^{-3}} = 1.41070\dots, \quad c_\emptyset - c_S = \sum_{p \in S} \frac{\log p}{p^3(p-1)(1 - p^{-2} - p^{-3})} \geq 0.$$

Then  $|E_S(s)| \leq 2B e^{-s/2}$  with  $B := \prod_p (1 + \frac{p^{-1/2}}{p-1} + \frac{p^{-1}}{p(p-1)}) < 3.73$ , uniformly in  $S$ . Moreover  $0 \leq c_\emptyset - c_S < 0.168$ , and the primes  $p > P$  in  $S$  contribute at most  $3.2 \sum_{p > P} \frac{\log p}{p^4} \leq \frac{3.2 \log P}{P^3}$  ( $P \geq 3$ ) to  $c_\emptyset - c_S$ .

(iv) If  $S = S_1 \sqcup S_2$ , then  $R_S(t) = \sum_{m \geq 1} \frac{\lambda_{S_2}(m)}{m} R_{S_1}(mt)$  where  $\lambda_{S_2}$  is multiplicative, supported on  $S_2$ -smooth integers, with  $\sum_m \lambda_{S_2}(m)/m = 1$  and  $\delta_{S_2} := \sum_{m \geq 2} |\lambda_{S_2}(m)|/m = \prod_{p \in S_2} (1 + \frac{2}{p^3(p-1)}) - 1$ . Consequently  $R_S(t) \leq R_{S_1}(t) + \frac{1}{2} \delta_{S_2} \sup R_{S_1}$  for all  $t$ .

*Proof.* (i) The summand  $\mu(r) G_S(k, r) / (r\varphi(r))$  factors over primes as a function of  $(v_p(k), v_p(r))$ , so the convolution is multiplicative. For  $p \notin S$  and  $a \geq 1$  the pairs  $(p^a, 1)$  and  $(p^{a-1}, p)$  contribute  $(1 - \frac{1}{p})$  and  $-\frac{1}{p(p-1)}(1 - \frac{1}{p})$ , total  $1 - \frac{1}{p} - \frac{1}{p^2}$ . For  $p \in S$ :  $(p^a, 1)$  contributes  $1 - \frac{1}{p}$ ;  $(p^{a-1}, p)$  is allowed only for  $a = 1$  and contributes  $-\frac{1}{p(p-1)}$ ; this gives  $f_S(p) = \frac{p-2}{p-1}$  and  $f_S(p^a) = \frac{p-1}{p}$  ( $a \geq 2$ ). The second formula follows from  $t^2 k w(krt) = \tilde{w}(nt)/(nr)$ ,  $n = kr$ . Absolute convergence:  $t^2 \sum_k k w(krt) = r^{-2} \sum_k \tilde{w}(krt)/k \leq r^{-2} \|\tilde{w}\|_\infty (L_\eta + 1)$  and  $\sum_r (r^2 \varphi(r))^{-1} < \infty$ .

(ii)  $g_S = f_S * \mu$ ; the values follow from (i).  $\frac{1}{p} + \frac{1}{p^2} \leq \frac{1}{p-1}$ . The Euler factor of  $\sum_d g_S(d)/d$  at  $p$  is  $1 - p^{-2} - p^{-3}$  for  $p \notin S$  and  $1 - \frac{1}{p(p-1)} + \frac{1}{p^3(p-1)} = 1 - p^{-2} - p^{-3}$  for  $p \in S$ . (iii)

With  $\mathcal{H}$  as in the proof of Lemma 6.9,  $F_S(s) = \sum_{d \leq x} \frac{g_S(d)}{d} \mathcal{H}(x/d)$ ,  $x = e^s \geq 1$ , and  $c_S = \gamma - \mathcal{E}^{-1} \sum_d g_S(d) \log d/d$ ; the error estimate is word for word that of Lemma 6.9 with  $\mu(d)/d^2$  replaced by  $g_S(d)/d$ , giving  $|E_S(s)| \leq 2x^{-1/2} \sum_d |g_S(d)| d^{-1/2} \leq 2Bx^{-1/2}$  by (ii). For the closed form,  $A_S(z) := \sum_d g_S(d) d^{-1-z} = \prod_p a_p(z)$  with  $a_p = 1 - (p^{-1} + p^{-2})X$  ( $p \notin S$ ) and  $a_p = 1 - \frac{X}{p-1} + \frac{X^2}{p(p-1)}$  ( $p \in S$ ),  $X = p^{-1-z}$ ; then  $c_S = \gamma + A'_S(0)/A_S(0)$ , and  $a'_p(0) = \frac{(p+1) \log p}{p^3}$ , resp.  $\frac{(p^2-2) \log p}{p^3(p-1)}$ , whose difference is  $\frac{\log p}{p^3(p-1)}$ , while  $a_p(0) = 1 - p^{-2} - p^{-3} \geq \frac{5}{8}$ . Numerically

$\sum_p \frac{\log p}{p^3(p-1)(1-p^{-2}-p^{-3})} = 0.16722\dots < 0.168$ , and  $B < 3.73$  (both CERTIFIED; Table 4). For the tail use  $p-1 \geq p/2$ . (iv)  $f_S = f_{S_1} * \lambda_{S_2}$ , where at  $p \in S_2$  the Euler factor of  $\lambda_{S_2}$  is the ratio of the two factors in (iii):  $\lambda_p(X) = (1 - \frac{X}{p-1} + \frac{X^2}{p(p-1)}) / (1 - \frac{p+1}{p^2} X)$ . Its coefficients are  $\lambda_0 = 1$ ,  $\lambda_1 = -\frac{1}{p^2(p-1)}$ ,  $\lambda_2 = \frac{p^3-p-1}{p^4(p-1)} > 0$ ,  $\lambda_a = (\frac{p+1}{p^2})^{a-2} \lambda_2 > 0$  ( $a \geq 2$ ), and  $\sum_a \lambda_a p^{-a} = \lambda_p(1/p) = 1$ ; hence  $\sum_{a \geq 1} |\lambda_a| p^{-a} = 2|\lambda_1|/p = \frac{2}{p^3(p-1)}$ . Substituting  $f_S(n) = \sum_{m|n} \lambda_{S_2}(m) f_{S_1}(j)$  into (6.11)

gives the identity, and  $R_S(t) - R_{S_1}(t) = \sum_{m \geq 2} \frac{\lambda_{S_2}(m)}{m} (R_{S_1}(mt) - R_{S_1}(t))$ . Let  $P$  and  $N$  be the sums of  $|\lambda_{S_2}(m)|/m$  over  $m \geq 2$  with  $\lambda_{S_2}(m) > 0$ , resp.  $< 0$ ; then  $P - N = 0$  and  $P + N = \delta_{S_2}$ , so  $P = N = \delta_{S_2}/2$ . As  $0 \leq R_{S_1} \leq M := \sup R_{S_1}$ , the positive terms are at most  $P(M - R_{S_1}(t))$  and the negative ones at most  $N R_{S_1}(t)$ ; the sum is at most  $\frac{1}{2} \delta_{S_2} M$ .  $\square$

The local count for the positive part.

**Lemma 6.20** (local count for the positive part). *Fix  $\varepsilon \in (0, 1)$ . Let  $Q \geq Q_1(\varepsilon)$ ,  $Q \leq K \leq Q^{2-\varepsilon}$ ,  $\kappa := Q^{-\varepsilon/4}$ ,  $\varsigma := \kappa/K$ . Then for every  $\theta \in \mathbb{T}$*

$$\begin{aligned} (\mu^\natural * k_\varsigma)(\theta) &\leq \mathcal{E} I_w Q^2 C_T^+(w) + \mathfrak{E}, \\ \mathfrak{E} &\leq \underbrace{6\|\tilde{w}\|_\infty \eta^{-2} Q^{2-3\varepsilon/4}}_{\text{isolated point}} + \underbrace{120 V_w Q^{2-\varepsilon/4} (1 + \log Q)^3}_{\text{spokes}} + \underbrace{3\|\tilde{w}\|_\infty (L_\eta + 1)}_{r>Q \text{ tail}}. \end{aligned}$$

There is no “far part”: the dual kernel  $k_\varsigma$  has compact support.

*Proof.* Choose  $u/v$  and  $y$  by Dirichlet’s theorem with  $V := Q^{1-\varepsilon/2}$  as in Proposition 6.4, fix  $u', v'$  with  $vu' - uv' = 1$ , and let  $S$  be the set of primes dividing  $v$ . Let  $k_* := Q/V + QV\varsigma$ . Since  $\varsigma = \kappa/K$ ,  $\kappa = Q^{-\varepsilon/4}$  and  $Q \leq K \leq Q^{2-\varepsilon}$ , we have  $\varsigma^{-1}k_* = QK/(V\kappa) + QV \leq Q^{2-\varepsilon/4} + Q^{2-\varepsilon/2} \leq 2Q^{2-\varepsilon/4}$  and  $k_* \leq Q^{\varepsilon/2} + Q^{1-3\varepsilon/4} \leq 2Q$ , so  $1 + \log^+ k_* \leq 2(1 + \log Q)$ .

*Decomposition.* Since  $w(er/Q) = 0$  for  $r > Q$ , (6.8) gives  $\mu_\Omega = \sum_{r \leq Q} \frac{\mu(r)}{\varphi(r)} \nu_r$  with the positive measures  $\nu_r := \sum_{e \geq 1, (e,r)=1} w(er/Q) \sum_{c \bmod e}^* \delta_{c/e}$  ( $r$  squarefree). Also  $\mu^m$  only sees  $m_Q(u) := (\sum_{\mu(r)=-1, r \leq Q} w(ur)/\varphi(r) - w(u))_+ \leq m(u)$ , since  $m(e/Q) = m_Q(e/Q)$  for integers  $e \geq 1$ . The function  $m_Q$  vanishes outside  $[\eta/Q, \frac{1}{2}]$  and has variation  $V_{m_Q} \leq V_w(1 + \sum_{r \leq Q} \frac{1}{\varphi(r)}) \leq 3V_w(1 + \log Q)$ , using  $\sum_{r \leq x} \frac{1}{\varphi(r)} = \sum_k \frac{\mu^2(k)}{k\varphi(k)} \mathcal{H}(x/k) \leq 1.95(1 + \log x)$ .

*The measure  $\mu^m$ .* Proposition 6.4 with  $\mathfrak{w} = m_Q$  gives  $(\mu^m * k_\varsigma)(\theta) = m(v/Q)k_0(y) + \int k_0(z - y)\rho_{m_Q, v}(z)dz + \varrho_m$  with  $|\varrho_m| \leq 4V_{m_Q}\varsigma^{-1}k_*(1 + \log^+ k_*) \leq 48V_w Q^{2-\varepsilon/4}(1 + \log Q)^2$ , and  $\rho_{m_Q, v}(z) \leq \rho_{m, v}(z) = Q^2 \mathcal{E} I_w R^m(t)$  with  $t = 1/(v|z|Q)$ , by (6.5) and (6.11).

*The measures  $\nu_r$ : spokes with a coprimality twist.* Fix squarefree  $r \leq Q$ . Exactly as in (6.6),

$$(\nu_r * k_\varsigma)(\theta) = \mathbf{1}[(v, r) = 1] w(vr/Q)k_0(y) + \sum_{k \neq 0} \sum_{\substack{m: (m,k)=1 \\ (q,r)=1, q:=v'k+vm \geq 1}} f_{k,r}(q),$$

$$f_{k,r}(q) := w(qr/Q)k_0\left(\frac{k}{qv} - y\right).$$

Let  $p \mid r$ . If  $p \mid v$  then  $q \equiv v'k \pmod{p}$  with  $p \nmid v'$ , so  $p \mid q \iff p \mid k$ : the whole spoke vanishes if  $p \mid (k, v)$ , and otherwise there is no condition at  $p$ . If  $p \nmid v$  and  $p \mid k$ , then  $p \mid q \iff p \mid m$ , already excluded by  $(m, k) = 1$ . If  $p \nmid vk$ , then  $p \mid q$  iff  $m$  lies in one residue class  $m_p \pmod{p}$  (possibly  $m_p \equiv 0$ , if  $p \mid v'$ ; since  $p \nmid k$  this does not interact with the condition  $(m, k) = 1$ ). Let  $r_k := \prod \{p \mid r : p \nmid vk\}$ . By inclusion–exclusion over  $d_1 \mid k$ ,  $d_2 \mid r_k$  (coprime, so the joint condition is one class modulo  $d_1 d_2$ ) and (6.7), the inner sum equals

$$\frac{G_S(|k|, r)}{v} \int_{\mathbb{R}} f_{k,r} + O^*(2^{\omega(k)} 2^{\omega(r)} \text{TV}(f_{k,r})), \quad \text{TV}(f_{k,r}) \leq \frac{2V_w}{\varsigma},$$

because  $\sum_{d_1 \mid k} \frac{\mu(d_1)}{d_1} \sum_{d_2 \mid r_k} \frac{\mu(d_2)}{d_2} = \frac{\varphi(|k|)}{|k|} \frac{\varphi(r_k)}{r_k}$ , which together with the vanishing rule is  $G_S(|k|, r)$  (Lemma 6.19); the variation bound is as in Proposition 6.4, since  $u \mapsto w(ru)$  has variation  $V_w$ . Only  $1 \leq |k| < k_*$  occur. The main terms are evaluated by the substitution of Proposition 6.4:  $\sum_{k \neq 0} \frac{G_S(|k|, r)}{v} \int f_{k,r} = \int k_0(z - y)\rho_r(z)dz$  with  $\rho_r(z) = Q^2 t^2 \sum_{k \geq 1} k G_S(k, r)w(krt)$ ,  $t = 1/(v|z|Q)$ .

*Summing over  $r$ .* The isolated points combine to  $\Omega(v)k_0(y)$ . The main terms combine to  $\int k_0(z - y) \sum_{r \leq Q} \frac{\mu(r)}{\varphi(r)} \rho_r(z)dz$ , and by Lemma 6.19(i),  $\sum_r \frac{\mu(r)}{\varphi(r)} \rho_r(z) = Q^2 \mathcal{E} I_w R_S(t)$ , while  $\sum_{r > Q} \frac{|\rho_r(z)|}{\varphi(r)} \leq Q^2 \|\tilde{w}\|_\infty (L_\eta + 1) \sum_{r > Q} \frac{\mu^2(r)}{r^2 \varphi(r)} \leq 3\|\tilde{w}\|_\infty (L_\eta + 1)$  (proof of Lemma 6.15); this is

the  $r > Q$  tail. The spoke errors sum to at most

$$\sum_{r \leq Q} \frac{\mu^2(r) 2^{\omega(r)}}{\varphi(r)} \cdot 2k_*(1 + \log^+ k_*) \cdot \frac{2V_w}{\varsigma} \leq 4(1 + \log Q)^2 \cdot 4V_w \cdot 2Q^{2-\varepsilon/4} \cdot 2(1 + \log Q),$$

using  $\sum_{r \leq x} \mu^2(r) 2^{\omega(r)} / \varphi(r) \leq (\sum_{d \leq x} 1/\varphi(d))^2 \leq 4(1 + \log x)^2$ . Together with  $\varrho_m$  this is  $\leq 120V_w Q^{2-\varepsilon/4} (1 + \log Q)^3$ .

*Conclusion.* The isolated point of  $\mu^\natural$  carries  $\Omega(v) + m(v/Q) \leq 6\|\tilde{w}\|_\infty \eta^{-2}$  (Lemma 6.15(a),(c)), times  $k_0(y) \leq \varsigma^{-1} = K/\kappa \leq Q^{2-3\varepsilon/4}$ . The total main term is  $\int k_0(z-y) Q^2 \mathcal{E} I_w [R_S(t) + R^m(t)] dz$  (up to the  $r > Q$  tail); as  $z \mapsto t = 1/(v|z|Q)$  is a diffeomorphism of each half-line onto  $(0, \infty)$ , the bracket is  $\leq \text{ess sup}_t R_S^+(t) \leq C_T^+(w)$  for a.e.  $z$ , and  $\int k_0 = 1$ .  $\square$

**Proposition 6.21** (sharp large sieve on  $Q$ -rough vectors). *Fix  $\varepsilon \in (0, 1)$  and  $w$ . As  $Q \rightarrow \infty$ , uniformly over intervals  $I$  with  $K \leq Q^{2-\varepsilon}$  integers,*

$$\lambda_{\max}(T_I^+) \leq \lambda_{\max}(T_I^\natural) \leq (C_T^+(w) + \vartheta'_Q)H, \\ \vartheta'_Q \ll \frac{V_w + \|\tilde{w}\|_\infty \eta^{-2}}{I_w} Q^{-\varepsilon/4} (\log Q)^3 + \left( Q^{-\varepsilon/2} + \frac{V_w}{I_w} Q^{-1/2} \right) C_T^+(w),$$

with an absolute implied constant. In particular  $y^* \Delta y \leq (C_T^+(w) + \vartheta'_Q)H \|y\|^2$  for all  $y$  supported on  $Q$ -rough integers of  $I$ . For the log-wide weights ( $V_w \leq 2\eta^{-2}$ ,  $\|\tilde{w}\|_\infty = 1$ ,  $I_w \asymp L_\eta$ ) the error is  $\ll \eta^{-2} L_\eta^{-1} Q^{-\varepsilon/4} (\log Q)^3$ .

*Proof.* Lemma 6.20 needs  $K \geq Q$ . If  $K < Q$ , apply the bound to an enclosing interval of  $Q$  integers:  $\lambda_{\max}$  of the restriction of a Hermitian form to a subinterval is at most its  $\lambda_{\max}$  on the larger interval. So we may assume  $Q \leq K \leq Q^{2-\varepsilon}$ ; combine Lemmas 6.17 and 6.20 with  $1 + \kappa^2 = 1 + Q^{-\varepsilon/2}$  and  $H = \mathcal{E} I_w Q^2 (1 + O(V_w I_w^{-1} Q^{-1/2}))$  (Lemma 6.1).  $\square$

In particular  $C_T^+(w) \geq 1$ : the diagonal entries of  $T_I^+$  equal  $k^+(0) = \mu_\Omega^+(\mathbb{T}) \geq \mu_\Omega(\mathbb{T}) = H$ , so  $H \leq \lambda_{\max}(T_I^+) \leq (C_T^+(w) + \vartheta'_Q)H$  for every  $Q$ , and  $\vartheta'_Q \rightarrow 0$ .

The constant tends to 1.

**Lemma 6.22** ( $C_T^+(w_\eta) \rightarrow 1$ ). *Let  $\tilde{w}(u) = h(\log \frac{1}{u})$  with  $h : \mathbb{R} \rightarrow [0, 1]$  vanishing on  $(-\infty, 0) \cup (L_\eta, \infty)$  and quasi-concave. Then*

$$\sup_S \text{ess sup}_t R_S(t) \leq 1 + \frac{c_\emptyset + 4B/\mathcal{E}}{I_w} \leq 1 + \frac{32.6}{I_w}.$$

If moreover  $h = \mathbf{1}_{[0, L_\eta]}$  (sharp), or  $h$  is Lipschitz with  $\|h'\|_\infty \leq \pi/L_\eta$  (log-smooth), then  $\sup_t R^m(t) \leq (3 + \frac{4}{3}\pi\sigma_1^- \cdot \mathbf{1}_{\text{smooth}})/(\mathcal{E} I_w) \leq 8.7/I_w$  (for  $L_\eta \geq 3$ ). Hence

$$C_T^+(w_\eta) \leq 1 + \frac{42}{L_\eta}, \quad C_T^+(w_\eta^{\text{sm}}) \leq 1 + \frac{83}{L_\eta},$$

and  $C_T^+(w_\eta), C_T^+(w_\eta^{\text{sm}}) \rightarrow 1$  as  $\eta \rightarrow 0$ .

*Proof.*  $R_S$ . As in Lemma 6.10,  $\mathcal{E} I_w R_S(t) = \int_0^1 \sum_{n: \log(1/nt) \in \{h > \lambda\}} \frac{f_S(n)}{n} d\lambda$ , and for  $0 < \alpha \leq \beta$ , by Lemma 6.19(iii) (applied at  $\log \beta$  and at the left limit at  $\log \alpha$ ),  $\sum_{\alpha \leq n \leq \beta} \frac{f_S(n)}{n} \leq \mathcal{E} \log(\beta/\alpha) + \mathcal{E} c_S^+ + 4B \leq \mathcal{E} \log(\beta/\alpha) + \mathcal{E} c_\emptyset + 4B$  (if  $\alpha < 1$  the sum is  $F_S(\log \beta) \leq \mathcal{E}(\log \beta + c_S) + 2B$ ). Integrating over  $\lambda$  as in Lemma 6.10 gives  $R_S \leq 1 + (c_\emptyset + 4B/\mathcal{E})/I_w$ , and  $c_\emptyset + 4B/\mathcal{E} < 1.411 + 31.14 < 32.6$  (by  $B < 3.73$  and the certified bounds  $c_\emptyset < 1.411$ ,  $\mathcal{E} > 0.47914$ ; Table 4). This bound is uniform in  $S$ .  $R^m$ .  $\mathcal{E} I_w R^m(t) \leq \sum_k \tilde{m}(kt)/k$ . By Lemma 6.15(c) and  $\|\tilde{w}\|_\infty \leq 1$  (as  $h \leq 1$ ),  $\sum_{k < \eta/t} \tilde{m}(kt)/k \leq 3(t/\eta)^2 \sum_{k < \eta/t} k \leq \frac{3}{2}(1 + t/\eta) \leq 3$  (the sum is empty if  $t \geq \eta$ ). By

Lemma 6.15(d),  $\tilde{m} = 0$  on  $[\eta, \infty)$  in the sharp case; in the Lipschitz case Lemma 6.15(e),(c) give  $\sum_{\eta/t \leq k \leq 1/(2t)} \tilde{m}(kt)/k \leq \frac{\pi}{L_\eta} \sigma_1^-(L_\eta + 1) \leq \frac{4}{3} \pi \sigma_1^-(L_\eta \geq 3)$ . Finally  $I_w = L_\eta$ , resp.  $L_\eta/2$ .  $\square$

Fixed  $\eta$ : a finite reduction and the numerical constants.

**Proposition 6.23** (finite reduction in  $S$ ). *Let  $P_0 := \{2, 3, 5, 7, 11, 13\}$  and  $M_w := \max_{S_1 \subset P_0} \sup_t R_{S_1}(t)$ . Then*

$$C_T^+(w) \leq \max_{S_1 \subset P_0} \operatorname{ess\,sup}_{t>0} (R_{S_1}(t) + R^m(t)) + \frac{1}{2} M_w \delta_{P_0},$$

$$\delta_{P_0} := \prod_{p>13} \left(1 + \frac{2}{p^3(p-1)}\right) - 1 < 5.83 \cdot 10^{-5}.$$

Moreover (“far field”), for  $T_1 \geq 1$ , almost every  $t < \eta/T_1$  and every  $S$ , with  $\tilde{w}(u) = h(\log \frac{1}{u})$ :

- (a)  $R_S(t) = 1 - (\mathcal{E}I_w)^{-1} \int E_S(s) d_s \tilde{w}(e^s t)$ , hence  $R_S(t) \leq 1 + \operatorname{TV}(h) \sup_{s \geq \log T_1} |E_S(s)| / (\mathcal{E}I_w)$ ;
- (b)  $R^m(t) \leq (\mathcal{E}I_w)^{-1} \left[ \frac{6}{\pi^2} I_m + \frac{18c_\varphi \|\tilde{w}\|_\infty}{\pi^2} T_1^{-2} + 2B_\varphi t^{1/2} V'_m \right]$ , where  $I_m := \int_0^\infty \tilde{m}(u) \frac{du}{u}$  and  $V'_m := \int_0^\infty u^{-1/2} |d\tilde{m}(u)|$ ; here  $V'_m \leq 2\eta^{-1/2} (1 + \sum_{\mu(r)=-1} r^{1/2}/(r^2\varphi(r))) \operatorname{TV}(h) < \infty$ , so the last term is  $O(T_1^{-1/2})$ .

*Proof.* Write  $S = S_1 \sqcup S_2$  with  $S_1 = S \cap P_0$  and apply Lemma 6.19(iv);  $R^m$  does not depend on  $S$ . (a) If  $t < \eta$  then  $\tilde{w}(e^s t) = 0$  for  $s < \log(\eta/t)$ , so the Stieltjes integral  $\int \tilde{w}(e^s t) dF_S(s)$  only involves  $s > 0$ , where  $dF_S = \mathcal{E} ds + dE_S$ ;  $\mathcal{E} \int \tilde{w}(e^s t) ds = \mathcal{E}I_w$ , and integrating by parts (valid for a.e.  $t$ , when the jump sets are disjoint; the boundary terms vanish) gives the formula, with  $s \geq \log(\eta/t) > \log T_1$  on the support. (b) Let  $F_\varphi(s) := \sum_{k \leq e^s} \varphi(k)/k^2 = \frac{6}{\pi^2}(s + c_\varphi) + E_\varphi(e^s)$  ( $s \geq 0$ ; Lemma 6.9). Then  $\sum_k \frac{\varphi(k)}{k^2} \tilde{m}(kt) = \int_{[0, \infty)} \tilde{m}(e^s t) dF_\varphi(s) = \frac{6}{\pi^2} \int_0^\infty \tilde{m}(e^s t) ds + \frac{6c_\varphi}{\pi^2} \tilde{m}(t) - \int_{[0, \infty)} E_\varphi(e^s) d_s \tilde{m}(e^s t)$  for a.e.  $t$ . The first term is  $\leq \frac{6}{\pi^2} I_m$ ;  $\tilde{m}(t) \leq 3\|\tilde{w}\|_\infty (t/\eta)^2 \leq 3\|\tilde{w}\|_\infty T_1^{-2}$  by Lemma 6.15(c), and  $c_\varphi > 0$ ; and  $|E_\varphi(e^s)| \leq 2B_\varphi e^{-s/2} = 2B_\varphi (t/u)^{1/2}$  at  $u = e^s t$ . For  $V'_m$ :  $\tilde{m} = (\Sigma - \tilde{w})_+$  with  $\Sigma(u) = \sum_{\mu(r)=-1} \tilde{w}(ur)/(r^2\varphi(r))$ ;  $(\cdot)_+$  does not increase variation, the  $r$ -th term has variation  $\operatorname{TV}(h)/(r^2\varphi(r))$  and lives on  $u \geq \eta/r$ , where  $u^{-1/2} \leq (r/\eta)^{1/2}$ ; the term  $\tilde{w}$  lives on  $u \geq \eta$ .  $\square$

The right-hand side of Proposition 6.23 is a finite computation (64 sets  $S_1$ , exact finite sums on  $t \geq \eta/T_1$ , the far-field bounds (a) and (b), the tabulated constants  $B, c_S$ ). Its evaluation for the sharp and log-smooth weights at  $\eta = 10^{-2}, 10^{-3}, 10^{-4}$  is in Table 6; for both weights  $\|\tilde{w}\|_\infty = 1$  ( $h$  takes values in  $[0, 1]$  and attains 1), so the  $T_1^{-2}$  term of (b) is  $\frac{18c_\varphi}{\pi^2} T_1^{-2}$ , as implemented (ancillary file `ctplus_arb.py`, with the logs `ctplus_arb*.log`; the resulting bounds and the exact rational values of  $p(C)$  at them are collected in `certify_ctplus_arb.log`). The constants  $\mathcal{E}, B, c_\emptyset, \delta_{P_0}$  and  $\sigma_0^-$  are enclosed in the same interval arithmetic. Every truncation, interpolation and far-field error is bounded by the inequalities above, and the finite sums are evaluated in interval (ball) arithmetic, so the resulting upper bounds are CERTIFIED; the limit statements (Lemma 6.22) need no computation.

*Interface with the time-integrated majorant.* We use the notation of Lemmas 5.5, 5.6, 5.9, 5.11:  $x_y(u)_n := y_n \widehat{1_J}(u - \log n)$  for a finitely supported vector  $y$ ,  $\delta := T^{-1/2}$ ,  $\kappa_T := T^{-1/4}$ ,  $\rho \in C_c^\infty((-2, 2))$  with  $0 \leq \rho \leq 1$  and  $\rho = 1$  on  $[-1, 1]$ , and

$$x_y^s(u)_n := x_y(u)_n \rho\left(\frac{\log n - u}{\delta}\right), \quad x_y^t(u) := x_y(u) - x_y^s(u).$$

These are linear in  $y$ ; in particular  $x_a = x_b + x_{a^\sharp}$  and  $x_a^s = x_b^s + x_{a^\sharp}^s$ . We write  $x_\sharp := x_{a^\sharp}$  and, for a positive semidefinite Hermitian matrix  $P$ ,  $\|x\|_P^2 := x^* P x$ ; this is the square of a seminorm, so

$$(6.12) \quad \|x + x'\|_P^2 \leq (1 + \kappa)\|x\|_P^2 + (1 + \kappa^{-1})\|x'\|_P^2 \quad (0 < \kappa \leq 1).$$

Recall that  $\mu^\sharp = \sum_{e \geq 1} \nu(e) \sum_{c \bmod e}^* \delta_{c/e}$  with  $\nu(e) := \Omega(e) + m(e/Q)$ . By Lemma 6.15(a)–(c),

$$(6.13) \quad 0 \leq \Omega^+(e) \leq \nu(e) \leq 6\|\tilde{w}\|_\infty \eta^{-2} \quad (e \geq 1), \quad \nu(e) = 0 \quad (e > Q),$$

because  $\Omega(e) = 0$  for  $e > Q$  and  $m(u) = 0$  for  $u > \frac{1}{2}$ ; hence

$$\|x\|_{T^\sharp}^2 = \sum_{e \leq Q} \nu(e) \sum_{c \bmod e}^* |S_x(c/e)|^2.$$

*The band constant.* Fix  $\varepsilon \in (0, \frac{1}{3})$ . Let  $C_{\text{band}} \geq 1$  be a constant such that  $\Lambda_{\text{mult}}(Q^{1+2\varepsilon}) \leq (C_{\text{band}} + o(1))H$  as  $Q \rightarrow \infty$ . Two admissible choices are  $C_{\text{band}} = C_w$  (Corollary 6.8 with  $1 - 2\varepsilon$  in place of  $\varepsilon$ ) and, without Lemma 6.5,  $C_{\text{band}} = 2w_{\max}/(\mathcal{E}I_w)$ : by (4.1) and  $\omega(q) \leq w_{\max}q/\varphi(q)$  one has  $\Lambda_{\text{mult}}(K) \leq w_{\max}(Q^2 + K) \leq 2w_{\max}Q^2$  for  $K \leq Q^2$ , and  $H = \mathcal{E}I_w Q^2(1 + o(1))$  by Lemma 6.1. Let  $C_{\max} := \max(C_{\text{band}}, C_T^+(w))$  (recall  $C_T^+(w) \geq 1$ , Proposition 6.21), and let  $\tilde{C}_\varepsilon : \mathbb{R} \rightarrow [1, C_{\max}]$  be smooth with

$$\begin{aligned} \tilde{C}_\varepsilon &= 1 \text{ on } (-\infty, 1 - 2\varepsilon], & \tilde{C}_\varepsilon &\geq C_{\text{band}} \text{ on } [1 - \tfrac{3}{2}\varepsilon, 1 + \tfrac{3}{2}\varepsilon], \\ \tilde{C}_\varepsilon &\geq C_T^+(w) \text{ on } [1 + \tfrac{1}{2}\varepsilon, \infty), & \tilde{C}_\varepsilon &= C_T^+(w) \text{ on } [1 + 2\varepsilon, \infty). \end{aligned}$$

**Proposition 6.24** (sharp time-integrated majorant; band device). *Assume the setting of Proposition 5.12 ( $\lambda < 2$ ,  $Y = X^{1+\varepsilon_1} \leq Q^{2-\varepsilon_1}$ ,  $\ell^{a_0} \leq T \leq \ell^{A_0}$ ,  $b = a - a^\sharp$  as in §5.7). For each fixed  $\varepsilon \in (0, \frac{1}{3})$  there is  $\vartheta_Q^* \rightarrow 0$  (depending on the fixed data, not on  $T$ ) such that*

$$(6.14) \quad \sum_{n,m} b_n b_m \Delta(n, m) \mathcal{K}(n, m) \leq (1 + \vartheta_Q^*)(1 + \kappa_T)^3 H \sum_n |b_n|^2 \tilde{C}_\varepsilon\left(\frac{\log n}{\ell}\right) \mathcal{K}(n, n) + \mathcal{E}_{\text{TI}},$$

where, with an implied constant depending only on the fixed data,

$$\mathcal{E}_{\text{TI}} \ll T^{3/4} H L^3 + T^{1/4} \eta^{-2} Q^{-\varepsilon_1} H |J| L^2 \ell + T^{1/4} Q^{-1/2} \ell^2 H |J| L + Q^{-1} = o(H |J| L^2 \ell).$$

Consequently (Lemma 5.7 with the smooth weight  $g(\cdot) \tilde{C}_\varepsilon(\cdot/\ell)$ , then  $\varepsilon, \varepsilon_4 \rightarrow 0$  by dominated convergence) the second-moment majorant of Proposition 5.12 holds with  $F = F_C$ ,  $C = C_T^+(w)$ .

*Proof.* Throughout,  $u$  ranges over  $\text{supp } g \subset (-L, L)$ , so  $e^u < X$ . By Lemma 5.9 the left side of (6.14) equals  $\int g(u) x_b(u)^* \Delta x_b(u) du$ , with  $g \geq 0$  and  $\int g = (\int \psi_L^2)^2 = a^2 L^2$ .

*Step 0: invisibility, before any localisation.* For  $\chi \in \mathcal{F}$  and every  $u$ ,  $\sum_n x_\sharp(u)_n \chi(n) = \int_J e^{itu} S_\chi[a^\sharp](t) dt \ll_A |J| Q^{-A}$  by Lemma 5.6 ( $J \subset [0, 2T]$ ). Also  $|\sum_n x_b(u)_n \chi(n)| \leq |J| \sum_n |b_n| \ll Q^3$ . Expanding  $|\alpha + \beta|^2 - |\alpha|^2$  and summing with the weights  $\omega_\chi$  ( $H \ll Q^2$ ),

$$(6.15) \quad x_b(u)^* \Delta x_b(u) \leq x_a(u)^* \Delta x_a(u) + Q^{-3} \quad (\text{all } u, Q \geq Q_0).$$

We now choose, for each  $u$  separately, which of the two forms to localise:  $y(u) := b$  if  $u \leq (1 + \varepsilon)\ell$  and  $y(u) := a$  if  $u > (1 + \varepsilon)\ell$ . Put  $K(u) := 5\delta e^u + 1$ ; by Lemma 5.11(ii) each  $x_y^s(u)$  is supported in an interval of at most  $K(u) \leq 5\delta X + 1 \leq Q^{2-\varepsilon_1}$  integers, contained in  $[1, Y]$ .

*Step 1: localisation.* By Lemma 5.11(i) with  $\kappa = \kappa_T$ ,  $x_y^* \Delta x_y \leq (1 + \kappa_T) x_y^{s*} \Delta x_y^s + (1 + \kappa_T^{-1}) x_y^{t*} \Delta x_y^t$  for  $y = y(u)$ .

*Step 2: tails.*  $x_y^t(u)$  is supported in  $[1, Y]$ , so  $x_y^{t*} \Delta x_y^t \leq \Lambda_{\text{mult}}(Y) \|x_y^t\|^2 \leq 2w_{\max} Q^2 \|x_y^t\|^2$  by (4.1) ( $Y \leq Q^2$ ). By Lemma 5.11(iv) and (5.10),  $\int g \|x_{y(u)}^t(u)\|^2 du \leq 8bL(\|a\|^2 + \|b\|^2)/\delta \ll T^{1/2} L^3$  (apply Lemma 5.11(iv) to  $y = a$  and to  $y = b$ , since  $y(u)$  changes with  $u$ ). With the factor  $1 + \kappa_T^{-1} \leq 2T^{1/4}$  and  $Q^2 \ll_w H$ , the tails contribute  $\ll T^{3/4} H L^3$  in total.

*Step 3: short parts in regimes (i) and (ii).* Let  $u \leq (1 + \varepsilon)\ell$ , so  $y = b$ .

(i) If  $u \leq (1 - \varepsilon)\ell$ , then  $K(u) \leq 5\delta Q^{1-\varepsilon} + 1 \leq Q^{1-\varepsilon/2}$ , and Lemma 5.10 gives  $x_b^{s*} \Delta x_b^s \leq c_i H \|x_b^s\|^2$  with  $c_i := 1 + 4w_{\max} Q^{2-\varepsilon/2}(1 + \ell)/H = 1 + O_w(Q^{-\varepsilon/2}\ell)$ , with no condition on  $T$ .

(ii) (the band) If  $(1 - \varepsilon)\ell < u \leq (1 + \varepsilon)\ell$ , then  $K(u) \leq 5\delta Q^{1+\varepsilon} + 1 \leq Q^{1+2\varepsilon}$ , and by the definition of  $C_{\text{band}}$ ,  $x_b^{s*} \Delta x_b^s \leq c_{ii} H \|x_b^s\|^2$  with  $c_{ii} := C_{\text{band}} + o(1)$ .

*Step 4: short parts in regime (iii),  $u > (1 + \varepsilon)\ell$ .* Here  $y = a$ . If  $(x_a^s(u))_n \neq 0$  then  $a_n \neq 0$  and  $\log n > u - 2\delta$ , so  $n$  is a prime power and  $n > Q^{1+\varepsilon} e^{-2\delta} > Q^{1+\varepsilon}/2$ . Split  $x_a^s = x_r + x_{\text{pp}}$ , where  $x_r$  is the restriction to  $Q$ -rough  $n$  and  $x_{\text{pp}}$  the restriction to  $n = p^k$  with  $p \leq Q$ ,  $k \geq 2$  (a prime  $p \leq Q$  is  $< Q^{1+\varepsilon}/2$ ). By (6.12) for  $\Delta$  with  $\kappa = \kappa_T$  and Lemma 6.17 for the  $Q$ -rough vector  $x_r$  (first line below); then by the identity  $x_r = x_b^s + (x_{\sharp}^s - x_{\text{pp}})$  and (6.12) for  $T^{\natural}$ , first with  $\kappa = \kappa_T \leq 1$  for this split and then with  $\kappa = 1$  for  $\|x_{\sharp}^s - x_{\text{pp}}\|_{T^{\natural}}^2 \leq 2\|x_{\sharp}^s\|_{T^{\natural}}^2 + 2\|x_{\text{pp}}\|_{T^{\natural}}^2$ ; and finally by  $2(1 + \kappa_T)(1 + \kappa_T^{-1}) \leq 4(1 + \kappa_T^{-1})$ ,

$$\begin{aligned} x_a^{s*} \Delta x_a^s &\leq (1 + \kappa_T) \|x_r\|_{T^{\natural}}^2 + (1 + \kappa_T^{-1}) x_{\text{pp}}^* \Delta x_{\text{pp}} \\ &\leq (1 + \kappa_T)^2 \|x_b^s\|_{T^{\natural}}^2 + 4(1 + \kappa_T^{-1}) (\|x_{\sharp}^s\|_{T^{\natural}}^2 + \|x_{\text{pp}}\|_{T^{\natural}}^2 + x_{\text{pp}}^* \Delta x_{\text{pp}}). \end{aligned}$$

(a) *Main term.*  $x_b^s(u)$  is supported in an interval of  $\leq K(u) \leq Q^{2-\varepsilon_1}$  integers. Put  $\varepsilon'_1 := \min(\varepsilon_1, \frac{1}{2}) \in (0, 1)$ ; then  $K(u) \leq Q^{2-\varepsilon'_1}$ , so Proposition 6.21 (with  $\varepsilon'_1$  in place of  $\varepsilon$ , which that proposition requires to lie in  $(0, 1)$ ) gives  $\|x_b^s\|_{T^{\natural}}^2 \leq c_{iii} H \|x_b^s\|^2$ ,  $c_{iii} := C_T^+(w) + \vartheta'_Q$ , where  $\vartheta'_Q \rightarrow 0$  is the error term of Proposition 6.21 for  $\varepsilon'_1$ . Note that  $x_b^s$  is *not* supported on  $Q$ -rough integers; this is harmless because  $T^{\natural} \geq 0$  is defined on all vectors.

(b) *The subtracted part  $x_{\sharp}^s$ .* Let  $R_{\max} := \max\{R_j : N_j \leq 2Y\} \leq (2Y)^{1-\varepsilon_3}/(QT)$ . For  $e \geq 1$  and  $(c, e) = 1$ , by the definitions of  $a^{\sharp}$ ,  $\Lambda_R$  and  $c_r$ , and (5.5),

$$(6.16) \quad S_{x_{\sharp}^s(u)}\left(\frac{c}{e}\right) = \int_J e^{itu} \sum_{j: R_j \geq 1} \sum_{r \leq R_j} \frac{\mu(r)}{\varphi(r)} \sum_{h \bmod r}^* \sum_n F_{j,t,u}(n) e\left(n\left(\frac{h}{r} + \frac{c}{e}\right)\right) dt.$$

*Levels  $R_{\max} < e \leq Q$ .* Let  $r \leq R_j < e$  and  $(h, r) = 1$ . If  $\beta := \frac{h}{r} + \frac{c}{e}$  were an integer, then  $re \mid he + cr$ , hence  $e \mid cr$  and, as  $(c, e) = 1$ ,  $e \mid r$ , contradicting  $e > r$ . So  $\beta \notin \mathbb{Z}$ , and since  $re\beta \in \mathbb{Z}$ ,  $\|\beta\| \geq 1/(re) \geq 1/(R_j Q)$ . By (5.6) ( $|t| \leq 2T$ ) and Lemma 5.5(a) with  $\Theta = T$  and (5.4),

$$\left| \sum_n F_{j,t,u}(n) e(n\beta) \right| \leq B_i N_j^{1/2} \left( \frac{TR_j Q}{N_j} \right)^i \leq B_i N_j^{1/2-i\varepsilon_3}.$$

Summing over  $h$  and  $r$  with the weights  $\mu(r)/\varphi(r)$  (total weight  $\sum_{r \leq R_j} \mu^2(r)\varphi(r)/\varphi(r) \leq R_j \leq N_j$ ), over the  $\leq 2 + \log_2 Y$  blocks with  $R_j \geq 1$  (for which  $N_j \geq QT$ ), and integrating over  $J$ ,

$$\left| S_{x_{\sharp}^s(u)}\left(\frac{c}{e}\right) \right| \leq B_i |J| (2 + \log_2 Y) (QT)^{3/2-i\varepsilon_3} \leq Q^{-4} \quad (i \geq 7/\varepsilon_3, Q \geq Q_0),$$

uniformly in  $u$ ,  $e \in (R_{\max}, Q]$  and  $c$ . By (6.13) these levels contribute at most  $6\|\tilde{w}\|_{\infty} \eta^{-2} Q^2 \cdot Q^{-8}$  to  $\|x_{\sharp}^s\|_{T^{\natural}}^2$ .

*Levels  $e \leq R_{\max}$ .* By (6.13) and the additive large sieve [27], [23, Thm. 7.7],

$$\sum_{e \leq Ec \bmod e} \sum^* |S_x(c/e)|^2 \leq (K + E^2) \|x\|^2$$

for  $x$  supported on  $K$  consecutive integers,

$$\sum_{e \leq R_{\max}} \nu(e) \sum_{c \bmod e}^* \left| S_{x_{\sharp}^s}\left(\frac{c}{e}\right) \right|^2 \leq 6\|\tilde{w}\|_{\infty} \eta^{-2} (K(u) + R_{\max}^2) \|x_{\sharp}^s\|^2 \leq 36\|\tilde{w}\|_{\infty} \eta^{-2} Q^{2-\varepsilon_1} \|x_{\sharp}^s\|^2,$$

since  $K(u) \leq 5\delta Y \leq Q^{2-\varepsilon_1}$  and  $R_{\max}^2 \leq 4Y^2/Q^2 \leq 4Q^{2-2\varepsilon_1}$ . By Lemma 5.11(iii) with  $c \equiv 1$ ,  $\mathcal{K}(n, n) \leq 2\pi|J|bL + O(1)$  and (5.10),  $\int g\|x_b^s\|^2 \leq \sum_n |a_n^\sharp|^2 \mathcal{K}(n, n) \ll |J|L \cdot L\ell$ . With  $Q^2 \ll_w H$ , the contribution of (b), including the factor  $(1 + \kappa_T) \cdot 4(1 + \kappa_T^{-1}) \leq 16T^{1/4}$  (Steps 1 and 4), is  $\ll T^{1/4}\eta^{-2}Q^{-\varepsilon_1}H|J|L^2\ell + Q^{-1}$ .

(c) *Prime powers*  $p^k$ ,  $p \leq Q$ ,  $k \geq 2$ . The prime powers  $n \leq y$  with  $k \geq 2$  number  $O(y^{1/2})$ , so by partial summation  $\sum_{n=p^k > Q^{1+\varepsilon/2}, k \geq 2} \Lambda(n)^2/n \ll Q^{-1/2}\ell^2$ . The vector  $x_{\text{pp}}(u)$  lives in an interval of  $\leq K(u) \leq Q^{2-\varepsilon_1}$  integers inside  $[1, Y]$ , so  $\|x_{\text{pp}}\|_{T^{\mathfrak{t}}}^2 \leq c_{\text{iii}}H\|x_{\text{pp}}\|^2$  (Proposition 6.21 with  $\varepsilon'_1$ , as in (a)) and  $x_{\text{pp}}^* \Delta x_{\text{pp}} \leq 2w_{\max}Q^2\|x_{\text{pp}}\|^2$  ((4.1)). By Lemma 5.11(iii),  $\int g\|x_{\text{pp}}\|^2 \leq \sum_{n=p^k > Q^{1+\varepsilon/2}, k \geq 2} |a_n|^2 \mathcal{K}(n, n) \ll |J|LQ^{-1/2}\ell^2$ . With the factor  $16T^{1/4}$  the contribution of (c) is  $\ll T^{1/4}Q^{-1/2}\ell^2H|J|L$ .

*Step 5: integration in  $u$ .* Combining (6.15) and Steps 1–4, for every  $u$  in regime (i), (ii) or (iii),

$$x_b(u)^* \Delta x_b(u) \leq (1 + \kappa_T)^3 c(u) H \|x_b^s(u)\|^2 + (\text{tail, (b), (c) terms at } u) + Q^{-3},$$

with  $c(u) = c_i, c_{\text{ii}}, c_{\text{iii}}$  respectively. Let  $n$  satisfy  $\rho((\log n - u)/\delta) \neq 0$ , i.e.  $|u - \log n| < 2\delta$ , and put  $\alpha := \log n/\ell$ ; for  $Q$  large,  $2\delta < \frac{1}{2}\varepsilon\ell$ . If  $\alpha \leq 1 - \frac{3}{2}\varepsilon$  then  $u < (1 - \varepsilon)\ell$ , so only (i) occurs; if  $1 - \frac{3}{2}\varepsilon < \alpha < 1 + \frac{1}{2}\varepsilon$  then  $u < (1 + \varepsilon)\ell$ , so only (i) or (ii) occurs; if  $1 + \frac{1}{2}\varepsilon \leq \alpha \leq 1 + \frac{3}{2}\varepsilon$  then  $u > (1 - \varepsilon)\ell$ , so only (ii) or (iii) occurs; if  $\alpha > 1 + \frac{3}{2}\varepsilon$  then only (iii) occurs. Comparing with the defining properties of  $\tilde{C}_\varepsilon$  (and  $\tilde{C}_\varepsilon \geq 1$ ), in every case

$$c(u) \leq (1 + \vartheta_Q^*) \tilde{C}_\varepsilon(\alpha), \quad \vartheta_Q^* := \max(c_i - 1, c_{\text{ii}} - C_{\text{band}}, \vartheta'_Q) \rightarrow 0.$$

Therefore, by Lemma 5.11(iii) applied to the bounded function  $c(u)$ ,  $\int g(u)c(u)\|x_b^s(u)\|^2 du \leq (1 + \vartheta_Q^*) \sum_n |b_n|^2 \tilde{C}_\varepsilon(\log n/\ell) \mathcal{K}(n, n)$ . Adding the tails (Step 2), (b), (c) and  $\int g \cdot Q^{-3} \ll L^2 Q^{-3} \leq Q^{-1}$  gives (6.14). Each term of  $\mathcal{E}_{\text{TI}}$  is  $o(H|J|L^2\ell)$  uniformly in  $T$ : the first is  $O(T^{-1/4})$  relative to it ( $|J| \asymp T$ ,  $L \asymp \ell$ ), and the others carry  $Q^{-\varepsilon_1}$  or  $Q^{-1/2}$  against  $T^{1/4} \leq \ell^{A_0/4}$ .

For the final sentence:  $F_\varepsilon^\sharp(\alpha) := \tilde{C}_\varepsilon(\alpha) \min(\alpha, 1 + \varepsilon_4)$  tends to  $F_C(\alpha)$  for every  $\alpha \neq 1$  as  $\varepsilon, \varepsilon_4 \rightarrow 0$ , and  $0 \leq F_\varepsilon^\sharp \leq 2C_{\max}$ , so  $\mathcal{Q}_{F_\varepsilon^\sharp}(f) \rightarrow \mathcal{Q}_{F_C}(f)$  for every admissible  $f$  by dominated convergence.  $\square$

*Remark 6.25* (order of limits; what is *not* used). (1) The order of limits is the one fixed in §2.3:  $Q \rightarrow \infty$  with  $(\eta, w, \lambda, \varepsilon_1, \theta, \psi, \varepsilon, \varepsilon_3)$  and the cut-offs  $\Upsilon_0, (\psi_j), \rho$  of §2.3 fixed; then  $\varepsilon, \varepsilon_3, \theta \rightarrow 0$  (so  $\varepsilon_4 = 2\varepsilon_3 \rightarrow 0$ ); then  $\lambda \rightarrow 2$ , with  $\varepsilon_1$  chosen in terms of  $\lambda$  so that  $\lambda(1 + \varepsilon_1) \leq 2 - \varepsilon_1$ , and  $\psi \rightarrow$  optimal, giving  $p(C_T^+(w))$  for the fixed weight  $w$  (Theorem 1.3); finally  $\eta \rightarrow 0$ , using Lemma 6.22 and the monotonicity and Lipschitz continuity of  $C \mapsto p(C)$  (Lemma 7.1; proof of Theorem 1.1). No weight depends on  $Q$ . (2) Lemma 5.10 is used only for windows of length  $K \leq Q^{1-\varepsilon/2}$ , which requires no condition on  $T$ . Using it up to  $u \leq \ell + 2\delta$  would require  $K \log K/Q \ll \eta^2$  with  $K \asymp Q/\sqrt{T}$ , i.e.  $T \gg \eta^{-4}(\log Q)^{2+o(1)}$ ; the band of width  $2\varepsilon$  carrying  $C_{\text{band}}$  avoids this, and it disappears in the limit  $\varepsilon \rightarrow 0$ . (3) Primes  $p \leq Q$  never enter the Toeplitz step: regime (iii) lives on  $n > Q^{1+\varepsilon/2}$ .

Proposition 6.24 is the form in which these statements enter: it replaces Hypothesis LS( $C$ ) in Proposition 5.13 by the profile  $\tilde{C}_\varepsilon$ , which equals 1 below  $1 - 2\varepsilon$ , is at most  $\max(C_{\text{band}}, C_T^+(w))$  on a band of width  $O(\varepsilon)$  around  $\alpha = 1$ , and equals  $C_T^+(w)$  above  $1 + 2\varepsilon$ ; here  $C_{\text{band}}$  is any  $O(1)$  large sieve constant for intervals of length  $Q^{1+2\varepsilon}$ , for instance the classical one. Three features of the argument should be noted. Invisibility (Lemma 5.6) is applied to the whole vector before any localisation. The Toeplitz identity is only applied to vectors supported on  $Q$ -rough integers  $n > Q^{1+\varepsilon/2}$ , so primes  $p \leq Q$  dividing a modulus never enter it. And the swap  $a \rightarrow b$  takes place inside the positive semidefinite form  $T^\sharp$ , which is defined on all vectors.

## 7. THE VARIATIONAL PROBLEM AND THE CERTIFIED CONSTANTS

**7.1. The extremal problem.** For  $C \geq 1$  let  $F_C$  be as in (1.3) and  $p(C) = 2 - \inf_f \mathcal{Q}_{F_C}(f)$  over even  $f \geq 0$  with  $\text{supp } f \subset [-1, 1]$  and  $\int f = 1$ . Evenness is needed because the window  $\psi$  must be even (§2.4). The functional is finite on  $L^2([-1, 1])$  and continuous in  $L^1 \cap L^2$ , since  $0 \leq F_C \leq C$ .

**Lemma 7.1.**  $p$  is nonincreasing, and  $p(C') \geq p(C) - (C' - C)$  for  $C' \geq C \geq 1$ .

*Proof.*  $F_{C'} - F_C = (C' - C)\mathbf{1}_{(1, 2]}$ , so  $\mathcal{Q}_{F_{C'}}(f) - \mathcal{Q}_{F_C}(f) = (C' - C) \iint_{|x-y|>1} f(x)f(y) \in [0, C' - C]$  when  $f \geq 0$  and  $\int f = 1$ .  $\square$

**Lemma 7.2** (Realisation by windows). *Let  $C \geq 1$ , let  $f \geq 0$  be even, supported in  $[-1, 1]$ , with  $f \in L^2$  and  $\int f = 1$ , and let  $\eta' > 0$ . There are  $\lambda < 2$  and a real even  $\psi \in C_c^\infty((-\frac{1}{2}, \frac{1}{2}))$ ,  $\psi \not\equiv 0$ , such that  $\mathcal{Q}_{F_C}(f_v) \leq \mathcal{Q}_{F_C}(f) + \eta'$ , where  $v = \psi^2$  and  $f_v(x) = v(x/\lambda)/(\lambda \int v)$ .*

*Proof.* For  $0 \leq F \leq 2C$  one has  $|\mathcal{Q}_F(f) - \mathcal{Q}_F(g)| \leq \|f - g\|_2(\|f\|_2 + \|g\|_2) + 2C\|f - g\|_1(\|f\|_1 + \|g\|_1)$ , so  $\mathcal{Q}_F$  is continuous on  $L^1 \cap L^2$ . First rescale:  $f_\lambda(x) = \frac{2}{\lambda}f(\frac{2x}{\lambda})$  is supported in  $[-\frac{\lambda}{2}, \frac{\lambda}{2}]$ , and  $\mathcal{Q}_{F_C}(f_\lambda) = \frac{2}{\lambda} \int f^2 + \iint f(s)f(s')F_C(\frac{\lambda}{2}|s-s'|) \rightarrow \mathcal{Q}_{F_C}(f)$  as  $\lambda \uparrow 2$ , by dominated convergence ( $F_C(\frac{\lambda}{2}r) \rightarrow F_C(r)$  for  $r \neq 1$ , and  $\{|s-s'| = 1\}$  is null). Next dilate  $f_\lambda$  very slightly and mollify it, obtaining a smooth even  $f_1 \geq 0$  with  $\text{supp } f_1 \subset (-\frac{\lambda}{2}, \frac{\lambda}{2})$ ,  $\int f_1 = 1$  and  $f_1$  close to  $f_\lambda$  in  $L^1 \cap L^2$ . Finally let  $\zeta \in C_c^\infty((-\frac{1}{2}, \frac{1}{2}))$  be even with  $0 \leq \zeta \leq 1$  and  $\zeta = 1$  on  $\lambda^{-1}\text{supp } f_1$ , and put  $\psi(s) = \zeta(s)(f_1(\lambda s) + \epsilon')^{1/2}$ , which is smooth since  $f_1 + \epsilon' > 0$ . Then  $v = \psi^2 = \zeta^2(f_1(\lambda \cdot) + \epsilon') \rightarrow f_1(\lambda \cdot)$  in  $L^1 \cap L^2$  as  $\epsilon' \rightarrow 0$ , hence  $f_v \rightarrow f_1$ . (This follows the approximation argument of [22, §11].)  $\square$

## 7.2. Certified values.

**Proposition 7.3** (Rational certificates). *Let  $f_n^{(C)}$  be the explicit even step function on  $n$  equal cells of  $[-1, 1]$  generated deterministically by the ancillary file `certify.py` (a floating-point solution of the discretised problem, symmetrised and rounded to integer heights; Appendix A). Evaluating  $\mathcal{Q}_{F_C}(f_n^{(C)})$  in exact rational arithmetic gives:*

| $C$                   | $n$  | $p(C) \geq$ | $(1 + p(C))/2 \geq$ | used in          |
|-----------------------|------|-------------|---------------------|------------------|
| 1                     | 1600 | 0.932282    | 0.966141            | Theorem 1.1      |
| 1.0434                | 1600 | 0.924182    | 0.962091            | Theorem 1.3(ii)  |
| 1.0911                | 1600 | 0.915550    | 0.957775            | Theorem 1.3(i)   |
| 1.2688 ( $\geq C_G$ ) | 400  | 0.885912    | 0.942956            | Theorem 1.2      |
| 1.2890                | 1600 | 0.882798    | 0.941399            | Theorem 1.3(iii) |
| 1.3055                | 1600 | 0.880293    | 0.940146            | Theorem 1.3(iii) |
| 1.3448                | 1600 | 0.874467    | 0.937233            | Theorem 1.3(iii) |

*Proof.* For a step function  $f$  with heights  $f_i$  on cells of width  $h = 2/n$ ,  $\mathcal{Q}_{F_C}(f) = h \sum_i f_i^2 + \sum_{i,j} f_i f_j \kappa_{i-j}$ , where  $\kappa_m = G(mh + h) - 2G(mh) + G(mh - h)$  and  $G$  is the even function with  $G'' = F_C(|\cdot|)$ ,  $G(0) = G'(0) = 0$ , namely  $G(x) = |x|^3/6$  for  $|x| \leq 1$  and  $G(x) = \frac{1}{6} + \frac{|x|-1}{2} + \frac{C}{2}(|x|-1)^2$  for  $1 < |x| \leq 2$ . The heights are integers divided by an exact normalising constant, so  $\mathcal{Q}_{F_C}(f)$  is an explicit rational number; the script computes it exactly and prints  $\lfloor 10^6(2 - \mathcal{Q}) \rfloor / 10^6$ . The step functions are positive (their minima are 0.29, 0.27, 0.25 and 0.19 in the first four rows). Any admissible  $f$  gives  $p(C) \geq 2 - \mathcal{Q}_{F_C}(f)$ .  $\square$

*Remark 7.4* (Optimality). For  $C = 1$  the functional is strictly convex on  $\{\int f = 1\}$ : if  $\int h = 0$  and  $\text{supp } h \subset [-1, 1]$ , then, since  $F_1(|x|) = 1 - (1 - |x|)_+$  on  $[-2, 2]$ ,

$$\mathcal{Q}_{F_1}(h) = \int h^2 - \iint h(x)h(y)(1 - |x - y|)_+ dx dy = \frac{1}{2\pi} \int |\hat{h}(\xi)|^2 \left(1 - \left(\frac{\sin(\xi/2)}{\xi/2}\right)^2\right) d\xi > 0$$

for  $h \neq 0$ . Since  $f$  and  $f(-\cdot)$  give the same value, convexity gives  $\mathcal{Q}_{F_1}(\frac{1}{2}(f + f(-\cdot))) \leq \mathcal{Q}_{F_1}(f)$ , so the restriction to even  $f$  is immaterial for  $C = 1$ ; by strict convexity a minimiser over all admissible  $f$ , if it exists, is unique and hence even. The discretised values at  $n = 400, 800, 1600$  are 0.9322820, 0.9322825, 0.9322826, so numerically  $p(1) = 0.932283$  to six decimals (only the lower bound is used). For  $1 < C \leq 1.6$  strict convexity on  $\{\int h = 0\}$  holds numerically for the discretised problem (so the symmetrisation argument applies there too): the smallest value of  $\mathcal{Q}_{F_C}(h)/\|h\|_2^2$  over  $\int h = 0$  is 0.4305 at  $C = 1$ , 0.2648 at  $C = 1.2688$  and 0.0553 at  $C = 1.603$  (Appendix A). So the displayed values are, to the stated accuracy, the best the method gives for these majorants. None of this is needed for the theorems.

### 7.3. Assembly and order of limits.

*Proof of Theorem 5.16.* Fix  $\varepsilon \in (0, \frac{1}{3})$  and let  $c_\varepsilon : \mathbb{R} \rightarrow [1, C]$  be smooth and nondecreasing with  $c_\varepsilon = 1$  on  $(-\infty, 1 - 2\varepsilon]$  and  $c_\varepsilon = C$  on  $[1 - \frac{3}{2}\varepsilon, \infty)$ . We check (5.11) for  $c = c_\varepsilon$  by rerunning the proof of Proposition 5.12, which uses Corollary 6.8 only through the bound  $\Lambda_{\text{mult}}(K) \leq (C_w + o(1))H$  for  $K \leq Q^{2-\varepsilon_1}$ ; we use  $\text{LS}(C)$  there instead. By Lemma 5.9 the left side of (5.11) is  $\int g(u)x_b(u)^* \Delta x_b(u) du$ ; split each  $x_b(u)$  as in Lemma 5.11 with  $\delta = T^{-1/2}$ ,  $\kappa = T^{-1/4}$ . The tails contribute  $O(T^{-1/4})$  relatively, as in Proposition 5.12. The short part  $x^s(u)$  lies in an interval of length  $\leq 5\delta e^u$ , which contains at most  $5\delta e^u + 1$  integers. If  $u \leq (1 - \varepsilon)\ell$  this number is  $\leq Q^{1-\varepsilon/2}$  and Lemma 5.10 gives the constant  $1 + o(1)$ ; if  $u > (1 - \varepsilon)\ell$  it is  $\leq 5\delta X + 1 \leq Q^{2-\varepsilon_1}$  (as  $e^u < X$  on  $\text{supp } g$ ) and  $\text{LS}(C)$  gives  $C + o(1)$ . Now if  $(x^s(u))_n \neq 0$  then  $|u - \log n| < 2\delta$ . Hence, if  $\log n/\ell < 1 - \frac{3}{2}\varepsilon$ , then  $u < (1 - \varepsilon)\ell$  for  $Q$  large and only the constant  $1 + o(1) \leq (1 + o(1))c_\varepsilon(\log n/\ell)$  occurs; otherwise the constant is at most  $\max(1, C) + o(1) = (1 + o(1))c_\varepsilon(\log n/\ell)$ . Lemma 5.11(iii) then gives (5.11). By Propositions 5.13 and 3.6, uniformly in  $T$ ,

$$N_0^s \geq (2 - 8\theta - (1 - 2\theta)\mathcal{Q}_{F_\varepsilon}(f_v) - o(1))N, \quad F_\varepsilon = c_\varepsilon \min(\cdot, 1 + \varepsilon_4),$$

where we used  $\mathfrak{M} = O(N)$  to absorb  $\ell^{-B}(H\mathfrak{M})^{1/2} = o(N)$ . Given  $\eta' > 0$ , choose an even admissible  $f$  with  $2 - \mathcal{Q}_{F_C}(f) \geq p(C) - \eta'$ . Then choose, in this order:  $\lambda < 2$  and  $\psi$  by Lemma 7.2, so that  $\mathcal{Q}_{F_C}(f_v) \leq \mathcal{Q}_{F_C}(f) + \eta'$ ; then  $\varepsilon_1$  with  $\lambda(1 + \varepsilon_1) \leq 2 - \varepsilon_1$ ; then  $\theta, \varepsilon, \varepsilon_3$  (hence  $\varepsilon_4$ ) so small that  $8\theta + |\mathcal{Q}_{F_\varepsilon}(f_v) - \mathcal{Q}_{F_C}(f_v)| \leq \eta'$ , which is possible by dominated convergence for the fixed  $v$  since  $F_\varepsilon \rightarrow F_C$  off  $\{1\}$  as  $\varepsilon, \varepsilon_4 \rightarrow 0$  and  $0 \leq F_\varepsilon \leq 2C$ . Letting  $Q \rightarrow \infty$  and then  $\eta' \rightarrow 0$  proves the claim for  $N_0^s$ ; the same choices give  $N_0^*$  and  $N_d$ .  $\square$

*Proof of Theorem 1.1.* Fix  $\varepsilon > 0$ ; here  $\varepsilon$  is the accuracy in the statement, and we write  $\varepsilon'$  for the band parameter of §2.3 and Proposition 6.24. By Lemma 6.22,  $C_T^+(w) \leq 1 + O(1/\log(1/\eta))$  for  $w \in \{w_\eta, w_\eta^{\text{sm}}\}$ , with an explicit constant; choose  $\eta_0$  with  $C_T^+(w) \leq 1 + \varepsilon/3$  for  $\eta \leq \eta_0$ , and fix  $\eta \leq \eta_0$ . Apply Proposition 6.24 with the band constant  $C_{\text{band}} = 2w_{\text{max}}/(\mathcal{E}I_w)$ , which comes from the classical large sieve (4.1); any bound  $O(H)$  suffices on the band, since the band disappears as  $\varepsilon' \rightarrow 0$ . Run the proof of Theorem 5.16 with Proposition 6.24 in place of  $\text{LS}(C)$ : hypothesis (5.11) holds with  $c = \tilde{C}_{\varepsilon'}$  (the factor  $(1 + \vartheta_Q^*)(1 + \kappa_T)^3$  is  $1 + o(1)$  uniformly in  $T$ , and  $\mathcal{E}_{\text{TI}} = o(H|J|L^2\ell)$ ), and  $\mathcal{Q}_{F_\varepsilon}(f_v) \rightarrow \mathcal{Q}_{F_{C_T^+(w)}}(f_v)$  as  $\varepsilon', \varepsilon_4 \rightarrow 0$  by dominated convergence, since  $\tilde{C}_{\varepsilon'} \min(\cdot, 1 + \varepsilon_4) \rightarrow F_{C_T^+(w)}$  off  $\{1\}$  and is bounded by  $2\max(C_{\text{band}}, C_T^+(w))$ . This gives  $\liminf_Q \inf_T N_0^s/N \geq p(C_T^+(w)) \geq p(1) - \varepsilon/3$  by Lemma 7.1, which applies since  $C_T^+(w) \geq 1$

(Proposition 6.21); and  $p(1) \geq 0.932282$  by Proposition 7.3. The same argument gives  $N_0^*$  and  $N_d$ .

This proof uses neither Lemma 6.5 nor Lemma 6.7 nor Corollary 6.8: from §6.1 it uses only the elementary Lemmas 6.1, 6.2, 6.9, Proposition 6.4, and the layer-cake argument in the proof of Lemma 6.10 (reused in Lemma 6.22), which enter through §6.2.  $\square$

*Proof of Theorem 1.3.* For the fixed weight  $w$ , the proof of Theorem 1.1 (with  $\eta$  fixed; §2.3) gives  $\liminf_Q \inf_T N_0^s/N \geq p(C_T^+(w))$ , and likewise  $p(C_G R_w)$  by Theorem 1.2. The upper bounds for  $C_T^+(w)$  and  $R_w$  are the computer-assisted inputs, certified in interval arithmetic (Table 6, Table 5; ancillary files `ctplus_arb.py` and `rw_arb.py`, whose results are collected in `certify_ctplus_arb.log`, and Lemma 6.11 for the far field of  $P_w$ ):  $C_T^+(w) \leq 1.090990$  ( $\eta = 10^{-3}$ ),  $C_T^+(w) \leq 1.043290$  ( $\eta = 10^{-4}$ ),  $R_w \leq 1.059816, 1.028831, 1.015859$ , and  $C_G < 1.268774$  (`certify_constants_arb.log`). By monotonicity of  $p$  and Proposition 7.3,  $C_T^+(w) \leq 1.0911$  gives  $p \geq 0.915550$  and  $C_T^+(w) \leq 1.0434$  gives  $p \geq 0.924182$ . For (iii):  $C_G R_w < 1.268774 \cdot 1.0599 < 1.3448$ ,  $1.268774 \cdot 1.0289 < 1.3055$  and  $1.268774 \cdot 1.0159 < 1.2890$ , and Proposition 7.3 gives  $0.874467, 0.880293$  and  $0.882798$ .  $\square$

## 8. REMARKS

**8.1. Limits of the method.** Given the certificate of §3 and a majorant of the shape  $F_C$ , the value  $p(C)$  is extremal (Remark 7.4); improvements must come from the large sieve side. Two barriers remain. First, the bandwidth: all our large sieve bounds are for intervals of length  $K \leq Q^{2-\varepsilon}$ . For  $K \gg Q^2$  the length term of the large sieve exceeds the family mass  $H \asymp Q^2$ : if  $x$  is the indicator of the progression  $n \equiv 1 \pmod{q_0}$  in an interval of length  $K$ , with  $q_0 \in [\eta Q, Q]$  and  $w(q_0/Q) > 0$ , then the characters modulo  $q_0$  alone give  $x^* \Delta x \geq \omega(q_0) \varphi^*(q_0) (K/q_0) \|x\|^2 (1+o(1))$ , so  $\Lambda_{\text{mult}}(K)/H \gg_w K/Q^2$  is unbounded. Hence no worst-case majorant of the form  $F \leq C$  exists on  $\alpha > 2$ ; a bound there would have to use the structure of the prime vectors. This is the analogue, in the  $q$ -aspect, of the obstruction at  $\alpha = 1$  for  $\zeta$ . Second, the constant  $p(1)$  is the value of the GUE form factor  $\min(\alpha, 1)$  on  $[0, 2]$ , and the method cannot exceed it without information on  $\alpha > 2$ . The reason is that under GRH the family form factor on  $(1, 2)$  is expected to equal 1 (this is what the asymptotic of [8] gives for their dyadic family), so no majorant lying below  $F_1$  on  $(1, 2)$  can be expected to hold. (Under GRH, [9] exceed  $p(1)$  for the low-lying statistic with test functions that are not supported in  $|\alpha| \leq 2$  but are nonpositive for  $|\alpha| \geq 2$ , together with the nonnegativity of the family pair-correlation function; see §8.8. We have not examined whether test functions of this kind can be used in the present certificate.)

**8.2. Which ingredient gives what.** The three devices of §1.5 enter separately, which is useful when checking the argument. All values below are exact rational certificates at bandwidth  $\lambda \rightarrow 2$  (ancillary files `certify_tiers.py` and `tiers.log`).

- (a) *Flattening, Lemma 6.5, no time localisation.* Using only the Schur product bound (5.3) with  $\Lambda_{\text{mult}}(Y) \leq (C_w + o(1))H$  and Lemma 5.7, the majorant is  $F = C_w \min(\alpha, 1)$ , and for log-wide weights one gets  $\liminf N_0^s/N \geq 0.788171 - \varepsilon$  for  $\eta \leq \eta_0(\varepsilon)$  (distinct  $\geq 0.894085 - \varepsilon$ ).
- (b) *Flattening and time localisation, only the classical large sieve.* With  $w = \mathbf{1}_{[\eta, 1]}$ , (4.1) gives  $\Lambda_{\text{mult}}(K) \leq (2/(\mathcal{E}(1 - \eta^2)) + o(1))H$  for  $K \leq Q^{2-\varepsilon}$ , which is  $\leq (4.175 + o(1))H$  for  $\eta \leq 0.01$ ; Theorem 5.16 with  $C = 4.175$  then gives  $\liminf N_0^s/N \geq 0.738236$  (distinct  $\geq 0.869118$ ; even certificate with 300 cells, `tiers.log`). This is a lower bound for the same constant  $p(2; F_{4.175})$  as the entry 0.738253 in the row  $\kappa = 0$  of the table after Theorem 1.4; that entry comes from a finer step function with 400 cells, which gives a slightly larger value. Neither Lemma 6.5 nor §6.2 is used.

- (c) *None of the three.* Bandwidth  $\lambda < 1$ , where  $\Delta$  is diagonally dominant, gives the Montgomery–Taylor value 0.6725, as in [22].

**8.3. The  $\zeta$  barrier.** For  $\zeta(s)$  the analogue of  $\Delta$  is the mean value  $\int_T^{2T} |\sum_{n \leq N} x_n n^{-it}|^2 dt = (T + O(N)) \|x\|^2$  of Montgomery–Vaughan [28]; its length term dominates as soon as  $N \gg T$ , i.e.  $\alpha > 1$  in units of  $\log T$ . The family replaces the  $T$  harmonics of one  $L$ -function by the  $\asymp Q^2$  harmonics  $\chi$ , while the height  $T$  stays polylogarithmic; this is why bandwidth 2 becomes available. Nothing in this paper improves the constant 0.6725 for  $\zeta$  (for which Wang [42] has recently obtained an improvement by  $6.66 \cdot 10^{-8}$ , by refining the inequality of [25]; see [43] for short intervals  $(T, T + T^\theta]$ ) or for a single modulus: in [22] the family modulo one prime  $q$  has only  $q$  harmonics, and the same barrier appears at  $\alpha = 1$ . For the weight-2 newforms of prime level  $N$ , [15] identifies the analogous obstruction: beyond bandwidth 1 the Kloosterman terms with modulus  $N$  produce main terms, whose evaluation would require strong information on primes in progressions to modulus  $N$ . In our family it is the average over the moduli  $q \leq Q$  that gives bandwidth 2; we make no claim about newforms.

**8.4. Individual  $L$ -functions and the weights.** The certificate is summed with nonnegative weights, so the theorem is a statement about a weighted average. Passing to an unweighted average over  $\eta Q \leq q \leq Q$  multiplies the bound for the proportion of uncertified zeros by  $\max \omega / \min \omega$ , which is  $\asymp \eta^{-2} \log \log Q$  for  $w_\eta$  (because of the factor  $q/\varphi(q)$ ), so it is useless even for fixed  $\eta$ . The weights are essential to the constants: for the dyadic weight  $w = u^{-2} \mathbf{1}_{[1/2, 1]}$  one has  $R_w = \pi^2/(6 \log 2) \approx 2.37$ , so the Gauss route gives  $C_w \approx 3.0$  and a much weaker certificate (for  $w = \mathbf{1}_{[1/2, 1]}$  it is worse:  $R_w = 4\pi^2/9 \approx 4.39$ ); the constants approach 1 only when the weights spread over many logarithmic scales. Nothing is said about individual characters; a positive proportion of characters with few simple critical zeros is not excluded.

**8.5. Higher zeros.** Heights polynomial in  $Q$  are treated in §9: uniformly for  $\ell^{a_0} \leq T \leq Q^{\kappa_1}$  the proportion is at least  $p(\bar{\lambda}_T; F_1) - \varepsilon$ , where  $\bar{\lambda}_T = (2 + \kappa)/(1 + \kappa)$  at  $T = Q^\kappa$  (Theorem 1.4). A time localisation at scale  $T^{-1/2}$ , as in §5.6, with the tails bounded through  $\Lambda_{\text{mult}}(Y) \leq 2w_{\max} Q^2$ , would reach only the support  $(2 + \kappa/2)/(1 + \kappa)$ ; the scale  $T^{-1+\varepsilon_5}$  with the tail bound of Lemma 9.13 reaches the ceiling  $\bar{\lambda}_T < 2$ , which is the limit of worst-case bounds (Proposition 9.30). So the polylogarithmic range is indeed the most favourable one for this method.

**8.6. Relation to [1, Remark 7.2].** The announcement [1, Remark 7.2] of the proportion 0.811 for primitive  $\chi \bmod q$ ,  $q \leq Q$ , with smooth weights  $(1 - q/Q)^2$  at bandwidth 3/2, using a weighted large sieve constant 1.263, gives no details. With the weights and method of this paper the corresponding computation gives a different constant: for the weight  $(1 - q/Q)^2$  per primitive character (without the factor  $q/\varphi(q)$  of (1.1)), the analogue of the signed Toeplitz constant  $C_T(w) = \sup_{S,t} R_S$  of Definition 6.18 is  $\approx 2.03$  (Table 7), and Theorem 5.16 with  $C = 2.03$  would give  $p(2.03) \approx 0.805$ . Both are floating-point values, not certified (2.03 is not a value of  $C_T^+$ , and no matching lower bound is proved), and neither is used in any proof. They refer to bandwidth 2 and to our normalisations, whereas the figure of [1] refers to bandwidth 3/2, so the two are not directly comparable; we do not pursue the comparison further.

**8.7. Is 93% plausible?** An unconditional 93% for a Dirichlet family is far above the unconditional proportions obtained by Levinson’s method with mollifiers [12, 39] (0.56 and 0.6044 simple critical zeros, in slightly different families and normalisations; 1/3 for the twists of a fixed  $\text{GL}_2$  form [11]). There is no conflict. The methods differ: Levinson’s method bounds critical zeros through mollified moments, whereas the certificate bounds simple critical zeros through a pair-correlation functional; in families the pair-correlation method has always given much larger proportions, but previously only under GRH [31, 8, 38, 9]. Within the certificate

itself, GRH was needed only to read the zero side as a positive sum over real ordinates (in [8] it is also used on the prime side; see §8.8); the inertia argument of [1] (Lemma 3.1) removes that need character by character, and off-line zeros simply fail to be certified. The prime side needs only upper bounds, which the large sieve supplies. Finally, the theorem allows a small weighted proportion of characters (for example those with exceptional zeros, or with many off-line zeros) to be arbitrary.

**8.8. Comparison with the GRH results.** Under GRH, [8] compute the family form factor asymptotically for  $|\alpha| < 2$ , and [38] optimises the test function among those supported in  $|\alpha| \leq 2$  to reach 0.9322. Chirre, Gonçalves and de Laat [9, Theorem 5] go further under GRH: for the weighted statistic of low-lying zeros of [8, 38] (with  $\Phi$  and  $\widehat{\Phi}(i \cdot)$  nonnegative) they obtain 0.9350, using test functions that are not supported in  $|\alpha| \leq 2$  but are nonpositive for  $|\alpha| \geq 2$ , so that the nonnegativity of the family pair-correlation function controls the range  $|\alpha| > 2$ , and optimising over them by semidefinite programming. So 0.9322 is not the best known GRH constant for that statistic. The present method recovers Sono's constant because only the upper half of the form factor enters the certificate, and that upper half is supplied unconditionally by flattening, time localisation and a large sieve with constant tending to 1. The prime-side GRH input of [8] (bounds for prime sums twisted by characters of small conductor) is not needed here: the family has conductors  $\geq \eta Q$  only, and on sifted vectors the Toeplitz identity of §6.2 expresses the family form exactly through a signed Farey measure, whose negative part is not evaluated but majorised by an explicit positive measure (Lemma 6.17).

## 9. ZEROS AT POLYNOMIAL HEIGHT

This section proves Theorem 1.4 and Corollary 1.5. The argument is that of Sections 2–7, with the three changes described at the end of §1.5: the log-conductor  $\ell_* = \log(QT)$  replaces  $\ell = \log Q$  wherever  $\ell$  measures the density of zeros, the bandwidth or the flattening; the time localisation is at scale  $T^{-1+\varepsilon_5}$  instead of  $T^{-1/2}$ ; and the tails of the localisation are bounded by Lemma 9.13 instead of the crude bound  $\Lambda_{\text{mult}}(Y) \leq 2w_{\max}Q^2$ , which needs  $Y \leq Q^2$ . On the zero side nothing new is needed. Subsection 9.5, which contains a sharp hybrid large sieve and the example showing that  $\bar{\lambda}_T$  is the limit of worst-case large sieve bounds, is not used in the proofs.

*Notation.* In this section  $\kappa$  (with subscripts 1, c or superscripts  $(j)$ ) always denotes a height exponent,  $T = Q^\kappa$ . The parameter of Lemma 5.11(i) and of (6.12), called  $\kappa$  there, is written  $\varkappa$  here; the quantity  $\kappa_T = T^{-1/4}$  of Proposition 6.24 does not occur, and its role is played by  $\varkappa_{\text{loc}} = T^{-\varepsilon_5/2}$  (9.5). The Toeplitz form  $T^\natural$  of §6.2 is unrelated to the height  $T$ . Without subscript,  $\delta$  denotes a localisation scale;  $\delta_\rho = \text{Re } \rho - \frac{1}{2}$  and  $\delta_0$  (§4.4) are unrelated. We write  $\bar{\lambda}_T$ ,  $\bar{\lambda}(\kappa)$  and  $\ell_*$  as in (1.4), and  $p(\bar{\lambda}; F)$  as in (1.5).

### 9.1. Setup at height $T$ .

*What is used unchanged.* We keep all notation of Sections 2–7; the objects of §§2–3 are formed with the bandwidth  $L = \lambda\ell_*$  of (9.4) below. The following results are used at height  $T$  without change. Each is a statement about a single character, about linear algebra, about the family at a fixed height, or about intervals of at most  $Q^{2-\varepsilon}$  integers, and none of their proofs uses an upper bound for  $T$ ; the second column gives the reason.

| Result                                                                | Why it holds verbatim at height $T \leq Q^{\kappa_1}$                                                                                       |
|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| (2.1), (2.2), Lemma 6.1                                               | masses of the family; no $T$ .                                                                                                              |
| Lemmas 2.3, 2.4                                                       | statements for every $L$ and every lattice offset.                                                                                          |
| (2.3), (2.4), (2.5), Lemma 2.6, Proposition B.1                       | for every $\chi \in \mathcal{F}$ and every $L$ ; the prime sum is over $n < e^L$ .                                                          |
| (2.7)–(2.9), Lemma 5.9                                                | for every bounded interval $J$ .                                                                                                            |
| Lemmas 3.1, 3.2, Proposition 3.3                                      | finite-dimensional linear algebra in any dimension $d$ (here $d \asymp T\ell_*$ may be a power of $Q$ ).                                    |
| Lemma 4.5                                                             | no $T$ .                                                                                                                                    |
| Lemma 4.7                                                             | the bad set depends only on $Q$ ; (4.5) is used only at heights $T_j \leq e^{1/2}Q^{1/4}$ .                                                 |
| Lemma 5.5, (5.6)                                                      | one smooth function; the proof of (5.6) uses only $\delta^{-1} \leq T$ and $ t  \leq 3T$ .                                                  |
| Lemma 5.10                                                            | no $T$ ; independent of the position of the interval.                                                                                       |
| Lemma 5.11(i)–(iii)                                                   | for every $\delta \in (0, \frac{1}{4}]$ and $\varkappa \in (0, 1]$ ((iv) holds but is not used).                                            |
| Hypothesis 5.15, Corollary 6.8, (4.1)                                 | statements about the family form on intervals of at most $Q^{2-\varepsilon}$ (resp. $K$ ) integers; no $T$ .                                |
| Lemma 6.10                                                            | a statement about the Farey constant $R_w$ ; no $T$ .                                                                                       |
| Lemmas 6.13, 6.15, 6.17, 6.20, 6.22, Proposition 6.21, (6.12), (6.13) | statements about $\Delta$ on $Q$ -rough vectors, the measure $\mu^\sharp$ , and intervals of at most $Q^{2-\varepsilon}$ integers; no $T$ . |
| (6.16)                                                                | an identity (expansion of $x_\sharp^s$ ), valid for every $T$ and $\delta$ .                                                                |
| Lemma 7.2, Proposition 7.3, Remark 7.4                                | the variational problem at support 2; generalised in §9.4.                                                                                  |

The proofs of Lemmas 4.1–4.3, 5.1, 5.2, 5.4, 5.6, 5.7 and of Propositions 3.6, 4.4, 4.6, 4.8, 5.12, 5.13, 6.24 and Theorem 5.16 are used with the changes stated in §§9.2–9.4; so are (5.4), (5.7) and (5.10), which are redone at height  $T$  in Lemma 9.1(S3), Lemma 9.11 and (9.10), respectively. We use the weighted form of (4.1): for  $x$  supported on  $K$  consecutive integers,

$$(9.1) \quad x^* \Delta x = \sum_{\chi \in \mathcal{F}} \omega_\chi \left| \sum_n x_n \chi(n) \right|^2 \leq w_{\max} \sum_{q \leq Q} \frac{q}{\varphi(q)} \sum_{\chi \bmod q}^* \left| \sum_n x_n \chi(n) \right|^2 \leq w_{\max} (Q^2 + K) \|x\|^2.$$

*Heights and cells.* Fix  $a_0 > 0$  and  $\kappa_1 > 0$ . Heights range over  $\ell^{a_0} \leq T \leq Q^{\kappa_1}$ ; thus  $\ell \leq \ell_* \leq (1 + \kappa_1)\ell$ , and  $\bar{\lambda}_T$  decreases from 2 towards  $\bar{\lambda}(\kappa_1) > 1$  as  $T$  increases. Uniformity in  $T$  is obtained on finitely many *cells*  $[\kappa^{(j)}, \kappa^{(j+1)}] \subset [0, \kappa_1]$  (§9.4). Throughout §§9.1–9.3 we fix a cell top  $\kappa_c \in (0, \kappa_1]$  and prove all statements uniformly for

$$(9.2) \quad \ell^{a_0} \leq T \leq Q^{\kappa_c}.$$

*Parameters and order of limits.* We fix, in this order:

- (P1)  $a_0, \kappa_1 > 0$ ;  $\eta$  and an admissible  $w$ ; a cell top  $\kappa_c \in (0, \kappa_1]$ ;
- (P2) a bandwidth  $\lambda \in (0, \bar{\lambda}(\kappa_c))$  and a real even window  $\psi \in C_c^\infty((-\frac{1}{2}, \frac{1}{2}))$ ,  $\psi \not\equiv 0$ ;
- (P3)  $\varepsilon_1 \in (0, \frac{1}{4})$  with  $\lambda(1 + \varepsilon_1) \leq \bar{\lambda}(\kappa_c) - \varepsilon_1$ ; then  $\theta \in (0, \frac{1}{4})$  and  $\varepsilon_3 \in (0, \frac{1}{4})$ ;
- (P4) the band parameter  $\varepsilon \in (0, \varepsilon_{\max}]$ , where  $\varepsilon_{\max} := \frac{1}{8(1+\kappa_1)}$ ;
- (P5) *only then* the localisation exponent

$$(9.3) \quad \varepsilon_5 := \frac{\min(\varepsilon, \varepsilon_1)}{4(1 + \kappa_1)};$$

- (P6) the constants  $B_1$  and  $B$  of §4.4, and the smooth cut-offs  $\Upsilon_0$  (depending on  $\varepsilon_1$ ), the dyadic partition of unity  $(\psi_j)$  with its derivative bounds  $(c_i)$ , and the localisation cut-off  $\rho$ , as in §2.3, which we take even.

All of these are fixed while  $Q \rightarrow \infty$ , and  $T$  varies in (9.2); all  $o(1)$  terms are uniform in  $T$ , and implied constants may depend on the fixed data (including  $\kappa_1$ ). The choice of  $\varepsilon_5$  after  $\varepsilon$  matters: it guarantees that the region where the family form has constant 1 reaches  $\alpha = 1 - \frac{3}{2}\varepsilon$  (Lemma 9.1(S2)). The final order of limits is: first  $Q \rightarrow \infty$ ; then  $\theta, \varepsilon, \varepsilon_3 \rightarrow 0$  (with  $\varepsilon_5$  following  $\varepsilon$ ); then  $\lambda \rightarrow \bar{\lambda}(\kappa_c)$  and  $\psi$  tending to an optimiser, with  $\varepsilon_1 = \varepsilon_1(\lambda)$ ; then the cells are refined; and only at the very end  $\eta \rightarrow 0$  (Theorem 1.4(a), (b)). The precise choices are made in §9.4. We put

$$(9.4) \quad L = \lambda \ell_*, \quad X = e^L = (QT)^\lambda, \quad Y = X^{1+\varepsilon_1}, \quad I = (T, 2T], \quad J = [(1+\theta)T, (2-\theta)T],$$

$$(9.5) \quad \delta := T^{-1+\varepsilon_5}, \quad \varkappa_{\text{loc}} := T^{-\varepsilon_5/2}, \quad K(u) := 5\delta e^u + 1.$$

Since  $T \geq \ell^{a_0} \rightarrow \infty$  we have  $\delta \leq \frac{1}{4}$  and  $\varkappa_{\text{loc}} \leq 1$  for  $Q$  large. With this  $L$ ,  $d = |J|L/(2\pi) + O(1) \asymp T\ell_*$ , which may be a power of  $Q$ .

*Size facts.*

**Lemma 9.1.** *Uniformly for  $T$  in (9.2) and  $Q \geq Q_0$  (depending on the fixed data):*

- (S1)  $Y \leq Q^2 T (QT)^{-\varepsilon_1} = Q^{2-\varepsilon_1} T^{1-\varepsilon_1}$ ; in particular  $Y/Q^2 \leq T(QT)^{-\varepsilon_1}$  and  $2Y < (QT)^2$ .
- (S2) For  $u < L$  (the support of  $g$ ),  $K(u) \leq Q^{1-\varepsilon/2}$  if  $u \leq (1-\varepsilon)\ell_*$ ;  $K(u) \leq Q^{5/4}$  if  $(1-\varepsilon)\ell_* < u \leq (1+\varepsilon)\ell_*$ ; and  $K(u) \leq Q^{2-\varepsilon_1/2}$  always.
- (S3)  $R_{\max} := \max\{R_j : N_j \leq 2Y\} \leq 2Y/(QT) \leq 2Q(QT)^{-\varepsilon_1}$ , and  $R_j \leq N_j^{1/2-\varepsilon_3}$  for every  $j$  with  $N_j \leq 2Y$ .
- (S4)  $1 + \varkappa_{\text{loc}}^{-1} \leq 2T^{\varepsilon_5/2} \leq 2Q^{\varepsilon_1/8}$ .

*Proof.* (S1) Write  $\kappa := \log T / \log Q \leq \kappa_c$ . Then  $\log Y/\ell = \lambda(1+\varepsilon_1)(1+\kappa) \leq (\bar{\lambda}(\kappa_c) - \varepsilon_1)(1+\kappa)$ , and  $\bar{\lambda}(\kappa_c)(1+\kappa) \leq \bar{\lambda}(\kappa)(1+\kappa) = 2+\kappa$  because  $\bar{\lambda}$  is decreasing. Hence  $\log Y/\ell \leq 2+\kappa-\varepsilon_1(1+\kappa)$ , which is the claim. Since  $\varepsilon_1 > 0$ ,  $2Y < (QT)^2$  for  $Q$  large.

(S2) As  $u < L$ ,  $e^u < X \leq Y$ . If  $u \leq (1-\varepsilon)\ell_*$ , then  $5\delta e^u \leq 5T^{-1+\varepsilon_5}(QT)^{1-\varepsilon} = 5Q^{1-\varepsilon}T^{\varepsilon_5-\varepsilon} \leq 5Q^{1-\varepsilon}$  because  $\varepsilon_5 \leq \varepsilon$ . On the band,  $5\delta e^u \leq 5Q^{1+\varepsilon}T^{\varepsilon+\varepsilon_5} \leq 5Q^{1+\varepsilon(1+\kappa_1)+\kappa_1\varepsilon_5}$ , and  $\varepsilon(1+\kappa_1) \leq \frac{1}{8}$ ,  $\kappa_1\varepsilon_5 \leq \frac{\varepsilon}{4} \leq \frac{1}{32}$ . Always,  $5\delta e^u \leq 5\delta Y \leq 5Q^{2-\varepsilon_1}T^{\varepsilon_5-\varepsilon_1} \leq 5Q^{2-\varepsilon_1}$  by (S1) and  $\varepsilon_5 < \varepsilon_1$ .

(S3)  $R_j = \lfloor N_j^{1-\varepsilon_3}/(QT) \rfloor \leq 2Y/(QT)$  for  $N_j \leq 2Y$ , and  $2Y/(QT) \leq 2Q(QT)^{-\varepsilon_1}$  by (S1). For the second claim,  $R_j \leq N_j^{1/2-\varepsilon_3} \cdot N_j^{1/2}/(QT) \leq N_j^{1/2-\varepsilon_3}$  as  $N_j \leq 2Y < (QT)^2$ .

(S4)  $T^{\varepsilon_5/2} \leq Q^{\kappa_1\varepsilon_5/2}$  and  $\kappa_1\varepsilon_5 \leq \varepsilon_1/4$ . □

**9.2. The zero side at polynomial height.** The per-character certificate of Proposition 3.3 holds for every  $\chi \in \mathcal{F}$  and every window, in any dimension  $d$ . With  $A_\chi$ ,  $E_\chi$ ,  $\hat{A}_\chi$ ,  $\hat{E}_\chi$  and  $\mathcal{M}_\chi = 4 \operatorname{tr} \hat{A}_\chi - \|\hat{A}_\chi\|_{\mathbb{F}}^2$  as in §3, now formed with  $I = (T, 2T]$  and the bandwidth (9.4), it gives for every  $\chi \in \mathcal{F}$

$$(9.6) \quad N_{0,\chi}^s \geq \mathcal{M}_\chi - 2N_\chi, \quad N_{0,\chi}^* \geq \mathcal{M}_\chi - 2N_\chi, \quad N_{d,\chi} \geq \frac{1}{2}(\mathcal{M}_\chi - N_\chi).$$

What changes at polynomial height is the analysis of the family sums: the normalisation, the pointwise bound, the trace and the exterior zeros. We follow the order of §4. Put

$$\mathcal{L}_*(t) := \log(Q(|t|+3)) \quad (t \in \mathbb{R}),$$

so that  $\#\{\rho : |\gamma - t| \leq 1\} \ll \mathcal{L}_*(t)$  for every  $\chi \in \mathcal{F}$  by (2.6), and  $\mathcal{L}_*(t) \leq \ell_* + O(1)$  for  $|t| \leq 3T$ .

*Weighted Riemann–von Mangoldt.*

**Lemma 9.2.** *Uniformly for  $T$  in (9.2),*

$$N = \frac{HT\ell_*}{2\pi} \left( 1 + O\left( \frac{1 + \log(1/\eta)}{\ell_*} + \frac{1}{T} \right) \right).$$

*Proof.* As in the proof of Lemma 4.1: with  $N^{\text{sym}}(t, \chi) = \frac{t}{\pi} \log \frac{qt}{2\pi e} + O(\log q(t+2))$  [13, Ch. 16], and since  $\mathcal{F}$  is closed under conjugation with  $\omega_{\bar{\chi}} = \omega_{\chi}$ , the identity  $N = \frac{1}{2} \sum_{\chi} \omega_{\chi} (N^{\text{sym}}(2T, \chi) - N^{\text{sym}}(T, \chi))$  is exact and gives  $N = \frac{T}{2\pi} \sum_{\chi} \omega_{\chi} (\log \frac{qT}{2\pi e} + 2 \log 2) + O(H\ell_*)$ . On  $\mathcal{F}$ ,  $\log(qT) = \ell_* + O(\log \frac{1}{\eta})$ .  $\square$

The terms  $\log T$  now belong to  $\ell_*$ ; in Lemma 4.1 they had to be  $o(\ell)$ .

*The pointwise bound and finite centres.*

**Lemma 9.3.** *Let  $\mathfrak{F}(t)^2 := \sum_{\chi \in \mathcal{F}} \omega_{\chi} \nu_{\chi}(t)^2$ . Then, for all real  $t$ ,*

$$\mathfrak{F}(t) \ll_w H^{1/2} \left( \mathcal{L}_*(t) + (1 + Y/Q^2)^{1/2} L \right).$$

*Proof.* By the digamma asymptotic,  $|\mu_{\chi}(t)| \ll \log(q(|t|+2)) \leq \mathcal{L}_*(t)$ . Since  $|P_{\chi}| \leq |S_{\chi}[a]|/\pi$ , twisting by  $n^{-it}$  does not change  $|a_n|$ , and  $a$  is supported on  $[1, Y]$ , (9.1) gives  $\sum_{\chi} \omega_{\chi} |P_{\chi}(t)|^2 \leq \pi^{-2} w_{\max}(Q^2 + Y) \sum_n a_n^2 \ll w_{\max} Q^2 (1 + Y/Q^2) L^2$ , because  $\sum_{n \leq Y} \Lambda(n)^2/n \leq \log Y (\log Y + O(1)) \ll L^2$ . As  $H \asymp_w Q^2$ , the claim follows.  $\square$

Compared with Lemma 4.2, the bound has lost the factor  $(1 + Y/Q^2)^{1/2} \leq (1 + T)^{1/2}$  (Lemma 9.1(S1)). It enters only the endpoint errors below, which carry a factor  $1/T$  that absorbs this loss. As in §4.2 we use the envelope  $\varpi_A(x) := C_A L(1 + L|x|)^{-A}$  of Lemma 2.3, so that  $|p_k(t)| \leq \varpi_A(t - \tau_k)$  for real  $t$ , and put  $\Xi_k(U) := \int_U \varpi_A(t - \tau_k) \mathfrak{F}(t) dt$  for measurable  $U \subset \mathbb{R}$ .

**Lemma 9.4.** *Let  $A > 5/2$ . Uniformly in  $T$  and in the lattice offset  $\tau_0$ ,*

$$\sum_{k \notin \mathbf{K}} \Xi_k(J)^2 + \sum_{k \in \mathbf{K}} (\Xi_k(\mathbb{R}) \Xi_k(J^c) + \Xi_k(J^c)^2) \ll_A H (\ell_*^2 + (1 + Y/Q^2) L^2),$$

and

$$\int_J \sum_{k \notin \mathbf{K}} \varpi_A(t - \tau_k)^2 \mathcal{L}_*(t) dt + \int_{J^c} \sum_{k \in \mathbf{K}} \varpi_A(t - \tau_k)^2 \mathcal{L}_*(t) dt \ll_A L \ell_*,$$

and the same with  $\mathcal{L}_*$  replaced by 1 and the bound  $L \ell_*$  by  $L$ .

*Proof.* This is the proof of Lemma 4.3 with  $\mathcal{L}(t) = \ell + \log(|t|+2)$  replaced by  $\mathcal{L}^b(t) := \mathcal{L}_*(t) + (1 + Y/Q^2)^{1/2} L$ , so that  $\mathfrak{F}(t) \ll H^{1/2} \mathcal{L}^b(t)$  by Lemma 9.3. The added term is constant in  $t$  and passes unchanged through the lattice sums. The two properties of  $\mathcal{L}$  used there hold for  $\mathcal{L}_*$ : for  $t \in J$ ,  $\mathcal{L}_*(t) \leq \log(Q(2T+3)) \ll \ell_*$ ; and for  $k \in \mathbf{K}$  we have  $|\tau_k| \leq 2T$ , so  $\mathcal{L}_*(\tau_k + x/L) \leq \ell_* + \log(3 + |x|) + O(1) \ll \ell_*(1 + |x|)^{1/2}$ . The lattice sums  $\sum_{k \notin \mathbf{K}} (1 + D_k)^{-\sigma} + \sum_{k \in \mathbf{K}} (1 + E_k)^{-\sigma} \ll_{\sigma} 1$  ( $\sigma > 1$ ) do not depend on  $T$ . The first claim follows with  $\ell^2$  replaced by  $(\ell_* + (1 + Y/Q^2)^{1/2} L)^2 \ll \ell_*^2 + (1 + Y/Q^2) L^2$ , and the second with  $\ell$  replaced by  $\ell_*$ .  $\square$

**Proposition 9.5** (Finite-centre replacement at height  $T$ ). *Uniformly in  $T$ ,*

$$(9.7) \quad \sum_{\chi \in \mathcal{F}} \omega_\chi \operatorname{tr} \widehat{G}_\chi = \sum_{\chi \in \mathcal{F}} \omega_\chi \int_J \nu_\chi(t) dt + O(H(1 + Y/Q^2)^{1/2}),$$

$$(9.8) \quad \sum_{\chi \in \mathcal{F}} \omega_\chi \operatorname{tr} ((G_\chi/L)^2) = \sum_{\chi \in \mathcal{F}} \omega_\chi \iint_{J^2} \Phi(t - t')^2 \nu_\chi(t) \nu_\chi(t') dt dt' + O(H(\ell_*^2 + (1 + Y/Q^2)L^2)).$$

The error in (9.7) is  $O(NT^{-1/2}/\ell_*)$ , and the error in (9.8), divided by  $a^2L^2$ , is  $O(N(T^{-1} + (QT)^{-\varepsilon_1})/\ell_*)$ ; both are  $o(N)$ .

*Proof.* The proof of Proposition 4.4, with  $A = 3$  and Lemmas 9.3 and 9.4 in place of Lemmas 4.2 and 4.3. For (9.7):  $\sum_\chi \omega_\chi \int e_J \nu_\chi \leq \int |e_J| H^{1/2} \mathfrak{F} \ll H \int |e_J| (\mathcal{L}_* + (1 + Y/Q^2)^{1/2} L) \ll H(L\ell_* + (1 + Y/Q^2)^{1/2} L^2)$  by Lemma 9.4; divide by  $aL^2$  and use  $\ell_*/L = 1/\lambda \ll 1$ . For (9.8), the difference is at most  $2a \sum_{k \notin K} \Xi_k(J)^2 + 2a \sum_{k \in K} (\Xi_k(\mathbb{R}) \Xi_k(J^c) + \Xi_k(J^c)^2)$ , and Lemma 9.4 applies. For the last sentence: by Lemma 9.2,  $N \asymp HT\ell_*$ , and by Lemma 9.1(S1),  $Y/Q^2 \leq T(QT)^{-\varepsilon_1}$ ; so the first error (already normalised, since  $\operatorname{tr} \widehat{G}_\chi = \operatorname{tr} G_\chi/(aL^2)$ ) is  $O(N(1+T)^{1/2}/(T\ell_*)) = O(NT^{-1/2}/\ell_*)$ , and the second divided by  $a^2L^2$  is  $O(H(1 + Y/Q^2)) = O(N(T^{-1} + (QT)^{-\varepsilon_1})/\ell_*)$ .  $\square$

*The trace.*

**Proposition 9.6.** *Uniformly in  $T$ ,  $\sum_{\chi \in \mathcal{F}} \omega_\chi \operatorname{tr} \widehat{G}_\chi = (1 - 2\theta)N + o(N)$ .*

*Proof. Gamma part.* By Stirling,  $\mu_\chi(t) = \frac{1}{2\pi} \log \frac{qt}{2\pi} + O(1/t)$  for  $t \geq 1$ , and on  $J$  this is  $\frac{\ell_*}{2\pi} + O(1 + \log \frac{1}{\eta})$  uniformly in  $\chi \in \mathcal{F}$ . Hence  $\sum_\chi \omega_\chi \int_J \mu_\chi = \frac{H|J|\ell_*}{2\pi} (1 + O(\frac{1 + \log(1/\eta)}{\ell_*})) = (1 - 2\theta)N(1 + o(1))$  by Lemma 9.2, since  $|J| = (1 - 2\theta)T$ . The terms  $\log T$  are part of  $\ell_*$  on both sides. *Prime part.* As in the proof of Proposition 4.6,  $|\int_J n^{\mp it} dt| \leq 2/\log n$  and the family first moment (Lemma 4.5), which does not involve  $T$ , bound it by  $\ll w_{\max} QY^{1/2} \log Y$ . Relative to  $N \asymp Q^2 T \ell_*$  this is  $\ll Y^{1/2} \log Y/(QT\ell_*) \ll T^{-1/2}(QT)^{-\varepsilon_1/2}$  by Lemma 9.1(S1). *Finite centres.* Apply (9.7).  $\square$

*Exterior zeros.* We keep the exceptional characters of §4.4: with  $\delta_0 := B_1 \log \ell/\ell$  and  $B_1$  as there, a character  $\chi \in \mathcal{F}$  is *bad* if  $L(s, \chi)$  has a zero  $\rho = \frac{1}{2} + \delta_\rho + i\gamma$  with  $|\delta_\rho| \geq \delta_0$  and  $|\gamma| \leq Q^{|\delta_\rho|/2}$ , and *good* otherwise. This definition involves only  $Q$ . By Lemma 4.7,

$$(9.9) \quad \#\{\chi \in \mathcal{F} \text{ bad}\} \ll Q^2 \ell^{-52}, \quad \sum_{\chi \text{ bad}} \omega_\chi \ll_w Q^2 \ell^{-51};$$

its proof applies Montgomery's zero-density theorem [26, Thm. 1], in the form (4.5), only at the heights  $T_j = Q^{\delta_{j+1}/2} \leq e^{1/2} Q^{1/4}$ . In particular (9.9) holds for every  $T$ , and no zero-density estimate at height  $T$  is needed.

**Proposition 9.7** (Exterior tail for good characters at height  $T$ ). *For every  $B > 0$ , uniformly for good  $\chi \in \mathcal{F}$  and  $T$  in (9.2),  $\|\widehat{E}_\chi\|_1 \ll_B \ell^{-B}$ .*

*Proof.* As in the proof of Proposition 4.8,  $\|E_\chi\|_1 \leq \sum_{\gamma \notin I} m_\rho \|u(z_\rho)\| \|u(\bar{z}_\rho)\|$ . We first bound  $e^{L|\delta_\rho|}$  for every zero  $\rho = \frac{1}{2} + \delta_\rho + i\gamma$  of a good  $\chi$ . Recall  $L = \lambda \ell_*$  with  $\lambda < 2$ . If  $|\delta_\rho| < \delta_0$ , then  $e^{L|\delta_\rho|} \leq (QT)^{2\delta_0} = \ell^{2B_1(1+\kappa)} \leq \ell^c$  with  $\kappa = \log T/\log Q$  and  $c := 2B_1(1 + \kappa_1)$ . If  $|\delta_\rho| \geq \delta_0$ , then  $|\gamma| > Q^{|\delta_\rho|/2} \geq \ell^{B_1/2} > 1$  because  $\chi$  is good, so

$$e^{L|\delta_\rho|} = Q^{\lambda|\delta_\rho|} T^{\lambda|\delta_\rho|} < |\gamma|^{2\lambda} T^{\lambda/2} \leq |\gamma|^4 T,$$

using  $|\delta_\rho| < \frac{1}{2}$ . Hence  $e^{L|\delta_\rho|} \leq \ell^c(1+|\gamma|)^4 T$  for every zero of a good  $\chi$ . By the second claim of Lemma 2.3 (any  $A > \frac{1}{2}$ ),

$$\|\widehat{E}_\chi\|_1 \ll_A \ell^c T \sum_{\gamma \notin I} m_\rho(1+|\gamma|)^4 (1+L \operatorname{dist}(\gamma, J))^{1-2A}.$$

For  $\gamma \notin I$  we have  $\operatorname{dist}(\gamma, J) \geq \theta T$ , and  $\operatorname{dist}(\gamma, J) \geq |\gamma|/2$  if  $|\gamma| > 4T+1$ . The zeros with  $|\gamma| \leq 4T+1$  number  $\ll T\ell_*$  and contribute  $\ll \ell^c T \cdot T\ell_* \cdot (5T)^4 (\theta TL)^{1-2A}$ . The zeros with  $|\gamma| > 4T+1$  contribute  $\ll \ell^c T \sum_{n \geq 4T} \mathcal{L}_*(n)(n+2)^4 (1+Ln/2)^{1-2A}$ . Both are  $\ll (1+\kappa_1)\ell^{c+1}\theta^{1-2A}T^{7-2A}$ , which is  $O(\ell^{-B})$  once  $A = A(B, a_0, \kappa_1)$  is large, since  $T \geq \ell^{a_0}$ .  $\square$

*Remark 9.8.* (1) Off-line zeros with ordinate in  $I$  are never bounded individually: they are handled by the inertia argument of Lemma 3.1, and their size enters only through  $\operatorname{tr} \widehat{G}_\chi$  and  $\|\widehat{G}_\chi\|_F$ , which the prime side bounds from above. Deleting characters with an off-line zero in  $I$  would fail at  $T = Q^\kappa$ , since density bounds cannot exclude one in each window. (2) If  $T \geq Q^{\kappa_0}$  for a fixed  $\kappa_0 > 0$ , the proof of Proposition 9.7 works for every  $\chi$ , good or bad:  $e^{L|\delta_\rho|} < (QT)^{\lambda/2} \leq T^{1+1/\kappa_0}$ , and  $A \geq 5+1/\kappa_0+B$  suffices. On that range Montgomery's theorem is not used at all. (3) The single window  $J$  of length  $\asymp T$  is essential for Proposition 9.7: it keeps every exterior zero at distance  $\geq \theta T$ . A partition of  $I$  into short sub-windows would require deleting pairs (character, sub-window) by a zero-density estimate at height  $T$ .

*The zero-side reduction.*

**Proposition 9.9.** *For every  $B > 0$ , uniformly for  $T$  in (9.2),*

$$\begin{aligned} N_0^s, N_0^* &\geq (2-8\theta)N - \mathfrak{M} - o(N) - O_B(\ell^{-B}(H\mathfrak{M})^{1/2}), \\ N_d &\geq \frac{1}{2}((3-8\theta)N - \mathfrak{M}) - o(N) - O_B(\ell^{-B}(H\mathfrak{M})^{1/2}). \end{aligned}$$

*Proof.* The proof of Proposition 3.6 (given at the end of §4), with Propositions 9.6 and 9.7 and (9.9) in place of Propositions 4.6, 4.8 and Lemma 4.7. Each eigenvalue  $x$  of  $\widehat{G}_\chi$  satisfies  $4x - x^2 \leq 4$ , so  $4 \operatorname{tr} \widehat{G}_\chi - \|\widehat{G}_\chi\|_F^2 \leq 4d$  for every  $\chi$ . For good  $\chi$  replace  $\widehat{A}_\chi$  by  $\widehat{G}_\chi$  at the cost of  $\|\widehat{E}_\chi\|_1$  (Proposition 9.7); for bad  $\chi$  use  $N_{0,\chi}^s \geq 0$ . The deletion costs  $4d \sum_{\text{bad}} \omega_\chi \ll T\ell_* \cdot Q^2 \ell^{-51} = o(N)$  by (9.9) and Lemma 9.2. The replacement costs  $\ll \ell^{-B}(H + (H\mathfrak{M})^{1/2})$  by Cauchy–Schwarz, and  $H = o(N)$ . Proposition 9.6 gives the main term  $4(1-2\theta)N - \mathfrak{M} - 2N$ . The same argument with the caps  $4d$  and  $2d$  gives  $N_0^*$  and  $N_d$ .  $\square$

### 9.3. The prime side at polynomial height.

*Decomposition and the non-ratio terms.* By (9.8),  $\mathfrak{M} = (a^2 L^2)^{-1}(\mathcal{M} + O(H(\ell_*^2 + (1+Y/Q^2)L^2)))$  with  $\mathcal{M} := \sum_\chi \omega_\chi \iint_{J^2} \Phi(t-t')^2 \nu_\chi(t) \nu_\chi(t') dt dt'$ , and the error divided by  $a^2 L^2$  is  $o(N)$ . As in (5.1), (5.2),

$$\mathcal{M} = M_{\mu\mu} + 2M_{\mu\Lambda} + M_{\Lambda\Lambda}^{\text{rat}} + M_{\Lambda\Lambda}^{\text{ss}}, \quad M_{\Lambda\Lambda}^{\text{rat}} = \frac{1}{2\pi^2} \sum_{n,m} a_n a_m \Delta(n, m) \mathcal{K}(n, m).$$

**Lemma 9.10.** *Uniformly for  $T$  in (9.2):*

- (a)  $M_{\mu\mu} = \frac{H|J|bL\ell_*^2}{2\pi} \left(1 + O\left(\frac{1 + \log(1/\eta)}{\ell_*}\right)\right) + O(H\ell_*^2);$
- (b)  $M_{\mu\Lambda} \ll w_{\max} Q L \ell_* Y^{1/2} \log Y$ , which is  $O(H|J|L\ell_*^2 \cdot T^{-1/2}(QT)^{-\varepsilon_1/2});$
- (c)  $M_{\Lambda\Lambda}^{\text{ss}} \ll w_{\max}(Q^2 + Y)L^2$ , which is  $O(H|J|L\ell_*^2 \cdot (T^{-1} + (QT)^{-\varepsilon_1})/\ell_*).$

*Proof.* (a) On  $J$ ,  $\mu_\chi(t) = \ell_*/(2\pi) + O(1 + \log \frac{1}{\eta})$  uniformly in  $\chi \in \mathcal{F}$ , and  $\iint_{J^2} \Phi(t - t')^2 = 2\pi bL|J| + O_\psi(1)$  (proof of Lemma 5.1). (b) Write  $\mu_\chi = A_q(t) + \chi(-1)B(t)$  as in the proof of Lemma 5.2; on  $J$ ,  $A_q \ll \ell_*$ ,  $B \ll 1$  and  $A'_q, B' \ll 1/T$ . For  $m_q(t') := \int_J \Phi(t - t')^2 A_q(t) dt$  we have  $\sup |m_q| \ll L\ell_*$ , and one integration by parts in  $t$  gives  $\int_J |m'_q| \ll L\ell_* + \|A'_q\|_\infty L|J| \ll L\ell_*$ . Hence  $|\int_J m_q(t') n^{\mp it'} dt'| \ll L\ell_*/\log n$ , and similarly for  $B$ . Lemma 4.5 gives the bound, and  $Y^{1/2} \log Y / (QT\ell_*) \ll T^{-1/2} (QT)^{-\varepsilon_1/2}$  by Lemma 9.1(S1). (c) The proof of Lemma 5.4 gives  $w_{\max}(Q^2 + Y)L^2$  verbatim (it uses (9.1) on  $[1, Y]$  and carries no factor  $|J|$ ). Relative to  $H|J|L\ell_*^2 \asymp_w Q^2 T L\ell_*^2$  this is  $\ll (1 + Y/Q^2)/(T\ell_*)$ , and (S1) applies.  $\square$

*Flattening.* The flattening of §5.7 is used with the same levels,

$$R_j := \lfloor N_j^{1-\varepsilon_3}/(QT) \rfloor, \quad \Lambda_R(n) = \sum_{r \leq R} \frac{\mu(r)}{\varphi(r)} c_r(n),$$

$$a_n^\sharp := \sum_{j: R_j \geq 1} \psi_j(n) \Upsilon(n) n^{-1/2} \Lambda_{R_j}(n), \quad b := a - a^\sharp,$$

where  $N_j = 2^j$  and  $(\psi_j)$  is the dyadic partition of unity of §5.7. Lemma 5.5 is a statement about one smooth function and does not involve  $T$ . The derivative bounds (5.6),  $|F_{j,t}^{(k)}(y)| + |F_{j,t,u}^{(k)}(y)| \leq B_k N_j^{-1/2-k} T^k$  for  $|t| \leq 3T$ , hold verbatim for  $\delta = T^{-1+\varepsilon_5}$ , since their proof only uses  $\delta^{-1} \leq T$ .

**Lemma 9.11** (Invisibility at height  $T$ ). *For every  $\chi \in \mathcal{F}$ , every  $|t| \leq 3T$  and every  $A > 0$ ,  $S_\chi[a^\sharp](t) \ll_A Q^{-A}$ , uniformly for  $T$  in (9.2). Consequently*

$$M_{\Lambda\Lambda}^{\text{rat}} = \frac{1}{2\pi^2} \sum_{n,m} b_n b_m \Delta(n, m) \mathcal{K}(n, m) + O(Q^{-1}).$$

*Proof.* The proof of Lemma 5.6 applies verbatim:  $n \mapsto c_r(n)\chi(n)$  has mean zero and period  $[q, r] \leq QR_j \leq N_j^{1-\varepsilon_3}/T$ , and Lemma 5.5(b) with  $\Theta = T$  gives  $|S_\chi[a^\sharp](t)| \leq B_i(2 + \log_2 Y)(QT)^{5/2-i\varepsilon_3}$ , which is  $\ll Q^{-A}$  for  $i \geq (A+3)/\varepsilon_3$  because  $QT \geq Q$  and  $\log Y \ll \ell_* \ll \ell$ . For the second claim we need a polynomial bound for  $\sum_n |b_n|$ , which in the derivation of (5.7) was  $\ll Q^3$ . Here  $a^\sharp$  is supported on all  $n \leq Y$ . For squarefree  $r$ ,  $|c_r(n)|/\varphi(r) = 1/\varphi(r/(r, n))$ , so  $|\Lambda_R(n)| \leq \sum_{d|n} \sum_{s \leq R} \mu^2(s)/\varphi(s) \leq 1.95 \tau(n)(1 + \log R)$ , and hence  $\sum_n |b_n| \leq \sum_n |a_n| + \sum_n |a_n^\sharp| \ll Y^{1/2} \log^2 Y \leq Q^{3+\kappa_1}$ . With  $\iint_{J^2} \Phi^2 \leq 2\pi bL|J| \ll Q^{\kappa_1+1}$  and  $H \asymp_w Q^2$  (2.2), the difference between the two forms is  $\ll Q^{2\kappa_1+7-A}$ ; take  $A = 2\kappa_1 + 8$ .  $\square$

**Lemma 9.12** (Flattened norm at height  $T$ ). *Let  $h : \mathbb{R} \rightarrow [0, \infty)$  be smooth with  $\|h^{(k)}\|_\infty \leq C_k \|h\|_\infty$  for all  $k \geq 1$ . Then, uniformly for  $T$  in (9.2),*

$$\sum_n |b_n|^2 h(\log n) \leq \int_{\mathbb{R}} \ell_* F_b(s/\ell_*) h(s) ds + O(\|h\|_\infty \ell_*), \quad F_b(\alpha) := \min(\alpha^+, 1 + \varepsilon_4), \quad \varepsilon_4 := 2\varepsilon_3.$$

*Proof.* Steps (1)–(5) of the proof of Lemma 5.7 apply verbatim with the following remarks. Step (4) needs  $R_j \leq N_j^{1/2-\varepsilon_3}$ , which is Lemma 9.1(S3). In step (3),  $|\Lambda_R(p^k)| \leq 3G(R)$  and  $G(R) \leq 1.95(1 + \log R) \ll \ell_*$ , and in step (5)  $G(R_j) \ll \ell_*$ , so the total error is  $\sum_j |\mathcal{E}_j| \ll \|h\|_\infty \ell_*$ . Only step (6), the density, changes. Recall that the density of the convexity majorant at  $y$  is  $\Upsilon(y)^2$  times a  $\psi_j$ -weighted average of  $\log y - G(R_j)$  over the blocks containing  $y$ , and that  $G(R) \geq \log(R+1)$ . If  $R_j \geq 1$ , then  $R_j \geq N_j^{1-\varepsilon_3}/(2QT)$ , and for  $y \leq 2N_j$ , using  $\log N_j \leq \log(2Y) \leq 2\ell_*$  (Lemma 9.1(S1)),

$$\log y - G(R_j) \leq \log y - \log R_j \leq \ell_* + \varepsilon_3 \log N_j + 2 \log 2 \leq (1 + 2\varepsilon_3)\ell_* + 2.$$

If  $R_j = 0$ , then  $N_j^{1-\varepsilon_3} < QT$ , so for  $y \leq 2N_j$ ,  $\log y \leq \log 2 + \ell_*/(1 - \varepsilon_3) \leq (1 + 2\varepsilon_3)\ell_* + 1$ . In all cases the density is at most  $\min(\log^+ y, (1 + \varepsilon_4)\ell_* + 2)$ . Summing over  $j$  as in step (7),  $\sum_n |b_n|^2 h(\log n) \leq \int_0^{\log Y} \min(s, (1 + \varepsilon_4)\ell_* + 2) h(s) ds + O(\|h\|_\infty \ell_*)$ , and the additive 2 contributes at most  $2\|h\|_\infty \log Y \ll \|h\|_\infty \ell_*$ .  $\square$

The factor  $1 + (2 \log T + 2)/\ell$  of Lemma 5.7 has disappeared into  $\ell_*$ : in units of the log-conductor, the flattened diagonal density is exactly  $\min(\alpha, 1)$ , up to  $\varepsilon_4$ . We also record, as in (5.10),

$$(9.10) \quad \|a\|^2 + \|a^\sharp\|^2 + \|b\|^2 \ll L^2,$$

since  $\|a\|^2 \leq \sum_{n \leq Y} \Lambda(n)^2/n \ll L^2$  and  $\|a^\sharp\|^2 \leq \sum_{j: R_j \geq 1} (G(R_j) \int \psi_j \Upsilon^2 \frac{dy}{y} + O(N_j^{-1})) \ll L\ell_*$  by step (4) of the proof of Lemma 5.7 with  $h \equiv 1$ .

*Time localisation at scale  $T^{-1+\varepsilon_5}$ .* Let  $\widehat{\mathbf{1}}_J(\xi) := \int_J e^{it\xi} dt$ , so  $|\widehat{\mathbf{1}}_J(\xi)| \leq \min(|J|, 2/|\xi|)$ , and for a finitely supported  $y$  and  $u \in \mathbb{R}$  put  $x_y(u)_n := y_n \widehat{\mathbf{1}}_J(u - \log n)$ . The factorisation of Lemma 5.9,

$$(9.11) \quad \sum_{n,m} y_n \overline{y_m} \Delta(n, m) \mathcal{K}(n, m) = \int_{\mathbb{R}} g(u) x_y(u)^* \Delta x_y(u) du,$$

is an exact identity for every bounded interval  $J$ . The short-interval bound of Lemma 5.10,

$$(9.12) \quad x^* \Delta x \leq (H + 4w_{\max} QK(1 + \log K)) \|x\|^2 \quad (x \text{ supported on } K \text{ consecutive integers}),$$

does not involve  $T$  and does not depend on the position of the interval. Lemma 5.11 holds for every  $\delta \in (0, \frac{1}{4}]$  and every value  $\varkappa \in (0, 1]$  of its parameter: with  $x_n^s := x_n \rho((\log n - u)/\delta)$  and  $x^t := x - x^s$ , (i)  $x^* \Delta x \leq (1 + \varkappa) x^{s*} \Delta x^s + (1 + \varkappa^{-1}) x^{t*} \Delta x^t$ ; (ii)  $x^s(u)$  is supported in an interval of length  $\leq 5\delta e^u$ , hence on at most  $K(u) = 5\delta e^u + 1$  integers; (iii)  $\int g(u) \|x^s(u)\|^2 c(u) du \leq \sum_n |y_n|^2 \mathcal{K}(n, n) \sup\{c(u) : |u - \log n| < 2\delta\}$  for bounded measurable  $c \geq 0$ . We use it with  $\delta = T^{-1+\varepsilon_5}$  and  $\varkappa = \varkappa_{\text{loc}} = T^{-\varepsilon_5/2}$  from (9.5). Part (iv) of Lemma 5.11,  $\int g \|x^t\|^2 \leq 8bL \|y\|^2/\delta$ , is still true, but it was combined in Propositions 5.12 and 6.24 with  $\Lambda_{\text{mult}}(Y) \leq 2w_{\max} Q^2$ , which is false once  $Y \gg Q^2$ . We replace this step by the following lemma.

**Lemma 9.13** (Tails by dyadic shells). *Let  $y$  be supported in  $[1, Y]$  with  $Y \geq 1$ , let  $\delta \in (0, \frac{1}{4}]$  and  $k_0 := \lceil \log_2(1/\delta) \rceil$ . For every measurable  $g_1 : \mathbb{R} \rightarrow [0, g_\infty]$ ,*

$$\int_{\mathbb{R}} g_1(u) x^t(u)^* \Delta x^t(u) du \leq 48 g_\infty w_{\max} (k_0 + 4) \left( \frac{Q^2 + 1}{\delta} + (k_0 + 3)Y \right) \|y\|^2.$$

*Proof. Shells.* For  $k \geq 0$  and each sign put  $S_k^\pm(u) := \{n \geq 1 : 2^k \delta \leq \pm(\log n - u) < 2^{k+1} \delta\}$  and  $x^{k,\pm}(u) := x^t(u) \mathbf{1}_{S_k^\pm(u)}$ . Since  $\rho = 1$  on  $[-1, 1]$ ,  $x^t(u)_n = 0$  unless  $|\log n - u| > \delta$ , so  $x^t(u) = \sum_{k,\pm} x^{k,\pm}(u)$ , a finite sum. On  $S_k^\pm(u)$ ,  $|x^t(u)_n| \leq |y_n| |\widehat{\mathbf{1}}_J(u - \log n)| \leq 2|y_n|/(2^k \delta)$ , as  $0 \leq 1 - \rho \leq 1$ .

*Lengths.*  $S_k^+(u) \subset [e^{u+2^k \delta}, e^{u+2^{k+1} \delta})$ , an interval of length  $e^{u+2^k \delta} (e^{2^k \delta} - 1) \leq n(e^{2^k \delta} - 1)$  for every  $n \in S_k^+(u)$ ; and  $S_k^-(u) \subset (e^{u-2^{k+1} \delta}, e^{u-2^k \delta}]$ , of length  $e^{u-2^k \delta} (1 - e^{-2^k \delta}) < n(e^{2^k \delta} - 1)$  for every  $n \in S_k^-(u)$ , since  $e^{u-2^k \delta} < n e^{2^k \delta}$  there. Intersecting with  $[1, Y]$ , the support of  $x^{k,\pm}(u)$  lies in an interval of at most  $K_k(n) := \min(Y, n(e^{2^k \delta} - 1)) + 1$  integers, for every  $n$  in that support. By (9.1),

$$x^{k,\pm}(u)^* \Delta x^{k,\pm}(u) \leq w_{\max} \sum_{n \in S_k^\pm(u)} (Q^2 + K_k(n)) \frac{4|y_n|^2}{(2^k \delta)^2}.$$

*Summing the shells.* Put  $c_k = 1$  for  $k \leq k_0$  and  $c_k = 2^{-(k-k_0)/2}$  for  $k > k_0$ , so that  $\sum_{k,\pm} c_k \leq 2(k_0 + 1) + 2 \sum_{j \geq 1} 2^{-j/2} < 2k_0 + 7$ . Since  $x \mapsto (x^* \Delta x)^{1/2}$  is a seminorm, Cauchy–Schwarz gives  $x^{t*} \Delta x^t \leq (2k_0 + 7) \sum_{k,\pm} c_k^{-1} x^{k,\pm*} \Delta x^{k,\pm}$ . For fixed  $n$  and  $k$ , the set of  $u$  with  $n \in S_k^+(u) \cup S_k^-(u)$  has measure  $2 \cdot 2^k \delta$ . Hence

$$\int g_1(u) x^t(u)^* \Delta x^t(u) du \leq 8g_\infty(2k_0 + 7)w_{\max} \sum_n |y_n|^2 \sum_{k \geq 0} c_k^{-1} \frac{Q^2 + K_k(n)}{2^k \delta}.$$

*The sums over  $k$ .* Recall  $2^{k_0-1}\delta < 1 \leq 2^{k_0}\delta$ . First,  $\sum_k c_k^{-1}/(2^k \delta) \leq 2/\delta + \sum_{j \geq 1} 2^{-j/2} \leq 3/\delta$ , because  $c_k^{-1}/(2^k \delta) = 2^{-(k-k_0)/2}/(2^{k_0}\delta) \leq 2^{-(k-k_0)/2}$  for  $k > k_0$  and  $\delta \leq \frac{1}{4}$ ; so the terms  $Q^2 + 1$  contribute at most  $3(Q^2 + 1)/\delta$ . Next, the terms  $K_k(n) - 1 \leq \min(Y, n(e^{2^k \delta} - 1))$ : if  $2^k \delta \leq 1$  then  $k \leq k_0$ ,  $c_k = 1$  and  $e^x - 1 \leq 2x$  on  $[0, 1]$  give  $K_k(n) - 1 \leq 2 \cdot 2^k \delta Y$ , so these  $k$  contribute at most  $2(k_0 + 1)Y$ ; if  $2^k \delta > 1$  then  $K_k(n) - 1 \leq Y$ , and these  $k$  contribute at most  $Y + \sum_{j \geq 1} 2^{-j/2} Y \leq 3.42Y$ . Altogether

$$\int g_1 x^{t*} \Delta x^t \leq 8g_\infty(2k_0 + 7)w_{\max} (3(Q^2 + 1)/\delta + (2k_0 + 6)Y) \|y\|^2,$$

which is at most the stated bound, since  $24(2k_0 + 7) \leq 48(k_0 + 4)$  and  $8(2k_0 + 7)(2k_0 + 6) \leq 48(k_0 + 4)(k_0 + 3)$ .  $\square$

**Corollary 9.14.** *Let  $y \in \{a, b\}$  and let  $x^t(u)$  be the tail of  $x_y(u)$  at scale  $\delta = T^{-1+\varepsilon_5}$ . Then, uniformly for  $T$  in (9.2),*

$$(1 + \varkappa_{\text{loc}}^{-1}) \int_{\mathbb{R}} g(u) x^t(u)^* \Delta x^t(u) du \ll_w (T^{-\varepsilon_5/2} \log T + Q^{-\varepsilon_1/2} \ell^2) H|J|L^2 \ell_* = o(H|J|L^2 \ell_*).$$

*Proof.* Apply Lemma 9.13 with  $g_1 = g$ ,  $g_\infty = g(0) = bL$ ,  $k_0 \leq \log_2 T + 1$  and  $\|y\|^2 \ll L^2$  (9.10); both  $a$  and  $b$  are supported in  $[1, Y]$ . The result is  $\ll T^{\varepsilon_5/2} w_{\max} L \log T (Q^2 T^{1-\varepsilon_5} + Y \log T) L^2$ . Relative to  $H|J|L^2 \ell_* \asymp_w Q^2 T L^2 \ell_*$  this is  $\ll T^{-\varepsilon_5/2} \log T + T^{\varepsilon_5/2} (\log T)^2 Y / (Q^2 T)$ , and  $Y/(Q^2 T) \leq (QT)^{-\varepsilon_1}$ ,  $T^{\varepsilon_5/2} \leq Q^{\varepsilon_1/8}$  by Lemma 9.1. Since  $T \geq \ell^{a_0}$ ,  $T^{-\varepsilon_5/2} \log T \rightarrow 0$  uniformly.  $\square$

The next lemma is an alternative to Lemma 9.13 without the logarithms; it uses Gallagher’s hybrid large sieve instead of shells. Either suffices for everything below.

**Lemma 9.15** (Tails by Plancherel and the hybrid large sieve). *Assume  $\rho$  is even, let  $y$  be supported in  $[1, Y]$ ,  $0 < \delta \leq 1$  and  $D := \max(2, 1/\delta)$ . Then*

$$\int_{\mathbb{R}} g(u) x^t(u)^* \Delta x^t(u) du \ll w_{\max} (\log \log Q) bL \left( \frac{Q^2}{\delta} + Y \right) \|y\|^2,$$

*with an implied constant depending only on  $\rho$ .*

*Proof.* Let  $r$  be the Schwartz function with  $\int r(t) e^{it\xi} dt = \rho(\xi)$  and  $r_\delta(t) := \delta r(\delta t)$ , so that  $\int r_\delta(t) e^{it\xi} dt = \rho(\xi/\delta)$ , and put  $k := \mathbf{1}_J - \mathbf{1}_J * r_\delta$ , whose Fourier transform is  $\widehat{\mathbf{1}}_J(\xi)(1 - \rho(\xi/\delta))$ . For a finitely supported  $c$  and every  $u$ ,  $\sum_n c_n \widehat{\mathbf{1}}_J(u - \log n)(1 - \rho((u - \log n)/\delta)) = \int k(t) e^{itu} \sum_n c_n n^{-it} dt$ . By Plancherel in  $u$ , with  $\chi$  inserted into  $c_n$ ,

$$\int_{\mathbb{R}} \left| \sum_n y_n \chi(n) \widehat{\mathbf{1}}_J(u - \log n) (1 - \rho(\frac{u - \log n}{\delta})) \right|^2 du = 2\pi \int_{\mathbb{R}} |k(t)|^2 |S_\chi[y](t)|^2 dt.$$

Using  $g \leq bL$ ,  $\omega_\chi \leq w_{\max} q/\varphi(q) \ll w_{\max} \log \log Q$  and summing over  $\chi$ ,  $\int g x^{t*} \Delta x^t \leq 2\pi bL \sum_\chi \omega_\chi \int |k|^2 |S_\chi[y]|^2$ . For  $t \in J$ ,  $|k(t)| = |\int_{\mathbb{R}} \delta(t-J) r|$ , and for  $t \notin J$ ,  $|k(t)| = |\int_{\delta(t-J)} r|$ ; in both cases  $|k(t)| \ll_M (1 + \delta \text{dist}(t, \partial J))^{-M}$ . Cover  $\mathbb{R}$  by consecutive intervals of length  $D$

indexed by their distance  $m$  (in units of  $D$ ) from  $\partial J$ ; there are at most four for each  $m$ , and  $|k|^2 \ll_M (1+m)^{-2M}$  on them. On each, Gallagher's hybrid large sieve [16, Theorem 3, (15)],  $\sum_{q \leq Q} \sum_{\chi \bmod q}^* \int_{t_0}^{t_0+D} |\sum_{n \leq Y} c_n \chi(n) n^{-it}|^2 dt \ll (Q^2 D + Y) \sum |c_n|^2$  (stated on  $[-T', T']$  with  $T' \geq 1$ ; translate by twisting  $c_n$  with  $n^{-it_0}$ ), gives the claim after summing over  $m$ .  $\square$

*Remark 9.16.* (1) Lemma 9.15 with  $\delta = T^{-1+\varepsilon_5}$  gives the tails as  $O(\log \log Q (T^{-\varepsilon_5} + (QT)^{-\varepsilon_1}))$  relative to  $H|J|L^2 \ell_*$ , without  $\log T$ . After the factor  $1 + \varkappa_{\text{loc}}^{-1} \leq 2T^{\varepsilon_5/2}$  of Corollary 9.14 this becomes  $O(\log \log Q (T^{-\varepsilon_5/2} + T^{\varepsilon_5/2} (QT)^{-\varepsilon_1}))$ , which is still  $o(1)$  uniformly, since  $T^{-\varepsilon_5/2} \leq \ell^{-a_0 \varepsilon_5/2}$  and  $T^{\varepsilon_5/2} \leq Q^{\varepsilon_1/8}$  (Lemma 9.1(S4)). Explicit constants for Gallagher's inequality are in [34, §6]. (2) The two error sources,  $(T\delta)^{-1}$  and  $Y/(Q^2 T)$ , are exactly the two ways in which the hybrid large sieve fails to be diagonal. With the crude tail bound of Propositions 5.12 and 6.24, the optimal scale is  $\delta = T^{-1/2}$  and the reach is only  $Y \leq Q^{2-\varepsilon} T^{1/2}$ , i.e. support  $(2 + \kappa/2)/(1 + \kappa)$  at  $T = Q^\kappa$  (cf. §8.5).

*The majorant with a large sieve constant.* Hypothesis 5.15 is a statement about the family form on intervals and does not involve  $T$ . By (9.1) and  $H = \mathcal{E}I_w Q^2(1+o(1))$  (Lemma 6.1),  $\text{LS}(C_{\text{cl}}(w))$  holds with  $C_{\text{cl}}(w) = w_{\text{max}}/(\mathcal{E}I_w)$ , and  $\text{LS}(C_G R_w)$  holds by Corollary 6.8.

**Proposition 9.17** (Time-integrated majorant at height  $T$ ). *Assume  $\text{LS}(C)$ . For  $\varepsilon \in (0, \varepsilon_{\text{max}}]$  let  $\tilde{C}_\varepsilon : \mathbb{R} \rightarrow [1, C]$  be smooth with  $\tilde{C}_\varepsilon = 1$  on  $(-\infty, 1 - 2\varepsilon]$  and  $\tilde{C}_\varepsilon = C$  on  $[1 - \frac{3}{2}\varepsilon, \infty)$ . Then, uniformly for  $T$  in (9.2),*

$$\sum_{n,m} b_n b_m \Delta(n, m) \mathcal{K}(n, m) \leq (1 + o(1)) H \sum_n |b_n|^2 \tilde{C}_\varepsilon \left( \frac{\log n}{\ell_*} \right) \mathcal{K}(n, n) + o(H|J|L^2 \ell_*).$$

*Proof.* By (9.11) the left side is  $\int g(u) x_b(u)^* \Delta x_b(u) du$ . At each  $u$  apply Lemma 5.11(i) with  $\delta$  and  $\varkappa = \varkappa_{\text{loc}}$ . The tails contribute  $o(H|J|L^2 \ell_*)$  by Corollary 9.14. If  $u \leq (1 - \varepsilon)\ell_*$ , the short part lies on at most  $K(u) \leq Q^{1-\varepsilon/2}$  integers (Lemma 9.1(S2)), and (9.12) gives the constant  $1 + 4w_{\text{max}} H^{-1} Q^{2-\varepsilon/2} (1 + \ell) = 1 + o(1)$ , with no condition on  $T$ . If  $u > (1 - \varepsilon)\ell_*$ , it lies on at most  $Q^{2-\varepsilon_1/2}$  integers, and  $\text{LS}(C)$  (with  $\varepsilon_1/2$  in place of  $\varepsilon$ ) gives  $C + o(1)$ . If  $(x^s(u))_n \neq 0$  then  $|u - \log n| < 2\delta$ ; since  $2\delta < \frac{1}{2}\varepsilon \ell_*$  for  $Q$  large,  $\log n / \ell_* < 1 - \frac{3}{2}\varepsilon$  forces  $u < (1 - \varepsilon)\ell_*$ . Hence the constant is at most  $(1 + o(1)) \tilde{C}_\varepsilon(\log n / \ell_*)$  for every  $n$  in the support, and Lemma 5.11(iii) gives the main term, with the factor  $1 + \varkappa_{\text{loc}} = 1 + o(1)$ .  $\square$

*The sharp majorant: the band device at polynomial height.* We now adapt Proposition 6.24. The objects of §6.2 are used without change: the Toeplitz identity for  $Q$ -rough vectors (Lemma 6.13), the positive measure  $\mu^\natural = \sum_{e \leq Q} \nu(e) \sum_{c \bmod e}^* \delta_{c/e}$  with  $0 \leq \nu(e) \leq 6\|\tilde{w}\|_\infty \eta^{-2}$  (6.13), the positive semidefinite Toeplitz form  $T^\natural$  with  $y^* \Delta y \leq y^* T^\natural y$  for  $y$  supported on  $Q$ -rough integers (Lemma 6.17), and the sharp bound of Proposition 6.21,

$$(9.13) \quad \lambda_{\text{max}}(T_I^\natural) \leq (C_T^+(w) + \vartheta'_Q) H \quad (\text{intervals } I \text{ of } K \leq Q^{2-\varepsilon'} \text{ integers, } \varepsilon' \in (0, 1)),$$

with  $\vartheta'_Q \rightarrow 0$  depending only on  $(\varepsilon', w)$ . None of them involves  $T$ . We write  $\|x\|_P^2 := x^* P x$  for positive semidefinite  $P$ , so that  $\|x + x'\|_P^2 \leq (1 + \varkappa) \|x\|_P^2 + (1 + \varkappa^{-1}) \|x'\|_P^2$  for  $0 < \varkappa \leq 1$  (6.12).

*The band constant and the profile.* Let  $C_{\text{band}} \geq 1$  be a constant with  $\Lambda_{\text{mult}}(Q^{5/4}) \leq (C_{\text{band}} + o(1))H$ ; by (9.1),  $C_{\text{band}} = 2w_{\text{max}}/(\mathcal{E}I_w)$  is admissible. Put  $C_{\text{max}} := \max(C_{\text{band}}, C_T^+(w))$  and let  $\tilde{C}_\varepsilon : \mathbb{R} \rightarrow [1, C_{\text{max}}]$  be smooth with  $\tilde{C}_\varepsilon = 1$  on  $(-\infty, 1 - 2\varepsilon]$ ,  $\tilde{C}_\varepsilon \geq C_{\text{band}}$  on  $[1 - \frac{3}{2}\varepsilon, 1 + \frac{3}{2}\varepsilon]$ ,  $\tilde{C}_\varepsilon \geq C_T^+(w)$  on  $[1 + \frac{1}{2}\varepsilon, \infty)$  and  $\tilde{C}_\varepsilon = C_T^+(w)$  on  $[1 + 2\varepsilon, \infty)$ , exactly as before Proposition 6.24.

**Proposition 9.18** (Sharp time-integrated majorant at height  $T$ ). *For each fixed  $\varepsilon \in (0, \varepsilon_{\max}]$  there is  $\vartheta_Q^* \rightarrow 0$ , depending on the fixed data but not on  $T$ , such that uniformly for  $T$  in (9.2)*

$$\sum_{n,m} b_n b_m \Delta(n, m) \mathcal{K}(n, m) \leq (1 + \vartheta_Q^*) (1 + \varkappa_{\text{loc}})^3 H \sum_n |b_n|^2 \tilde{C}_\varepsilon \left( \frac{\log n}{\ell_*} \right) \mathcal{K}(n, n) + \mathcal{E}_{\text{TI}}^H,$$

where

$$\mathcal{E}_{\text{TI}}^H \ll \left( T^{-\varepsilon_5/2} \log T + Q^{-\varepsilon_1/2} \ell^2 + \eta^{-2} T^{\varepsilon_5/2} Q^{-\varepsilon_1/2} + T^{\varepsilon_5/2} Q^{-1/2} \right) H |J| L^2 \ell_* + Q^{-1} = o(H |J| L^2 \ell_*).$$

*Proof.* We follow the five steps of the proof of Proposition 6.24, with  $\ell$  replaced by  $\ell_*$  in the regime boundaries,  $\delta = T^{-1+\varepsilon_5}$ ,  $\varkappa = \varkappa_{\text{loc}} = T^{-\varepsilon_5/2}$  in place of  $\kappa_T$ , and the tails bounded by Corollary 9.14. Throughout,  $u$  ranges over  $\text{supp } g \subset (-L, L)$ , so  $e^u < X$ .

*Step 0 (invisibility before localisation).* For  $\chi \in \mathcal{F}$  and every  $u$ ,  $\sum_n x_{\sharp}(u)_n \chi(n) = \int_J e^{itu} S_\chi[a^\sharp](t) dt \ll_A |J| Q^{-A}$  by Lemma 9.11 ( $J \subset [0, 2T]$ ), where  $x_{\sharp} := x_{a^\sharp}$ ; and  $|\sum_n x_a(u)_n \chi(n)| \leq |J| \sum_n |a_n| \leq Q^{3+2\kappa_1}$ . Since  $x_a = x_b + x_{\sharp}$ , expanding the square and summing with the weights  $\omega_\chi$  gives, for  $A = A(\kappa_1)$  large,

$$(9.14) \quad x_b(u)^* \Delta x_b(u) \leq x_a(u)^* \Delta x_a(u) + Q^{-3} \quad (\text{all } u, Q \geq Q_0).$$

For each  $u$  we localise  $y(u) := b$  if  $u \leq (1+\varepsilon)\ell_*$  and  $y(u) := a$  if  $u > (1+\varepsilon)\ell_*$ . By Lemma 9.1(S2), every  $x_y^s(u)$  lies on at most  $K(u) \leq Q^{2-\varepsilon_1/2}$  integers of  $[1, Y]$ .

*Step 1 (localisation).* By Lemma 5.11(i) with  $\varkappa = \varkappa_{\text{loc}}$ ,  $x_y^* \Delta x_y \leq (1 + \varkappa_{\text{loc}}) x_y^{s*} \Delta x_y^s + (1 + \varkappa_{\text{loc}}^{-1}) x_y^{t*} \Delta x_y^t$  for  $y = y(u)$ .

*Step 2 (tails).* The tail integrands are nonnegative, so  $\int g x_{y(u)}^{t*} \Delta x_{y(u)}^t \leq \int g (x_a^{t*} \Delta x_a^t + x_b^{t*} \Delta x_b^t)$ , and Corollary 9.14 bounds the tails, with the factor  $1 + \varkappa_{\text{loc}}^{-1}$ , by  $\ll (T^{-\varepsilon_5/2} \log T + Q^{-\varepsilon_1/2} \ell^2) H |J| L^2 \ell_*$ .

*Step 3 (regimes (i) and (ii);  $u \leq (1+\varepsilon)\ell_*$ ,  $y = b$ ).* (i) If  $u \leq (1-\varepsilon)\ell_*$  then  $K(u) \leq Q^{1-\varepsilon/2}$  and (9.12) gives  $x_b^{s*} \Delta x_b^s \leq c_i H \|x_b^s\|^2$  with  $c_i = 1 + O_w(Q^{-\varepsilon/2} \ell)$ . (ii) If  $(1-\varepsilon)\ell_* < u \leq (1+\varepsilon)\ell_*$  then  $K(u) \leq Q^{5/4}$ , and  $x_b^{s*} \Delta x_b^s \leq c_{ii} H \|x_b^s\|^2$  with  $c_{ii} = C_{\text{band}} + o(1)$ .

*Step 4 (regime (iii);  $u > (1+\varepsilon)\ell_*$ ,  $y = a$ ).* If  $(x_a^s(u))_n \neq 0$  then  $n$  is a prime power with  $\log n > u - 2\delta$ , so  $n > (QT)^{1+\varepsilon} e^{-2\delta} > Q^{1+\varepsilon}/2 > Q$ . Split  $x_a^s = x_r + x_{\text{pp}}$ , where  $x_r$  is the restriction to  $Q$ -rough  $n$  and  $x_{\text{pp}}$  the restriction to  $n = p^k$  with  $p \leq Q$ ,  $k \geq 2$  (a prime  $p \leq Q$  is  $< Q^{1+\varepsilon}/2$ ). Exactly as in Step 4 of the proof of Proposition 6.24, the seminorm inequality for  $\Delta$  with  $\varkappa = \varkappa_{\text{loc}}$ , Lemma 6.17 for the  $Q$ -rough vector  $x_r$ , the identity  $x_r = x_b^s + (x_{\sharp}^s - x_{\text{pp}})$  and the seminorm inequality for  $T^\natural$  (with  $\varkappa = \varkappa_{\text{loc}}$ , then with  $\varkappa = 1$ ) give

$$x_a^{s*} \Delta x_a^s \leq (1 + \varkappa_{\text{loc}})^2 \|x_b^s\|_{T^\natural}^2 + 4(1 + \varkappa_{\text{loc}}^{-1}) (\|x_{\sharp}^s\|_{T^\natural}^2 + \|x_{\text{pp}}\|_{T^\natural}^2 + x_{\text{pp}}^* \Delta x_{\text{pp}}).$$

(a) *Main term.*  $x_b^s(u)$  lies on at most  $K(u) \leq Q^{2-\varepsilon_1/2}$  integers, so (9.13) with  $\varepsilon' = \min(\varepsilon_1/2, \frac{1}{2})$  gives  $\|x_b^s\|_{T^\natural}^2 \leq c_{iii} H \|x_b^s\|^2$ ,  $c_{iii} := C_T^+(w) + \vartheta'_Q$ . The vector  $x_b^s$  is not supported on  $Q$ -rough integers; this is harmless since  $T^\natural \geq 0$  is defined on all vectors.

(b) *The subtracted part.* By (6.16), for  $(c, e) = 1$ ,

$$S_{x_{\sharp}^s(u)} \left( \frac{c}{e} \right) = \int_J e^{itu} \sum_{j: R_j \geq 1} \sum_{r \leq R_j} \frac{\mu(r)}{\varphi(r)_h} \sum_{h \bmod r}^* \sum_n F_{j,t,u}(n) e \left( n \left( \frac{h}{r} + \frac{c}{e} \right) \right) dt.$$

*Levels*  $R_{\max} < e \leq Q$ . For  $r \leq R_j < e$  and  $(h, r) = 1$ ,  $\xi_{h,c} := \frac{h}{r} + \frac{c}{e} \notin \mathbb{Z}$  and  $\|\xi_{h,c}\| \geq 1/(R_j Q)$ , exactly as in the proof of Proposition 6.24. By (5.6) with  $|t| \leq 2T$  and Lemma 5.5(a) with  $\Theta = T$ , using  $TR_j Q \leq N_j^{1-\varepsilon_3}$ ,  $|\sum_n F_{j,t,u}(n) e(n \xi_{h,c})| \leq B_i N_j^{1/2-i\varepsilon_3}$ . Summing over  $h, r$  (total weight  $\leq R_j \leq N_j$ ), over the  $\leq 2 + \log_2 Y$  blocks with  $R_j \geq 1$  (so  $N_j \geq QT$ ), and integrating over

$J$ ,  $|S_{x_{\sharp}^s(u)}(c/e)| \leq B_i |J| (2 + \log_2 Y) (QT)^{3/2 - i\varepsilon_3} \leq Q^{-4}$  for  $i \geq 8/\varepsilon_3$  and  $Q \geq Q_0$  (as  $|J| \leq QT$ ), uniformly in  $u$ ,  $e \in (R_{\max}, Q]$  and  $c$ . These levels contribute at most  $6\|\tilde{w}\|_{\infty} \eta^{-2} Q^2 \cdot Q^{-8}$  to  $\|x_{\sharp}^s\|_{T^{\natural}}^2$ . *Levels*  $e \leq R_{\max}$ . By the additive large sieve  $\sum_{e \leq E} \sum_{c \bmod e}^* |S_x(c/e)|^2 \leq (K + E^2) \|x\|^2$  [23, Thm. 7.7] for  $x$  on  $K$  consecutive integers, and Lemma 9.1(S2), (S3),

$$\sum_{e \leq R_{\max}} \nu(e) \sum_{c \bmod e}^* \left| S_{x_{\sharp}^s} \left( \frac{c}{e} \right) \right|^2 \leq 6\|\tilde{w}\|_{\infty} \eta^{-2} (K(u) + R_{\max}^2) \|x_{\sharp}^s\|^2 \leq 30\|\tilde{w}\|_{\infty} \eta^{-2} Q^{2-\varepsilon_1/2} \|x_{\sharp}^s\|^2.$$

By Lemma 5.11(iii) with  $c \equiv 1$ ,  $\mathcal{K}(n, n) \leq 2\pi |J| bL + O(1)$  and (9.10),  $\int g \|x_{\sharp}^s\|^2 \leq \sum_n |a_n^{\sharp}|^2 \mathcal{K}(n, n) \ll |J| L \cdot L \ell_*$ . With the factor  $(1 + \varkappa_{\text{loc}}) \cdot 4(1 + \varkappa_{\text{loc}}^{-1}) \leq 16T^{\varepsilon_5/2}$  from Steps 1 and 4 and  $Q^2 \ll_w H$ , the contribution of (b) is  $\ll \eta^{-2} T^{\varepsilon_5/2} Q^{-\varepsilon_1/2} H |J| L^2 \ell_* + Q^{-1}$ , and  $T^{\varepsilon_5/2} Q^{-\varepsilon_1/2} \leq Q^{-3\varepsilon_1/8}$  by (S4).

(c) *Prime powers*  $p^k$ ,  $p \leq Q$ ,  $k \geq 2$ . By partial summation  $\sum_{n=p^k > Q^{1+\varepsilon/2}, k \geq 2} \Lambda(n)^2/n \ll Q^{-1/2} \ell^2$ . As  $x_{\text{pp}}(u)$  lies on at most  $K(u) \leq Q^{2-\varepsilon_1/2}$  integers,  $\|x_{\text{pp}}\|_{T^{\natural}}^2 \leq c_{\text{iii}} H \|x_{\text{pp}}\|^2$  and  $x_{\text{pp}}^* \Delta x_{\text{pp}} \leq 2w_{\max} Q^2 \|x_{\text{pp}}\|^2$ , and  $\int g \|x_{\text{pp}}\|^2 \ll |J| L Q^{-1/2} \ell^2$ . With the factor  $16T^{\varepsilon_5/2}$  the contribution is  $\ll T^{\varepsilon_5/2} Q^{-1/2} \ell^2 H |J| L \ll T^{\varepsilon_5/2} Q^{-1/2} H |J| L^2 \ell_*$ .

*Step 5 (integration in  $u$ ).* Combining (9.14) and Steps 1–4, for  $u$  in regime (i), (ii) or (iii),  $x_b(u)^* \Delta x_b(u) \leq (1 + \varkappa_{\text{loc}})^3 c(u) H \|x_b^s(u)\|^2 + (\text{tail, (b), (c) terms at } u) + Q^{-3}$ , with  $c(u) = c_i, c_{\text{ii}}, c_{\text{iii}}$  respectively. If  $\rho((\log n - u)/\delta) \neq 0$  then  $|u - \log n| < 2\delta < \frac{1}{2}\varepsilon \ell_*$  for  $Q$  large; with  $\alpha := \log n / \ell_*$ , the case analysis of Step 5 of the proof of Proposition 6.24 applies verbatim with  $\ell$  replaced by  $\ell_*$ : if  $\alpha \leq 1 - \frac{3}{2}\varepsilon$  only (i) occurs; if  $1 - \frac{3}{2}\varepsilon < \alpha < 1 + \frac{1}{2}\varepsilon$  only (i) or (ii); if  $1 + \frac{1}{2}\varepsilon \leq \alpha \leq 1 + \frac{3}{2}\varepsilon$  only (ii) or (iii); if  $\alpha > 1 + \frac{3}{2}\varepsilon$  only (iii). Comparing with the defining properties of  $\tilde{C}_{\varepsilon}$ ,  $c(u) \leq (1 + \vartheta_Q^*) \tilde{C}_{\varepsilon}(\alpha)$  with  $\vartheta_Q^* := \max(c_i - 1, c_{\text{ii}} - C_{\text{band}}, \vartheta_Q')^+ \rightarrow 0$ , and Lemma 5.11(iii) gives the main term. Adding Step 2, (b), (c) and  $\int g \cdot Q^{-3} \ll L^2 Q^{-3} \leq Q^{-1}$  proves the proposition; each error is  $o(H |J| L^2 \ell_*)$  uniformly in  $T$  by Lemma 9.1(S4) and  $T \geq \ell^{a_0}$ .  $\square$

*The second moment.*

**Proposition 9.19.** *Let  $c : \mathbb{R} \rightarrow [1, \infty)$  be smooth, and suppose that, uniformly for  $T$  in (9.2),*

$$(9.15) \quad \sum_{n,m} b_n b_m \Delta(n, m) \mathcal{K}(n, m) \leq (1 + o(1)) H \sum_n b_n^2 c \left( \frac{\log n}{\ell_*} \right) \mathcal{K}(n, n) + o(H |J| L^2 \ell_*).$$

Put  $F(\alpha) = c(\alpha) \min(\alpha, 1 + \varepsilon_4)$  and  $f_v(x) = v(x/\lambda)/(\lambda\alpha)$ , which is supported in  $[-\frac{\lambda}{2}, \frac{\lambda}{2}]$ . Then, uniformly for  $T$  in (9.2),  $\mathfrak{M} \leq (1 - 2\theta) \mathcal{Q}_F(f_v) N + o(N)$ .

*Proof.* The proof of Proposition 5.13 with  $\ell$  replaced by  $\ell_*$ . By (2.9),  $\mathcal{K}(n, n) \leq 2\pi |J| g(\log n) + O(1)$ ; Lemma 9.12 with  $h(s) = c(s/\ell_*) g(s)$  (supported in  $|s| < L$ ;  $\|h^{(i)}\|_{\infty} \ll_i \|h\|_{\infty}$  since  $g^{(i)} \ll_i L^{1-i}$ ) and the substitution  $s = L\sigma$ ,  $g(L\sigma) = L(v * v)(\sigma)$ , give with Lemma 9.11

$$\mathcal{M}_{\Lambda\Lambda}^{\text{rat}} \leq \frac{H |J| L^2 \ell_*}{\pi} (1 + o(1)) \int_0^1 F(\lambda\sigma) (v * v)(\sigma) d\sigma + o(H |J| L^2 \ell_*).$$

Adding Lemma 9.10,  $\mathcal{M} \leq \frac{H |J| L^2 \ell_*}{2\pi} (b + 2\lambda \int_0^1 F(\lambda\sigma) (v * v)(\sigma) d\sigma) + o(H |J| L^2 \ell_*)$ . Divide by  $a^2 L^2$  and use Proposition 9.5,  $L = \lambda \ell_*$ ,  $|J| = (1 - 2\theta)T$  and  $N = (1 + o(1)) H T \ell_*/(2\pi)$  (Lemma 9.2):  $\mathfrak{M} \leq (1 - 2\theta) N a^{-2} (b/\lambda + 2 \int_0^1 F(\lambda\sigma) (v * v)(\sigma) d\sigma) + o(N)$ , and the bracket divided by  $a^2$  is  $\mathcal{Q}_F(f_v)$ , as in the proof of Proposition 5.13.  $\square$

**Theorem 9.20** (Conditional form at polynomial height). *Let  $w$  be admissible and suppose  $\text{LS}(C)$  holds for  $\mathcal{F}(Q, w)$  for some  $C \geq 1$ . Then, for fixed  $a_0, \kappa_1 > 0$ ,*

$$\liminf_{Q \rightarrow \infty} \inf_{\ell^{a_0} \leq T \leq Q^{\kappa_1}} \left( \frac{N_0^s}{N} - p(\bar{\lambda}_T; F_C) \right) \geq 0,$$

*likewise for  $N_0^*$ , and for  $N_d$  with  $\frac{1}{2}(1 + p(\bar{\lambda}_T; F_C))$ .*

The proof is given in §9.4.

**9.4. The variational problem at support  $\bar{\lambda}$ , and the proofs.** Throughout this subsection  $F = F_C$  with  $C \geq 1$  (restricted to  $[0, \bar{\lambda}]$ ), so that  $F$  is nondecreasing and  $0 \leq F \leq C$ .

**Lemma 9.21** (Monotonicity and continuity in  $\bar{\lambda}$ ). *For  $1 \leq \bar{\lambda}' \leq \bar{\lambda} \leq 2$ ,  $p(\bar{\lambda}'; F) \leq p(\bar{\lambda}; F) \leq p(\bar{\lambda}'; F) + 2(\bar{\lambda}/\bar{\lambda}' - 1)$ .*

*Proof.* The first inequality holds because the admissible classes in (1.5) are nested. For the second, note  $p(\bar{\lambda}; F) \geq p(1; F) = p(1; F_1) > 0$ , because  $F_C(\alpha) = \alpha$  on  $[0, 1]$ . Given  $0 < \eta' < p(\bar{\lambda}; F)$  take  $f$  admissible at support  $\bar{\lambda}$  with  $\mathcal{Q}_F(f) \leq 2 - p(\bar{\lambda}; F) + \eta' \leq 2$ , put  $s = \bar{\lambda}'/\bar{\lambda} \leq 1$  and  $f_s(x) = s^{-1}f(x/s)$ , which is admissible at support  $\bar{\lambda}'$ . Then  $\int f_s^2 = s^{-1} \int f^2$  and  $\iint f_s(x)f_s(y)F(|x-y|) = \iint f(x)f(y)F(s|x-y|) \leq \iint f(x)f(y)F(|x-y|)$ , since  $F$  is nondecreasing. As  $F \geq 0$ ,  $\int f^2 \leq \mathcal{Q}_F(f) \leq 2$ , so  $\mathcal{Q}_F(f_s) \leq \mathcal{Q}_F(f) + 2(s^{-1} - 1)$ . Let  $\eta' \rightarrow 0$ .  $\square$

**Lemma 9.22.** *For  $C' \geq C \geq 1$  and  $\bar{\lambda} \in [1, 2]$ ,  $p(\bar{\lambda}; F_{C'}) \geq p(\bar{\lambda}; F_C) - (C' - C)$ .*

*Proof.*  $F_{C'} - F_C = (C' - C)\mathbf{1}_{(1, \bar{\lambda}]}$  on  $[0, \bar{\lambda}]$ , and  $\iint_{|x-y|>1} f(x)f(y) \leq (\int f)^2 = 1$  for admissible  $f$ .  $\square$

**Lemma 9.23** (Realisation by windows at support  $\bar{\lambda}$ ). *Let  $\bar{\lambda} \in (1, 2]$ , let  $f \geq 0$  be even, in  $L^2$ , with  $\int f = 1$  and  $\text{supp } f \subset [-\frac{\bar{\lambda}}{2}, \frac{\bar{\lambda}}{2}]$ , and let  $\eta' > 0$ . There are  $\lambda < \bar{\lambda}$  and a real even  $\psi \in C_c^\infty((-\frac{1}{2}, \frac{1}{2}))$ ,  $\psi \not\equiv 0$ , with  $\mathcal{Q}_{F_C}(f_v) \leq \mathcal{Q}_{F_C}(f) + \eta'$ , where  $v = \psi^2$  and  $f_v(x) = v(x/\lambda)/(\lambda \int v)$ .*

*Proof.* First rescale: with  $s = \lambda/\bar{\lambda} < 1$ ,  $f_s(x) = s^{-1}f(x/s)$  is supported in  $[-\frac{\lambda}{2}, \frac{\lambda}{2}]$ , and the computation in the proof of Lemma 9.21 gives  $\mathcal{Q}_{F_C}(f_s) \leq \mathcal{Q}_{F_C}(f) + (s^{-1} - 1) \int f^2$ , which tends to  $\mathcal{Q}_{F_C}(f)$  as  $\lambda \uparrow \bar{\lambda}$ . Then dilate slightly, mollify and take a square root exactly as in the proof of Lemma 7.2;  $\mathcal{Q}_{F_C}$  is continuous on  $L^1 \cap L^2$  since  $0 \leq F_C \leq C$ .  $\square$

*Cells.* Given  $\eta' > 0$ , choose  $0 = \kappa^{(0)} < \kappa^{(1)} < \dots < \kappa^{(m)} = \kappa_1$  with

$$(9.16) \quad 2 \left( \frac{\bar{\lambda}(\kappa^{(j)})}{\bar{\lambda}(\kappa^{(j+1)})} - 1 \right) \leq \eta' \quad (0 \leq j < m),$$

which is possible because  $\bar{\lambda}$  is continuous and decreasing on  $[0, \kappa_1]$ . If  $\log T / \log Q \in [\kappa^{(j)}, \kappa^{(j+1)}]$  then  $\bar{\lambda}(\kappa^{(j+1)}) \leq \bar{\lambda}_T \leq \bar{\lambda}(\kappa^{(j)})$ , so by monotonicity and Lemma 9.21

$$(9.17) \quad p(\bar{\lambda}_T; F) \leq p(\bar{\lambda}(\kappa^{(j)}); F) \leq p(\bar{\lambda}(\kappa^{(j+1)}); F) + \eta'.$$

For  $T \leq \ell^{A_0}$  we have  $\log T / \log Q \rightarrow 0$ , so the polylogarithmic range lies in the first cell.

*Proofs.*

*Proof of Theorem 9.20.* Fix  $\eta' > 0$  and cells as in (9.16). Fix a cell  $[\kappa^{(j)}, \kappa^{(j+1)}]$  and apply §§9.1–9.3 with  $\kappa_c := \kappa^{(j+1)}$ ; write  $\bar{\lambda}_c := \bar{\lambda}(\kappa_c)$ . Choose, in this order:

- (a) an even admissible  $f$  at support  $\bar{\lambda}_c$  with  $2 - \mathcal{Q}_{F_C}(f) \geq p(\bar{\lambda}_c; F_C) - \eta'$ ;
- (b)  $\lambda < \bar{\lambda}_c$  and  $\psi$  by Lemma 9.23, so that  $\mathcal{Q}_{F_C}(f_v) \leq \mathcal{Q}_{F_C}(f) + \eta'$ ;
- (c)  $\varepsilon_1$  with  $\lambda(1 + \varepsilon_1) \leq \bar{\lambda}_c - \varepsilon_1$ ;

- (d)  $\theta, \varepsilon \leq \varepsilon_{\max}$  and  $\varepsilon_3$  (hence  $\varepsilon_4 = 2\varepsilon_3$ ) so small that  $8\theta + |\mathcal{Q}_{F_\varepsilon}(f_v) - \mathcal{Q}_{F_C}(f_v)| \leq \eta'$ , where  $F_\varepsilon := \tilde{C}_\varepsilon \min(\cdot, 1 + \varepsilon_4)$  with  $\tilde{C}_\varepsilon$  as in Proposition 9.17; this is possible by dominated convergence for the fixed  $v$ , since  $F_\varepsilon \rightarrow F_C$  off  $\{1\}$  and  $0 \leq F_\varepsilon \leq 2C$ ;  
(e) only then  $\varepsilon_5$  by (9.3).

Note that  $F_\varepsilon$  does not depend on  $\varepsilon_5$ , and for each fixed  $\varepsilon_5 > 0$  every error term in §§9.2–9.3 tends to 0 as  $Q \rightarrow \infty$ , uniformly for  $\ell^{a_0} \leq T \leq Q^{\kappa_c}$ . By Proposition 9.17, hypothesis (9.15) holds with  $c = \tilde{C}_\varepsilon$ , so Proposition 9.19 gives  $\mathfrak{M} \leq (1 - 2\theta)\mathcal{Q}_{F_\varepsilon}(f_v)N + o(N)$ . In particular  $\mathfrak{M} = O(N)$ , so  $\ell^{-B}(H\mathfrak{M})^{1/2} = o(N)$ , and Proposition 9.9 gives, uniformly in  $T$ ,

$$\frac{N_0^s}{N} \geq 2 - 8\theta - (1 - 2\theta)\mathcal{Q}_{F_\varepsilon}(f_v) - o(1) \geq 2 - \mathcal{Q}_{F_C}(f_v) - \eta' - o(1) \geq p(\bar{\lambda}_c; F_C) - 3\eta' - o(1).$$

By (9.17),  $N_0^s/N - p(\bar{\lambda}_T; F_C) \geq -4\eta' - o(1)$  uniformly for  $\log T / \log Q \in [\kappa^{(j)}, \kappa^{(j+1)}]$ ,  $T \geq \ell^{a_0}$ . There are finitely many cells; let  $Q \rightarrow \infty$  and then  $\eta' \rightarrow 0$ . The same choices give  $N_0^*$ , and  $N_d$  from the second inequality of Proposition 9.9.  $\square$

*Proof of Theorem 1.4(b), (c).*  $\text{LS}(C_{\text{cl}}(w))$  holds by (9.1) and Lemma 6.1, and  $\text{LS}(C_G R_w)$  by Corollary 6.8; apply Theorem 9.20. For the log-wide weights,  $R_w \rightarrow 1$  (Lemma 6.10) and Lemma 9.22 give  $p(\bar{\lambda}; F_{C_G R_w}) \geq p(\bar{\lambda}; F_{C_G}) - C_G(R_w - 1)$ .  $\square$

*Proof of Theorem 1.4(a).* Fix  $\varepsilon > 0$ . By Lemma 6.22,  $C_T^+(w_\eta) \leq 1 + 42/\log(1/\eta)$  and  $C_T^+(w_\eta^{\text{sm}}) \leq 1 + 83/\log(1/\eta)$  for  $\eta \leq e^{-3}$ ; choose  $\eta_0$  with  $C_T^+(w) \leq 1 + \varepsilon/3$  for  $\eta \leq \eta_0$ , and fix  $\eta \leq \eta_0$ . This choice involves neither  $T$  nor  $\kappa_1$  (it is the choice made in the proof of Theorem 1.1). Here  $\varepsilon$  is the accuracy in the statement; we write  $\varepsilon'$  for the band parameter of (P4) and Proposition 9.18 (so  $\varepsilon_5 = \min(\varepsilon', \varepsilon_1)/(4(1 + \kappa_1))$ ). Run the proof of Theorem 9.20 with Proposition 9.18 in place of Proposition 9.17,  $C = C_T^+(w)$  and the profile  $\tilde{C}_{\varepsilon'}$  of Proposition 9.18 (with band constant  $C_{\text{band}} = 2w_{\max}/(\mathcal{E}I_w)$ ): hypothesis (9.15) holds, since  $(1 + \vartheta_Q^*)(1 + \varkappa_{\text{loc}})^3 = 1 + o(1)$  uniformly in  $T$  and  $\mathcal{E}_{\text{TI}}^H = o(H|J|L^2\ell_*)$ ; and  $\mathcal{Q}_{F_{\varepsilon'}}(f_v) \rightarrow \mathcal{Q}_{F_{C_T^+(w)}}(f_v)$  as  $\varepsilon', \varepsilon_4 \rightarrow 0$  by dominated convergence, where  $F_{\varepsilon'} := \tilde{C}_{\varepsilon'} \min(\cdot, 1 + \varepsilon_4) \rightarrow F_{C_T^+(w)}$  off  $\{1\}$  and  $0 \leq F_{\varepsilon'} \leq 2C_{\max}$ . This gives the conclusion of Theorem 9.20 with  $C = C_T^+(w)$ , and Lemma 9.22 gives  $p(\bar{\lambda}; F_{C_T^+(w)}) \geq p(\bar{\lambda}; F_1) - \varepsilon/3$ . The same argument gives  $N_0^*$  and  $N_d$ .  $\square$

*Proof of Corollary 1.5.* Theorem 1.4(c) with  $w = \mathbf{1}_{[1/2, 1]}$  (admissible with any  $\eta < \frac{1}{2}$ ),  $C_{\text{cl}}(w) = 8/(3\mathcal{E}) \leq 5.5657$  by the certified bound  $\mathcal{E} \geq 0.47914533$  (Table 3), the monotonicity of  $p(\bar{\lambda}; F_C)$  in  $C$  and  $\bar{\lambda}$ , and the dyadic column of the table after Theorem 1.4.  $\square$

**Proposition 9.24** (Certified values). *For each entry of the table after Theorem 1.4 there is an explicit even step function  $f \geq 0$  on  $n$  equal cells of  $[-\frac{\bar{\lambda}}{2}, \frac{\bar{\lambda}}{2}]$ ,  $\bar{\lambda} = \bar{\lambda}(\kappa)$  ( $n = 400$ ), with  $\int f = 1$ , generated deterministically by the ancillary file `certify_hybrid.py` (`certify_dyadic.py` for the dyadic column), such that  $2 - \mathcal{Q}_{F_C}(f)$  is at least the tabulated value. The value  $\mathcal{Q}_{F_C}(f)$  is computed in exact rational arithmetic. Hence the tabulated values are lower bounds for  $p(\bar{\lambda}; F_C)$ , and  $(3 - \mathcal{Q}_{F_C}(f))/2$  for  $\frac{1}{2}(1 + p(\bar{\lambda}; F_C))$ .*

*Proof.* As in the proof of Proposition 7.3, with cells of width  $h = \bar{\lambda}/n$ :  $\mathcal{Q}_{F_C}(f) = h \sum_i f_i^2 + \sum_{i,j} f_i f_j \Gamma_{i-j}$  with  $\Gamma_m = G(mh + h) - 2G(mh) + G(mh - h)$ , where  $G$  is the even function with  $G'' = F_C(|\cdot|)$  and  $G(0) = G'(0) = 0$ :  $G(x) = |x|^3/6$  for  $|x| \leq 1$  and  $G(x) = \frac{1}{6} + \frac{|x|-1}{2} + \frac{C}{2}(|x| - 1)^2$  for  $1 < |x| \leq \bar{\lambda}$ . The script solves the discretised problem in floating point (by the KKT system, or by a nonnegative quadratic program when the unconstrained solution is not positive), symmetrises, rounds the heights to integers, renormalises exactly, and evaluates  $\mathcal{Q}_{F_C}$  exactly; it prints  $\lfloor 10^6(2 - \mathcal{Q}) \rfloor / 10^6$ . Any admissible  $f$  gives  $p(\bar{\lambda}; F_C) \geq 2 - \mathcal{Q}_{F_C}(f)$ . For

the columns (a)–(c) the same script with  $n = 800$  gives values larger by at most  $9 \cdot 10^{-6}$  (`log certify_hybrid_n800.log`). As checks that are not part of the proof, an independent exact evaluator, which computes  $\mathcal{Q}_{F_C}(f)$  from the autocorrelation of  $f$  instead of the second differences of  $G$ , gives the same rationals for all step functions of the table, and two independent optimisers with  $n = 600$  cells give values larger by at most  $6 \cdot 10^{-6}$ .  $\square$

**9.5. A sharp hybrid large sieve and the ceiling.** This subsection is not used in the proofs of Theorem 1.4 and Corollary 1.5. It isolates the mechanism of Proposition 9.18 for the simplest time weight, and shows that  $\bar{\lambda}_T$  is the limit of worst-case bounds. For a bounded interval  $J$  and a finitely supported  $y$  put  $x(u)_n := y_n \widehat{\mathbf{1}_J}(u - \log n)$ .

**Lemma 9.25.**  $\sum_{\chi \in \mathcal{F}} \omega_\chi \int_J |S_\chi[y](t)|^2 dt = \frac{1}{2\pi} \int_{\mathbb{R}} x(u)^* \Delta x(u) du.$

*Proof.* For fixed  $\chi$  the function  $\mathbf{1}_J(t) S_\chi[y](t)$  has Fourier transform  $\sum_n y_n \chi(n) \widehat{\mathbf{1}_J}(u - \log n)$ ; apply Plancherel, multiply by  $\omega_\chi$  and sum.  $\square$

**Theorem 9.26** (Sharp hybrid large sieve on  $Q$ -rough vectors). *Fix  $\varepsilon \in (0, 1)$  and an admissible  $w$ . There is  $Q_1(\varepsilon, w)$  such that for  $Q \geq Q_1$ , every bounded interval  $J$ , every  $\delta \in (0, \frac{1}{4}]$ , every  $\varkappa \in (0, 1]$  and every  $y$  supported on  $Q$ -rough integers of  $[1, Y]$  with  $9\delta Y + 1 \leq Q^{2-\varepsilon}$ ,*

$$\begin{aligned} \sum_{\chi \in \mathcal{F}} \omega_\chi \int_J |S_\chi[y](t)|^2 dt &\leq (1 + \varkappa)(C_T^+(w) + \vartheta'_Q) H|J| \|y\|^2 \\ &\quad + (1 + \varkappa^{-1}) \frac{24}{\pi} w_{\max}(k_0 + 4) \left( \frac{Q^2 + 1}{\delta} + (k_0 + 3)Y \right) \|y\|^2, \end{aligned}$$

where  $k_0 = \lceil \log_2(1/\delta) \rceil$  and  $\vartheta'_Q \rightarrow 0$  is the error term of Proposition 6.21 with  $\varepsilon$  there.

*Proof.* By Lemma 9.25 and Lemma 5.11(i), the left side is at most  $(2\pi)^{-1} \int ((1 + \varkappa)x^{s*} \Delta x^s + (1 + \varkappa^{-1})x^{t*} \Delta x^t) du$ . Only  $\delta \leq \frac{1}{4}$  is used below. The vector  $x^s(u)$  is supported on the  $Q$ -rough  $n \leq Y$  with  $|\log n - u| < 2\delta$ . If this set is nonempty then  $e^{u-2\delta} < Y$ , and it lies in an interval of length  $\leq e^u(e^{2\delta} - e^{-2\delta}) \leq 5\delta e^{1/2} Y$ , hence on at most  $9\delta Y + 1 \leq Q^{2-\varepsilon}$  integers. By Lemma 6.17 and (9.13) (after enlarging the interval to at least  $Q$  integers),  $x^{s*} \Delta x^s \leq (C_T^+(w) + \vartheta'_Q) H \|x^s(u)\|^2$ , and  $(2\pi)^{-1} \int \|x^s(u)\|^2 du \leq (2\pi)^{-1} \int \|x(u)\|^2 du = |J| \|y\|^2$  by Plancherel. The tail is Lemma 9.13 with  $g_1 \equiv 1/(2\pi)$ , which gives the coefficient  $48/(2\pi) = 24/\pi$ .  $\square$

**Corollary 9.27.** *Fix  $\varepsilon \in (0, 1)$  and  $w$ . Uniformly for  $|J| \geq 16$  and  $y$  supported on  $Q$ -rough integers of  $[1, Y]$  with  $Y \leq Q^{2-\varepsilon} |J|^{1-\varepsilon/9}$ ,*

$$\sum_{\chi \in \mathcal{F}} \omega_\chi \int_J |S_\chi[y](t)|^2 dt \leq (C_T^+(w) + \vartheta'_Q + O_{w,\varepsilon}(|J|^{-\varepsilon/5})) H|J| \|y\|^2.$$

*Proof.* Take  $\delta = |J|^{-1+\varepsilon/2}$ , which is at most  $16^{-1/2} = \frac{1}{4}$  as  $|J| \geq 16$  and  $\varepsilon < 1$ , so that  $9\delta Y + 1 \leq Q^{2-\varepsilon} |J|^{-\varepsilon/2} + 1 \leq Q^{2-\varepsilon}$ . Relative to  $H|J| \|y\|^2$  the tail term of Theorem 9.26 is  $\ll_w \log |J| (|J|^{-\varepsilon/2} + \log |J| Q^{-\varepsilon} |J|^{-\varepsilon})$ , using  $H \asymp_w Q^2$ . Call this quantity  $E$  and choose  $\varkappa := \min(\sqrt{E}, 1)$ , as Theorem 9.26 requires  $\varkappa \leq 1$ . If  $E \leq 1$ , the right side of Theorem 9.26 is  $(C_T^+(w) + \vartheta'_Q + O_w(\sqrt{E})) H|J| \|y\|^2$  and  $\sqrt{E} \ll_{w,\varepsilon} |J|^{-\varepsilon/5}$ . Since  $E \rightarrow 0$  as  $|J| \rightarrow \infty$ , uniformly in  $Q$ ,  $E > 1$  only for  $|J|$  bounded in terms of  $w$  and  $\varepsilon$ ; then  $\varkappa = 1$  gives a bound  $O_w(1) H|J| \|y\|^2$ , and the loss is absorbed into  $O_{w,\varepsilon}(|J|^{-\varepsilon/5})$ .  $\square$

With  $J = [(1+\theta)T, (2-\theta)T]$  and  $Y = (QT)^\alpha$ , the condition  $Y \leq Q^{2-\varepsilon} T^{1-\varepsilon}/9$  is  $\alpha < \bar{\lambda}_T - O(\varepsilon)$ . This is the reach of Theorem 1.4.

**Corollary 9.28** (General vectors). *Without the roughness assumption, the conclusion of Theorem 9.26 holds with  $(C_T^+(w) + \vartheta'_Q)H$  replaced by  $\Lambda_{\text{mult}}(Q^{2-\varepsilon})$ . In particular, for  $Y \leq Q^{2-\varepsilon}|J|^{1-\varepsilon}/9$  and  $|J| \geq 16$ ,*

$$\sum_{q \leq Q} \frac{q}{\varphi(q)} \sum_{\chi \bmod q}^* \int_J |S_\chi[y](t)|^2 dt \leq (1 + O(Q^{-\varepsilon} + |J|^{-\varepsilon/5})) Q^2 |J| \|y\|^2.$$

*Proof.* In the proof of Theorem 9.26 replace Lemma 6.17 and (9.13) by  $\Lambda_{\text{mult}}(9\delta Y + 1) \leq \Lambda_{\text{mult}}(Q^{2-\varepsilon})$ ; for the second statement use the unweighted form of (4.1),  $(Q^2 + K)$  with  $K \leq Q^{2-\varepsilon}$ . Lemma 9.13 and its proof hold verbatim for the form  $\sum_{q \leq Q} \frac{q}{\varphi(q)} \sum_{\chi \bmod q}^* |\sum_n x_n \chi(n)|^2$  in place of  $x^* \Delta x$ , with  $w_{\max} = 1$ , since they use only (9.1); this family (all  $q \leq Q$ ) is not of the form  $\mathcal{F}(Q, w)$  with  $w$  admissible.  $\square$

The second statement of Corollary 9.28 is a hybrid multiplicative large sieve with leading constant  $1+o(1)$  in the restricted range  $N \leq Q^{2-\varepsilon}T^{1-\varepsilon}$ . It follows in a few lines from localisation, and we regard it as standard; we claim no novelty for it. Gallagher's inequality [16, Theorem 3] holds for all  $N$  with an unspecified absolute constant, and Ramaré [34, §6] gives explicit constants (the  $Q^2$ -coefficient per unit length in  $t$  is about 2.06 there). Displays of the form  $\leq (Q^2T+N)\|a\|^2$  for  $\int_{-T}^T$  in the literature must be read as  $\ll$ ; the statement above uses the true length  $|J|$  and is consistent at  $Q = 1$ ,  $y = \delta_1$ .

*Remark 9.29.* (1) A tail-free proof of Theorem 9.26 and Corollary 9.28 (for the  $\int_J$  form) takes a nonnegative majorant  $M = |G|^2 \geq \mathbf{1}_J$  with  $\widehat{G}$  supported in  $[-\frac{\delta}{2}, \frac{\delta}{2}]$  and  $\int M = |J|(1+o(1))$ ; by Plancherel  $\sum_\chi \omega_\chi \int M |S_\chi[y]|^2$  is exactly an average of the family form over vectors supported on intervals of about  $\delta N$  integers. Lemma 9.13 is still needed for the second moment of §9.3, whose time weight  $\Phi(t-t')^2 \mathbf{1}_J(t) \mathbf{1}_J(t')$  is not monotone in the window. (2) Averaging over heights does not increase the constant  $C_T^+(w)$ ; this is Theorem 9.26. We expect that it does not decrease it either:  $C_T^+(w)$  is a supremum of local densities near  $\theta \in \mathbb{Z}$  which vary on the scale  $1/Q$ , much coarser than the smoothing scale  $T/N \leq Q^{-1-\varepsilon'}$  in the band above  $\alpha = 1$ . This is heuristic and is not used.

**Proposition 9.30** (The ceiling). *Let  $w$  be admissible. There are  $c_w > 0$  and, for  $Q$  large, a prime  $q_0 \in [\eta Q, Q]$  with  $w(q_0/Q) \geq c_w$ . Let  $0 < c \leq \frac{1}{10}$ , let  $J$  be an interval with  $|J| \geq 2/c$  and midpoint  $t_0$ , let  $N \geq 1$  and let  $I \subset [N, 2N]$  be an interval of  $K := \lceil 2N/(c|J|) \rceil$  integers, with  $K \geq 2q_0$ . For  $y_n := n^{it_0} \mathbf{1}_{[n \equiv 1 \bmod q_0]} \mathbf{1}_I(n)$ ,*

$$\sum_{\chi \in \mathcal{F}} \omega_\chi \int_J |S_\chi[y](t)|^2 dt \gg_{w,c} \frac{N}{Q^2 |J|} \cdot H |J| \|y\|^2.$$

*Proof.* Since  $w$  has bounded variation and  $\int uw > 0$ , there is an interval  $[a', b'] \subset [\eta, 1]$  on which  $w \geq c_w > 0$  (take a point of continuity where  $w > 0$ ); by the prime number theorem there is a prime  $q_0 \in [a'Q, b'Q]$  for  $Q$  large. Then  $\omega(q_0) \geq c_w$ , and every nonprincipal character modulo  $q_0$  is primitive, so there are  $\varphi^*(q_0) = q_0 - 2$  of them, each with  $\chi(n) = 1$  on the progression. For  $|t - t_0| \leq cN/K$ , which lies inside  $J$  as  $cN/K \leq c^2|J|/2$ , the phase  $(t - t_0) \log n$  varies by at most  $|t - t_0|K/N \leq c$  over  $n \in I$ , so  $|S_\chi[y](t)| = |\sum_{n \in I, n \equiv 1 \bmod q_0} n^{-i(t-t_0)}| \geq (1-c)(K/q_0 - 1)$ . Hence the characters modulo  $q_0$  alone contribute  $\geq c_w(q_0 - 2) \cdot 2(cN/K)(1-c)^2(K/q_0 - 1)^2 \gg_{w,c} NK/q_0$ , and all other terms are nonnegative. Since  $\|y\|^2 \leq K/q_0 + 1$  and  $H \asymp_w Q^2$ , the claim follows.  $\square$

A vector at  $n \asymp N = (QT)^\alpha$  with  $|J| \asymp T$  has  $N/(Q^2|J|) \asymp (QT)^{\alpha-\bar{\lambda}_T}$ , which is unbounded for  $\alpha > \bar{\lambda}_T$ . So no bound of the form “hybrid form  $\leq CH|J|\|y\|^2$  for all  $y$ ” can hold beyond  $\bar{\lambda}_T$ . The vector above is not  $Q$ -rough; Remark 9.31 gives a prime-supported version. Going beyond

$\bar{\lambda}_T$  would therefore require the structure of the prime vector, i.e. information on prime pairs; this is the  $q$ -aspect counterpart of the obstruction at  $\alpha = 1$  for  $\zeta$  (§§8.1, 8.3).

*Remark 9.31* (Prime-supported vectors). The threshold persists for vectors supported on primes, hence on  $Q$ -rough integers, up to a factor  $\log(QT)$  in  $N$ . On a residue class  $n \equiv a \pmod{q_0}$ ,  $(a, q_0) = 1$ , every character  $\chi \pmod{q_0}$  is constant,  $\chi(n) = \chi(a)$ , so  $|S_\chi[y]|$  is the same as for  $\chi(a) = 1$ . Take  $y_n := n^{it_0} \mathbf{1}[n \text{ prime}, n \equiv a \pmod{q_0}] \mathbf{1}_I(n)$ . By the prime number theorem and the pigeonhole principle over the  $\varphi(q_0)$  classes  $a$  and the  $\lfloor N/K \rfloor$  disjoint blocks of  $K$  consecutive integers in  $[N, 2N]$ , there are  $a$  and  $I$  with  $m \geq (1 + o(1))K/(\varphi(q_0) \log N)$  such primes, which exceed  $N > Q$  and are therefore  $Q$ -rough. The proof of Proposition 9.30, with the  $m$  primes in place of the progression, gives the ratio  $\gg q_0 N m / (K Q^2 |J|) \gg N / (Q^2 |J| \log N)$ , provided  $m \geq 2$ . With  $N = (QT)^\alpha$  and  $|J| \asymp T$  this is unbounded for  $\alpha > \bar{\lambda}_T$  (the factor  $\log N \asymp \log(QT)$  does not affect the exponent). The fundamental lemma of sieve theory is not needed.

## 9.6. Remarks on polynomial height.

*Limits of the method at height  $T$ .* Proposition 9.30 shows that the support  $\bar{\lambda}_T$  is the limit of any argument that uses the family second moment only through worst-case large-sieve bounds. Within that limit,  $p(\bar{\lambda}_T; F_1)$  is the value of the GUE form factor  $\min(\alpha, 1)$  at support  $\bar{\lambda}_T$  in the Montgomery–Taylor problem, and for  $C = 1$  the extremal problem is strictly convex at every support, by the argument of Remark 7.4 (which uses only  $F_1(|x|) = 1 - (1 - |x|)_+$ ); so, up to discretisation, the sharp constants in the table after Theorem 1.4 are the best the method gives. For  $\zeta$ , and for one  $L$ -function at height  $T \rightarrow \infty$ , the analogue of  $\Delta$  is the mean value  $\int_T^{2T} |\sum_{n \leq N} x_n n^{-it}|^2 dt = (T + O(N)) \|x\|^2$  [28], whose length term dominates for  $N \gg T$ , i.e. beyond  $\alpha = 1$  (§8.3); this is the limit  $\kappa \rightarrow \infty$  of  $\bar{\lambda}(\kappa)$ . As  $\kappa \rightarrow \infty$  the sharp constant decreases to  $p(1; F_1) = 0.6725 \dots$ ; for instance  $p(\bar{\lambda}(50); F_1) \approx 0.6856$  and  $p(\bar{\lambda}(100); F_1) \approx 0.6792$  (floating point, ancillary file `kappa_curve.log`; not certified). The value 0.6725 is the one obtained for a single fixed primitive character as  $T \rightarrow \infty$  by Alpöge and Furman [1, Theorem B].

*Super-polynomial heights.* (Sketch; not claimed.) The exterior-zero estimate extends to all  $T \geq Q^{\kappa_0}$  for fixed  $\kappa_0 > 0$  (Remark 9.8(2), with  $e^{L|\delta_\rho|} \leq T^{1+1/\kappa_0}$ ), and below bandwidth 1 the prime side needs no parameter depending on  $\kappa_1$ : if  $\lambda(1 + \varepsilon_1) \leq 1 - \varepsilon_1$ , then  $Y \leq (QT)^{1-\varepsilon_1}$ , only regime (i) occurs, every  $R_j$  vanishes (so  $b = a$ , and neither Lemma 9.11 nor §6 is used), and the tails of Corollary 9.14 are uniform in all  $T \geq \ell^{a_0}$  once  $\varepsilon_5 \leq 2\varepsilon_1$ . This suggests that the Montgomery–Taylor value 0.6725 holds at every height for the weighted family, which would be the family analogue of [1, Theorem B]. We have not checked the uniformity of the rest of the zero side and of the assembly in that range, and we make no claim.

*A single modulus.* (Heuristic; not used.) For the family of characters modulo one prime  $q$  at height  $T = q^\theta$ , the average over heights together with the characters modulo  $q$  should give bandwidth 1 in units of  $\log(qT)$  (through Gallagher’s single-modulus hybrid sieve [16, Theorem 2] and exact orthogonality on intervals shorter than  $q$ ), and hence 0.6725 at every polynomial height. Under GRH, Yıldırım [46] evaluated an average over all pairs of characters modulo  $q$  of the pair correlation of their zeros up to height  $T$ ; as quoted in [24], his asymptotic holds in the corresponding range, i.e. up to bandwidth 1 in units of  $\log(qT)$ . As we read [21, Remark 2.4], its bandwidth  $1/(1 + \theta)$  comes from orthogonality modulo  $q$  alone.

*Which ingredient gives what.* Without the Toeplitz route and the Gauss transfer, the classical large sieve alone gives Theorem 1.4(c): at least 0.738 for  $T \leq Q^2$  and 0.714 for  $T \leq Q^{10}$  with  $w = \mathbf{1}_{[\eta, 1]}$ ,  $\eta \leq 0.0146$  (the table after Theorem 1.4). Its optimal test function does not use the full support once  $C$  is large, which is why this column is nearly constant for  $\kappa \leq 2$ .

Without flattening as well (i.e. with  $b$  replaced by  $a$ , so that the diagonal density is  $\alpha$  instead of  $\min(\alpha, 1)$  and the majorant is  $\alpha$  on  $[0, 1]$  and  $C\alpha$  above 1), the same certificates give 0.7342 for  $\kappa \leq 2$  and 0.7142 at  $\kappa = 10$  with  $C = 4.175$  (`certify_hybrid_n400_rerun.log`, shape `Calpha`). Bandwidth below 1 alone gives the Montgomery–Taylor value 0.6725. The discussion of §8.4 applies unchanged at polynomial height; already the family  $Q/2 \leq q \leq Q$  with the weights  $q/\varphi(q)$  gives more than  $2/3$  at every polynomial height: more than 0.71 for  $T \leq Q^{10}$  by Corollary 1.5, and at least  $p(\bar{\lambda}_T; F_C) - o(1) \geq p(1; F_C) - o(1) = 0.6725 \dots - o(1)$  for  $T \leq Q^{\kappa_1}$  and every fixed  $\kappa_1$  by Theorem 1.4(c) with  $C = C_{\text{cl}}(\mathbf{1}_{[1/2, 1]})$ , since  $\bar{\lambda} \mapsto p(\bar{\lambda}; F_C)$  is nondecreasing and  $p(1; F_C) = p(1; F_1)$  ( $F_C = F_1$  on  $[0, 1]$ ).

#### AI-USE STATEMENT

This paper was produced with substantial use of AI. The mathematical arguments, the numerical computations and the text were generated by instances of Claude Opus 5.5 (Anthropic) working as coordinated agents: one set of agents developed the arguments (building on [1, 22]), separate agents acted as independent hostile referees of each component, and the referees’ requested repairs were incorporated. The numerical certificates were computed by AI-written scripts that are included as supplementary material and can be rerun, and the Lean formalisations of Theorems 1.1 and 1.4(a) described in §A.4 were also written by Claude agents. The extension to polynomial height in §9 was produced in the same way: two agents ported the zero side and the prime side of the argument, and three further agents acted as independent hostile referees of the two ports and of the standalone draft from which §9 was ported; their requested repairs were incorporated. The author initiated and directed this project, designed its verification process, and reviewed the results and the verification record, but did not independently check every step of the written proofs; the two headline results, Theorem 1.1 and Theorem 1.4(a), are checked by the Lean kernel as described in §A.4. The author takes full responsibility for the content of this paper, including any errors, and welcomes independent verification and corrections. Section 1.4 states precisely what is claimed, and §A.4 lists what has been checked by machine and what rests on the written proofs alone.

#### APPENDIX A. NUMERICAL COMPUTATIONS

This appendix lists the computations behind the numerical constants. It is separate from the proofs. The numerical facts used in the proofs of Theorems 1.1 and 1.2 are the exact rational evaluations of Proposition 7.3, the upper bound  $C_G \leq 1.2688$  (the margin is  $2.6 \cdot 10^{-5}$ ), and the elementary inequalities of Table 4 (for instance  $\sigma_0^- < 1$ , used in Lemma 6.15(d),(e)); all of them are CERTIFIED. Theorem 1.3 additionally uses computer-assisted upper bounds for  $R_w$  and  $C_T^+(w)$ , and  $C_G < 1.268774$ . Theorem 1.4 and Corollary 1.5 use the exact rational evaluations of Proposition 9.24 and the certified bound  $\mathcal{E} \geq 0.47914533$  of Table 3. We label numerical statements as follows: *PROVED* means established in the text (a closed form or a proved inequality); *CERTIFIED* means established by a computation in exact rational or interval arithmetic, so that no rounding error is involved; *COMPUTED* means evaluated in floating point, with truncation errors bounded analytically where stated but with rounding errors not controlled. Only PROVED and CERTIFIED statements are used in proofs.

**A.1. Certificates.** The ancillary file `certify.py` solves the discretised problem for  $\mathcal{Q}_{F_C}$  in floating point (the discretised shape- $F_C$  functional is numerically convex for  $C \leq 1.6$ , see below, so the KKT system is solved directly and the positivity of the solution is checked), symmetrises, rounds the step heights to integers, renormalises exactly, and evaluates  $\mathcal{Q}_{F_C}$  in rational arithmetic. Output: `certify.log` for  $n = 400, 800, 1600$ . The minimum over  $\int h = 0$  of  $\mathcal{Q}_{F_C}(h)/\|h\|^2$  is computed in `convexity.py` (`convexity.log`): 0.4305 ( $C = 1$ ), 0.4040 (1.0434),

| Quantity                                      | Value                                   | Status / source                                                                                                               |
|-----------------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| $\mathcal{E} = \prod_p (1 - p^{-2} - p^{-3})$ | 0.4791453444                            | certified<br>$\mathcal{E} \in [0.47914533, 0.47914535]$<br>(interval arithmetic); two<br>independent evaluations              |
| $C_G = 6/(\pi^2 \mathcal{E})$                 | 1.2687739                               | certified $C_G < 1.26877393$<br>(interval arithmetic); finite- $Q$<br>check $W/H = 1.26877$ at<br>$Q = 2 \cdot 10^5$          |
| $p(1)$                                        | $\geq 0.932282$ ( $\approx 0.9322826$ ) | exact rational certificate; ODE<br>and FEM agree                                                                              |
| $p(C_G)$                                      | $\geq 0.885912$ ( $\approx 0.8859177$ ) | exact rational certificate at<br>$C = 1.2688 \geq C_G$                                                                        |
| $R_w, C_w$ for log-wide weights               | Table 5                                 | computed (essential suprema);<br>certified upper bounds for the<br>log-smooth weight at<br>$\eta = 10^{-3}, 10^{-4}, 10^{-5}$ |
| $C_T^+(w)$ for fixed $\eta$                   | Table 6                                 | upper bounds certified (interval<br>arithmetic) via Proposition 6.23                                                          |

TABLE 3. Numerical constants used or quoted in the text; see also Table 4.

0.3748 (1.0911), 0.2648 (1.2688), 0.2173 (1.3448), 0.0553 (1.603),  $-0.0689$  (1.8). An independent Euler–Lagrange solution and an independent finite-element solution give  $p(C_G) = 0.8859177$  and  $p(1) = 0.9322826$  (sanity checks only; these scripts are not included in the ancillary files, and nothing in the proofs depends on them). For Theorem 1.4 the ancillary files `certify_hybrid.py` and `certify_dyadic.py` do the same at the support  $\bar{\lambda}(\kappa)$ ,  $\kappa \in \{0, \frac{1}{4}, \frac{1}{2}, 1, 2, 3, 5, 10\}$  (Proposition 9.24; the table after Theorem 1.4 uses  $n = 400$ , logs `certify_hybrid_n400_rerun.log` and `certify_hybrid_dyadic_n400.log`; the log `certify_hybrid_n800.log` records the finer step functions with  $n = 800$ ); `classical_constant_check.py` checks  $C_{\text{cl}}(\mathbf{1}_{[\eta, 1]}) \leq 4.175$  for  $\eta \leq 0.0146$  and  $8/(3\mathcal{E}) \leq 5.5657$  exactly, and `kappa_curve.py` computes the (uncertified) curve  $\kappa \mapsto p(\bar{\lambda}(\kappa); F_C)$  quoted in §9.6.

**A.2. Farey and Toeplitz constants.**  $R_w$  is the essential supremum of the explicit function  $P_w$  of (6.1), evaluated exactly on the open intervals between breakpoints (Remark 6.6).  $C_T(w) = \sup_{S,t} R_S(t)$  is the supremum of the explicit local density of the Toeplitz measure near rationals  $u/v$  with  $\{p \mid v\} = S$ ; for the weights with  $A(q) = q/\varphi(q)$  the supremum is attained at  $S = \emptyset$  in all tested cases, including non-primorial  $S$ . As a sanity check (the scripts are not included in the ancillary files, and no proof depends on it): for weights where finite- $Q$  effects are small, the top eigenvalue of the finite Toeplitz matrix agrees with the predicted supremum to 4–6 digits and its eigenvector concentrates at the predicted rational and scale (e.g.  $(1 - u)^2$  with  $q/\varphi(q)$ : 1.56528 at  $Q = 10^4$  against 1.5653 predicted; an independent reimplement obtains 1.565240, 1.565259 at  $Q = 10^3, 2 \cdot 10^3$ ). The log-wide weights with small  $\eta$  cannot be tested directly at feasible  $Q$  because the isolated-point term is  $O(Q^{\alpha-2}\eta^{-2})$ . For fixed  $\eta$ , Proposition 6.23 reduces  $C_T^+(w)$  to a finite computation over the 64 sets  $S_1 \subset \{2, \dots, 13\}$ , a far-field bound and an explicit slack; its evaluation (Table 6) bounds every truncation, interpolation and far-field error by the proved inequalities of Proposition 6.23 and Lemma 6.19, and evaluates all finite sums in ball arithmetic (Arb, through python-flint, at 64-bit precision), with every maximum and comparison taken on exact rational endpoints (ancillary file `ctplus_arb.py`; its `README.md` lists the steps). The same method bounds  $R_w$  (ancillary file `rw_arb.py`), with the far-field bound

of Lemma 6.11 (Remark 6.12). Hence the fixed- $\eta$  upper bounds for  $C_T^+(w)$  and  $R_w$  used in Theorem 1.3 are CERTIFIED: for the log-smooth weight  $C_T^+(w) \leq 1.090990$  ( $\eta = 10^{-3}$ ) and  $C_T^+(w) \leq 1.043290$  ( $\eta = 10^{-4}$ ), and  $R_w \leq 1.059816, 1.028831, 1.015859$  ( $\eta = 10^{-3}, 10^{-4}, 10^{-5}$ ). The suprema themselves are enclosed to within  $3 \cdot 10^{-7}$ ; for instance at  $\eta = 10^{-4}$ ,  $\sup_t R_\emptyset(t) \in [1.0432587, 1.0432589]$ , and the slack  $\frac{1}{2}\delta_{P_0}M_w \leq 3.1 \cdot 10^{-5}$  accounts for almost all of the gap to the certified bound. An independent implementation that shares no code (ancillary files `xcheck_*.py`) reproduces all six maxima to ten digits. The whole computation takes about two minutes on six cores.

**A.3. Sanity checks of the prime side.** At moderate  $Q$  (weights  $(1-u)^2q/\varphi(q)$ ,  $Q = 1000$ – $3000$ ) the Gauss-transfer inequality holds in every run, and the additive top eigenvalue reproduces  $R_w = 1.2633$ . On short intervals the multiplicative constant is 1.0000, 1.0008, 1.025, 1.10, 1.21 for  $K = Q^{0.3}, Q^{0.5}, Q^{0.7}, Q^{0.85}, Q^{1.0}$  ( $Q = 1000$  and  $2000$ ), illustrating Lemma 5.10. The Rayleigh quotients of actual flattened prime vectors,  $F(b)/(H\|b\|^2)$ , lie between 0.57 and 0.94 at all tested sizes (up to  $Q = 3 \cdot 10^5$ ) and increase towards 1 from below; the invisibility defect of  $a^\sharp$  is below  $10^{-4}$ . These checks are consistent with, but not used in, the proofs; their scripts are not included in the ancillary files.

**A.4. What has been checked by machine.** Three kinds of evidence are distinct: numerical certificates, a formal proof in Lean, and the written proofs of this paper. Theorem 1.1 is supported by all three, and so is Theorem 1.4(a). Theorems 1.2 and 1.3 rest on the written proofs and the numerical certificates, and so do Theorem 1.4(b), (c) and Corollary 1.5.

*Numerical statements.* Every numerical statement used in a proof is CERTIFIED by a script in the ancillary files, whose output log is shipped with it: Proposition 7.3 (`certify.py`), the fixed- $\eta$  constants of Theorem 1.3 (`ctplus_arb.py`, `rw_arb.py`, `certify_ctplus_arb.py`) and the arithmetic constants of Table 4 (`certify_constants_arb.py`), and Proposition 9.24 (`certify_hybrid.py`, `certify_dyadic.py`). The maxima behind  $C_T^+(w)$  were also reproduced by an independent implementation that shares no code with these scripts (`xcheck_*.py`).

*Formal proof of Theorem 1.1.* The ancillary directory `lean/` contains a Lean 4 project (toolchain v4.33.0-rc2) built on Mathlib and on the Lean formalisation of [1] (project `zeta23` of <https://github.com/anthropics/formal-math>); the exact revisions are pinned in `lean/lake-manifest.json`. From the latter it uses Weil’s explicit formula (Proposition B.1), zero-counting estimates for Dirichlet  $L$ -functions, Stirling-type bounds for  $\Gamma'/\Gamma$ , a prime number theorem with error term, and the linear algebra behind Lemma 3.2. The Lean theorem `Families.thmMain` proves `Families.thmMain.Statement`, the Lean statement of Theorem 1.1, including the bound  $p(1) \geq 0.932282$  (the row  $C = 1$  of Proposition 7.3, evaluated by the Lean kernel). It has no hypotheses, and `#print axioms Families.thmMain` reports only Lean’s standard axioms `propext`, `Classical.choice` and `Quot.sound`; in particular nothing it depends on uses `sorry` or `native_decide`. The formal proof follows the written one and assumes none of its steps. Where it deviates, it uses weaker inputs; for instance, it uses the large sieve (4.1) with a larger absolute constant, and a zero-density estimate of the form (4.5) with a smaller exponent  $c$ , proved in Lean for heights  $T' \leq Q$ , the only range used in the proof of Lemma 4.7. The script `lean/scripts/audit.sh` repeats these checks: it scans every declaration of the project for axioms and `sorry`, and it requires that `Families.thmMain` have no hypotheses, have type `thmMain.Statement`, and depend only on the three standard axioms.

*What the formal proof does not show.* The machine checks the Lean statement, not Theorem 1.1 as displayed in §1. That the two agree is a correspondence of definitions: the family and the weights (1.1), (1.2); the zero counts, with multiplicity, and the conditions “simple” and  $\beta = 1/2$ ; the constant (1.3); the heights  $\ell^{a_0} \leq T \leq \ell^{A_0}$ ; and the order of the quantifiers and

limits. The file `lean/STATEMENTS.md` sets out this correspondence, definition by definition, for a human reader. In the Lean statement the  $\liminf$  is written without division by  $N$ : for every  $\delta > 0$  and all  $Q \geq Q_0(\delta)$ ,  $N_0^s \geq (p(1) - \varepsilon - \delta)N$ , and similarly for  $N_0^*$ ; for  $N_d$  the Lean bound is  $N_d \geq ((1 + p(1))/2 - \varepsilon/2 - \delta)N$ , slightly stronger than the displayed  $-\varepsilon$ .

*Formal proof of Theorem 1.4(a).* The Lean project in the ancillary directory `lean/` contains a second library, `FamiliesH`, next to the library `Families` of Theorem 1.1; it imports `Families` without modifying it, with the same toolchain and pinned revisions. The Lean theorem `Families.Hybrid.thmH` proves `Families.Hybrid.thmH.Statement`, the Lean statement of Theorem 1.4(a), including the lower bounds in column (a) of the table after Theorem 1.4 for  $\kappa = 1, 2, 3, 5, 10$  (evaluated by the Lean kernel on the step functions with 400 cells). It has no hypotheses, and `#print axioms Families.Hybrid.thmH` reports only `propext`, `Classical.choice` and `Quot.sound`; in particular nothing it depends on uses `sorry` or `native_decide`. The formal proof follows §9 and assumes none of its steps. Where it deviates, it uses weaker inputs or another route for a single estimate; for instance, it uses the large sieve (4.1) with a larger absolute constant, as for Theorem 1.1, a direct bound for the prime part of the trace in Proposition 9.6 that does not use Proposition 9.5, and a bound for the mixed term of Lemma 9.10(b) that holds under the hypothesis (9.15) of Proposition 9.19, which is where that lemma is used. The project also proves that its statement implies `Families.thmMain.Statement`. The script `lean/scripts/audit.sh` repeats these checks for both headlines, `Families.thmMain` and `Families.Hybrid.thmH`.

*What this formal proof does not show.* As for Theorem 1.1, the machine checks the Lean statement, not Theorem 1.4(a) as displayed in §1. That the two agree is again a correspondence of definitions, now also for the heights  $\ell^{a_0} \leq T \leq Q^{\kappa_1}$ , the support  $\bar{\lambda}_T$  of (1.4), the constant  $p(\bar{\lambda}; F_1)$  of (1.5), and the order of the quantifiers ( $\eta_0$  does not depend on  $a_0$  and  $\kappa_1$ ). The file `lean/STATEMENTS-H.md` sets out this correspondence, definition by definition. In the Lean statement the target is evaluated at each height: for every  $\delta > 0$  and all  $Q \geq Q_0(\delta)$ ,  $N_0^s \geq (p(\bar{\lambda}_T; F_1) - \varepsilon - \delta)N$  for all  $\ell^{a_0} \leq T \leq Q^{\kappa_1}$ , and similarly for  $N_0^*$ ; for  $N_d$  the Lean bound is  $N_d \geq ((1 + p(\bar{\lambda}_T; F_1))/2 - \varepsilon/2 - \delta)N$ , slightly stronger than the displayed  $-\varepsilon$  (as for Theorem 1.1).

*Not formalised.* Theorems 1.2, 1.3 and 5.16 are stated in the Lean project (as propositions) but not proved there, and `Families.thmMain` does not depend on them. Also not formalised are Lemma 6.5, Corollary 6.8, Lemmas 6.9, 6.10 and 6.11, and Propositions 5.12 and 6.23; the formal proof of Theorem 1.1 does not use them. These results rest on the written proofs, on the cited literature and, for Theorem 1.3, on the certified numerics. Some ingredients of Theorem 1.2 are proved in Lean: Lemma 6.7, the row  $C = 1.2688$  of Proposition 7.3, and  $\mathcal{E} \geq 0.47914$ , hence  $C_G \leq 1.2688$ . The file `lean/STATUS.md` lists the status of each result in the Lean project. Theorem 1.4(b), (c), Corollary 1.5, Theorem 9.20 for general  $C$ , and the results of §9.5 (which the proofs do not use) are not stated in the Lean project; they rest on the written proofs and, for the columns (b), (c) and the dyadic column of the table after Theorem 1.4, on the certified numerics.

**A.5. Numerical constants.** Table 4 lists the arithmetic constants that occur in the proofs; Table 5 the Gauss-transfer constants  $R_w$ ,  $C_w = C_G R_w$  (Lemma 6.5); Table 6 the sharp constants  $C_T^+(w)$  at fixed  $\eta$  (Proposition 6.23) with the certified values of  $p(C)$ ; Table 7 other weights, for comparison. All suprema in  $t$  are *essential* suprema (Remark 6.6). For step weights they are evaluated exactly on the open intervals between consecutive breakpoints  $t \in \{\eta/n, 1/n\}$ ; for the log-smooth weight on a grid in  $\log(1/t)$ , plus a proved interpolation allowance (in the certified computation, a second-order bound using that  $R'_S$  is Lipschitz between grid points; see the ancillary `README.md`).

| constant                                                                       | value                 | where used                                     | status / source                                   |
|--------------------------------------------------------------------------------|-----------------------|------------------------------------------------|---------------------------------------------------|
| $\mathcal{E} = \prod_p (1 - p^{-2} - p^{-3})$                                  | 0.4791453...          | $H, C_G, R_S$                                  | CERTIFIED,<br>$\in [0.47914533,$<br>$0.47914535]$ |
| $C_G = 6/(\pi^2 \mathcal{E})$                                                  | $\approx 1.2687739$   | $W/H \rightarrow C_G$ (Lemma 6.1)              | CERTIFIED,<br>$< 1.26877393$                      |
| $c_\varphi = \gamma - \zeta'(2)/\zeta(2)$                                      | 1.1471767...          | Lemma 6.9                                      | closed form;<br>CERTIFIED                         |
| $B_\varphi = \zeta(3/2)/\zeta(3)$                                              | 2.17325...            | $ E_\varphi(y)  \leq 2B_\varphi y^{-1/2}$      | closed form;<br>PROVED bound                      |
| $4B_\varphi \pi^2/6$                                                           | 14.2994...            | $R_w \leq 1 + 14.31/I_h$<br>(Lemma 6.10)       | PROVED;<br>CERTIFIED $\leq 14.31$                 |
| $\zeta(2)\zeta(3)/\zeta(6) = \sum_d \mu^2(d)/(d\varphi(d))$                    | 1.943596...           | tail bounds, $\sum_{r \leq x} 1/\varphi(r)$    | closed form;<br>CERTIFIED $\leq 1.95$             |
| $\sum_r \mu^2(r)/(r^2 \varphi(r))$                                             | 1.33978...            | Lemma 6.15(a)                                  | CERTIFIED, $< 1.34$                               |
| $\sigma_0^- = \sum_{\mu(r)=-1} 1/(r^2 \varphi(r))$                             | 0.32114...            | Lemma 6.15(d)                                  | CERTIFIED, $< 0.33$                               |
| $\sigma_1^- = \sum_{\mu(r)=-1} \log r/(r^2 \varphi(r))$                        | 0.26313...            | Lemma 6.15(e),<br>Lemma 6.22                   | CERTIFIED, $< 0.27$                               |
| $c_\emptyset = \gamma + \sum_p \frac{(p^{-2}+p^{-3}) \log p}{1-p^{-2}-p^{-3}}$ | 1.4107008...          | first-order excess,<br>Lemma 6.19(iii)         | CERTIFIED, $< 1.411$                              |
| $\max_S (c_\emptyset - c_S) = \sum_p \frac{\log p}{p^3(p-1)(1-p^{-2}-p^{-3})}$ | 0.16722...            | Lemma 6.19(iii)                                | CERTIFIED, $< 0.168$                              |
| $B = \prod_p (1 + \frac{p^{-1/2}}{p-1} + \frac{p^{-1}}{p(p-1)})$               | 3.7254...             | $ E_S(s)  \leq 2Be^{-s/2}$ , all $S$           | CERTIFIED, $< 3.73$ ;<br>PROVED bound             |
| $c_\emptyset + 4B/\mathcal{E}$                                                 | $< 32.6$              | $\sup_S R_S \leq 1 + 32.6/I_w$                 | PROVED (from the<br>rows above)                   |
| $\delta_{P_0} = \prod_{p>13} (1 + \frac{2}{p^3(p-1)}) - 1$                     | $5.824 \cdot 10^{-5}$ | reduction to<br>$S_1 \subset \{2, \dots, 13\}$ | CERTIFIED,<br>$< 5.83 \cdot 10^{-5}$              |

TABLE 4. Arithmetic constants. CERTIFIED entries are enclosed in interval arithmetic with explicit tail bounds (ancillary file `certify_constants_arb.py`, output `certify_constants_arb.log`);  $\mathcal{E}$ ,  $c_\emptyset$ ,  $B$ ,  $\delta_{P_0}$  and  $\sigma_0^-$  are also enclosed by `ctplus_arb.py`, and  $\mathcal{E} \geq 0.47914$  (hence  $C_G \leq 1.2688$ ) is also checked in the Lean formalisation (Section A.4).

| weight                                                                         | $\eta$    | $R_w$ (ess sup)              | pointwise sup | $C_w = C_G R_w$   | $p(C_w)$        |
|--------------------------------------------------------------------------------|-----------|------------------------------|---------------|-------------------|-----------------|
| sharp $u^{-2} \mathbf{1}_{[\eta,1]}$                                           | $10^{-1}$ | 1.480065                     | 1.508641      | 1.8779            | –               |
|                                                                                | $10^{-2}$ | 1.247814                     | 1.249243      | 1.5832            | –               |
|                                                                                | $10^{-3}$ | 1.166030                     | 1.166125      | 1.4794            | –               |
|                                                                                | $10^{-4}$ | 1.124548                     | 1.124555      | 1.4268            | –               |
|                                                                                | $10^{-5}$ | 1.099642                     | 1.099643      | 1.3952            | $\approx 0.867$ |
| log-smooth $u^{-2} \sin^2(\pi \frac{\log u}{\log \eta}) \mathbf{1}_{[\eta,1]}$ | $10^{-3}$ | $\leq 1.059816$              | –             | $\leq 1.3447$     | $\geq 0.874482$ |
|                                                                                | $10^{-4}$ | $\leq 1.028831$              | –             | $\leq 1.3054$     | $\geq 0.880308$ |
|                                                                                | $10^{-5}$ | $\leq 1.015859$              | –             | $\leq 1.2889$     | $\geq 0.882814$ |
| $(1-u)^2$                                                                      | –         | 1.2633                       | –             | 1.603             | $\geq 0.841248$ |
| $(1-u)^6$                                                                      | –         | 1.1720                       | –             | 1.487             | $\approx 0.855$ |
| any log-wide, $\eta \rightarrow 0$                                             | –         | $\rightarrow 1$ (Lemma 6.10) | –             | $\rightarrow C_G$ | $\geq 0.885912$ |

TABLE 5. Gauss-transfer constants. Sharp and  $(1-u)^k$  rows: COMPUTED (ancillary file `rw_ess.log`; the  $(1-u)^k$  rows from independent computations that are not included). The limit  $R_w \rightarrow 1$  is PROVED. Log-smooth rows: CERTIFIED upper bounds in interval arithmetic (ancillary files `rw_arb.py`, `rw_arb_lsmooth*.log`; far field by Lemma 6.11); the enclosures are  $R_w \in [1.0598154, 1.0598156]$ ,  $[1.0288302, 1.0288304]$ ,  $[1.0158583, 1.0158585]$ , and  $C_G R_w \leq 1.344667, 1.305354, 1.288895$ , so the bounds  $R_w \leq 1.0599, 1.0289, 1.0159$  of Theorem 1.3(iii) hold.  $p$ -values marked  $\geq$  are CERTIFIED by exact rational evaluation of an explicit even step window at the tabulated value of  $C$  (a certified upper bound in the log-smooth rows, `certify_ctplus_arb.log`; other rows: `certify.log`); those marked  $\approx$  are interpolated ( $dp/dC \approx -0.16$ ).

| weight                       | $\eta$    | ess sup $_t R_\emptyset \in$ | sup $R^+$ over $\mu^-$ region $\leq$ | $C_T^+(w) \leq$              | $p(C_T^+) \geq$ | distinct $\geq$ |
|------------------------------|-----------|------------------------------|--------------------------------------|------------------------------|-----------------|-----------------|
| sharp                        | $10^{-2}$ | [1.304866, 1.304867]         | 1.149557                             | 1.30491                      | 0.880382        | 0.940191        |
|                              | $10^{-3}$ | [1.204189, 1.204190]         | 1.099476                             | 1.20423                      | 0.896219        | 0.948109        |
|                              | $10^{-4}$ | [1.153159, 1.153160]         | 1.074595                             | 1.15320                      | 0.904739        | 0.952369        |
| log-smooth                   | $10^{-2}$ | [1.236095, 1.236097]         | 1.136613                             | 1.23614                      | 0.891060        | 0.945530        |
|                              | $10^{-3}$ | [1.090957, 1.090958]         | 1.045532                             | 1.09099                      | 0.915569        | 0.957784        |
|                              | $10^{-4}$ | [1.043258, 1.043259]         | 1.028502                             | 1.04329                      | 0.924202        | 0.962101        |
| either, $\eta \rightarrow 0$ | –         | $\rightarrow 1$              | –                                    | $\rightarrow 1$ (Lemma 6.22) | 0.932282        | 0.966141        |

TABLE 6. The sharp constant  $C_T^+(w)$  at fixed  $\eta$  (Proposition 6.23); every entry is CERTIFIED in interval (ball) arithmetic (ancillary files `ctplus_arb.py`, `ctplus_arb*.log`, `certify_ctplus_arb.log`). Column 3: an enclosure of  $\text{ess sup}_t R_\emptyset(t)$ ; the maximum over  $S_1$  is at  $S_1 = \emptyset$  in every case (sharp: at  $t \rightarrow \eta^+$ ; log-smooth: at  $t/\eta = 5.81, 8.21, 9.84$ ). Column 4: an upper bound for  $\max_{S_1} \sup(R_{S_1} + R^m)$  over  $t < \eta$  (sharp) resp.  $t < 3\eta$  (log-smooth), where the majorant  $R^m$  lives; it is below column 3, so the positive part does not change the constant. Column 5 is the reduction of Proposition 6.23 over all 64 sets  $S_1 \subset \{2, 3, 5, 7, 11, 13\}$ , with both far-field bounds and the slack  $\frac{1}{2}\delta_{P_0} M_w \leq 3.9 \cdot 10^{-5}$ , rounded up to five decimals. Columns 6–7: exact rational lower bounds for  $p(C)$  at the value in column 5 ( $n = 1600$  cells). The last row is PROVED (Lemma 6.22) with  $p(1) \geq 0.932282$  CERTIFIED. An earlier double-precision evaluation (ancillary file `ctplus.py`) gave the same column 5 for the sharp weight, and 1.23617, 1.09101, 1.04331 for the log-smooth weight.

| weight (per-character $\omega(q)$ )           | $\sup_{S,t} R_S$                    | $p$             |
|-----------------------------------------------|-------------------------------------|-----------------|
| $(1-u)^2 q/\varphi(q)$                        | 1.5653                              | $\approx 0.846$ |
| $(1-u)^6 q/\varphi(q)$                        | 1.3757                              | $\approx 0.870$ |
| $(1-u)^2$ (no $q/\varphi(q)$ ; $A \equiv 1$ ) | 2.028 ( $S \rightarrow$ primorials) | $\approx 0.805$ |

TABLE 7. Other weights, for comparison only. Column 2: the signed constant  $\sup_{S,t} R_S$ , COMPUTED in floating point and compared with top eigenvalues of the finite Toeplitz matrices to 5–6 digits; the positive part was not examined for these narrow weights, where  $\mu^-$  is large, so these are *not* values of  $C_T^+$ , and they are not certified. Column 3:  $p$  at the value in column 2, from a floating-point solution of the discretised problem (1.3) (bandwidth 2; for the last row two solvers give 0.8048 and 0.8050). The last row, with weight  $(1 - q/Q)^2$  per primitive character, is discussed in §8.6. These computations are not included in the ancillary files, and no proof depends on them.

## APPENDIX B. PROOFS OF THE PORTED LEMMAS

This appendix proves two auxiliary statements whose arguments follow [1, 22] and which are used in §§2–4: Weil’s explicit formula on the test class (Lemma 2.6) and the rank–trace inequality (Lemma 3.2). The other statements adapted from [22] (Lemmas 2.4, 3.1, 4.1, 4.3, Propositions 3.3, 4.4) are proved where they are stated. Apart from the classical facts about Dirichlet  $L$ -functions quoted from [13], the arguments are self-contained.

**B.1. The explicit formula.** For a primitive character  $\chi$  modulo  $q \geq 3$  with parity  $\mathfrak{a} = \mathfrak{a}_\chi$  let  $\xi(s, \chi) := (q/\pi)^{(s+\mathfrak{a})/2} \Gamma(\frac{s+\mathfrak{a}}{2}) L(s, \chi)$ . This is an entire function whose zeros are exactly the nontrivial zeros of  $L(s, \chi)$ , with multiplicity, all in  $0 < \operatorname{Re} s < 1$ ; and  $\xi(s, \chi) = \epsilon(\chi) \xi(1-s, \bar{\chi})$  with  $|\epsilon(\chi)| = 1$ , where  $\bar{\chi}$  has the same parity [13, Chs. 9, 12]. Writing  $\psi_\Gamma = \Gamma'/\Gamma$ , we get

$$(B.1) \quad \frac{\xi'}{\xi}(s, \chi) = \frac{1}{2} \log \frac{q}{\pi} + \frac{1}{2} \psi_\Gamma\left(\frac{s+\mathfrak{a}}{2}\right) + \frac{L'}{L}(s, \chi), \quad \frac{\xi'}{\xi}(s, \chi) = -\frac{\xi'}{\xi}(1-s, \bar{\chi}).$$

**Proposition B.1** (Weil’s explicit formula; cf. [23, Thm. 5.12], [22, Prop. 4.1]). *Let  $\chi$  be a primitive character modulo  $q \geq 3$ , let  $h \in C_c^\infty(\mathbb{R})$  be complex valued, and put  $H(z) := \int_{\mathbb{R}} h(u) e^{izu} du$  ( $z \in \mathbb{C}$ ). Then*

$$\sum_{\rho} m_{\rho} H\left(\frac{\rho - 1/2}{i}\right) = \int_{\mathbb{R}} H(t) \mu_{\chi}(t) dt - \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} (\chi(n) h(\log n) + \bar{\chi}(n) h(-\log n)),$$

where  $\rho$  runs over the nontrivial zeros of  $L(s, \chi)$  with multiplicities  $m_{\rho}$  and  $\mu_{\chi}$  is as in (2.3). The zero sum and the integral converge absolutely, and the prime sum is finite.

*Proof.* Let  $\operatorname{supp} h \subset [-U, U]$ . Integrating by parts  $k$  times,

$$(B.2) \quad |H(x + iy)| \leq \|h^{(k)}\|_1 e^{U|y|} |x + iy|^{-k} \quad (k \geq 0),$$

so  $H$  is entire and decays faster than any power of  $|x|$ , uniformly in every horizontal strip. Put  $\tilde{H}(s) := H((s - \frac{1}{2})/i)$ . For  $s = \sigma + i\tau$  we have  $(s - \frac{1}{2})/i = \tau - i(\sigma - \frac{1}{2})$ , so  $\tilde{H}$  decays faster than any power of  $|\tau|$  uniformly for  $-\frac{1}{2} \leq \sigma \leq \frac{3}{2}$ . By (2.6) and (B.2),  $\sum_{\rho} m_{\rho} |\tilde{H}(\rho)| < \infty$ .

*Contour.* For  $-1 \leq \sigma \leq 2$  and  $|t| \geq 2$ ,  $\frac{L'}{L}(s, \chi) = \sum_{|\gamma-t| < 1} (s - \rho)^{-1} + O(\log(q|t|))$  [13, Ch. 16]. By (2.6), at most  $O(\log(qV))$  zeros have  $|\gamma| \in [V-1, V+2]$ , so for every  $V \geq 2$  there is  $V' \in [V, V+1]$  with  $|V' - |\gamma|| \gg 1/\log(qV)$  for all zeros; then  $\frac{L'}{L}(\sigma \pm iV', \chi) \ll \log^2(qV)$  for  $-\frac{1}{2} \leq \sigma \leq \frac{3}{2}$  (this is the standard choice of height in the proof of the explicit formula for  $\psi(x, \chi)$ , [13, Ch. 19]). Since also  $\psi_\Gamma(w) \ll \log(2 + |w|)$  for  $|\operatorname{Im} w| \geq 1$ , (B.1) gives  $\frac{\xi'}{\xi}(\sigma \pm iV', \chi) \ll \log^2(qV)$  there. Integrate  $\tilde{H}(s) \frac{\xi'}{\xi}(s, \chi)$  over the positively oriented boundary of the rectangle with vertices  $-\frac{1}{2} \pm iV'$ ,  $\frac{3}{2} \pm iV'$ . No zero lies on the boundary, and the residue theorem gives  $2\pi i \sum_{|\gamma| < V'} m_{\rho} \tilde{H}(\rho)$ . By (B.2) with  $k = 2$ , the horizontal sides contribute  $O(V^{-2} \log^2(qV))$ . Letting  $V \rightarrow \infty$ ,

$$(B.3) \quad \sum_{\rho} m_{\rho} \tilde{H}(\rho) = I_+ - I_-, \quad I_{\pm} := \frac{1}{2\pi i} \int_{(c_{\pm})} \tilde{H}(s) \frac{\xi'}{\xi}(s, \chi) ds, \quad c_+ = \frac{3}{2}, \quad c_- = -\frac{1}{2},$$

where both integrals converge absolutely, since  $\frac{\xi'}{\xi}(c_{\pm} + i\tau, \chi) \ll \log(q(2 + |\tau|))$  on these lines by (B.1).

*Prime terms.* On  $\operatorname{Re} s = \frac{3}{2}$ ,  $-\frac{L'}{L}(s, \chi) = \sum_n \Lambda(n) \chi(n) n^{-s}$  converges absolutely, and by (B.1) on  $\operatorname{Re} s = -\frac{1}{2}$ ,  $\frac{\xi'}{\xi}(s, \chi) = -\frac{1}{2} \log \frac{q}{\pi} - \frac{1}{2} \psi_\Gamma(\frac{1-s+\mathfrak{a}}{2}) + \sum_n \Lambda(n) \bar{\chi}(n) n^{s-1}$ . For  $n \geq 2$  substitute  $s = \frac{1}{2} + iz$ : on  $\operatorname{Re} s = \frac{3}{2}$  the variable  $z$  runs over  $\operatorname{Im} z = -1$  from left to right, and  $ds = i dz$ ; moving the line to  $\operatorname{Im} z = 0$  is allowed by (B.2), because  $|n^{-iz}| = n^{\operatorname{Im} z} \leq 1$  for  $-1 \leq \operatorname{Im} z \leq 0$ .

On  $\operatorname{Re} s = -\frac{1}{2}$  the variable  $z$  runs over  $\operatorname{Im} z = +1$ , and we move it to  $\operatorname{Im} z = 0$  using  $|n^{iz}| = n^{-\operatorname{Im} z} \leq 1$  for  $0 \leq \operatorname{Im} z \leq 1$ . By Fourier inversion,

$$\begin{aligned} \frac{1}{2\pi i} \int_{(3/2)} \tilde{H}(s) n^{-s} ds &= \frac{n^{-1/2}}{2\pi} \int_{\mathbb{R}} H(t) e^{-it \log n} dt = \frac{h(\log n)}{\sqrt{n}}, \\ \frac{1}{2\pi i} \int_{(-1/2)} \tilde{H}(s) n^{s-1} ds &= \frac{n^{-1/2}}{2\pi} \int_{\mathbb{R}} H(t) e^{it \log n} dt = \frac{h(-\log n)}{\sqrt{n}}. \end{aligned}$$

Interchanging sum and integral, the prime parts of  $I_+$  and of  $-I_-$  are

$$-\sum_n \Lambda(n) \chi(n) n^{-1/2} h(\log n) \quad \text{and} \quad -\sum_n \Lambda(n) \bar{\chi}(n) n^{-1/2} h(-\log n).$$

Only  $n \leq e^U$  contribute.

*Gamma terms.* The function  $\psi_\Gamma(\frac{s+\mathfrak{a}}{2})$  is holomorphic for  $\operatorname{Re} s > 0$ , and  $\psi_\Gamma(\frac{1-s+\mathfrak{a}}{2})$  for  $\operatorname{Re} s < 1$  (their poles are real); for  $|\tau| \geq 1$  both are  $\ll \log(2 + |\tau|)$  in the strip  $-\frac{1}{2} \leq \sigma \leq \frac{3}{2}$ . By (B.2) we may move the remaining parts of  $I_+$  and of  $-I_-$  to the line  $\operatorname{Re} s = \frac{1}{2}$ , where  $s = \frac{1}{2} + it$ ,  $\tilde{H}(s) = H(t)$  and  $ds = i dt$ :

$$\begin{aligned} \frac{1}{2\pi} \int_{\mathbb{R}} H(t) \left( \frac{1}{2} \log \frac{q}{\pi} + \frac{1}{2} \psi_\Gamma \left( \frac{\frac{1}{2} + \mathfrak{a} + it}{2} \right) \right) dt \\ + \frac{1}{2\pi} \int_{\mathbb{R}} H(t) \left( \frac{1}{2} \log \frac{q}{\pi} + \frac{1}{2} \psi_\Gamma \left( \frac{\frac{1}{2} + \mathfrak{a} - it}{2} \right) \right) dt = \int_{\mathbb{R}} H(t) \mu_\chi(t) dt, \end{aligned}$$

because  $\psi_\Gamma(\bar{w}) = \overline{\psi_\Gamma(w)}$ . This integral converges absolutely since  $\mu_\chi(t) \ll \log(q(2 + |t|))$ . Inserting the prime and gamma terms in (B.3), and noting  $\tilde{H}(\rho) = H((\rho - \frac{1}{2})/i)$ , proves the proposition.  $\square$

*Proof of Lemma 2.6.* Let  $f_k(u) := \psi_L(u) e^{-i\tau_k u}$  and  $h := f_k * f_l \in C_c^\infty(\mathbb{R})$ . Then  $\operatorname{supp} h \subset \operatorname{supp} \psi_L + \operatorname{supp} \psi_L$ , a compact subset of  $(-L, L)$ , and  $H(z) = \int h(u) e^{izu} du = \widehat{\psi_L}(z - \tau_k) \widehat{\psi_L}(z - \tau_l) = p_k(z) p_l(z)$ . The zero side of Proposition B.1 is  $G_{\chi,kl}$  by (2.5). Since  $h(\pm \log n) = 0$  for  $n \geq X = e^L$  and  $\Upsilon(n) = 1$  for  $n \leq X$ , the prime side is  $-\sum_n a_n (\chi(n) h(\log n) + \bar{\chi}(n) h(-\log n))$ . As  $2\pi h(\pm \log n) = \int H(t) n^{\mp it} dt$  and  $a_n$  is real, this equals  $-\frac{1}{2\pi} \int H(t) (S_\chi[a](t) + \overline{S_\chi[a](t)}) dt = \int H(t) P_\chi(t) dt$  (a finite sum). Hence  $G_{\chi,kl} = \int p_k p_l (\mu_\chi + P_\chi)$ . The right side is symmetric in  $k, l$  and real, because  $p_k$  is real on  $\mathbb{R}$  (Lemma 2.4) and  $\mu_\chi, P_\chi$  are real. Only  $n < X$  enter, so the value does not depend on  $\Upsilon_0$ .  $\square$

## B.2. The rank–trace inequality.

*Proof of Lemma 3.2.* We follow [22, Lemma 8.2]; see [1, Lemma 3.2] for a proof via von Neumann’s trace inequality. Let  $d$  be the size of the matrices,  $r_0 := \operatorname{rank} P \leq r$  and  $b_0 := n_+(R) \leq b$ . Since the right side decreases in  $r$  and  $b$ , it suffices to prove the inequality with  $(r_0, b_0)$ . Let  $p_1 \geq \dots \geq p_{r_0} > 0$  be the nonzero eigenvalues of  $P$ , with orthonormal eigenvectors  $e_1, \dots, e_{r_0}$ , and let  $q_1 \leq \dots \leq q_d$  be the eigenvalues of  $R$ , with orthonormal eigenvectors  $f_1, \dots, f_d$ .

*Ky Fan step.* Fix  $k \leq r_0$  and put  $\alpha_j := \sum_{i \leq k} |\langle e_i, f_j \rangle|^2$ . These are the diagonal entries, in the basis  $(f_j)$ , of the orthogonal projection onto  $\operatorname{span}(e_1, \dots, e_k)$ , so  $0 \leq \alpha_j \leq 1$  and  $\sum_j \alpha_j = k$ . As

$(q_j)$  is nondecreasing,

$$\begin{aligned} S_k &:= \sum_{i \leq k} \langle Re_i, e_i \rangle = \sum_j q_j \alpha_j = \sum_{j \leq k} q_j + \sum_{j > k} q_j (\alpha_j - 1) + \sum_{j > k} q_j \alpha_j \\ &\geq \sum_{j \leq k} q_j + q_k \left( \sum_j \alpha_j - k \right) = \sum_{j \leq k} q_j. \end{aligned}$$

*Abel summation.* With  $p_{r_0+1} := 0$ ,

$$\mathrm{tr}(PR) = \sum_{i \leq r_0} p_i \langle Re_i, e_i \rangle = \sum_{k \leq r_0} (p_k - p_{k+1}) S_k \geq \sum_{k \leq r_0} (p_k - p_{k+1}) \sum_{j \leq k} q_j = \sum_{i \leq r_0} p_i q_i.$$

*Pairing.* Using  $\|P + R\|_F^2 = \sum_i p_i^2 + 2 \mathrm{tr}(PR) + \sum_j q_j^2$ ,  $\mathrm{tr} P = \sum_i p_i$ ,  $\mathrm{tr} R = \sum_j q_j$  and  $b_0 = \#\{j : q_j > 0\}$ ,

$$\|P + R\|_F^2 - (2 \mathrm{tr} P - r_0 + 4 \mathrm{tr} R - 4b_0) \geq \sum_{i \leq r_0} \phi(p_i, q_i) + \sum_{j > r_0} (q_j^2 - 4q_j + 4 \cdot \mathbf{1}_{q_j > 0}),$$

where  $\phi(p, q) := p^2 + 2pq - 2p + 1 + q^2 - 4q + 4 \cdot \mathbf{1}_{q > 0}$ . Each unpaired term is  $\geq 0$ : it is  $q^2 - 4q \geq 0$  if  $q \leq 0$  and  $(q - 2)^2$  if  $q > 0$ . For  $p > 0$  and  $q = -x \leq 0$ ,  $\phi(p, q) = (p - 1)^2 + x^2 + (4 - 2p)x$ , which is  $\geq 0$  if  $p \leq 2$ , while for  $p > 2$  its minimum over  $x \geq 0$  is  $(p - 1)^2 - (p - 2)^2 = 2p - 3 > 0$ . For  $q > 0$ , with  $\sigma := p + q$ ,  $\phi(p, q) = \sigma^2 - 2p - 4q + 5 \geq \sigma^2 - 4\sigma + 5 > 0$ . Hence the left side is  $\geq 0$ .  $\square$

## REFERENCES

1. Levent Alpöge and Ralph Furman, *More than two thirds of the zeta zeros are simple and on the critical line*, 2026, arXiv:2608.13637v2.
2. Stephan Baier, *On the large sieve with sparse sets of moduli*, J. Ramanujan Math. Soc. **21** (2006), no. 3, 279–295, arXiv:math/0512228.
3. Siegfried Alan C. Baluyot, Daniel Alan Goldston, Ade Irma Suriajaya, and Caroline L. Turnage-Butterbaugh, *An unconditional Montgomery theorem for pair correlation of zeros of the Riemann zeta-function*, Acta Arith. **214** (2024), 357–376, arXiv:2306.04799.
4. Michael A. Bennett, Greg Martin, Kevin O’Bryant, and Andrew Rechnitzer, *Counting zeros of Dirichlet  $L$ -functions*, Math. Comp. **90** (2021), no. 329, 1455–1482, arXiv:2005.02989.
5. Florin P. Boca, Cristian Cobeli, and Alexandru Zaharescu, *A conjecture of R. R. Hall on Farey points*, J. Reine Angew. Math. **535** (2001), 207–236.
6. Florin P. Boca and Alexandru Zaharescu, *The correlations of Farey fractions*, J. London Math. Soc. (2) **72** (2005), no. 1, 25–39, arXiv:math/0404114.
7. E. Bombieri and H. Davenport, *On the large sieve method*, Abhandlungen aus Zahlentheorie und Analysis. Zur Erinnerung an Edmund Landau (1877–1938) (P. Turán, ed.), VEB Deutscher Verlag der Wissenschaften, Berlin, 1968, pp. 9–22.
8. Vorrapan Chandee, Yoonbok Lee, Sheng-Chi Liu, and Maksym Radziwiłł, *Simple zeros of primitive Dirichlet  $L$ -functions and the asymptotic large sieve*, Q. J. Math. **65** (2014), no. 1, 63–87, arXiv:1211.6725.
9. Andrés Chirre, Felipe Gonçalves, and David de Laat, *Pair correlation estimates for the zeros of the zeta function via semidefinite programming*, Adv. Math. **361** (2020), 106926, 22 pp., arXiv:1810.08843.
10. Brian Conrey, Henryk Iwaniec, and Kannan Soundararajan, *Asymptotic large sieve*, 2011, arXiv:1105.1176; AIM preprint 2011-7.
11. Brian Conrey, Chung-Hang Kwan, Yongxiao Lin, and Caroline L. Turnage-Butterbaugh, *Critical zeros and unconditional mean value theorems for twisted  $\mathrm{PGL}(2)$  and  $\mathrm{PGL}(3)$   $L$ -functions*, 2026, arXiv:2607.00282.
12. J. Brian Conrey, Henryk Iwaniec, and Kannan Soundararajan, *Critical zeros of Dirichlet  $L$ -functions*, J. Reine Angew. Math. **681** (2013), 175–198, arXiv:1105.1177.
13. Harold Davenport, *Multiplicative number theory*, third ed., Graduate Texts in Mathematics, vol. 74, Springer-Verlag, New York, 2000, Revised and with a preface by Hugh L. Montgomery.
14. George Dickinson, *Zeros of Dirichlet  $L$ -functions near the critical line*, Mathematika **70** (2024), no. 1, Paper No. e12239, arXiv:2211.06264.

15. Giacomo Fabbian, *Simple zeros on the critical line for weight two newforms of prime level, on harmonic average, without the Generalized Riemann Hypothesis*, Zenodo preprint, 2026, Concept DOI 10.5281/zenodo.23025217; published 29 September 2026.
16. P. X. Gallagher, *A large sieve density estimate near  $\sigma = 1$* , Invent. Math. **11** (1970), no. 4, 329–339.
17. Ramūnas Garunkštis and Julija Paliulionytė, *Pair correlation of zeros of Dirichlet  $L$ -functions and proportion of simple zeros*, Glas. Mat. Ser. III **60** (2025), no. 2, 207–227.
18. D. A. Goldston, *On Bombieri and Davenport’s theorem concerning small gaps between primes*, Mathematika **39** (1992), no. 1, 10–17.
19. D. A. Goldston and C. Y. Yıldırım, *Higher correlations of divisor sums related to primes I: triple correlations*, Integers **3** (2003), Paper A5, 66 pp., arXiv:math/0111212.
20. S. Graham, *An asymptotic estimate related to Selberg’s sieve*, J. Number Theory **10** (1978), no. 1, 83–94.
21. Zhixu Hua and Xiufan Yang, *Gevrey localization for simple and distinct zeros in a prime-modulus Dirichlet family*, 2026, arXiv:2609.27808v1; companion to arXiv:2608.16034.
22. ———, *Simple and distinct zeros in a prime-modulus Dirichlet family from near-microscopic to polylogarithmic heights*, 2026, arXiv:2608.16034v2.
23. Henryk Iwaniec and Emmanuel Kowalski, *Analytic number theory*, American Mathematical Society Colloquium Publications, vol. 53, American Mathematical Society, Providence, RI, 2004.
24. Neelam Kandhil, Alessandro Languasco, and Pieter Moree, *Pair correlation of zeros of Dirichlet  $L$ -functions: a possible path towards the conjectures of Chowla, Elliott–Halberstam and Montgomery*, Math. Ann. **394** (2026), no. 2, Paper No. 43, arXiv:2411.19762.
25. Youness Lamzouri, *A new proof that more than  $2/3$  of the zeros of the Riemann zeta function are simple and on the critical line*, 2026, arXiv:2609.02882v2.
26. H. L. Montgomery, *Zeros of  $L$ -functions*, Invent. Math. **8** (1969), no. 4, 346–354.
27. H. L. Montgomery and R. C. Vaughan, *The large sieve*, Mathematika **20** (1973), no. 2, 119–134.
28. ———, *Hilbert’s inequality*, J. London Math. Soc. (2) **8** (1974), 73–82.
29. Hugh L. Montgomery, *The pair correlation of zeros of the zeta function*, Analytic Number Theory (St. Louis Univ., St. Louis, Mo., 1972), Proc. Sympos. Pure Math., vol. 24, Amer. Math. Soc., Providence, RI, 1973, pp. 181–193.
30. ———, *Distribution of the zeros of the Riemann zeta function*, Proceedings of the International Congress of Mathematicians (Vancouver, B.C., 1974), Vol. 1, Canad. Math. Congress, 1975, pp. 379–381.
31. Ali E. Özlük, *On the  $q$ -analogue of the pair correlation conjecture*, J. Number Theory **59** (1996), no. 2, 319–351.
32. Fredrik Prüzeliu, *Two thirds of the zeros of Dirichlet  $L$ -functions at polylogarithmic height are simple and on the critical line, on average*, Zenodo preprint, version 2.4.1-draft, August 2026, Concept DOI 10.5281/zenodo.21980224; version DOI 10.5281/zenodo.22071915 (23 August 2026).
33. Olivier Ramaré, *Arithmetical aspects of the large sieve inequality*, Harish-Chandra Research Institute Lecture Notes, vol. 1, Hindustan Book Agency, New Delhi, 2009, With the collaboration of D. S. Ramana.
34. ———, *An explicit density estimate for Dirichlet  $L$ -series*, Math. Comp. **85** (2016), no. 297, 325–356.
35. ———, *The large sieve through Parseval, large sifted sets are regular*, 2026, arXiv:2609.25879v1.
36. ———, *The weighted large sieve through Parseval*, 2026, arXiv:2609.25885v1.
37. Atle Selberg, *On the zeros of Riemann’s zeta-function*, Skr. Norske Vid.-Akad. Oslo I **1942** (1942), no. 10, 1–59.
38. Keiju Sono, *A note on simple zeros of primitive Dirichlet  $L$ -functions*, Bull. Aust. Math. Soc. **93** (2016), no. 1, 19–30.
39. ———, *Zeros of Dirichlet  $L$ -functions on the critical line*, J. Number Theory **271** (2025), 348–388, arXiv:2105.07422.
40. R. C. Vaughan, *Moments for primes in arithmetic progressions, I*, Duke Math. J. **120** (2003), no. 2, 371–383.
41. ———, *Moments for primes in arithmetic progressions, II*, Duke Math. J. **120** (2003), no. 2, 385–403.
42. Biao Wang, *Proportions of the non-trivial zeros of the Riemann zeta function*, 2026, arXiv:2609.24167v1.
43. ———, *Simple critical zeros and distinct zeros of the Riemann zeta-function in short intervals*, 2026, arXiv:2609.07918.
44. D. Wolke, *On the large sieve with primes*, Acta Math. Acad. Sci. Hungar. **22** (1971), no. 1–2, 239–247.
45. Xiaosheng Wu, *Distinct zeros and simple zeros for the family of Dirichlet  $L$ -functions*, Q. J. Math. **67** (2016), no. 4, 757–779, arXiv:1206.1679.
46. C. Yalçın Yıldırım, *The pair correlation of zeros of Dirichlet  $L$ -functions and primes in arithmetic progressions*, Manuscripta Math. **72** (1991), no. 1, 325–334.